

$$S = - N k_B \log \left[\frac{N}{V} \left(\frac{1}{2 \pi m k_B T} \right)^{3/2} \right] + \frac{3}{2} N k_B$$

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \vec{\nabla}_r f + \vec{a} \cdot \vec{\nabla}_v f = \left(\frac{\partial f}{\partial t} \right)_{coll.}$$

$$\dot{x} = \sigma (y - x)$$

$$\dot{y} = \rho x - y - xz$$

$$\dot{z} = -\beta z + xy$$

$$\frac{\partial f_n}{\partial t} = \{H_n, f_n\} + \sum_{i=1}^n \int d^3r_{n+1} d^3p_{n+1} \frac{\partial U(\vec{r}_i - \vec{r}_{n+1})}{\partial \vec{r}_i} \cdot \frac{\partial f_{n+1}}{\partial \vec{p}_i}$$

$$|BES\rangle = \prod_k (u_k + \sqrt{v_k} c_{k+}^\dagger c_{-k+}^\dagger) |vac\rangle$$

$$\rho \left(\frac{\partial}{\partial t} + \vec{u} \cdot \vec{\nabla} \right) \vec{u} = \rho \frac{\vec{F}}{m} - \vec{\nabla} P + \eta \nabla^2 \vec{u} + \left(\frac{\eta}{3} + S \right) \vec{\nabla} (\vec{\nabla} \cdot \vec{u})$$

$$R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R = \frac{8\pi G}{c} T^{\mu\nu}$$

$$Q_N(V,T) = \frac{1}{N! h^{3N}} \int d^3N p \, d^3N r \, \exp \left(-\beta \sum_i \frac{p_i^2}{2m} - \beta \sum_{i<j} v_{ij} \right)$$

$$\mathcal{L}(\chi) = \frac{1}{2} \partial^\mu \phi(\chi) \partial_\mu \phi(\chi) - \frac{1}{2} m^2 \phi^2(\chi)$$

$$\mathcal{L}_{QED} = \bar{\psi}(i\not{\partial} - m)\psi + i q \bar{\psi} \gamma^\mu A_\mu \psi - \frac{1}{4} F^{\mu\nu} F_{\mu\nu}$$

$$\rho = \frac{e^{-\beta \mathcal{H}}}{Z}$$

$$\left[-\frac{\hbar^2}{2m} \nabla_i^2 + v_{ion}(\vec{r}_i) + e^2 \sum_{j \neq i} \int d^3r_j \frac{|\phi_j(\vec{r}_j)|^2}{|\vec{r}_i - \vec{r}_j|} \right] \phi_i(\vec{r}_i) - e^2 \sum_{j \neq i} \int d^3r_j \frac{\phi_i^*(\vec{r}_j) \phi_j(\vec{r}_j)}{|\vec{r}_i - \vec{r}_j|} \phi_j(\vec{r}_i) = \varepsilon_i \phi_i(\vec{r}_i)$$

$$\frac{d\vec{P}}{dt} + \frac{d\vec{G}}{dt} = \oint_S (\vec{T}^{em} \cdot \vec{n}) dS$$

$$Z = \int \mathcal{D} \phi(\vec{x}) \, e^{-(\mathcal{H} - \int d^d x \, h(\vec{x}) \phi(\vec{x})) / T}$$

$$\mathcal{L} = \frac{\epsilon_0}{2} \vec{E} \cdot \vec{E} + \frac{\epsilon_0}{2\mu_0} \vec{B} \cdot \vec{B} + \vec{J} \cdot \vec{A} - eV$$

$$\varepsilon(\vec{q},\omega) = 1 - \frac{e^2}{\epsilon_0 q^2 V} \sum_{\vec{k},\sigma} \frac{f_{\vec{k}+\vec{q}} - f_{\vec{k}}}{\varepsilon_{\vec{k}+\vec{q}} - \varepsilon_{\vec{k}} + \hbar\omega + i\eta}$$

$$\frac{df(v,t)}{dt} = \frac{d}{dv} \left(v \sigma f(v,t) \right) + \frac{1}{2} \alpha \frac{d^2}{dv^2} f(v,t)$$

$$F_{LG} = \int d^3x \left[\frac{1}{2} |\psi|^2 + g \left(\vec{\nabla} - \frac{2\pi}{\hbar c} \vec{A} \right)^2 + \frac{1}{2} g |\psi|^4 \right] + \frac{1}{8\pi\mu_0} \int d^3x (\vec{\nabla} \times \vec{A})^2 - \frac{1}{4\pi} \int d^3x \vec{H} \cdot (\vec{\nabla} \times \vec{A})$$

PhD Thesis

$$\Psi(\vec{r},t) = \sqrt{m c(t)} \, e^{iS(\vec{r},t)/\hbar}$$

$$\frac{\partial n}{\partial t} = D \nabla^2 n$$

Optical Properties of Two-dimensional Semiconductors: Excitonic and Polaritonic Effects

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} [g_{\sigma\mu,\nu} + g_{\sigma\nu,\mu} - g_{\mu\nu,\sigma}]$$

Pedro Ninhos

$$S = k_B \log \Omega$$

$$\oint_C f(z) \, dz = 0$$

Pedro Ninhos

Optical Properties of Two-dimensional Semiconductors: Excitonic and Polaritonic Effects

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*A thesis submitted in fulfilment of the requirements
for the degree of Doctor of Philosophy*

in the

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Mads Clausen Institute

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Dedicated to family and friends

*Em memória dos meus avós,
Maria Natália Silvestre Lopes Correia*

José Manuel Pessoa Correia^e

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”

*“Estar vivo é o contrário de estar
morto.”*

— **Lili Canêças**, Ao vivo na TV
(Socialite)

ABSTRACT

In this thesis, we use a combination of analytical and numerical methods to evaluate the excitonic and optical effects in isolated or nanostructured monolayered semiconductors or insulators. Systems in two dimensions capable of presenting many-body effects, through strong optical responses and highly dispersive surface-confined light-matter hybrids, caused a shift in the attention of the scientific community to understand their properties and how to tailor them for applications in nanophotonics, polaritonics, optoelectronics, optical sensors, among others. Their capacity in supporting long-lived excitons at room temperatures makes them viable for fabricating devices that could operate outside cryogenic conditions.

This thesis proposes ways of manipulating material properties by engineering the experimental setup, providing means of predicting the optical properties of two-dimensional materials with a bandgap more efficiently. For that, we guide the reader through the theoretical concepts that we need to move forward the ideas herein explored. The starting point is the classical formulation of electromagnetic phenomena in solids, including the application of the formalism to two-dimensional materials. Next, we introduce the formalism of quantum mechanics to describe the physics of solid matter at the microscopic level, to provide insights the classical formalism does not offer, and to prepare the reader for the rest of the dissertation. In particular, we provide the necessary background to describe the concept of the exciton, which is central in this work. Afterward, the theory behind the excitons is formulated, making it possible to study optical effects in two-dimensional materials caused by excitonic effects.

To choose the right material with practical applications in mind is not trivial. Furthermore, it has been seen before in the literature that monolayer hexagonal boron nitride (hBN) presents strong optical resonances in the ultraviolet. We propose a way to engineer the optical properties of this material by setting up a nanostructured environment, where the monolayer is deposited in a substrate in contact with a metallic grating. The idea is to change the electronic and optical properties of hBN and how they interplay, by subjecting it to an electrostatic periodic potential. In this way, we can change the bandgap as we like, enabling changing the excitonic and optical properties as well. By tuning the potential, we can make hBN optically useful in a wider range of frequencies down to the mid-infrared,

increase the oscillation strength of the optical resonances, enhance its optical absorption, and make it promising for application in nanophotonics and polaritonics. All of this, using always the same material.

Finally, we look at the problem of electrostatic screening in single-layer semiconductors and insulators from a strict two-dimensional perspective. We reformulate the problem of calculating numerically the dielectric function the way it is done in the existing literature, by reducing the dimensionality of the approach, from three to two dimensions. The theory applied to monolayer hBN produces results that agree satisfactorily with the ones reported in previous works, and the gain in numerical efficiency is notable. Then, the problem is reformulated once more; we go from a strict two-dimensional description to a quasi-two-dimensional one. By taking the thickness of the monolayer into account, we improve the agreement between our results and the ones we compared with previously. This is allowed by a mixed-representation of the dielectric function, defining it in a combination of the two-dimensional reciprocal space with the one-dimensional space perpendicular to the layer. The code developed is analyzed and validated, this time for a different material, monolayer molybdenum disulfide, MoS₂. Once more, the results from our approach agree very well with more sophisticated ones, validating our theory.

RESUMÉ

Denne afhandling omhandler analytisk og numeriske metoder til beskrivelse af excitoner og optiske egenskaber af isolerede eller nanostrukturerede to-dimensionelle materialer, så som krystallinske halvledere og isolatorer. To-dimensionelle materialer hvor mangeelektron effekter viser sig, ved stærk lys-stof vekselvirkning og polaritoner med dispersive adfærd, tiltrak stor interesse i at forstå deres egenskaber, og hvordan de kan bruges til nye eksperimenter og moderne anvendelser i nanoplasmonik, fotonik, optoelektroniske komponenter, optisk sensorer, eller andre. Deres evne til at understøtte langlivede excitoner ved stuetemperatur gør dem levedygtige til fremstilling af komponenter, som kan fungere uden for kryogene forhold.

Denne afhandling foreslår måder at manipulere de elektroniske og optiske egenskaber af materialerne ved nye eksperimenter metoder til at bedre forstå, kontrollere og modellere den optiske respons af to-dimensionelle halvledere og isolatorer. Derfor, udruste vi læseren med de nødvendige teoretiske koncepter til at studere og løse problemerne vi fokuserer på i denne afhandling. Vi begynder med den klassiske formulering af de elektromagnetiske fænomener i faststoffer, og hvordan vi kan bruge den til at beskrive elektrodynamikken af to-dimensionelle materialer. Bagefter introducerer vi en mikroskopisk beskrivelse af faststoffer ved hjælp af kvantemekanik, hvilket vil give læseren grundviden for forståelsen af resten af afhandlingen. Med koncepterne fra kvantemekanik, kan læseren være introduceret til definitionen af exciton, hovedemnet af dette projekt. Derefter formulerer vi teorien bag excitoner til at studere de optiske effekter i to-dimensionelle halvledere og isolatorer, dem der kommer fra excitonen. Undersøgelsen gennemført i dette PhD projekt præsenteres lige efter.

Det er ikke nemt at vælge de rigtige materialer som passer bedre til praktiske applikationer. Desuden, er det vidste i litteraturen, at monolags hexagonal bornitrid, hBN, har stærk optisk resonanser i det ultraviolette (UV) del af det elektromagnetiske spektrum. Vi foreslår et måde til at ændre den optiske respons af dette materiale med et nanostruktureret lag, over et kvartssubstrat og med en vekslende bias. Formålet er, at manipulere krystallens energibåndstruktur og de optiske egenskaber, og hvordan de afhænger af hinanden med et en-dimensionelt periodisk potentiale. På denne måde kan man ændre materialets energigab,

og ændre også dens excitoniske og optiske effekter. Idet kan man vælge energigabet, kan hBN bruges til applikationer med UV-lys eller med lys af mindre frekvenser, e.g. infrarødt lys. Desuden, med et mindre energigab får hBN stærkere optisk resonanser, forbedret optisk absorption, og bliver det et interessant og alsidigt material til applikationer i nanofotonik.

Endelig, kigger vi på problemet af statisk skærmning in monolags halvledere og isolatorer fra et to-dimensionelle perspetiv. De numeriske metoder til den mikroskopiske dielektriske funktion som man kan find i litteraturen omformuleres. Vi bruger ikke et tre-dimensionale formulering, vi bruger et to-dimensionelle formulering af den dielektriske respons i stedet. Vi bruger vores teori til at forudsige de dielektriske egenskaber af monolags hBN og sammenligner med resultaterne fra litteraturen. Konklusionen er, at vores teori kan bruges til to-dimensionelle materialer med gode resultater, og dem fra litteraturen kan reproduceres, hvilket er bemærkelsesværdigt. Derudover, omformulerer vi vores teori til at inkludere materialets tykkelse i den to-dimensionelle dielektriske funktion. Det er muligt med en mikroskopiske dielektriske funktion som afhænger af begge bølgevektoren og af koordinaten i z-aksen. Vi bruger denne quasi-2D teori til at beregne den dielektriske funktion af hBN igen, og konklusionen er, at vores resultater er endnu bedre en før. Teorien har også fordelelen, at de numeriske implementering er meget effektiv. Til sidste, analyserer vi hvor meget effektiv er vores kode og validerer den. Denne gang bruger vi vores teori til at beregne den statisk dielektriske funktion af et andet material, monolags molybdædisulfid, MoS_2 . Fra resultaterne kan vi sige igen, at vores teori er så god som de andre teorier i litteraturen, der normalt er mere sofistikerede og kræver mere computerkraft end vores.

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LIST OF PUBLICATIONS

Publications published during the thesis work:

- P. Ninhos, A. J. Uría-Álvarez, C. Tserkezis, N. Asger Mortensen and J. J. Palacios, "*Microscopic screening theory for excitons in two-dimensional materials: A bridge between effective model and ab initio descriptions*", in preparations (2025)
- V. G. M. Duarte, P. Ninhos, C. Tserkezis, N. A. Mortensen, and N. M. R. Peres, "*Tunable exciton polaritons in biased bilayer graphene*", Physical Review B **111**(7), 075411 (2025)
- P. Ninhos, C. Tserkezis, N. A. Mortensen, and N. M. R. Peres, "*Tunable exciton polaritons in band-gap engineered hexagonal boron nitride*", ACS Nano **18**(31), 20751 (2024)

Publications published prior to the thesis work:

- M. A. N. Araújo, P. Ninhos, and P. Ribeiro, "*Symmetry-induced even/odd parity in charge and heat pumping*", Physical Review B **103**(17), 174307 (2021)

ACRONYMS

hBN	Hexagonal Boron Nitride (<i>pp.</i> <i>xix</i> , 4, 5, 39, 44–48, 64, 72, 73, 77, 85, 93–97, 101, 103, 104, 106, 111, 114, 118–121, 124–126, 128, 129, 136–139, 141, 143–145, 147–152, 156, 157, 168, 169, 171–175, 180, 181)
vdW	van der Waals (<i>pp.</i> 94, 95, 154, 156)
1D	One Dimensional (<i>pp.</i> 40, 77, 93, 95, 96, 157)
2D	Two-Dimensional (<i>pp.</i> 2–5, 7, 8, 10, 13–16, 18, 21, 22, 24, 25, 28–30, 44, 50, 61, 65, 67, 68, 70, 78, 83–86, 90, 93–96, 104, 122, 124, 126–131, 133–136, 139, 141–145, 147, 150–159, 163, 167–169, 172, 173, 176, 177, 179–182, 184, 185, 194, 197)
3D	Three-Dimensional (<i>pp.</i> 3, 18, 39, 47, 65, 67, 83, 84, 86, 90, 129, 130, 133, 134, 153, 154, 157, 167, 177, 184, 185, 195, 198)
API	Application Programming Interface (<i>p.</i> 158)
BCS	Bardeen–Cooper–Schrieffer (<i>pp.</i> 36, 51)
BSE	Bethe–Salpeter Equation (<i>pp.</i> 5, 68, 69, 73, 75, 77, 79, 80, 82, 86, 93, 118, 120, 130, 132, 135, 136, 148, 149, 157, 173, 177, 180)
BZ	Brillouin Zone (<i>pp.</i> 39, 41, 44–47, 58, 61, 73, 76, 80, 83, 97, 129–132, 137–139, 141, 142, 145, 146, 151, 159, 160, 162–173, 177, 185, 191, 195)
DFT	Density Functional Theory (<i>pp.</i> 32, 37, 46, 49, 50, 52, 57, 69, 78, 81, 101, 118, 132, 136, 137, 145, 148, 158, 170, 171, 173)
EM	Electromagnetic (<i>pp.</i> 2, 4, 7–9, 19–21, 26–28, 93, 123–126, 180)
GGA	Generalized Gradient Approximation (<i>pp.</i> 46, 50, 101, 136)
HF	Hartree–Fock (<i>pp.</i> 37, 57, 61, 78, 81, 136)
IPA	Independent-Particle Approximation (<i>pp.</i> 89, 90, 119, 149, 152, 174–176)

IR	Infrared (<i>pp. 44, 95</i>)
LDA	Local-Density Approximation (<i>pp. 50, 136, 145, 148, 173</i>)
LRA	Local-Response Approximation (<i>pp. 16, 21, 22, 24</i>)
OOP	Object-Oriented Programming (<i>p. 158</i>)
QEH	Quantum Electrostatic Heterostructure (<i>pp. 154, 156, 157, 173</i>)
RPA	Random-Phase Approximation (<i>pp. 54–57, 78, 88, 127–129, 133, 134, 154, 179, 200, 201</i>)
SOC	Spin–orbit Coupling (<i>pp. 48, 71, 89, 163, 169–171, 191</i>)
SPP	Surface Plasmon–polariton (<i>pp. 21, 25, 26</i>)
STM	Scanning Tunneling Microscopy (<i>p. 2</i>)
TB	Tight-Binding (<i>pp. 41–46, 48–50, 77, 78, 81–83</i>)
TDDFT	Time-Dependent Density Functional Theory (<i>p. 69</i>)
TDSE	Time-Dependent Schrödinger Equation (<i>pp. 33, 87</i>)
TE	Transverse Electric (<i>pp. 23, 25, 123</i>)
TISE	Time-Independent Schrödinger Equation (<i>pp. 33–36, 68</i>)
TM	Transverse Magnetic (<i>pp. 23–26, 123–125</i>)
TMD	Transition Metal Dichalcogenide (<i>pp. 67, 169</i>)
TRS	Time-Reversal Symmetry (<i>pp. 131, 191, 192</i>)
UV	Ultra Violet (<i>p. 95</i>)

INTRODUCTION

From the time of its most primal form until today, humankind has always been fascinated by the surrounding world. Since millions of years ago that the human being uses nature itself in their favor, taking advantage of the plethora of different types of material properties for variate use cases. One does not have to be a scientist to realize that by combining materials with certain characteristics, it is possible to induce phenomena that were not present in the individual components. A great example in the history of humanity is the discovery of magnets [1], which dates back to ancient Greece. One isolated magnet, although seemingly uninteresting, can: attract or repel other magnets; turn other materials that are originally nonmagnetic into magnets when they are in proximity, like iron; point north or south if left moving freely. Has the reader ever wondered, what is the actual definition of north and south? These opposite directions, which guided explorers in their journeys across the seas, are defined by the Earth's magnetic field. Quite fascinatingly, the planet we inhabit is a gigantic magnet itself moving around in outer space. At some point, a tool taking advantage of these facts was invented; the compass. To better illustrate the importance of having different materials at our disposal, we can also give the example of a metal, copper. Naturally occurring in nature, it is commonly used nowadays in communications and electronics due to its excellent conducting properties. Just like other metals, such as silver or gold, it has a shiny appearance almost like a mirror. This is due to its light-reflecting properties for wavelengths within the visible range of the electromagnetic spectrum.

The scientific field that studies these effects that nature exhibits, from the subatomic scale to the longest cosmic distances one can think of, is called physics. Having countless disciplines within itself, the field that studies the physics of materials is called condensed matter physics. It studies all kinds of material properties, to name a few, electronic, optical, magnetic, thermal, mechanical, among others. What we find surprising is that within only one subfield of physics, it offers explanations for phenomena related to each one of these properties. Surprisingly, it can describe the interplay between them, as one can see when diving deep into the field of condensed matter. Not only can it explain the behavior of materials we can see to the naked eye in our daily lives, but it can also explain exotic phenomena that we have no intuition for from our day-to-day experiences. In certain extreme

conditions, matter can exhibit peculiar behaviors, where the elementary constituents talk to each other and cooperate to form new phases of matter with collective nature, other than gases, liquids, and solids. For instance, when cooled down to cryogenic temperatures, the isotope helium-4 can realize a state called the Bose–Einstein condensate [2], where atoms populate the lowest-energy state, forming a collective condensate that brings the fluid viscosity and the entropy to zero. In these conditions, we can say that helium-4 behaves like a superfluid [2, 3]. Another exotic example is the superconducting state [4], realized by certain metals and some alloys at very low temperatures, where the electrons in the material cooperate with each other via interactions mediated by excitations of the crystalline lattice, the phonons. The electrons experience an effective attractive interaction (which is almost unthinkable), and pair among themselves to form the so-called Cooper pairs. This results in the charge carriers having an electron charge of twice the elementary one, and the resistivity of the material drops to zero [3, 4]¹.

The reader might have realized, by all the different things that condensed matter physics encompasses that one has to decide what to focus on. In the advent of the growing attention paid to two-dimensional (2D) materials since the isolation of a single graphene layer in 2004 [6], 2D materials have become a hot topic of research with a big traction in scientific research. Physical systems in two dimensions have become a playground for both theorists and experimentalists, since those materials can display a very intricate interplay between material properties that one would expect to be unrelated. Again, we have to further narrow down our focus. In this thesis, we deal with the electronic and optical properties of 2D materials and how they interrelate. The work is narrowed down even more to semiconductors and insulators, which have poor conducting properties, contrarily to metals. More specifically, we will study how the electronic properties of these materials can be tailored for displaying emergent optical effects, and also how their electronic properties affect the material response to external electromagnetic (EM) perturbations. An interesting feature of semiconductors and insulators is that they have an energy gap separating the occupied electronic states from the unoccupied ones. This is why they are bad in conducting electricity; the presence of an energy gap, called the bandgap, in the electronic structure confers inertia to the charge carriers, which are not allowed to move freely in the material without an energy cost. However, this does not mean that they do not respond to external EM perturbations for energies where electronic states are not allowed. This comes as quite surprising, as one would expect to not see anything happening to the system, unless we excite it with energies above the bandgap. Let us explain this better. In the scanning tunneling microscopy (STM) experiment of Ref. [7] with the 2D semiconductor molybdenum diselenide (MoSe_2), the (DC differential) conductance spectrum was measured. Inside the bandgap, where no conducting states are allowed, the conductance measured over a voltage range corresponding to the bandgap was zero. In the very same experiment, the optical characterization of the same material through photoluminescence measurements showed a

¹Just on a passing note, a superconductor is not the same as an ideal metal. In a superconductor, the magnetic field is expelled from its inside; this is called the Meissner effect [5].

sharp peak inside the bandgap. What this effect shows is that for insulating materials there is no analogue to the free-charge carrier picture in metals to explain the experimental results. The independent-particle framework for metals does not explain these effects. Henceforth, we can conclude that electronic correlations and interactions are very relevant in these materials. The optical measurements observed in Ref. [7] are an imprint of many-particle effects with no single-particle counterpart. The sharp peak can be explained taking into account the interaction between the electron excited to the bands above the bandgap, and the absence of negative charge it left in the Fermi sea; the hole. Having opposite electrical charges, they speak to each other through an attractive Coulomb interaction. As a result of their attraction, they form a bound state—the exciton—conferring to the material a nontrivial optical structure inside the bandgap, as evidenced from the pronounced optical resonances they show at sub-gap frequencies. Out of curiosity, we can also mention that the exciton, being a composite particle composed of two fermions, can be perceived as a single bosonic particle. Naturally, physicists wondered if excitons could undergo Bose–Einstein condensation. Theoretically predicted in 1962 [8], the excitonic Bose–Einstein condensate was unambiguously observed experimentally quite recently in 2022 [9].

In the early stages of the quantum theory of solids in the last century, an approximate picture to describe excitonic effects was proposed in 1937 by Gregory H. Wannier [10], and further developed in 1938 by Nevill F. Mott [11–13]. Due to the fast progress in the literature of quantum mechanics and condensed matter physics in the past decades, more accurate descriptions of the exciton have appeared, especially for the 2D case [14, 15]. For a matter of fact, among all the composite particles that are studied in condensed matter, the exciton is one of the first to be confirmed experimentally, not in a 2D material, but in the bulk semiconductor gallium arsenide (GaAs) [16, 17]. The interest in excitonic effects in 2D materials is quite recent. Due to the abrupt technological advances in this century, it did not take long for experimentalists to isolate 2D semiconductors after the isolation of graphene monolayers in 2004 [6]. As far as we know, the very first single-layer 2D semiconductor to be isolated was molybdenum disulfide (MoS_2) in 2010 [18], in a collaboration involving the initial discoverers of graphene in Ref. [6]. In the same year, it was experimentally confirmed to be a direct bandgap semiconductor in Ref. [19]. Shortly after, in the same material strong photoluminescence effects were experimentally observed [20]. The appeal of realizing excitons in 2D materials comes from the possibility of monitoring excitonic effects at room temperature [21, 22]. Actually, the experimental work for MoSe_2 in Ref. [7] cited above was performed at room temperature. Excitons in 2D materials are long-lived and can be sustained at room temperatures. Typically, they present binding energies—the energy required to dissociate the electron–hole pair—in a range from a fraction of one electron-Volt to a couple of electron-Volts, generically more than binding energies of three-dimensional (3D) excitons. That is because the screening in 2D materials is less effective than in the 3D case. But this also makes it harder to accurately capture the excitonic effects in the 2D case, in which the excitonic energy levels are far from following the hydrogenic Rydberg series [23–25]. In three dimensions,

the excitonic states and energies in general agree very well with the hydrogenic Rydberg series, because it is sufficient to take a constant for the relative permittivity of the medium for the screened Coulomb potential. In two dimensions, and as we will see later on in this work, the screening is highly nonlocal [24, 26], and properly accounting for this nonlocality is imperative. Nonetheless, there are plenty of techniques by now to address the excitonic problem in 2D materials [15]. In the first instance, one can use an approximate linear model for the dielectric function (not a constant) in momentum space; the Rytova–Keldysh model [15, 26], possible to be extended through other auxiliary models for the permittivity [27]. Although the model is quite simple, it has proved sufficient in some cases [15, 28, 29], and as we will see later on, it is also quite useful to understand excitons in fairly simple materials, such as monolayer hBN [28, 30]. However, in more complicated systems, such as monolayer MoS₂, one cannot work with simple analytical expressions to model the material, as oftentimes analytical expressions for the material do not exist (unless several approximations are introduced). To better capture the fine details of a material and its dielectric properties, one has to resort to numerical techniques. Usually done using numerical packages implementing first-principles methods [31–36], which are very exquisite, but simultaneously very computationally heavy. Therefore, implementing more efficient numerical methods is a need. We will see later on this work, how one can avoid incurring in high computational efforts by addressing the excitonic problem and the screening within a 2D computational framework. Another interesting aspect of excitons in two dimensions is that they can couple to surface EM modes and form the light–matter hybrid called exciton–polariton [21, 22]. This very dispersive radiation mode presents high field confinement, deeming it applicable to optical nanosensing.

In this context, this thesis introduces the reader to the concepts necessary to understand excitons in two dimensions, before we present the investigation performed during the PhD studies. This dissertation splits into two parts, across the next five chapters. In Chapters 2, 3, and 4, we go through the theoretical aspects and concepts that are necessary to grasp the research herein reported. Essentially, they consist of a thorough literature review, along which we tease the reader for the research outcomes in Chapters 5 and 6. In these two chapters, we adapt the theory and the results obtained from our studies to this dissertation. A brief outline of each chapter follows. In Chapter 2, we introduce the classical electrodynamics formalism. Therein, key concepts pertaining to electromagnetic phenomena are introduced from a classical point of view. A classical picture is more than enough to introduce concepts that will be touched upon again in the thesis. Aspects of electromagnetic phenomena in two dimensions are discussed as well, mainly the concept of polariton. In Chapter 3, we introduce the basics of the quantum mechanical formalism for crystalline solids. All requirements for the physics that follows in subsequent chapters are covered. For the research performed, some of them are necessary only for Chapter 6. Specifically, the quantum picture for the dielectric function is highlighted. In Chapter 4, we extend the concepts from the previous one, which are essentially related to single-particle properties of solids, to many-body properties of solids. The emphasis is on semiconductors

and insulators, the systems that present strong excitonic effects. We show how to determine the excitonic states and energies in different ways, in increasing degrees of complexity and difficulty; from the Wannier equation formalism, to the Bethe–Salpeter equation (BSE) formalism. The concepts introduced this far are applied in Chapter 5. We study in this chapter a nanostructured hBN monolayer, where its electronic and optical properties are tuned by applying an external electrostatic potential along one direction. We address the renormalized band dispersion, the excitonic states, the excitonic optical conductivity, the optical absorption, and the emergent exciton–polaritons, by this order. Finally, in Chapter 6, we present a novel formulation and numerical implementation of the dielectric function in a strict 2D framework. The implementation is done on top of the **Xatu** code, whose first version can be consulted in Ref. [37]. The results obtained are benchmarked and compared with those in the literature. For instance, we are able to recover the Rytova–Keldysh limit of the dielectric function from the numerical microscopic one. This last one provides a way of computing the effective strict 2D dielectric function beyond the low momentum limit. The formalism is extended to a quasi-2D one, which is no longer strictly 2D, but accounts for the monolayer thickness. This formalism is applied to the monolayer case, and the new results are compared with those in the literature once again. Besides the theoretical aspects of the problem and showing its results, details about the computational implementation in the code we developed are provided and analyzed. As a case study to test our code, we compute the screening, the excitons, and the optical conductivity for monolayer MoS_2 . The dissertation finishes with Chapter 7, in which we go through the main developments and conclusions of this work and discuss ideas for future work.

CLASSICAL ELECTRODYNAMICS OF CONTINUOUS MEDIA

In this chapter we give an overview of concepts from classical electrodynamics that will serve as a basis for the physical phenomena we will study later on, mainly those pertaining to classical effects observable at macroscopic scales in solids. We give particular emphasis to two-dimensional (2D) materials at the interface between two different media, as 2D materials have been shown to present enticing physics and peculiar electromagnetic (EM) properties. Even though with a solely classical description of matter we cannot grasp the fine details of 2D matter, classical electrodynamics provides insightful tools to describe the propagation of light in matter and how these two interact. We hope that this introduction to basic concepts in electromagnetism will suffice for the reader expecting a thesis as self-contained as possible. Whenever further details or derivations could break the flow of the text, we refer to standard literature on this subject [38–40].

We start the chapter with the celebrated Maxwell’s equations in continuous media, a set of equations governing, among others, the behavior of the EM field in space and time. In the same section we present the boundary conditions of the EM field at the interface between two contiguous media. Next, we describe how the EM field in matter relates to the external applied fields through the so-called constitutive relations. The central quantity will be the dielectric function, a quantity characteristic to each material. We consider different limits, the static limit and the local limit, in separate case studies, namely the 2D dielectric and the Drude metal, respectively. For the curious reader, other examples in different contexts are briefly mentioned. The first two sections serve as the basis for what is to come in the third and final section of the chapter: the polariton. To introduce the reader to the concept of polariton, we need first a framework for EM wave propagation in matter. The concept of polariton is finally introduced, along with the derivation of the dispersion relation for these modes at a surface between two different materials, taking into account the surface optical response. Again, we take the opportunity to mention plasmons, but now coupled to light to form the plasmon–polariton. We finish the chapter with a discussion on

the Fresnel coefficients with the same setup, in the situation where the incident radiation is normal to the planar surface and linearly polarized. We generalize to the case where the interstitial surface, consisting of a 2D material, has its own response. These coefficients will allow us to estimate the capacity of the surface in absorbing, reflecting and transmitting radiation.

2.1 Fundamentals of Classical Electrodynamics

2.1.1 Maxwell's equations

The four central equations of classical electrodynamics, which govern the dynamics of the EM field and all EM phenomena in continuous media, are called the macroscopic Maxwell equations, and in their local form they read [38, 39]

$$\nabla \cdot \mathbf{D}(\mathbf{r}, t) = \rho_f(\mathbf{r}, t), \quad \text{Gauss' law} \quad (2.1a)$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t}, \quad \text{Faraday's law} \quad (2.1b)$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \quad \text{Inexistence of magnetic monopoles} \quad (2.1c)$$

$$\nabla \times \mathbf{H}(\mathbf{r}, t) = \mathbf{J}_f(\mathbf{r}, t) + \frac{\partial \mathbf{D}(\mathbf{r}, t)}{\partial t}. \quad \text{Ampère–Maxwell's law} \quad (2.1d)$$

In the previous system of equations, the vector fields $\mathbf{E}$ and $\mathbf{B}$ are called the electric, and magnetic induction fields, respectively. We will simply denote the latter by magnetic field. These two fields are the fundamental physical quantities in electromagnetism with a physical meaning, unlike the vector fields $\mathbf{D}$ and $\mathbf{H}$, called electric displacement and magnetic excitation fields, respectively, that are auxiliary identities with no physical interpretation. The fields $\mathbf{D}$ and $\mathbf{H}$ are introduced to make implicit in Maxwell's equations the properties of the matter the EM field penetrates in. Nonetheless, they are strictly related to the electric and magnetic fields, as we will see. The last two quantities to define are the free charge (volume) density ρ_f and the free current (surface) density $\mathbf{J}_f$, that together with the bound charge density and the bound current density form the total charge and current densities inside the material, respectively. The free charge and current densities are not independent, as they are related through the continuity equation

$$\frac{\partial \rho_f}{\partial t} + \nabla \cdot \mathbf{J}_f = 0, \quad (2.2)$$

encoding the conservation of free electric charge, which can be seen by taking the divergence of Ampère–Maxwell's law. The free charge and current densities can in turn be split into an external part, which is the one externally applied on the material, and an induced part resulting from the material response to the external perturbation. The continuity equation is also verified by the bound charge and bound current densities, and consequently,

also by its counterpart total quantities. As said above, the fundamental fields $\mathbf{E}$ and $\mathbf{B}$ are connected to the fields $\mathbf{D}$ and $\mathbf{H}$, respectively, through a set of two equations called constitutive relations given by

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad (2.3)$$

and¹

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}), \quad (2.4)$$

where $\mathbf{P}$ is the bound volume polarization density and $\mathbf{M}$ the bound volume magnetization density, and the parameters ε_0 and μ_0 are the vacuum electric permittivity and the vacuum magnetic permeability, respectively. Within the scope of this work, we study only materials with no magnetic response at optical frequencies nor with permanent magnetization, so that $\mathbf{M}(\mathbf{r}, t) = \mathbf{0}$ and $\mathbf{B}(\mathbf{r}, t) = \mu_0 \mathbf{H}(\mathbf{r}, t)$ will always hold.

With all these equations at hand, one can, in principle, solve any problem in electrodynamics in a continuous medium. One would just need to know how in fact the material properties determine the polarization density, and also the magnetization density in case the material is magnetic. However, before we dwell on material properties, we will see in the next section how the global form of Maxwell's equations can give insight into solving problems in electrodynamics where the system under study involves more than one material and their interfaces. From a technical point of view, Maxwell's equations in their local form are not useful in these cases where one has material discontinuities, as taking spatial derivatives is not possible (or introduces infinities). In that situation, one then needs to set proper boundary conditions to find the EM field at all points in space and at all time instants.

2.1.2 Boundary conditions at an interface

In this section we consider an infinite planar interface between two different media, say medium 1 and medium 2. This separation in space may introduce discontinuities in the vector fields $\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$ and $\mathbf{H}$, which can be determined by applying pill-box arguments to Maxwell's equations in their integral form [38, 40]. By doing so, we can relate the fields $\mathbf{E}_i$, $\mathbf{D}_i$, $\mathbf{B}_i$ and $\mathbf{H}_i$ on each side $i = 1, 2$ of the interface by the following boundary conditions:

$$\mathbf{n}_{12} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0} \quad \text{and} \quad (2.5a)$$

$$\mathbf{n}_{12} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{j}_s, \quad (2.5b)$$

for the tangential components of the electric and magnetic excitation fields, respectively, where $\mathbf{j}_s$ is a sheet current density;

¹This asymmetry in the constitutive relations is in fact due to a historical mistake, as it was believed that it was actually the vector field $\mathbf{H}$ that had physical meaning instead of the $\mathbf{B}$ one.

$$\mathbf{n}_{12} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_s \quad \text{and} \quad (2.6a)$$

$$\mathbf{n}_{12} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0, \quad (2.6b)$$

for the normal components of the electric displacement and magnetic fields, respectively, where σ_s is a surface charge density, and $\mathbf{n}_{12}$ is the versor normal to the interface pointing from medium 1 to medium 2. The quantities σ_s and $\mathbf{j}_s$ account for any charge flow that might exist at the interface. If it consists simply of a surface separating two different media, then in principle $\sigma_s = 0$ and $\mathbf{j}_s = \mathbf{0}$. However, if on the interface we have a material of arbitrary small thickness, or if the interface is itself the existing material separating two semi-infinite regions, then in general we have to account for the physics of its optical response and cannot ignore σ_s nor $\mathbf{j}_s$. If surface charge and current densities are not null, then naturally they satisfy the continuity equation $\nabla \cdot \mathbf{j}_s + \partial_t \sigma_s = 0$. A striking example is that of graphene, either by itself or at a boundary between a dielectric and a metal [41]. In this work, we focus on dielectric 2D materials, also termed insulators, and on their optical properties. These can be quantified by the dielectric function, or as we shall explain in the next section, by the conductivity in an equivalent way.

2.2 Material Properties: Classical Picture

The material properties that up until now we have been hiding in the fields $\mathbf{D}$ and $\mathbf{P}$, are encoded in the constitutive relation above between the total electric field $\mathbf{E}$ and the response of the material to it quantified by $\mathbf{P}$. This polarization density results from the charges inside the material rearranging themselves to generate new local bound dipoles. This relation is in the most general case functional and non-linear. Notwithstanding this, herein we restrict ourselves to the linear response regime, where the polarization density $\mathbf{P}$ depends linearly on the electric field $\mathbf{E}$. We will see how in this section.

2.2.1 Dielectric function and conductivity

In the linear relation between the total electric field and the polarization density, the proportionality constant is given in terms of the electric susceptibility χ_e through

$$\mathbf{P} = \varepsilon_0 \overset{\leftrightarrow}{\chi}_e \mathbf{E}, \quad (2.7)$$

assuming that the material can be anisotropic, such that the susceptibility is in fact a second-rank tensor (and presuming also that there is no permanent polarization $\mathbf{P}_0$ contributing to $\mathbf{P}$). As we admit that there is no magnetic response, we can overlook the other constitutive relation for the magnetic field². If the material is isotropic, then $\overset{\leftrightarrow}{\chi}_e = \chi_e \mathbb{1}$, with $\mathbb{1}$ being the identity matrix. Then, for the electric displacement field within linear response approximation and for an anisotropic material we can write

²Otherwise, we would have to account for the magnetic susceptibility, χ_m .

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 \left(\mathbf{1} + \overset{\leftrightarrow}{\chi}_e \right) \mathbf{E} \equiv \varepsilon_0 \overset{\leftrightarrow}{\varepsilon} \mathbf{E}, \quad (2.8)$$

where we defined the relative permittivity tensor $\overset{\leftrightarrow}{\varepsilon} = \mathbf{1} + \overset{\leftrightarrow}{\chi}_e$. The proportionality constant $\overset{\leftrightarrow}{\varepsilon}$ is a characteristic quantity of the material, and it is in general a tensorial response function. This means that the field at the point $\mathbf{r}$ and at the time t depends on all the other points $\mathbf{r}'$ in space and all the other previous instants of time $t' < t$. One can readily see that the previous constitutive relation hides an intricate dependence of the displacement field on the electric field. The operation between the permittivity (we omit from now onwards the word relative) tensor and the electric field is the convolution operation in both space and time coordinates

$$\mathbf{D}(\mathbf{r}, t) = \int d\mathbf{r}' \int_{-\infty}^t dt' \overset{\leftrightarrow}{\varepsilon}(\mathbf{r}, \mathbf{r}'; t, t') \mathbf{E}(\mathbf{r}', t'), \quad (2.9)$$

where the permittivity is now a (tensorial) response function, i.e., a two-point function in real space and a two-point function in time. In this expression, we have assumed that the material extends to infinity in all directions, omitting the integration region in the space integral. From now on, whenever the real space integration region is not explicit in the notation, that will mean that the integral is over the entire space/infinite material, either in two or three dimensions. If the integration region is bounded, then we will accordingly explicit the integration region in the integral symbol. Without knowing any details about the permittivity, we can promptly impose causality which implies that $\varepsilon(\mathbf{r}, \mathbf{r}'; t, t') = 0$ if $t' > t$ (let us omit the tensor nature of the permittivity for now). The integral in Eq. (2.9) above could then be taken up to $+\infty$ (this is useful to take the Fourier transform in time, as we shall do just now). Furthermore, if the system enjoys *continuous* time translational invariance³, then the permittivity depends only on the time difference $t - t'$. This property is extremely useful, as per the convolution theorem, we can perform a Fourier transform in a single variable of time $t - t'$, so that the permittivity and the electric field operate in the frequency domain through a mere multiplication at a single frequency ω :

$$\mathbf{D}(\mathbf{r}, \omega) = \varepsilon_0 \int d\mathbf{r}' \varepsilon(\mathbf{r}, \mathbf{r}'; \omega) \mathbf{E}(\mathbf{r}', \omega). \quad (2.10)$$

If we now further assume that the material is homogeneous, which is a reasonable approximation at the macroscopic scale, then the system enjoys *continuous* spatial translational invariance, and the permittivity becomes a function solely of the difference $\mathbf{r} - \mathbf{r}'$. One can show that if this is true, then we can Fourier transform the permittivity from real to reciprocal space keeping only one wavevector (or momentum, in the language of solid state

³This assumption is only valid if we treat the external probe or driving field as a perturbation to the system, so small that the response is that of a material in (dynamic) equilibrium, and its intrinsic properties are time-independent. If the probe or field is strong enough, then it can drive the system to a non-equilibrium state, and in that case we have to account for the full dynamics of the composed system material+perturbation [42]. Even if the external field is time periodic and the system still enjoys time translation invariance, albeit discrete, we must separate the two time dependences t and t' . As stated above we work solely within a perturbative regime. Henceforth, this is not a big concern for us.

physics) variable $\mathbf{k}$, and through direct application of the convolution theorem once more, we can write

$$\mathbf{D}(\mathbf{k}, \omega) = \varepsilon_0 \varepsilon(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega), \quad (2.11)$$

which simplifies immensely the relation between the displacement field and the electric field.

The dielectric function, a central quantity in this text, in reciprocal space $\varepsilon(\mathbf{k}, \omega)$ is in general complex valued and a very complicated function of $\mathbf{k}$ and ω , and determining the full dependence on both momentum and frequency is a herculean task. With the tools provided by classical electrodynamics, one can only hope to work with ad-hoc models for the dielectric function, usually using material parameters obtained experimentally. We provide a few case examples of models commonly found in the literature in the next sections.

Before we proceed, it is worthy to remind the reader that in electromagnetism we can consider charge currents as a result from an application of an external electric field. The relation between the electric field and the induced current is quantified by the famous (local and generalized) Ohm's law, that when assuming time and space translational invariance⁴, and isotropy as well, reads

$$\mathbf{J}_{\text{ind}}(\mathbf{r}, t) = \int d\mathbf{r}' \int_{-\infty}^t dt' \sigma(\mathbf{r} - \mathbf{r}'; t - t') \mathbf{E}(\mathbf{r}', t'), \quad (2.12)$$

where σ is the conductivity of the material (that can assume a tensorial form, just like the permittivity), and accounts for the excitation of free charges due to the applied electric field. It is a non-null conductivity that leads to losses by Joule effect (apart from the particular case of superconductivity, where a system conducts electricity without any losses). We introduce this quantity here, as it can be useful in electrodynamics to account simultaneously for a material's dielectric response and its conductivity in a single effective dielectric function. Let us then rewrite Maxwell's equations (2.1) in reciprocal space and frequency domain by Fourier transforming them (or equivalently, assuming a dependence of the form $\mathbf{F}(\mathbf{r}, t) = \mathbf{F}_0 e^{-i(\omega t - \mathbf{k} \cdot \mathbf{r})}$ for all the fields)

$$i\mathbf{k} \cdot \mathbf{D}(\mathbf{k}, \omega) = \rho_f(\mathbf{k}, \omega), \quad (2.13a)$$

$$i\mathbf{k} \times \mathbf{E}(\mathbf{k}, \omega) = -i\omega \mathbf{B}(\mathbf{k}, \omega), \quad (2.13b)$$

$$i\mathbf{k} \cdot \mathbf{B}(\mathbf{k}, \omega) = 0, \quad (2.13c)$$

$$i\mathbf{k} \times \mathbf{H}(\mathbf{k}, \omega) = \mathbf{J}_f(\mathbf{k}, \omega) - i\omega \varepsilon_0 \varepsilon(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega), \quad (2.13d)$$

and focus on Ampère–Maxwell's law alongside Ohm's law written in its Fourier transformed version as $\mathbf{J}_{\text{ind}}(\mathbf{k}, \omega) = \sigma(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega)$, we can rewrite the last equation as

⁴Until the time when it will become relevant, we will not explicitly say that the invariance in space is continuous. In the context of solid state physics, translational invariance is discrete instead.

$$\begin{aligned}
 \mathbf{k} \times \mathbf{H}(\mathbf{k}, \omega) &= \mathbf{J}_{\text{ext}}(\mathbf{k}, \omega) + \sigma(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega) - i\omega\varepsilon_0\varepsilon(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega) \\
 &= \mathbf{J}_{\text{ext}}(\mathbf{k}, \omega) - i\omega\varepsilon_0 \left[\varepsilon(\mathbf{k}, \omega) + i \frac{\sigma(\mathbf{k}, \omega)}{\omega\varepsilon_0} \right] \mathbf{E}(\mathbf{k}, \omega) \equiv \\
 &\equiv \mathbf{J}_{\text{ext}}(\mathbf{k}, \omega) - i\omega\varepsilon_0\varepsilon_{\text{eq}}(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega),
 \end{aligned}$$

where through the last equality we defined the equivalent permittivity⁵

$$\varepsilon_{\text{eq}}(\mathbf{k}, \omega) = \varepsilon(\mathbf{k}, \omega) + i \frac{\sigma(\mathbf{k}, \omega)}{\omega\varepsilon_0}. \quad (2.14)$$

This suggests redefining the total electric displacement field as $\mathbf{D} \rightarrow \mathbf{D}_{\text{tot}} = \mathbf{D} + i\omega^{-1} \mathbf{J}_{\text{ind}}$. In this way, equation (2.13a) can be rewritten in terms of only the external charge density, $\mathbf{k} \cdot \mathbf{D}_{\text{tot}}(\mathbf{k}, \omega) = \rho_{\text{ext}}(\mathbf{k}, \omega)$.

With only the external charge and current densities appearing explicitly in Maxwell's equations and a total displacement field incorporating the induced current, we can define a total electric field as $\mathbf{E}_{\text{tot}} = \mathbf{D}_{\text{tot}}/\varepsilon_0$ and interpret the dielectric function as the ratio between the external field and the total one: $\varepsilon(\mathbf{k}, \omega) = |\mathbf{E}_{\text{ext}}(\mathbf{k}, \omega)|/|\mathbf{E}_{\text{tot}}(\mathbf{k}, \omega)|$. This interpretation is the cornerstone of the theory around the dielectric function. It is actually introduced like this in introductory electromagnetism courses, appearing usually as a single constant relative permittivity denoted by ε_r in the denominator in Coulomb's law for the electrostatic force. In bulk materials, it is typically sufficient to speak of the relative permittivity as a constant of the material. Unfortunately (or the opposite), it is in general a very complicated function of space and time, although working in reciprocal space seems to ease things. This is a striking feature of 2D materials, where the dependence on $\mathbf{k}$ and ω cannot be ignored, even encoding rich physics such as spatial dispersion [43] (nontrivial $\mathbf{k}$ -dependence), temporal dispersion [44] (nontrivial ω -dependence), screening in crystalline solids [45] (the main subject of Chapter 6) and their nontrivial absorption spectra [46–49], plasmons in metals and 2D materials such as graphene [6, 41], the effect called Landau damping in the context of kinetic theory in plasma physics [50], among others.

To familiarize a reader that has not extensive experience on the dielectric function, we provide in the next subsections two key examples of models for the dielectric function, one in the static limit $\omega \rightarrow 0$, and the other one in the local limit $\mathbf{k} \rightarrow \mathbf{0}$. Isolating these two limits and considering the dependence on $\mathbf{k}$ and ω separately is enough to realize how intricate this object can be.

2.2.1.1 Static limit in a 2D material

In this section we take a 2D material as an example, as this thesis is concerned only with such materials, and we keep our focus for now on the static and long-wavelength limits of

⁵The function $\varepsilon(\mathbf{k}, \omega)$ on the right-hand side has quite often in the literature a subscript writing $\varepsilon_{\infty}(\mathbf{k}, \omega)$, standing for the contribution coming from the ionic background together with bound polarization charges. In this work we consider typically $\varepsilon_{\infty}(\mathbf{k}, \omega) = 1$.

the dielectric function. The main goal is to derive a minimal model that somehow captures the effect of electrostatic screening felt by test point-like charges inside the material. We want then to find a function $\varepsilon(\mathbf{k})$ for which the total electrostatic potential between two test charges is $\phi(\mathbf{k}) = v_c(\mathbf{k})/\varepsilon(\mathbf{k})$, where $v_c(\mathbf{k})$ is the Coulomb potential in reciprocal space that the two charges would feel remaining static in vacuum. We say that ϕ is the screened Coulomb potential and v_c is the bare Coulomb potential. We will present a derivation adapted from a work by P. Cudazzo et al. [26].

In real space, the Coulomb potential is given by

$$v_c(\mathbf{r}) = \frac{e}{4\pi\varepsilon_0|\mathbf{r}|}, \quad (2.15)$$

with e being the elementary charge (electron charge in absolute value)⁶. To find the screened potential felt by a test charge e in the material, we can make use of Maxwell's equations that we already introduced in Section 2.1.1, in particular Gauss' law (2.1a) in the form

$$\nabla \cdot \mathbf{D}(\mathbf{r}) = \rho_{\text{ext}}(\mathbf{r}), \quad (2.16)$$

where we dropped the subscript “tot” for the displacement field, and the external charge density is that of a single point-like elementary charge, that without loss of generality can be placed at the origin. Since the charge has no dimensions, we can simply write $\rho_{\text{ext}}(\mathbf{r}) = e\delta(\mathbf{r})$, and for the displacement field we write $\mathbf{D}(\mathbf{r}) = \varepsilon_0\mathbf{E}(\mathbf{r}) + \mathbf{P}^{2D}(\mathbf{r})$, where $\mathbf{P}^{2D}$ is a 2D polarization vector. Assuming that the electric susceptibility is local (i.e., $\chi_e(\mathbf{r}, \mathbf{r}') \propto \delta(\mathbf{r} - \mathbf{r}')$) and the 2D material is isotropic, the polarizability reads $\mathbf{P}^{2D}(\mathbf{r}) = \varepsilon_0\chi_e^{2D}\mathbf{E}_{||}(\mathbf{r})|_{z=0}$, that is, the polarization is proportional to the in-plane component of the total electric field computed along the plane of the material with Cartesian equation $z = 0$, and χ_e^{2D} is a 2D susceptibility. Substituting the electric field by the negative gradient of the potential in Gauss' law and the charge density by its Dirac delta like expression, we obtain the following Poisson equation

$$\nabla \cdot [\nabla\phi(\mathbf{r}) + \chi_e^{2D}\nabla_{||}\phi(\mathbf{r}_{||}, z=0)] = -\frac{e}{\varepsilon_0}\delta(\mathbf{r}), \quad (2.17)$$

where the in-plane electric field contributing to the polarizability is given in terms of an in-plane gradient operation, $\nabla_{||}$, acting on the electrostatic potential computed at $z = 0$ and at any position vector $\mathbf{r}_{||}$ on the plane. Within the approximation of zero thickness, we can work with an effective 2D electrostatic potential in momentum space $\phi^{2D}(\mathbf{q})$, defined by the total electrostatic potential computed at the plane of the material, that is, $\phi^{2D}(\mathbf{q}) \equiv \phi(\mathbf{q}, z=0)$. To find an expression for such a potential, one Fourier transforms

⁶In this text we avoid using Gaussian units, adopting the SI system instead. The Coulomb potential is to be understood in electron-Volt (eV), the reason why we have e in the numerator instead of e^2 . We call it potential even though it has units of energy. Throughout this text potential and energy are interchangeable terms when speaking of the Coulomb interaction.

the previous equation, keeping the variables for in-plane and off-plane momentum, q and κ respectively, separate as:

$$(q^2 + \kappa^2) \phi(\mathbf{q}, \kappa) + \chi_e^{2D} q^2 \phi^{2D}(\mathbf{q}) = \frac{e}{\varepsilon_0}. \quad (2.18)$$

Upon solving the equation above for $\phi(\mathbf{q}, \kappa)$, doing the inverse Fourier transform in the κ variable back to the variable z and setting $z = 0$, we find an equation for the quantity $\phi^{2D}(\mathbf{q})$, that after being isolated reads

$$\phi^{2D}(\mathbf{q}) = \frac{e}{2\varepsilon_0 q (1 + r_0 q)}, \quad \text{defining} \quad r_0 \equiv \varepsilon_0 \chi_e^{2D}. \quad (2.19)$$

We call the parameter r_0 the screening length parameter, or simply screening parameter, and its value depends only on the material. This formula contains the expression for the 2D Fourier transform of the bare Coulomb potential given by⁷

$$v_c(q) = \frac{e}{2\varepsilon_0 q}, \quad (2.20)$$

but corrected with a factor of $1 + r_0 q$. We can then define an effective momentum-resolved dielectric function, $\varepsilon_{\text{RK}}(q) \equiv v_c(q)/\phi^{2D}(\mathbf{q}) = 1 + r_0 q$, called Rytova–Keldysh dielectric function [51], which accounts for screening in 2D materials. Having previously assumed that the material is isotropic, then it depends only on the magnitude of the in-plane momentum $q = |\mathbf{q}|$. It actually coincides with the static macroscopic dielectric function in 2D materials in the high-wavelength (small q) limit. It carries the usual interpretation of being the ratio between the external potential $\phi^{\text{ext}}(\mathbf{r}) = \phi_0^{\text{ext}}(\mathbf{q})e^{i\mathbf{q}\cdot\mathbf{r}}$ and the total (or screened) potential $\phi^{\text{tot}}(\mathbf{r}) = \phi_0^{\text{tot}}(\mathbf{q})e^{i\mathbf{q}\cdot\mathbf{r}}$, which has a contribution from the external potential and another one from the potential induced by the external perturbation. We actually introduced this interpretation in Section 2.2.1 in terms of fields instead of potentials, but the idea is the same. The resulting effective Coulomb potential is called Rytova–Keldysh potential [51].

To find the real space resolved Rytova–Keldysh potential, we do the 2D inverse Fourier transform of (2.19) to get

$$V_{\text{RK}}(r) = \frac{e}{4\pi\varepsilon_0} \frac{\pi}{2r_0} \left[\mathbf{H}_0\left(\frac{r}{r_0}\right) - Y_0\left(\frac{r}{r_0}\right) \right], \quad (2.21)$$

where r is a purely in-plane distance, $\mathbf{H}_0$ is the Struve special function of order zero [52], and Y_0 is the Bessel special function of the second kind of order zero. To get a feeling for how this potential behaves in real space compared to the bare Coulomb potential, we can make a plot of both potentials as a function of r .

In Figure 2.1, we compare the two potentials as a function of the distance r , the bare Coulomb potential $v_c(r)$ and the Rytova–Keldysh potential $V_{\text{RK}}(r)$ as per (2.21). For large enough distances, the Coulomb potential and the Rytova–Keldysh potential become closer

⁷We remark that this expression does not have units of energy, which can be traced back to the used definition of Fourier transform in this chapter. That can be fixed by simply dividing by the area of the total system. It poses not a problem, as long as the final calculated quantity of interest has the right dimensions.

and closer, meaning that for very large distances the electrostatic potential is not screened at all, and the attractive or repulsive behavior of two test charges is just the same as in vacuum. However, for very small distances we see that the potentials have different degrees of divergence, with the divergence of the Rytova–Keldysh potential being logarithmic (to see it just take the small r limit in expression (2.21)).

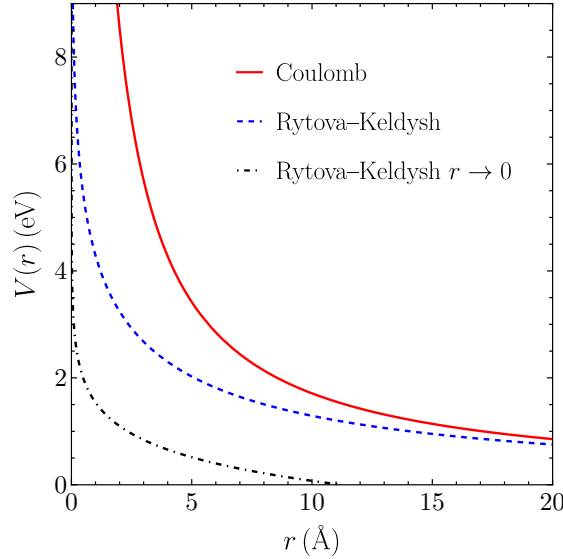


Figure 2.1: Bare Coulomb and Rytova–Keldysh potentials in real space. Comparison between the bare Coulomb potential (red continuous line) and the Rytova–Keldysh potential (dashed blue line) as a function of the in-plane distance r . The dotdashed black curve represents the small r limit of the Rytova–Keldysh potential in Eq. (2.21). The value $r_0 = 10 \text{ Å}$ was used.

The Rytova–Keldysh potential is a model that captures well enough the electrostatic interaction in 2D materials, in particular insulating and semiconducting ones. Whenever it comes to the interaction between charges in these materials, this is the model interaction one should use to make accurate predictions. How the screening parameter r_0 can be estimated will only be left for later, when the necessary tools for doing so are provided. For now, this picture is just a classical model with an estimated parameter. Nonetheless, it can already capture the nontrivial dependence of the dielectric function on the wavevector. We consider next the opposite limit, where the dielectric function depends only on the frequency, but not on the wavevector.

2.2.1.2 Local limit

In this section we provide another model for the dielectric function where we only account for the dependence on the frequency and take the long-wavelength limit $\mathbf{k} \rightarrow \mathbf{0}$. This approximation is called the local-response approximation (LRA), which consists in admitting that the response function of the material, the permittivity or conductivity, is short-ranged, so that the induced field at a certain point in space depends only on the external field at that same location. Mathematically, we can write correspondingly for the permittivity and

conductivity $\varepsilon(\mathbf{r} - \mathbf{r}', \omega) = \delta(\mathbf{r} - \mathbf{r}')\varepsilon^{\text{LRA}}(\omega)$ and $\sigma(\mathbf{r} - \mathbf{r}', \omega) = \delta(\mathbf{r} - \mathbf{r}')\sigma^{\text{LRA}}(\omega)$. In this approximation, the displacement field is simply given by

$$\mathbf{D}(\mathbf{r}, \omega) = \varepsilon_0 \varepsilon^{\text{LRA}}(\omega) \mathbf{E}(\mathbf{r}, \omega), \quad (2.22)$$

that together with Ohm's law

$$\mathbf{J}_{\text{ind}}(\mathbf{r}, \omega) = \sigma^{\text{LRA}}(\omega) \mathbf{E}(\mathbf{r}, \omega) \quad (2.23)$$

form the necessary constitutive relations.

Ignoring nonlocal effects is valid if the wavelength of the optical excitation is much larger than the characteristic length scales of the material. One may speak of the typical distance between atoms in a solid, or the Fermi wavelength (the inverse of the Fermi wavevector), which are quite small, usually in the order of the angstrom. In the context of nanophotonics and polaritonics, the typical length scale is of the order of the nanometer, a bit close to the frontier separating classical physics from quantum physics. Nonetheless this approximation has revealed to be quite accurate in many contexts [41, 53, 54].

Just for illustrative purposes, we give a typical example of a model for the dielectric function, commonly used to explain and study the optical properties of metals. It is called the (AC) Drude model [55], often introduced in the form of a conductivity that reads

$$\sigma(\omega) = \frac{\omega_p^2 \varepsilon_0}{\gamma - i\omega}, \quad (2.24)$$

where ω_p is the plasma frequency⁸, a parameter related to the natural frequency of free electron oscillations in a plasma, and γ is a relaxation or scattering rate, with $\tau = 1/\gamma$ giving the timescale of consecutive collisions between the electrons and the ions in the lattice, being of the order of 10^{-14} s in typical metals. Together with the interrelation between the conductivity and the permittivity (2.14), we can write the equivalent Drude dielectric function as

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma}, \quad (2.25)$$

where we omitted the subscript “eq” on the left-hand side, and used $\varepsilon(\omega) = 1$ for the bound polarization contribution to the dielectric function. This last statement is called the jellium approximation. We skip here the derivation of this model and refer the curious reader to literature where one can find it. For instance, possible texts are Refs. [41, 56, 57].

Even though we skip its derivation, we can say that this model is constructed solely from classical considerations and introduces the parameter τ in a ad-hoc way, somehow related to fine details of the material at the microscopic scale we cannot assess (yet). Although this model is widely used in the literature to describe simple metals (alkali ones,

⁸The definition of plasma frequency is $\omega_p = \sqrt{e^2 n_e / \varepsilon_0 m}$, where n_e is the volume density of electrons in the material or plasma, and m is the mass of the free electrons.

for instance) at low enough frequencies, it is bad in describing the optical properties of dielectric materials that have characteristic resonant frequencies, or phenomena associated with resonant frequencies that metals might display at higher frequencies.

To describe the electrodynamics of non-metallic systems, we can use models based on Lorentzian oscillators. Once more we skip the derivation and refer the reader to the literature, for instance, Ref. [38]. The Lorentz model, based on the classical harmonic oscillator with a damping parameter, for the dielectric function reads

$$\varepsilon(\omega) = 1 + \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\Gamma_j\omega}, \quad (2.26)$$

where the sum is over all the oscillators, f_j is the oscillation strength, ω_j the resonant frequency and Γ_j the damping parameter⁹. Although we do not work with this model in this thesis, it is useful to motivate the dynamical dielectric function (or more precisely, conductivity) we will find later on. Instead of having a quadratic dependence on the frequency, we will find a linear dependence on the frequency as

$$\sigma(\omega) = -i\sigma_0 \sum_j \left[\frac{f_j}{\omega_j - \omega - i\Gamma_j} + \frac{f_j^*}{\omega_j + \omega + i\Gamma_j} \right], \quad (2.27)$$

where σ_0 is a characteristic conductivity, and once more f_j and Γ_j are oscillation strengths and damping rates, respectively. At this moment, we have to be satisfied with the fact that these parameters are only adjustable parameters that can be obtained from fitting to experimental data. We can say, though, that these parameters hide very rich physics to be unveiled in the next chapters, just like in the case of Section 2.2.1.1, where we introduced the adhoc parameter r_0 . We beg then for the reader's patience, as further clarifications will come in the next chapters.

Finding a simple enough analytical expression for the dielectric function that encompasses both nonlocal and dynamical effects is a daunting task, but it can still be done for a couple of cases. One striking example is the Lindhard function, the dielectric function for the three-dimensional (3D) homogeneous noninteracting electron gas, named after the Danish physicist Jens Lindhard, who derived the function in 1954 [58]. For the reader more inclined to hydrodynamical and kinetic theories of 2D materials and plasmas, we refer to Refs. [50, 59], where one can find mesoscopic approaches to the dielectric and plasmonic properties of Dirac materials, for which analytical or semi-analytical expressions for the dielectric function can be found. Another thorough text on plasmonic properties of Dirac materials that is worth mentioning is the one by P. A. D. Gonçalves and N. M. R. Peres, Ref. [41]. One more example is the hydrodynamic Drude model [60, 61], a nonlocal extension of the Drude model from the hydrodynamic equations of motion of the electron gas, seen as a charged fluid moving in the ionic positive background of a metal.

⁹The way we wrote it here, it hides parameters of the material that are common to all oscillators in the oscillator strength. For brevity, we keep the model as simple as possible.

Using only concepts introduced in this chapter, in the next section we turn to a key topic within the realm of electrodynamics that has been gaining traction in the scientific community, pertaining to the interaction of light with matter, still through the lens of classical physics. We shall present a light–matter hybrid, the quasiparticle called polariton.

2.3 Fundamentals of Polaritonics

In this part of the chapter, we dwell on polaritons: a polariton is a light–matter hybrid, resulting from the coupling of a dipolar matter excitation occurring in the plane that is an interface between two media, with the surrounding EM field. The result is an EM mode that propagates in-plane but is ultra-confined along the off-plane direction. These modes can be predicted by simply trying a specific *ansatz* solution on Maxwell’s equations (2.1) describing the situation above, complemented by the boundary conditions dictated by (2.5) and (2.6), and the constitutive relations (2.3) and (2.4) to close the problem. Before doing so, we will warm up by deriving the wave equation for the EM field in matter and introduce useful concepts for later on.

2.3.1 Electromagnetic wave propagation in matter

It is well known that light propagates in vacuum at a very specific speed $c = 1/\sqrt{\varepsilon_0\mu_0}$, like a wave, where the electric field and the magnetic field oscillate in phase at a given frequency, ω , and with a given wavevector $\mathbf{k}$ giving the direction of propagation. The electric field $\mathbf{E}$, the magnetic field $\mathbf{B}$ and the wavevector $\mathbf{k}$ form a direct trihedron, that is, they verify orthogonality conditions akin to the three Cartesian axes. These transverse oscillations can be predicted by Maxwell’s equations (2.1) upon taking the cross product between $\mathbf{k}$ and Faraday’s law (2.13b) from the left, and doing the same to Ampère–Maxwell’s law (2.13d) to write

$$\mathbf{k} \times [\mathbf{k} \times \mathbf{E}(\mathbf{k}, \omega)] = -\omega \mathbf{k} \times \mathbf{B}(\mathbf{k}, \omega), \quad (2.28a)$$

$$\mathbf{k} \times [\mathbf{k} \times \mathbf{B}(\mathbf{k}, \omega)] = -i\sigma(\mathbf{k}, \omega) \mathbf{k} \times \mathbf{E}(\mathbf{k}, \omega) - \omega \varepsilon_0 \mu_0 \varepsilon(\mathbf{k}, \omega) \mathbf{k} \times \mathbf{E}(\mathbf{k}, \omega). \quad (2.28b)$$

By combining these two equations and using the vector identity of the double cross product $\mathbf{a} \times [\mathbf{b} \times \mathbf{c}] = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$, we arrive at

$$k^2 \mathbf{F}(\mathbf{k}, \omega) - \mathbf{k}(\mathbf{k} \cdot \mathbf{F}(\mathbf{k}, \omega)) = -\omega^2 \varepsilon_0 \mu_0 \varepsilon_{\text{eq}}(\mathbf{k}, \omega) \mathbf{F}(\mathbf{k}, \omega), \quad (2.29)$$

for both fields $\mathbf{F} = \mathbf{E}, \mathbf{B}$, and we used the definition for the equivalent permittivity (2.14), while assuming that there are no external currents influencing the dynamics of the material ($\mathbf{J}_{\text{ext}} = \mathbf{0}$). Now, if we assume that the oscillation of the EM field is transversal, then $\mathbf{k} \cdot \mathbf{E} = \mathbf{k} \cdot \mathbf{B} = 0$, and the equations above simplify to

$$\left[k^2 - \varepsilon_{\text{eq}}(\mathbf{k}, \omega) \frac{\omega^2}{c^2} \right] \mathbf{F}(\mathbf{k}, \omega) = \mathbf{0}, \quad \mathbf{F} \in \{\mathbf{E}, \mathbf{B}\}. \quad (2.30)$$

The reader might have recognized the previous equation as the wave equation but written in the $(\mathbf{k}, \omega)$ domain. It presents a non-trivial solution $\mathbf{F} \neq \mathbf{0}$ if the quantity in brackets is zero, therefore $\mathbf{k}$ and ω have to verify

$$k^2 = \varepsilon_{\text{eq}}(\mathbf{k}, \omega) \frac{\omega^2}{c^2}. \quad (2.31)$$

Relationships of the previous kind, relating the frequency of the wave with the wavevector, are called dispersion relations. In this case, the wave pertains to light, so it is an optical dispersion relation. We will face another type of dispersion relation in the next chapter, pertaining to propagating electrons in a crystal. We further note that the wavevector $\mathbf{k}$, that can also be called the wavenumber, is in general complex, as well the permittivity $\varepsilon_{\text{eq}}(\mathbf{k}, \omega)$. Writing the wavenumber in terms of its real and imaginary parts as $k = k' + i\kappa$, when looking at modes oscillating in space as $\exp(ikx)$, then we have $\exp(ikx) = \exp(ik'x) \exp(-\kappa x)$. Therefore, the inverse of the imaginary part of the wavenumber $1/\kappa$ has the interpretation of a penetration length scale of the wave into the medium. The parameter κ is then called the extinction coefficient, and $1/\kappa$ the skin depth¹⁰.

By identifying the phase velocity of the wave as $v = \omega/k$, then we have a definition for the quantity $c/v = \sqrt{\varepsilon_{\text{eq}}(\mathbf{k}, \omega)} \equiv n$, the refractive index of the medium. It is in this context that we speak of dispersion, both spatial or temporal, coming from the dependence of the refractive index in $\mathbf{k}$ or ω , respectively. In dispersive media, if the wave is a superposition of harmonics, then it actually makes no sense of speaking in the velocity of the wave, as there are two distinct velocities. There is the phase velocity $v_p = \omega/k$ and the group velocity $v_g = \partial\omega/\partial k$.¹¹ The phase velocity is the speed with which the phase of the wave-front propagates. The group velocity is the speed with which the wave envelope travels, and can be smaller or greater than the phase velocity, in a normal or anomalous dispersive medium, respectively. To summarize, we can say that in a material the dispersion of propagating EM waves can be in general different from that of free light in vacuum, $\omega = kc$.

Looking back at equation (2.29), and looking now for longitudinal oscillations with $\mathbf{k} \parallel \mathbf{E}$ for which $\mathbf{k} \cdot \mathbf{E} \neq 0$, then we find instead

$$\varepsilon_{\text{eq}}(\mathbf{k}, \omega) \frac{\omega^2}{c^2} \mathbf{F}(\mathbf{k}, \omega) = \mathbf{0}, \quad \mathbf{F} \in \{\mathbf{E}, \mathbf{B}\}. \quad (2.32)$$

This last equation admits a non-trivial solution $\mathbf{F} \neq \mathbf{0}$ only for¹²

$$\varepsilon_{\text{eq}}(\mathbf{k}, \omega) = 0. \quad (2.33)$$

We will immediately interpret the previous condition. Omitting the subscript “eq”, let us rewrite the constitutive relation for the electric displacement field $\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} =$

¹⁰We discard off course the unphysical $\kappa < 0$ solutions.

¹¹We could have been more rigorous, since these quantities are real, but the refractive index and the wavenumber are complex in general.

¹²We remind the reader that we have been considering materials with no magnetic susceptibility, such that the relative permeability is identically one in the entire $(\mathbf{k}, \omega)$ domain.

$\varepsilon_0 \varepsilon \mathbf{E} = \mathbf{0}$, the last equality by virtue of (2.33). This means for the electric field $\mathbf{E} = -\mathbf{P}/\varepsilon_0$, that is, the electric field in the material stems only from the polarization field—the only way for $\mathbf{E}$ to admit oscillating modes when $\mathbf{D}$ is zero, is by having ε equal to zero too. These modes are freestanding and derive from the collective oscillation of free electrons in the material, typically a metal. They are called plasmonic modes, or plasmons, and were first introduced by David Pines and David Bohm in 1952 [62] and underpin the entire field of plasmonics [41, 53, 59, 61, 63].

The relation (2.33) can be seen as the dispersion relation of the plasmon, that in general is implicitly defined and can only be solved numerically to get $\omega = \omega(\mathbf{k})$ explicitly. However, if the expression of the permittivity is simple enough, then we can find an analytical expression for the dispersion relation of transverse and longitudinal modes. In the example of a bulk, local and Drude metal without the background contribution, with $\varepsilon_\infty = 1$, the permittivity is given by $\varepsilon(\omega) = 1 - \omega_p^2/\omega^2$, so that light and plasmon modes have by dispersion relations $\omega^2 = \omega_p^2 + k^2 c^2$ and $\omega = \omega_p$, respectively.

The plasmon, being a dipolar excitation of the material, can couple to the transverse EM mode, for instance, along the interface separating a dielectric and a metal, to generate a polariton. Then we would be talking of the so-called surface plasmon–polariton (SPP). We derive the dispersion relation of the SPP in the next section, but considering instead the more general case where the interface itself has its own optical response, quantified through an optical conductivity $\sigma(\omega)$, in the LRA limit. Then, the dispersion relation of the SPP can be obtained *a posteriori* by setting $\sigma = 0$.

2.3.2 Dispersion relation of polaritonic modes

In this section we derive the dispersion relation of surface polaritons, in the case where we have a responsive 2D material, separating two different media with permittivities ε_1 and ε_2 , possibly frequency dependent. The situation is depicted in Figure 2.2, where the substrate, in gray, is material 2, with permittivity ε_2 , while the material on top is material 1, with permittivity ε_1 . At the interface there is a 2D material represented in light blue, with its own optical response that we quantify, by convenience, through its conductivity $\sigma(\omega)$, ω being the frequency of the polariton. The polariton propagates in the x direction with wavevector $\mathbf{q} = q\hat{\mathbf{x}}$ that is confined on the xy plane, decaying exponentially in the z direction away from it.

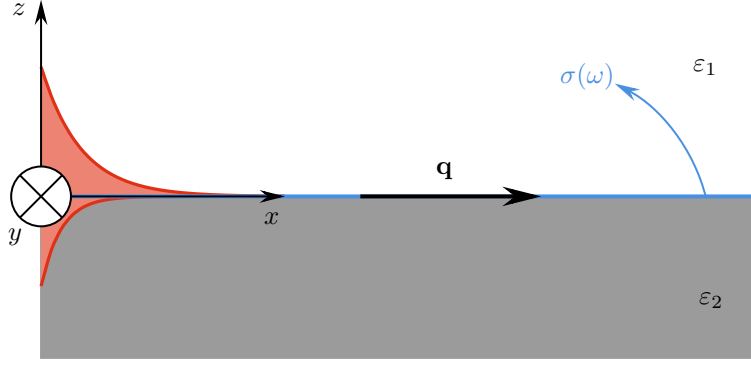


Figure 2.2: Schematic representation of a surface polariton. A polariton with wavevector $\mathbf{q}$ propagating along a 2D material separating two bulk media, medium 1 on top as the cladding material, and medium 2 underneath acting as a substrate. Each medium $j = 1, 2$ has its own permittivity $\varepsilon_j(\omega)$, while the 2D material has its own conductivity $\sigma(\omega)$. The regions shaded in red represent the exponentially decaying amplitude of the polariton along the z direction. In medium 2, assuming it has a higher permittivity than medium 1, the polariton is more confined in comparison with medium 1.

The starting point is assuming a plane wave dependence on x and t for the electric field $\mathbf{E}$ and the magnetic excitation field $\mathbf{H}$ (or magnetic field for short). So we write

$$\mathbf{F}(\mathbf{r}, t) = \mathbf{f}(z) e^{i(qx - \omega t)}, \quad \mathbf{F} \in \{\mathbf{E}, \mathbf{H}\}, \quad (2.34)$$

where the lowercase $\mathbf{f}$ is $\mathbf{e}$ in the case of the electric field or $\mathbf{h}$ in the case of the magnetic field, if we factor out the z dependence from the (x, t) dependence for both fields; the system is y invariant. To obtain an equation governing the z dependence, we take the curl of Faraday's law (2.1b) and use Ampère–Maxwell's law (2.1d), with both laws in frequency domain. Furthermore, we work with the total displacement field $\mathbf{D}$ defined in Section 2.2.1 and with no contribution coming from external currents such that $\mathbf{J} = \mathbf{0}$. We can then write for the electric field the so-called Helmholtz equation

$$\begin{aligned} \nabla^2 \mathbf{E}(\mathbf{r}, \omega) + \frac{\omega^2 \varepsilon(\omega)}{c^2} \mathbf{E}(\mathbf{r}, \omega) &= \mathbf{0} \Leftrightarrow \\ \Leftrightarrow (\nabla^2 + k_0^2) \mathbf{E}(\mathbf{r}, \omega) &= \mathbf{0}, \end{aligned} \quad (2.35)$$

by defining $k_0^2 \equiv \omega^2 \varepsilon(\omega)/c^2$, and neglecting nonlocal effects in the material (either 1 or 2), so that the permittivity appearing is only frequency dependent. This allows to transform the spatial convolution of the permittivity with the electric field into a mere multiplication, such that $\mathbf{D}(\mathbf{r}, \omega) = \varepsilon(\omega) \mathbf{E}(\mathbf{r}, \omega)$ as already explained previously (see Section 2.2.1.2). Otherwise, we would have to take spatial derivatives of the response function. This is equivalent to taking in (2.29) the LRA limit $\mathbf{k} \rightarrow \mathbf{0}$ of $\varepsilon_{\text{eq}}(\mathbf{k}, \omega)$, and converting the square of the wavenumber k^2 back into the Laplacian operator in real space (up to a sign). Replacing the electric field $\mathbf{E}$ by expression (2.34), we obtain the ordinary differential equation

$$\frac{d^2 \mathbf{e}(z)}{dz^2} + (k_0^2 - q^2) \mathbf{e}(z) = \mathbf{0}. \quad (2.36)$$

Now what is left is to relate the components of the electric field with the components of the magnetic field. This is achieved by using again Faraday's law (2.1b), and Ampère–Maxwell's law (2.1d) in frequency domain with no external current sources, but for the uppercase electric field $\mathbf{E}$ and the magnetic excitation field $\mathbf{H}$, not the undercase fields dependent only on z . We rewrite those two laws in terms of Cartesian coordinates omitting the frequency dependence in the material's permittivity, using y invariance so that $\partial_y = 0$ and replacing $\partial_x \rightarrow iq$ to write

$$\partial_z E_y = -i\omega\mu_0 H_x, \quad \text{Faraday's law: } x \text{ component}; \quad (2.37a)$$

$$\partial_z E_x - iqE_z = i\omega\mu_0 H_y, \quad \text{Faraday's law: } y \text{ component}; \quad (2.37b)$$

$$iqE_y = i\omega\mu_0 H_z, \quad \text{Faraday's law: } z \text{ component}; \quad (2.37c)$$

$$\partial_z H_y = i\omega\varepsilon_0 \varepsilon E_x, \quad \text{Ampère–Maxwell's law: } x \text{ component}; \quad (2.37d)$$

$$\partial_z H_x - iqH_z = -i\omega\varepsilon_0 \varepsilon E_y, \quad \text{Ampère–Maxwell's law: } y \text{ component}; \quad (2.37e)$$

$$iqH_y = -i\omega\varepsilon_0 \varepsilon E_z, \quad \text{Ampère–Maxwell's law: } z \text{ component}. \quad (2.37f)$$

The previous set of equations admits two types of solutions, the transverse magnetic (TM) or p -polarized modes, and the transverse electric (TE) or s -polarized modes. In TM modes the electric field is in-plane and the magnetic field is off-plane, while in TE modes the magnetic field is the one in the plane and the electric field is off-plane. While TE modes present in general light-like dispersion, therefore displaying weak field confinement, TM modes are typically more interesting, being able to achieve higher field confinement. It is a custom to classify polariton modes in these two types of polarizations since not all systems can host both types of polarization. Figure 2.3 represents the orientation of the electric and magnetic fields with respect to the polariton wavevector for the two distinct situations. We will focus first on the TM mode, corresponding to the right-hand side of the figure below.

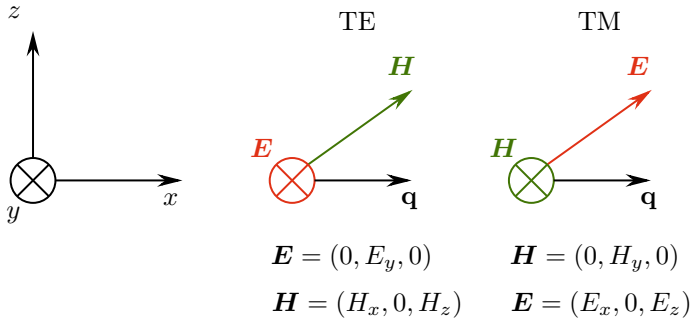


Figure 2.3: Schematic representation of TE and TM modes. Vectorial representation of the TE polarized and TM polarized polaritonic modes. The magnetic field is represented in green, the electric field in red, and the polariton wavevector in black.

For TM modes, the equations in (2.37) read

$$\begin{aligned} \partial_z E_x - iqE_z &= i\omega\mu_0 H_y, \\ \partial_z H_y &= i\omega\varepsilon\varepsilon_0 E_x, \\ iqH_y &= -i\omega\varepsilon\varepsilon_0 E_z. \end{aligned} \quad (2.38)$$

From this set of equations, upon elimination of the electric field components, we obtain the equation

$$\frac{\partial^2 H_y}{\partial z^2} + \left(\frac{\omega^2 \varepsilon}{c^2} - q^2 \right) H_y = 0. \quad (2.39)$$

According to equation (2.34), assuming that (the spatial dependence of) the magnetic field is of the form

$$H_y(x, z) = Ae^{iqx} e^{-\kappa|z|}, \quad (2.40)$$

we obtain for the extinction coefficient

$$\kappa = \sqrt{q^2 - \varepsilon \frac{\omega^2}{c^2}}. \quad (2.41)$$

We consider separately the bulk materials in the two different domains, the one over the interface with relative permittivity ε_1 and the one below with relative permittivity ε_2 , as in Figure 2.2. From inspection of (2.38), the spatially dependent part of the electric and magnetic fields write ($j = 1$ for $z > 0$ and $j = 2$ for $z < 0$)

$$H_{jy}(x, z) = A_j e^{iqx} e^{(-1)^j \kappa_j z}, \quad (2.42a)$$

$$E_{jx}(x, z) = -i(-1)^j A_j \frac{\kappa_j}{\varepsilon_j \varepsilon_0 \omega} e^{iqx} e^{(-1)^j \kappa_j z}, \quad \text{and} \quad (2.42b)$$

$$E_{jz}(x, z) = -A_j \frac{q}{\varepsilon_j \varepsilon_0 \omega} e^{iqx} e^{(-1)^j \kappa_j z}. \quad (2.42c)$$

By continuity of the in-plane component of the electric field at the interface $z = 0$, as per (2.5a) we have

$$E_{1x}(z = 0) = E_{2x}(z = 0) \Leftrightarrow A_2 = -\frac{\varepsilon_2 \kappa_1}{\varepsilon_1 \kappa_2} A_1. \quad (2.43)$$

The parallel component of the magnetic field, however, is discontinuous. Recalling condition (2.5b);

$$\mathbf{n}_{12} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{j}_s \Leftrightarrow \mathbf{H}_{1,\parallel} - \mathbf{H}_{2,\parallel} = -\mathbf{j}_s \times \mathbf{n}_{12}. \quad (2.44)$$

The sheet current density $\mathbf{j}_s$ relates to the electric field $\mathbf{E}$ through Ohm's law, where the conductivity is that of the 2D material, σ . In the frequency domain, and within the LRA,

Ohm's law is as simple as $\mathbf{j}_s(\mathbf{r}, \omega) = \sigma(\omega)\mathbf{E}(\mathbf{r}, \omega)$. For TM modes, and taking into account the tensor nature of the conductivity, the right-hand side of the previous equation reads

$$\mathbf{j}_s \times \mathbf{n}_{12} = \sigma_{xx} E_x \hat{\mathbf{y}}. \quad (2.45)$$

Rewriting (2.44), using (2.43) and simplifying

$$\begin{aligned} H_{1t}(z=0) - H_{2t}(z=0) &= -\sigma_{xx} E_x(z=0) \Leftrightarrow \\ A_1 - A_2 &= -\sigma_{xx} i \frac{\kappa_1}{\varepsilon_1 \varepsilon_0 \omega} A_1 \Leftrightarrow \\ 1 + \frac{\varepsilon_2 \kappa_1}{\varepsilon_1 \kappa_2} &= -i \sigma_{xx} \frac{\kappa_1}{\varepsilon_1 \varepsilon_0 \omega} \Leftrightarrow \\ \frac{\varepsilon_1}{\kappa_1} + \frac{\varepsilon_2}{\kappa_2} + i \frac{\sigma_{xx}}{\varepsilon_0 \omega} &= 0, \end{aligned} \quad (2.46)$$

We have finally obtained the dispersion relation of TM polarized polaritons (omitting the ω -dependence for clarity)

$$\frac{\varepsilon_1}{\kappa_1} + \frac{\varepsilon_2}{\kappa_2} + i \frac{\sigma_{xx}}{\varepsilon_0 \omega} = 0. \quad (2.47)$$

We can see that TM polarized polaritons are supported in the range of frequencies for which $\text{Im}\{\sigma_{xx}(\omega)\} > 0$ (condition that results in positive values for the real part of the extinction coefficients). If the real part of the conductivity is finite (non-zero), that is, if there is loss, then the equation holds on the complex plane, and either the frequency or the wavenumber has to be complex.

An analogous procedure for TE modes would lead to

$$\kappa_1 + \kappa_2 - i\omega\mu_0\sigma_{yy} = 0, \quad (2.48)$$

the dispersion relation for TE modes. We can see that the system supports TE modes when $\text{Im}\{\sigma_{yy}(\omega)\} < 0$, and if the real part of the conductivity is finite, then the dispersion relation holds on the complex plane, just like for TM modes. Analytically, both dispersion relations (2.47) and (2.48) can only be written implicitly, so we have to solve them numerically to obtain the dispersion relations in its explicit form $\omega(q)$.

We consider the particular case of TM polarization and no 2D material response, $\sigma(\omega) = 0$, so that the interface is simply the boundary separating a dielectric and a metal. In that case, we finally arrive at the dispersion relation for SPPs

$$\frac{\varepsilon_1}{\kappa_1} + \frac{\varepsilon_2}{\kappa_2} = 0. \quad (2.49)$$

With material 1 being a dielectric, and material 2 being a metal, SPPs can be supported for frequency windows where ε_1 and ε_2 have opposite signs [41, 53]. This can be achieved in common lossless Drude metals, pertaining to the $\gamma = 0$ case in equation (2.25) enabling $\varepsilon_2(\omega) < 0$, while for most insulators ε_1 can be taken as a positive constant for all frequencies.

Figure 2.4 shows the dispersive behavior of SPPs at a dielectric–metal interface, where the dielectric is air, and the metal is a local Drude metal whose permittivity is given by Eq. (2.25). In panel **a**, for the lossless case, one can notice the very distinct behavior of the SPP dispersion relation in the bottom branch in blue, and of the wave propagating inside the metal in the top branch from the linear dispersion curve $\omega(q) = qc/\sqrt{\varepsilon_1}$, which corresponds to light propagating in air. We can notice a region of forbidden frequencies opening a gap between the two branches. In the top branch in red, the low-frequency behavior of the SPP is light-like, which contrasts with the high-frequency behavior where the polariton can achieve very high wavenumbers, meaning we can realize really high EM field confinement in this arrangement.

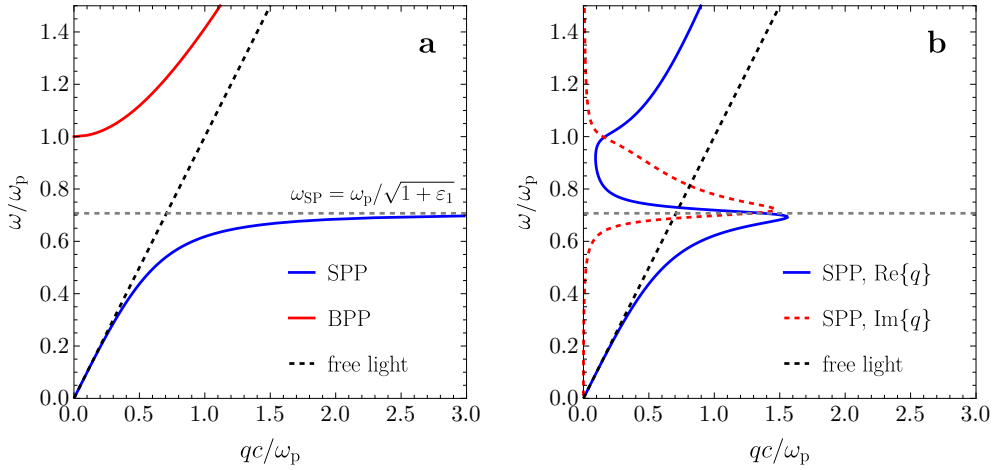


Figure 2.4: Dispersion relation of surface plasmon–polaritons at a dielectric–metal interface. Dispersion relation $\omega(q)$ of SPPs at a dielectric–metal interface for a Drude metal with $\varepsilon_\infty = 1$ in Eq. (2.25), and a dielectric with $\varepsilon_1 = 1$. Panel **a** is for a collisionless Drude model with $\gamma = 0$. The dashed black line corresponds to light propagating in vacuum, the top branch, in red, corresponds to free light propagating in the (bulk) metal, and the bottom branch, in blue, corresponds to the SPP dispersion relation, given by solving Eq. (2.47). The gray dashed horizontal line corresponds to the dispersion relation of the TM polarized mode in the limit $c \rightarrow \infty$, also called the nonretarded limit. Panel **b** is for a Drude metal with finite damping rate $\gamma = 0.2\omega_p$. The blue full line represents the real part of the wavenumber, while the blue dashed one the imaginary part.

The interesting properties of this EM mode can in principle be exploited for EM field manipulation at very small length scales and promote strong light–matter interactions. However, and quite unfortunately, one cannot ignore Ohmic losses, especially in metals, henceforth we would say that the situation depicted in panel **b** is more realistic. In the lossy case, the theoretical value for the highest wavenumber we could possibly achieve is more comparable to that of free light (but still higher). Moreover, the fact that the permittivity is complex results in a wavenumber with non-negligible imaginary part [64], that actually peaks at approximately the same frequency where the real part peaks. It tells us that, in practice, a high-field confinement means also high losses. Even though we now have EM modes in the frequency region that was previously forbidden, these modes are evanescent

and do not propagate over long distances. Recall that the decay length of an evanescent wave is given by the inverse of the imaginary part of the wavenumber, as discussed in Section 2.3.1.

It is pertinent here to make the following technical remark. In the analysis made above, for dispersion relations that may differ from that of free light and in the case the permittivity is complex, we could have assumed instead that the frequency ω is complex, while the wavenumber q is kept real.¹³ In theory both approaches are valid, but if we are to compare theory with experiments, then we have to be aware of what choice would be significant. For instance, if we assume that the frequency is complex, then it means the polariton would be a propagating wave, but decaying in time. We should not expect to observe this effect in a continuous wave experiment, for instance. To observe it, an experiment where the system is probed by short-lived light pulses is more adequate. For a continuous wave experiment the complex valued wavenumber picture would offer more reliable predictions.

We will see later on in this work another kind of surface polaritons, where the polar matter excitation is one characteristic of insulators and semiconductors, the so-called exciton. The coupling of the exciton with the EM field is logically called exciton–polariton. The exciton is a quasiparticle that results from the optical excitation of materials that present an electronic bandgap. These can be classified as semiconductors if the bandgap is smaller than or of the order of the environment thermal energy dictated by the external temperature, or insulators if the bandgap is large compared with the thermal energy. Before we can understand these concepts (bandgap, insulators/semiconductors) and describe this exciton quasiparticle, and the consequent coupling with light, we need to dive deeper into the field of quantum physics, with particular emphasis on solid-state matter in the context of condensed matter physics. That is what the next chapter is devoted to, introduce the reader to the concepts of quantum mechanics that are necessary to address the excitations of materials at length scales where the tools of classical physics are deemed insufficient. The excitations of matter using a quantum description of materials is then the subject of the subsequent chapters.

Before we conclude the chapter and leave the realm of classical physics, we show that it can still offer more insight into how light and matter interact. In the next section we will see how the laws of classical electrodynamics allow us to estimate the amount of radiation that can be absorbed, reflected or transmitted at an interface between two domain regions when light traverses from one region to the other. It will culminate with Fresnel’s coefficients, but in the more general case where the interface has its own optical response.

¹³The reader might wonder if we could even consider complex frequency and wavenumber at the same time. One may do so, but that increases the dimension of the space we work with without necessity, making it harder to understand the physics and visualize the results. It is thus sufficient to consider in turns a real valued frequency or a real valued wavenumber “cuts” of that higher dimensional and more abstract space.

2.3.3 Material absorption, reflection and transmission

The purpose of this section is to derive an expression for the amount of EM radiation that is absorbed by a 2D material when impinged by a linearly polarized plane wave field. The configuration is that of Figure 2.5, with an incoming plane wave propagating from above in medium 1 with normal incidence on the interface, with the 2D material reflecting and transmitting part of the incident radiation. Using Maxwell's equations and the boundary conditions at the interface once more, we will derive the Fresnel coefficients in the particular case of normal incidence, but in the more general form where the surface separating the two media has its own optical response σ (we will omit for now the frequency dependence of the conductivity). Omitting the time dependence, we write for the incident electric field $\mathbf{E}_{\text{inc}} = A_i e^{-ik_i z} \hat{\mathbf{x}}$, for the reflected electric field $\mathbf{E}_r = A_r e^{ik_r z} \hat{\mathbf{x}}$, and for the transmitted electric field $\mathbf{E}_t = A_t e^{-ik_t z} \hat{\mathbf{x}}$. Capital A stands for field amplitude. Using Faraday's law in reciprocal space (2.13b), the incident, reflected and transmitted magnetic fields read $\mathbf{H}_{\text{inc}} = -k_i/\omega\mu_0 A_i e^{-ik_i z} \hat{\mathbf{y}}$, $\mathbf{H}_r = k_i/\omega\mu_0 A_r e^{ik_r z} \hat{\mathbf{y}}$, and $\mathbf{H}_t = -k_t/\omega\mu_0 A_t e^{-ik_t z} \hat{\mathbf{y}}$, respectively, with $k_i = k_r$. Boundary conditions at the interface $z = 0$ given by (2.5a) and (2.5b), where the subscript “2” denotes the transmitted field, while the subscript “1” denotes the total field in region 1, incident plus reflected, demand that

$$A_i + A_r = A_t, \quad (2.50a)$$

$$-\frac{k_i}{\omega\mu_0} A_i + \frac{k_i}{\omega\mu_0} A_r + \frac{k_t}{\omega\mu_0} A_t = -\sigma A_t. \quad (2.50b)$$

Note that in the right-hand side of the second equation we could equally write $A_i + A_r$ instead of A_t , due to the continuity of the tangential component of the electric field. Using the identity $k = n\omega/c$ and solving the previous system of equations, we can define the Fresnel coefficients for reflection, r , and transmission, t , as

$$r \equiv \frac{A_r}{A_i} = \frac{n_1 - n_2 - \sigma/(c\varepsilon_0)}{n_1 + n_2 + \sigma/(c\varepsilon_0)} \quad \text{and} \quad (2.51a)$$

$$t \equiv \frac{A_t}{A_i} = \frac{2n_1}{n_1 + n_2 + \sigma/(c\varepsilon_0)}. \quad (2.51b)$$

The Fresnel coefficients are not yet a measure of the amount of radiation that the 2D material reflects or transmits, they are only relative field amplitudes. In order to calculate the amount of radiation reflected and transmitted, and consequently absorbed, we actually still need to calculate the relative field intensities. We can define the intensity of the EM field as the magnitude of the Poynting vector $\mathbf{S}$, defined by $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ [38, 54], averaged in time over one complete oscillation. As we are considering only plane waves composed of a single harmonic frequency $\omega = ck/n$, we can write for the time-averaged Poynting vector [54]

$$\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}\{\mathbf{E} \times \mathbf{H}^*\} = \frac{|\mathbf{E}|^2}{2\omega\mu_0} \text{Re}\{\mathbf{k}\}, \quad (2.52)$$

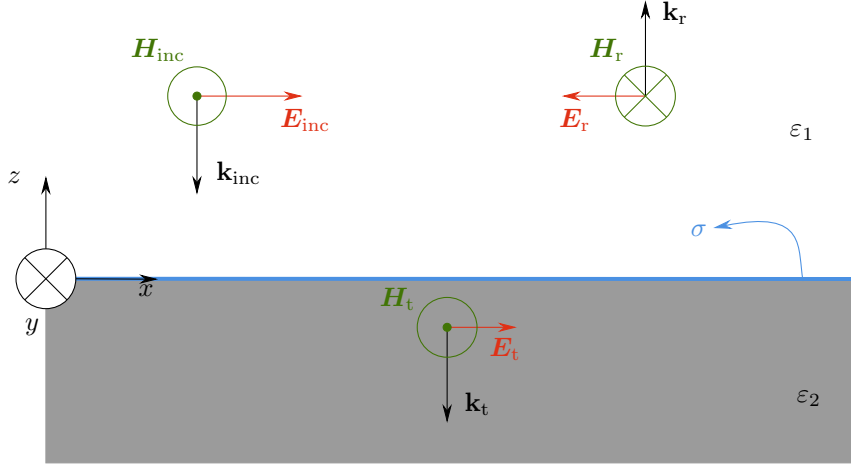


Figure 2.5: Normal incidence of a plane wave on an interface between two dielectrics. A plane wave with wavevector $\mathbf{k}_{\text{inc}}$ and frequency ω , travelling through medium 1 with permittivity ε_1 , impinges on the surface of a 2D material, in blue, deposited on top of a dielectric substrate with permittivity ε_2 .

where, to get the second equality, we made use once more of Faraday's law, and also the vector identity for the double cross product already mentioned in Section 2.3.1. For the incident and reflected radiation fields, respectively, we have $\mathbf{S}_{\text{inc}} = -\frac{1}{2}c\varepsilon_0|\mathbf{E}_{\text{inc}}|^2\text{Re}\{n_1\}\hat{\mathbf{z}}$ and $\mathbf{S}_r = \frac{1}{2}c\varepsilon_0|\mathbf{E}_r|^2\text{Re}\{n_1\}\hat{\mathbf{z}}$, while for the transmitted field we have $\mathbf{S}_t = -\frac{1}{2}c\varepsilon_0|\mathbf{E}_t|^2\text{Re}\{n_2\}\hat{\mathbf{z}}$. We can finally define the relative amount of radiation that is reflected and transmitted at the interface, respectively, as

$$\mathcal{R} = \frac{\langle|\mathbf{S}_r|\rangle}{\langle|\mathbf{S}_{\text{inc}}|\rangle} = \frac{|\mathbf{E}_r|^2}{|\mathbf{E}_{\text{inc}}|^2} = |r|^2 \quad \text{and} \quad (2.53a)$$

$$\mathcal{T} = \frac{\langle|\mathbf{S}_t|\rangle}{\langle|\mathbf{S}_{\text{inc}}|\rangle} = \frac{\text{Re}\{n_2\}}{\text{Re}\{n_1\}} \frac{|\mathbf{E}_t|^2}{|\mathbf{E}_{\text{inc}}|^2} = \frac{\text{Re}\{n_2\}}{\text{Re}\{n_1\}} |t|^2, \quad (2.53b)$$

where the expressions for r and t can be found in equations (2.51a) and (2.51b), respectively. We are in place now to compute the amount of radiation that is absorbed by the 2D material, the optical absorption $\mathcal{A}_{\text{opt}}$, through the identity

$$\mathcal{A}_{\text{opt}} = 1 - \mathcal{R} - \mathcal{T} = 1 - |r|^2 - \frac{\text{Re}\{n_2\}}{\text{Re}\{n_1\}} |t|^2. \quad (2.54)$$

Maxwell's equations valid in continuous domains, together with the boundary conditions for boundaries between domains and the constitutive relations, form a closed system of equations, with which we could tackle any problem in electrodynamics involving 2D materials; at least in principle, and if not analytically, then numerically. It happens that to make realistic predictions and expect agreement with experiments, we would need to know how the dielectric function, or equivalently the conductivity, depends on the frequency and wavevector.¹⁴ We have seen two examples in the Section 2.2.1 of models for the dielectric

¹⁴Apart from non-linear phenomena, where higher-order susceptibilities have to be defined, and it no longer makes sense speaking of a single permittivity or conductivity.

function that have an easy analytical expression. However, at this point we have no insight on the validity of these models, or where exactly do they come from, apart from classical explanations. More specifically, there is no way for us to obtain the values of the adjustable parameters used in those models besides from experimental measurements. In the next chapter, we introduce the reader to the principles of quantum mechanics, which are absolutely necessary to attack the problem of electrodynamics in materials at the quantum level, with focus on crystalline solids. Doing so, we will be able to have a better feeling for the dielectric and optical properties of materials. In the class of 2D materials we are interested in, insulators and semiconductors, having a quantum description for their optical properties is essential, as it is the only way in which we can predict the exciton, responsible for the material optical properties. After reading the next two chapters, the reader should be able to understand how a microscopic description of crystals allows properly quantifying the permittivity and the conductivity of periodic solids. These type of solids are the subject of the next chapter.

QUANTUM MECHANICS OF ELECTRONS IN CRYSTALLINE SOLIDS

We have seen in the previous chapter that, even though classical models explain many phenomena in electrodynamics fairly well, both qualitatively and quantitatively, classical physics cannot predict the precise values of their respective parameters, nor can it explain why they work so well for certain materials, while failing to describe others. To have a deeper understanding of these models and their regimes of validity, or to faithfully predict material properties in general, one has to adopt a microscopic description of the systems we want to study. This means that we have to look at the chemical composition of the material and at the spatial arrangement of the atoms making it up. One has to look at the detailed physics at length scales of the order of the angstrom, where classical physics as we saw it in the last chapter is insufficient.

The tools that allow us to surpass the shortcomings of classical physics are provided by quantum mechanics. It is said that the father of quantum mechanics is Max Planck, the German physicist who solved the problem of the ultraviolet catastrophe; an open problem in physics until then. Quantum mechanics allowed solving other open problems in physics. Such an example was knowing why the electron does not fall into the nucleus of the atom in its orbital motion. It was the danish physicist Niels Bohr, who came up with an atomic model that explained the stability of the atoms and their discrete energy levels. A pioneer in the quantum mechanics of solids was Arnold Sommerfeld, who formulated a theory of his own for metals [65].

With the machinery provided by the principles and laws of quantum mechanics, many techniques have been developed since last century to approach different quantum systems. We keep our discussion narrowed to solids with a periodic atomic structure, called crystals, until the end of this thesis, and leave all other types of systems (amorphous solids, such as glasses, crystalline solids with defects, quasicrystals, few-particle systems, etc.) out of the discussion.

In the field of solid state physics, it is customary to separate the calculations to predict all kinds of material properties into two main groups; (i) those based on first-principles,

also called *ab-initio* methods, which are usually very computational intensive, but with a more fundamental starting point; (ii) second-principles calculations, which start from a more phenomenological stand point, but are computationally more accessible. In *ab-initio* calculations, by providing only the chemical species and how the atoms bond among each other in the material (sort of, not exactly), we can get the lengths of the chemical bonds, and different material properties, for instance electronic, magnetic, and thermal, among others. The calculations based on second principles, while much less computationally demanding, they depart from some information that we already know about our system, like model parameters obtained experimentally or via *ab-initio* calculations. In this thesis, *ab-initio* approaches will be discussed on a passing note, as those can easily materialize into a thesis in itself, but are nonetheless very relevant for our work, which is essentially based in second-principles calculations.

We start the chapter by introducing the Schrödinger equation, the workhorse of basic quantum mechanics, alongside the concept of the Hamiltonian operator. Next, we particularize the study to crystals, by writing the full many-body Hamiltonian of a crystal, followed by key approximations to reduce the complexity of the problem. Afterwards, we define the concepts of (direct) lattice and reciprocal lattice, and discuss single-particle properties of electrons in periodic solids, such as Bloch states and the band structure; these define the dispersive behavior of electronic waves in materials. First-principle methods at the mean-field level, such as density functional theory (DFT), are discussed in brief as well. We extend the discussion on first-principle methods to the many-particle description of electrons in solids, provided by the machinery of many-body perturbation theory. With all these concepts understood, the reader should be capable of grasping the microscopic dielectric function, reserved for the last section of this chapter. The concept of dielectric function, introduced in the previous chapter on a classical footing, will be therein revisited through the lens of quantum mechanics.

3.1 The Hamiltonian of a Crystal

3.1.1 The Schrödinger equation

In classical mechanics, we follow all the degrees of freedom of a system, whose dynamics is governed by the Lagrangian function in the Lagrangian formalism, or equivalently by the Hamiltonian function in the Hamiltonian formalism. Both formalisms provide a set of equations governing the time evolution of the degrees of freedom of the system, that together with the initial conditions determine the state of the system at any time in the future in the configuration space (Lagrangian formalism), or in the phase space (Hamiltonian formalism) [66]. When the length scale of a system is too small to be considered classic, and is thus deemed quantum, it is not possible to follow the exact trajectory of the system. The state of the system can no longer be identified with a set of coordinates and (canonical) momenta with precise values at any time instant. Instead, the state of the system is a

vector, vector state, or simply state living in a Hilbert space, that we can denote by a ket $|\Psi\rangle$ in the famous bracket notation. The corresponding bra is written as $\langle\Psi|$, and it acts on kets in the same Hilbert space, in the sense that the bra $\langle\Psi|$ acting on a ket $|\Phi\rangle$ gives the complex number $\langle\Psi|\Phi\rangle$, which is the inner product between the states $|\Psi\rangle$ and $|\Phi\rangle$. A Hilbert space is thus any vector linear space endowed with an inner product¹. We can call an inner product also a scalar product as a synonym. That same state can evolve in time, and the time evolution is governed by the time-dependent Schrödinger equation (TDSE), which reads [68, 69]

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle, \quad (3.1)$$

where $\hbar = h/2\pi$ is the reduced Planck constant, and $\hat{H}$ is the Hamiltonian operator, no longer a function, of the system. The hat above the symbol H is used to stress that we are dealing with an operator, not a function. We use the term operator to denote a linear transformation that maps a vector in a Hilbert space to another vector in another Hilbert space; eventually, both Hilbert spaces coincide. It is the generalization of a linear transformation from linear algebra. At some point we shall drop the hat for ease of notation.

If in the TDSE the Hamiltonian has no explicit dependence on time, then we can easily integrate Eq. (3.1) and write a formal solution for the state as

$$|\Psi(t)\rangle = e^{-i\hat{H}(t-t_0)/\hbar} |\Psi(t_0)\rangle \equiv \hat{U}(t) |\Psi(t_0)\rangle, \quad (3.2)$$

if we know the state of the system $|\Psi(t_0)\rangle$ at the initial time t_0 , and where we defined the time-evolution operator $\hat{U}(t) = \exp\{-i\hat{H}(t-t_0)\}$. We can therefore factor out the time dependence of the state, which is encoded in the operator $\hat{U}(t)$ ². Now we assume that the time-independent part of the state of the system, $|\Psi(t_0)\rangle$, can be written as a linear combination of basis states $|\phi_n\rangle$, which are eigenstates of the Hamiltonian with eigenvalues E_n , such that $|\Psi(t_0)\rangle = \sum_n c_n |\phi_n\rangle$, with $\hat{H} |\phi_n\rangle = E_n |\phi_n\rangle$. We may also assume that the basis set $\{|\phi_n\rangle\}$ is orthonormal and complete. A set of basis states being complete means that any vector state in the Hilbert space can be written as a linear combination of the states $|\phi_n\rangle$. The coefficients c_n can then be computed by projecting $\langle\phi_m|\Psi(t_0)\rangle = c_m$. However, to know the coefficients c_n , we need to know the eigenstates and the eigenvalues of the Hamiltonian operator, which can be found by solving the eigenvalue problem

$$\hat{H} |\phi_n\rangle = E_n |\phi_n\rangle, \quad (3.3)$$

which is called the time-independent Schrödinger equation (TISE). The eigenvalues E_n , being eigenvalues of the Hamiltonian operator, can be interpreted as energies and called

¹For us physicists, it suffices to say that a Hilbert space is a vector space with an inner product, which is not pathological. For mathematicians, we have to add the condition of the space being complete for the distance induced by the (norm induced by the) inner product. The curious reader can consult Ref. [67] for more details.

²If the Hamiltonian depends explicitly on time, we can still find an expression for the time-evolution operator, albeit more complicate. It is usually left out from introductory textbooks on quantum mechanics. But one can consult Ref. [69] for details.

eigenenergies. Solving the TISE is then a key problem in quantum mechanics, whenever $\partial_t \hat{H} = 0$. If the problem is simple enough that the Hamiltonian has a finite matrix representation, in other words, if the Hilbert space is finite-dimensional, then it is a typical eigenvalue/eigenvector problem from linear algebra. On the contrary, if the Hilbert space is infinite-dimensional, then the TISE cannot be solved analytically, apart from a small set of examples, like the particle in a box, the quantum harmonic oscillator, the hydrogen atom [68], among others. Henceforth, since the birth of quantum mechanics, physicists have developed many techniques to study different systems, mainly depending on the aspect of the Hamiltonian. In analogy with classical mechanics, in many cases the Hamiltonian of a system is the sum of two contributions; one coming from the kinetic energy of the particle or particles involved, $\hat{T}$, and the other from the potential energy $\hat{V}$. In the next subsection, we try to write an expression for the Hamiltonian of a crystal as $\hat{H} = \hat{T} + \hat{V}$.

3.1.2 The many-body Hamiltonian of a crystal

In a solid, we have many particles composing the quantum system, namely the heavy nuclei of the atoms composing the material, the core electrons tightly bound to the nuclei and the outer valence electrons, the last determining most of the material properties at energy scales typical in condensed matter systems at low-temperatures, ranging from the meV up to the eV. We will consider that the core electrons act simply to screen the potential felt by the valence electrons from the nuclei, treating them as positively charged ions. If the valence number of the atom is Z , then the ion charge is $+Ze$, where e is the elementary charge. The total Hamiltonian of a solid is then the sum of several contributions as

$$\hat{H} = \hat{T}_e + \hat{T}_{\text{ion}} + \hat{V}_{e-\text{ion}} + \hat{V}_{e-e} + \hat{V}_{\text{ion-ion}}, \quad (3.4)$$

where $\hat{T}_e$ is the sum of the kinetic energies coming from the valence electrons, $\hat{T}_{\text{ion}}$ is total kinetic energy of the ions, $\hat{V}_{e-\text{ion}}$ is the potential energy coming from the Coulomb interaction between the outer electrons and the ionic nuclei, $\hat{V}_{e-e}$ is the potential energy coming from the pairwise interaction among the electrons, and finally $\hat{V}_{\text{ion-ion}}$ is the potential energy coming from the interaction among the ions. If we write each contribution using math notation, then, the total many-body Hamiltonian, dropping all the hats and ignoring relativistic effects, is³

$$H = \sum_i \frac{p_i^2}{2m} + \sum_I \frac{P_I^2}{2M_I} + \sum_{i,I} \frac{-Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2} \sum_{I \neq J} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|}, \quad (3.5)$$

if we write m for the electron mass, $\mathbf{r}_i$ and $\mathbf{p}_i$ for the position and momentum of the i -th electron, respectively, while $\mathbf{R}_I$, $\mathbf{P}_I$, M_I and Z_I are the I -th ion position, momentum, mass, and number of valence electrons, respectively. The factor of 1/2 in the last two terms is

³For a moment, we work with units such that $e^2/4\pi\epsilon_0 = 1$. We recover this factor later on.

used to avoid double counting, and the sums are over all pairs of particles (both pairs (i, j) and (j, i) are included in the sum; the same for the ions). This Hamiltonian is written in terms of the real space coordinates of the particles, meaning that in the TISE it acts on a function depending on all the coordinates, called the wavefunction. If the problem is that of a single particle that can be found in the state $|\psi\rangle$, then the wavefunction is defined by $\psi(\mathbf{r}) \equiv \langle \mathbf{r} | \psi \rangle$, with $\langle \psi | \psi \rangle = \int d\mathbf{r} |\psi(\mathbf{r})|^2 = 1$ ⁴. Obviously, we can see that solving for the eigenstates and eigenenergies of this Hamiltonian is a herculean task; some approximations are needed if we want to even aspire to describe the physics of this system. Not only that, but even if we could solve this problem in practice, the wavefunction would have so many variables (of the order of 10^{23}), that there would not be anything useful we could do with so much information. In the next subsections, we simplify our problem until we can approximately describe the physics it contains. We can start already by saying that we work under low enough temperatures and small enough energies, such that the system we are dealing with is in fact a solid—the ions are bound to their neighbors through strong chemical bonds, making the whole material rigid [56].

3.1.3 Born–Oppenheimer approximation

In this chapter, we are concerned with the eigenstate with the lowest energy possible, called the ground state (of the crystal). A possible way to find the ground state, $|\text{GS}\rangle$, of any system, is by applying the variational principle—for a Hamiltonian $\hat{H}$ acting on states on a given Hilbert space $\mathcal{H}$, then for every state $|\psi\rangle \in \mathcal{H}$ the ground-state energy verifies $E_{\text{GS}} \leq \langle \psi | \hat{H} | \psi \rangle$ [68]. The variational method consists in admitting a trial state $|\Psi\rangle$ depending on a given set of system parameters. In a crystal, let us say that the parameters are the ion positions $\{\mathbf{R}_I\}$. Then, we compute the expectation value of the Hamiltonian $\langle \Psi | H | \Psi \rangle \equiv E$, and minimize it with respect to the parameters by solving $\partial_{\mathbf{R}_I} E = 0$ for all $\mathbf{R}_I$. This is the philosophy of the so-called *ab-initio* methods, which will be discussed on passing in this work. Among other properties to be discussed later in this chapter, they permit finding the equilibrium positions of the atomic nuclei. We will simply assume that the arrangement of the ions in the solid is known, for instance, from X-ray experiments [56, 70].

To further narrow down the types of systems under consideration, we will only treat solids where the arrangement of the ions forms a periodic structure, the crystal. Other types of solids can be found in nature, such as amorphous solids, like glasses, or solid materials with rotational symmetry but no translation symmetry, the quasicrystals [71]. Herein, the ions occupy positions in the crystalline structure, also called the lattice, such that the potential felt by the electrons $\hat{V}_{e\text{-ion}}(\mathbf{r})$ from the ions is a periodic function with the periodicity of the lattice. We shall define more precisely what a lattice is in Section 3.2. In this work, we will assume that the ionic degrees of freedom are frozen, where the

⁴What we are saying, is that the state of the system can be written in a representation of basis functions belonging to the Hilbert space of all the square-integrable complex valued functions defined in real space. Being square-integrable means that the integral of the absolute value squared over their domain is one.

ions are stationary, forming a background of positive charge. This is the adiabatic or Born–Oppenheimer approximation [56, 70, 72]. Within this approximation, the electronic and ionic degrees of freedom decouple, and the problem of the electronic structure and excitations can be treated separately from finding the ground state of the lattice and its excitations, also called phonons. Interactions between electrons and phonons are thus disregarded in this work, as throughout this text electronic excitations occur at energy scales higher than those where lattice vibrations are relevant⁵. Therefore, the ions will remain static in their equilibrium positions at all times, and their role is just to act as a periodic potential on the valence electrons that form the electron gas. In the next subsection, we apply one more approximation—the independent electron approximation.

3.1.4 Noninteracting limit

Upon ignoring in the full many-body Hamiltonian (3.5) the contributions from the kinetic energy of the ions and their mutual interaction, and further ignoring the electron–electron interactions, we arrive at the Hamiltonian

$$H = \sum_i \frac{p_i^2}{2m} + V_{e-\text{ion}}(\mathbf{r}). \quad (3.6)$$

If we write the potential term in the previous expression as a sum over all electrons as $V_{e-\text{ion}}(\mathbf{r}) = \sum_i v(\mathbf{r}_i)$ (with v including a sum over all ions), then we have reduced the full many-body problem of Eq. (3.5) to a sum of single-particle Hamiltonians that are uncoupled, which is a noninteracting problem of identical particles. In that case, the problem encompassing all the electrons is separable. The wavefunction including all $\mathcal{N}$ electrons, $\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{\mathcal{N}})$, can be written as a product of all the single-particle wavefunctions $\psi_i(\mathbf{r}_i)$ as $\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{\mathcal{N}}) = \psi_1(\mathbf{r}_1)\psi_2(\mathbf{r}_2)\dots\psi_{\mathcal{N}}(\mathbf{r}_{\mathcal{N}})$, with each single wavefunction satisfying the TISE

$$\left[\frac{p^2}{2m} + v(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r}), \quad (3.7)$$

where ϵ_i represents the energy of a single electron, and the total energy of the system is simply the sum of all energies ϵ_i . We also omit the quantum number for spin, since it is not relevant for the discussion.

In fact, to account for the statistics of the electrons, which are fermions, the wavefunction of all electrons must be antisymmetric upon permutation of any two electrons. Say we permute electron i with electron j , then wavefunction must obey $\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots, \mathbf{r}_{\mathcal{N}}) = -\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots, \mathbf{r}_{\mathcal{N}})$. Therefore, the total wavefunction is the Slater determinant of all single-particle wavefunctions [68], which is the determinant of the matrix that has

⁵An example where the electron–phonon interaction [73] is important is in the Bardeen–Cooper–Schrieffer (BCS) theory of superconductivity; the microscopic theory that explains why some materials can conduct electricity without dissipation at very low temperatures (or very high pressures). For the curious reader, one can consult Refs [74] and [75] for in-depth technical treatments, or [76] for a not as technical discussion.

by entries $\psi_i(\mathbf{r}_j)$, with $i, j = 1, \dots, \mathcal{N}$, along with a factor of $1/\sqrt{\mathcal{N}!}$ for normalization⁶. The noninteracting wavefunctions can then serve as a basis for the interacting problem. With them, we can estimate the ground-state energy of the interacting electron gas in a crystalline solid by using the Slater determinant as a variational wavefunction. For that, one computes the expectation value of the interacting Hamiltonian

$$H = \sum_i \left[\frac{p_i^2}{2m} + v(\mathbf{r}_i) \right] + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \quad (3.8)$$

for the Slater determinant, and minimize it with respect to the single-particle wavefunctions $\psi_i(\mathbf{r}_j)$. This is called the Hartree–Fock (HF) method [76, 77], commonly introduced in introductory condensed matter physics courses as a first attempt in describing the ground state of the interacting electron gas⁷. With the HF method, we transform the interacting problem into an effective single-particle problem described by a Schrödinger equation of the type (3.7), where each electron feels an effective potential due to all the other electrons. There is a much more sophisticated way of doing so, within the framework of DFT, the foundation of *ab-initio* methods to calculate the electronic, mechanic, magnetic, thermal, etc., properties of extended or molecular systems. The electronic properties are quantified by the electronic dispersion—the variation in the electron energy when its momentum $\mathbf{k}$ varies. The electronic dispersion has been hinted at in the previous chapter, in analogy with the photonic dispersion. We can also call the electronic dispersion in an extended solid as electronic structure. Within quantum mechanics, subatomic particles have a dual particle–wave behavior. Therefore, we can perceive traveling electrons as propagating waves, with their wavevector $\mathbf{k}$ given by the de Broglie relation $\mathbf{p} = \hbar\mathbf{k}$. Traditionally, in solid state physics the quantities “momentum” and “wavevector” are used interchangeably, with $\mathbf{k}$ being the predominant notation (quite commonly putting $\hbar = 1$). Therefore, by abuse of language and notation, from here onwards, the symbols $\mathbf{p}/\mathbf{q}/\mathbf{k}$ might denote wavevector instead of momentum without notice, and vice-versa. After all, the conversion between these quantities is simply a multiplication factor. In the next section, we thoroughly discuss the concept of electronic dispersion in crystalline materials.

3.2 Single-Particle Properties of Electrons in Crystals

In the previous section, we have touched upon the concept of electronic dispersion in materials, which pertains to the nontrivial dependence of an electron energy on momentum. Before we address that problem, there are still some concepts we need to clarify. The first is the concept of the lattice. It can be defined as an infinite, but discrete, set of points in

⁶As an example, the determinantal wavefunction for a two-particle system is given by $\Psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2!}} [\psi_1(\mathbf{r}_1)\psi_2(\mathbf{r}_2) - \psi_1(\mathbf{r}_2)\psi_2(\mathbf{r}_1)]$.

⁷A very first attempt in assuming the symmetrized wavefunction, mentioned at the end of the paragraph right before Eq. (3.7). That is called the Hartree method, which is faulty, as it does not account for exchange nor correlation effects.

space, such that all points are equivalent. Regardless of where one stands in the lattice, the landscape is the same. Let us look at an example, the triangular lattice, depicted in Fig. 3.1(a) below. As one can see, this set of points has discrete translational symmetry. We call any translation vector $\mathbf{R}$ that leaves the lattice invariant a lattice vector. In other words, a lattice vector connects any two points in the lattice. We call the smallest possible lattice vectors the primitive lattice vectors. In two dimensions, we need two primitive lattice vectors in order to span the entire lattice. In Fig. 3.1(a), the red vectors denoted by $\mathbf{a}_1$ and $\mathbf{a}_2$ are a possible choice for the primitive lattice vectors. Evidently, the choice is not unique. Therefore, any lattice vector can be written as an integer linear combination of these vectors. Mathematically, we write

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2, \quad \text{with } n_1, n_2 \in \mathbb{Z}. \quad (3.9)$$

The shaded gray area in the same figure represents a portion of the lattice, such that upon all possible translations by a lattice vector, we can generate the entire crystal. We call such a region a unit cell. The unit cell is, of course, not unique—we can choose different shapes and sizes, as long as the whole lattice can be generated by translations of the tile. In Fig. 3.1(a), we display a tile with the smallest possible area. Such a tile is called a primitive unit cell. If a unit cell has a larger area than the primitive one, it can be called a conventional unit cell, and is usually chosen to highlight the symmetries of the crystal.

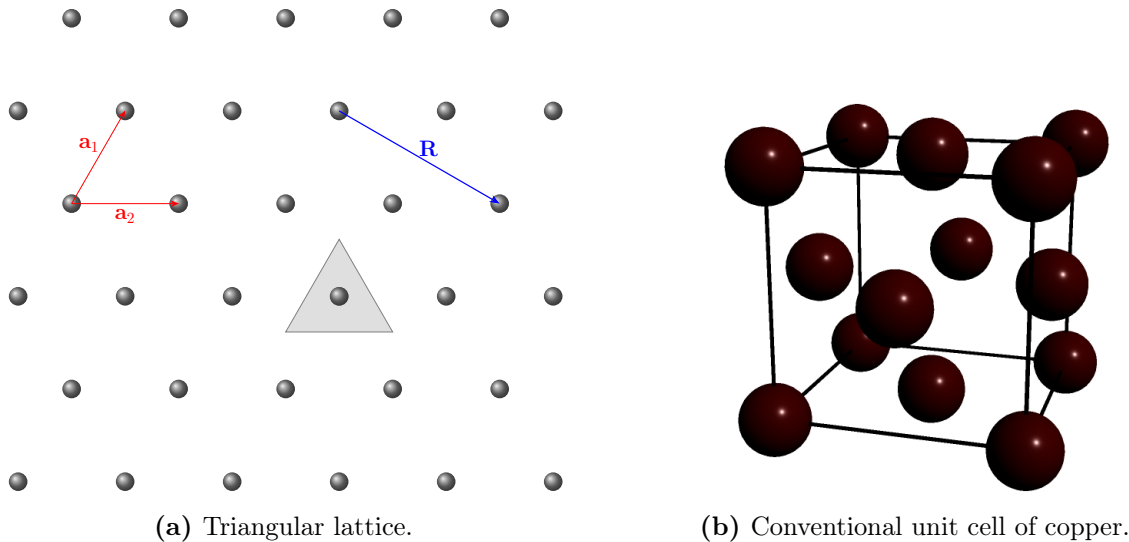


Figure 3.1: Triangular lattice and the unit cell of copper. Subfigure **a** is a schematic of the triangular lattice. The vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are primitive lattice vectors, and $\mathbf{R} = -\mathbf{a}_1 + 2\mathbf{a}_2$ is a (nonprimitive) lattice vector. The shaded gray area represents a possible primitive unit cell, and the lattice extends to infinity along the plane of the page. Subfigure **b** displays the conventional unit cell of copper, which is the region contained inside the cube delimited by the black lines. This subfigure was made using the `Asymptote` programming language, with the help of the `asy-atoms` package [78].

We have defined a lattice as a set of points that are all equivalent. An example that can be found in nature is copper, the chemical element with symbol Cu, which is a metal with

lots of applications in industry, has the conventional unit cell represented in Fig. 3.1(b). Its crystalline structure is called the face-centered cubic lattice, one of the possible fourteen Bravais lattices in three-dimensional (3D) [70]. Nonetheless, we can find in nature many crystalline materials composed of more than one chemical element, therefore we cannot say that all points are equivalent in those materials. We can even have a crystal composed of a single chemical element that cannot be classified as a lattice in the way we defined it. A striking example is graphene [6]; it is not a lattice according to our definition. In the literature, lattices as we defined them are commonly called Bravais lattices⁸. A lattice which is not a Bravais lattice, can be seen as a Bravais lattice plus a basis, or motif, where the unit cell is that of the underlying Bravais lattice, but including the atomic sites that “spoil” the equivalence of all points in the lattice. As an example, there is the hexagonal or honeycomb lattice, where the atomic sites are arranged in a hexagonal pattern. It can be seen as two interpenetrated triangular lattices, each one forming a sublattice. Graphene and hexagonal boron nitride (hBN) can be found, as monolayers, in this configuration, which is depicted in Fig. 3.2. The latter material is very relevant in this work, as we shall see later.

As lattices enjoy translation symmetry, we can guess that momentum is a good quantum number to describe electronic states in these materials. Therefore, it is useful to have a description of the lattice in terms of wavevectors and set a framework to write periodic functions in Fourier series. In other words, we need an equivalent to the lattice in reciprocal space. Let us define the reciprocal lattice as the set of all vectors $\mathbf{G}$ such that $e^{i\mathbf{G}\cdot\mathbf{R}} = 1$, for any lattice vector $\mathbf{R}$ [56, 57, 70, 74]. In two dimensions, any reciprocal lattice vector can be written as

$$\mathbf{G} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2, \quad \text{with } m_1, m_2 \in \mathbb{Z}, \quad (3.10)$$

where $\mathbf{b}_1$ and $\mathbf{b}_2$ are primitive vectors of the reciprocal lattice. They can be obtained by solving the set of equations of the form $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$, with $i, j = 1, 2$. In this way, given the expression for any lattice vector (3.9) and any reciprocal lattice vector (3.10), the condition $e^{i\mathbf{G}\cdot\mathbf{R}} = 1$ is always fulfilled. The real space lattice is composed in general by a Bravais lattice and a basis, but it is always the direct underlying Bravais lattice defining the reciprocal lattice.

We define now the first Brillouin zone (BZ), or simply BZ, as the set of points $\mathbf{k}$ such that $|\mathbf{k}| < |\mathbf{k} - \mathbf{G}|$ for all $\mathbf{G} \neq \mathbf{0}$ [56, 57]. Explained in another way, let us fix a primitive unit cell of the reciprocal lattice, and set the origin inside that cell, say at $\mathbf{k} = \mathbf{G} = \mathbf{0}$. Then, if any point $\mathbf{k}$ in that cell is closer to the origin than to any other point in the reciprocal lattice equivalent to the origin, then that cell is the BZ [77]. Any momentum in the reciprocal space can then be defined by another momentum in the BZ plus a reciprocal lattice vector. We can say that the momentum conservation in a crystal conserves the

⁸We gave two examples in Fig. 3.1, among others. However, there are not infinitely many, since the total number of possible Bravais lattices is finite. In two dimensions there are only five, while in three dimensions there are in total fourteen.

momentum modulo a vector of the reciprocal lattice. Therefore, we can speak of crystal momentum as the relevant quantum number in crystals, instead of momentum [56]. We do not wish for the language used in this thesis to be too pedantic, in detriment of clarity. By (one more) abuse of language, let us speak of momentum when we are in fact dealing with crystal momentum.

In this way, if we are working with a periodic function $f(\mathbf{r})$ with the periodicity of the direct lattice, then it can be expanded in Fourier series as $f(\mathbf{r}) = \sum_{\mathbf{G}} f_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}}$, with the coefficients given by $f_{\mathbf{G}} = 1/\mathcal{V}_{\text{UC}} \int_{\text{UC}} \exp\{i\mathbf{G} \cdot \mathbf{r}\} f(\mathbf{r}) d\mathbf{r}$, an integral over the unit cell, hence the symbol UC for the domain of integration, which has volume $\mathcal{V}_{\text{UC}}$. These concepts will be key for understanding the Bloch theorem and its importance, to be explained in the next subsection.

3.2.1 Bloch Theorem

We have at this point all the necessary tools to state and understand Bloch theorem⁹, named after the physicist Felix Bloch. It has different formulations and ways to be proved. The formal proofs are here omitted, but we provide a short motivation for its statements below. Until it becomes relevant for the discussion, we shall ignore the spin of the electron.

Theorem (Bloch Theorem) *Let H be a Hamiltonian with a potential $V(\mathbf{r})$ periodic in space, $\psi_{\mathbf{k}}(\mathbf{r})$ the eigenfunction of an electron with momentum $\mathbf{k}$ and $\epsilon(\mathbf{k})$ its eigenenergy. Let $\mathbf{R}$ be any lattice vector. Then, the following statements hold and are equivalent:*

- (1) *The wavefunction can be written in terms of a modulated plane wave $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}}(\mathbf{r})$, with the function $u_{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}}(\mathbf{r} + \mathbf{R})$ having the periodicity of the lattice;*
- (2) *The wavefunction $\psi_{\mathbf{k}}(\mathbf{r})$ satisfies $\psi_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}} \psi_{\mathbf{k}}(\mathbf{r})$.*

The eigenfunction $\psi_{\mathbf{k}}(\mathbf{r})$ is known as the Bloch wavefunction, while its periodic part $u_{\mathbf{k}}(\mathbf{r})$ is called the Bloch function. This theorem can be proved by translational symmetry arguments and basic concepts of group theory [72, 76].

We have studied in this subsection the properties of electronic eigenstates in a crystal subject to a periodic potential. The dynamics of the electrons are governed by the non-interacting single-particle Hamiltonian in Eq. (3.6). For that reason, the properties here discussed are said to be at the single-particle level. The state the electron with momentum $\mathbf{k}$ occupies is a single-particle state, also called a Bloch state in the context of solids. Bloch states will serve as a basis for the interacting problem, as we will see in the next chapter. In fact, for a single momentum $\mathbf{k}$, we can have several energy states the electrons can populate. If we label them by an integer n , then a general Bloch state in bracket notation

⁹Its original form is actual for one dimensional (1D) systems, or equivalently, for ordinary differential equations, and is due to the mathematician Gaston Floquet. The very same mathematician of the Floquet formalism of time-periodic Hamiltonians in quantum mechanics [79].

reads $|\psi_{n,\mathbf{k}}\rangle$, or more concisely, $|n, \mathbf{k}\rangle$. Then, the allowed energies are also labeled in the same way as $\epsilon_{n\mathbf{k}}$. For a fixed n , we call the entire set of energies $\epsilon_{n\mathbf{k}}$ with $\mathbf{k}$ varying in the BZ an energy band, or just a band. Then, the set of all bands $\{\epsilon_{n\mathbf{k}}\}$ is called the (electronic) band structure of the material. These bands define the dispersive behavior of electron waves in a crystal. The ground state of the $\mathcal{N}$ -electron system is obtained by filling the bands, starting from the bands with lower energies, until there are no valence electrons left. In the following subsection, we discuss different methods to calculate the band structure.

3.2.2 Electronic band structure: tight-binding and *ab-initio* methods

One central problem in condensed matter physics is to calculate the electronic band structure of a given solid material. This means calculating all the bands at every point $\mathbf{k}$ in the BZ. An electron in a specific band n with crystal momentum $\mathbf{k}$ has kinetic energy $\epsilon_{n\mathbf{k}}$. In analogy with optics, we can call the $\mathbf{k}$ dependence of the energy electronic dispersion, with the wave being now electronic instead of electromagnetic. From here onwards, when referring to the band structure $\{\epsilon_{n\mathbf{k}}\}$, we will always implicitly mean electronic band structure, as there are bands of other kinds (phononic, photonic as we have seen in the previous chapter, etc.). Once the band structure, a ground state characteristic of the material, is determined, one is invited to do post-calculations to find other ground state or excited state properties. But first, we need to better understand the concepts related to the bands of a crystalline solid.

3.2.2.1 Tight-binding method

There is a number of different descriptions for material band structures. Two simple ones are the nearly free electron model and the tight-binding (TB) model. In the first, one assumes the electron moves freely from atom to atom, so that it can be found in a plane wave state with parabolic dispersion, and the crystalline structure is seen as a small perturbation. The electronic dispersion in the case of the nearly free electron model is simply given by $\epsilon_{\mathbf{G}}(\mathbf{k}) = \hbar^2|\mathbf{k} - \mathbf{G}|^2/2m$, with the reciprocal lattice vectors $\mathbf{G}$ labeling the bands. This model is particularly good for standard metals, like aluminum [80], that have a compact or close-packed crystal structure.

In the opposite limit, where the crystal is not compact and the atoms are further apart, electrons have more difficulty in moving from atomic site to atomic site, and then we say that they are tightly-bound to the atoms. We usually associate this type of crystals with materials with poor conducting properties, those presenting a bandgap—the range of forbidden energies for which propagating electronic states are not allowed¹⁰. These can be classified as insulators or semiconductors, depending on how the size of the bandgap compares with the thermal energy, as we have explained before in Section 2.3.2.

¹⁰This is in analogy with the concept of the photonic bandgap in the context of photonics. We have seen a photonic dispersion with a bandgap in Fig. 2.4

Within this formalism, the Bloch wavefunctions are linear combinations of the atomic orbitals, instead of a superposition of plane waves. This is called the linear combination of atomic orbitals method or approximation. In math notation, we write for the wavefunction of an electron in the Bloch state of band n with momentum $\mathbf{k}$

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \phi_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i), \quad (3.11)$$

where the sum in $\mathbf{R}$ is over the vectors of the lattice (not over all the atomic sites), or over all N unit cells, i is an index that labels the atomic species at the position $\mathbf{t}_i$ in the motif, and α is an index that labels the type of the atomic orbital ($1s, 2s, 2p, 3s, \dots$). Usually, we sum only over the atomic orbitals in the outer shells of the atom that are populated by the valence electrons. This is justifiable, as the valence electrons are the only ones that participate in the physics we are studying, as explained in the beginning of Section 3.1.2.

We introduce now the concept of Bloch Hamiltonian, a very useful one in the TB method, and in this work overall. We start by rewriting the Bloch state in Eq. (3.11)

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \Phi_{i\alpha}(\mathbf{k}, \mathbf{r}), \quad (3.12)$$

where we defined the new function $\Phi_{i\alpha}(\mathbf{k}, \mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{R}} \phi_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i)$. The Bloch state (3.12) is an eigenstate of the Hamiltonian with energy ϵ (omitting band and momentum indices for now):

$$H\psi_{n,\mathbf{k}}(\mathbf{r}) = \epsilon \psi_{n,\mathbf{k}}(\mathbf{r}) \Leftrightarrow \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} H\Phi_{i\alpha}(\mathbf{k}, \mathbf{r}) = \epsilon \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \Phi_{i\alpha}(\mathbf{k}, \mathbf{r}). \quad (3.13)$$

Now, we multiply by $\Phi_{j\beta}^*(\mathbf{k}, \mathbf{r})$ and integrate in space over the volume of the material $\mathcal{V}$ as

$$\sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \underbrace{\int_{\mathcal{V}} d\mathbf{r} \Phi_{j\beta}^*(\mathbf{k}, \mathbf{r}) H\Phi_{i\alpha}(\mathbf{k}, \mathbf{r})}_{H_{j\beta,i\alpha}(\mathbf{k})} = \epsilon \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \underbrace{\int_{\mathcal{V}} d\mathbf{r} \Phi_{j\beta}^*(\mathbf{k}, \mathbf{r}) \Phi_{i\alpha}(\mathbf{k}, \mathbf{r})}_{S_{j\beta,i\alpha}(\mathbf{k})} \quad (3.14)$$

to obtain the eigenvalue problem

$$\sum_{i\alpha} H_{j\beta,i\alpha}(\mathbf{k}) C_{i\alpha}^{n\mathbf{k}} = \epsilon \sum_{i\alpha} S_{j\beta,i\alpha}(\mathbf{k}) C_{i\alpha}^{n\mathbf{k}} \Leftrightarrow H(\mathbf{k}) \mathbf{C}^{n\mathbf{k}} = \epsilon S(\mathbf{k}) \mathbf{C}^{n\mathbf{k}}, \quad (3.15)$$

where we defined the arrays $\mathbf{C}^{n\mathbf{k}}$ with entries $(\mathbf{C}^{n\mathbf{k}})_{i\alpha} = C_{i\alpha}^{n\mathbf{k}}$, and the matrices $H(\mathbf{k})$ and $S(\mathbf{k})$ with matrix elements

$$H_{j\beta,i\alpha}(\mathbf{k}) = \langle \Phi_{j\beta}(\mathbf{k}, \mathbf{r}) | H | \Phi_{i\alpha}(\mathbf{k}, \mathbf{r}) \rangle \quad \text{and} \quad S_{j\beta,i\alpha}(\mathbf{k}) = \langle \Phi_{j\beta}(\mathbf{k}, \mathbf{r}) | \Phi_{i\alpha}(\mathbf{k}, \mathbf{r}) \rangle, \quad (3.16)$$

respectively. We call the S matrix the overlap matrix, and for the matrix elements we can write

$$\begin{aligned}
 S_{j\beta,i\alpha}(\mathbf{k}) &= \langle \Phi_{j\beta}(\mathbf{k}, \mathbf{r}) | \Phi_{i\alpha}(\mathbf{k}, \mathbf{r}) \rangle = \frac{1}{N} \int_{\mathcal{V}} d\mathbf{r} \sum_{\mathbf{R}'/\mathbf{R}} e^{-i\mathbf{k} \cdot (\mathbf{R}' - \mathbf{R})} \phi_{\beta}^*(\mathbf{r} - \mathbf{R}' - \mathbf{t}_j) \phi_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) = \\
 &= \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \langle \phi_{j\beta}(\mathbf{r}) | \phi_{i\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) \rangle = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} s_{j\beta,i\alpha}(\mathbf{R}),
 \end{aligned} \tag{3.17}$$

where from the 1st to the second line we fixed $\mathbf{R}' = \mathbf{0}$, which is valid since both sums are over all vectors of the Bravais lattice. In matrix form, we can write

$$S(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} s(\mathbf{R}). \tag{3.18}$$

Herein, we typically work in a basis of orthogonal orbitals, such that any two different orbitals do not overlap. This holds in particular for the case where the localized orbitals are not atomic orbitals, but Wannier functions instead, which are orthogonal by construction¹¹. This makes $S(\mathbf{k}) = \mathbf{1}$, simplifying the eigenvalue problem (3.15). Following an analogous procedure for the Hamiltonian, one obtains

$$H(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} H(\mathbf{R}), \tag{3.19}$$

where $\mathbf{k}$ denotes crystal momentum as usual, the sum is over all vectors of the Bravais lattice $\mathbf{R}$, and $H(\mathbf{R})$ is the real space Hamiltonian at the site $\mathbf{R}$, commonly called a Fock matrix, and has matrix elements $H_{j\beta,i\alpha}(\mathbf{R}) = \langle \phi_{j\beta}(\mathbf{r} - \mathbf{t}_j) | H | \phi_{i\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) \rangle$. These matrix elements are called transfer integrals. If $\mathbf{R} \neq \mathbf{0}$, then they are a measure for how likely it is for an electron to hop from one orbital to another one in a different unit cell, and are called hopping parameters in the jargon of the TB approach. The matrix $H(\mathbf{k})$ presented just now is the so-called Bloch Hamiltonian. Its eigenvalues form the band structure $\{\epsilon_{n\mathbf{k}}\}$, and its eigenvectors are the arrays (or vectors) $\mathbf{C}^{n\mathbf{k}}$, whose elements are usually called tight-binding coefficients.

The Bloch Hamiltonian deserves a pertinent technical remark, related to the sum in Eq. (3.19) and the argument of the imaginary exponential. In Eq. (3.19) we use the Fock matrices summing over unit cells with positions $\mathbf{R}$. There is, in fact, an alternative definition for the Bloch Hamiltonian, where the sum is over all lattice sites¹². The first case is called the “periodic” form or the “lattice-gauge” form, as it has the periodicity of the reciprocal lattice. It means that $H(\mathbf{k}) = H(\mathbf{k} + \mathbf{G})$ for any reciprocal lattice vector $\mathbf{G}$, but it is not unique, as it depends on the choice of the unit cell. The alternative case is called the “canonical” form, or the “atomic-gauge” form, and it is unique, but in general $H(\mathbf{k}) \neq H(\mathbf{k} + \mathbf{G})$ for a reciprocal lattice vector $\mathbf{G}$. If there is only one atom per unit cell,

¹¹The Wannier functions result from transforming, by linear superpositions, the basis of the Bloch wavefunctions (3.11) into a basis of localized functions [81].

¹²The code we will work with later on in Chapter 6 supports both types of Bloch Hamiltonians. This is why we want to introduce these concepts now.

which is the case of Bravais lattices, the two forms coincide, but if the unit cell contains at least two non-equivalent sites, one can still go from one gauge to the other through a unitary transformation. A unitary transformation U , is a transformation that verifies $U^{-1} = U^\dagger$, where the dagger symbol $\dagger$ stands for transpose and complex conjugate. One can also say that $U^\dagger$ is the Hermitian conjugate of U . Nonetheless, in all cases in this work the results do not change with the chosen gauge; problems may arise, though, when dealing with geometric phases or topological invariants [82]. We present now a simple TB model for a key two-dimensional (2D) material in this work, monolayer hBN.

Monolayer hexagonal boron nitride: a minimal tight-binding model

In its multilayer form, hBN is a rather interesting material. For instance, it is hyperbolic in the mid-infrared (IR)—besides its refractive index being anisotropic, the components of the permittivity tensor pertaining to the off-plane and in-plane directions have opposite signs [83]. As a consequence, its isofrequency surfaces, the surfaces in momentum space over which the frequency is constant, are hyperboloids instead of spheres as in isotropic materials. The isofrequency surfaces, or contours, help visualise what happens to the polaritonic dispersion, which is relevant for developing applications in nanophotonics. This makes hBN a promising material for photonic applications in the mid-IR, such as hyperlensing. In this same range, it presents hyperbolic phonon–polaritons [84], which have been already experimentally observed in nanostructures [85], and even in its monolayer form [86, 87]. In similarity with the plasmon–polariton discussed in the previous chapter, the phonon–polariton is the result of the hybridization between the excitation of the lattice, the phonon, and the photon. When it comes to transport properties, it is a wide-bandgap semiconductor, with an indirect gap of around 5.955 eV [88]. However, in its monolayer limit, the bandgap is direct [89] and slightly higher, around 6.08 ± 0.2 eV [90]. We emphasize that these are experimental observations.

Theoretically, monolayer hBN can be described by a minimal TB model with staggered on-site energies and a single hopping amplitude between nearest-neighbours. This material is composed of boron and nitrogen atoms, alternated in a lattice arranged in hexagons, like in Fig. 3.2a below. In the language of Section 3.2, it is a triangular lattice with a basis, having the unit cell two atoms (shaded gray area in Fig. 3.2a). A typical choice for its BZ is the one depicted in Fig. 3.2b, with a hexagonal shape.

In Fig. 3.2a, if a is the length of the chemical bonds, we have for the primitive lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}), \quad \text{and} \quad \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \quad (3.20)$$

and the nearest-neighbour positions with respect to the blue sites are given by

$$\boldsymbol{\delta}_1 = a(1, 0), \quad \boldsymbol{\delta}_2 = \frac{a}{2}(-\sqrt{3}, -1), \quad \text{and} \quad \boldsymbol{\delta}_3 = \frac{a}{2}(-\sqrt{3}, 1), \quad (3.21)$$

which are represented by the green vectors, and the primitive reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ in blue in Fig 3.2b are given by

$$\mathbf{b}_1 = \left(\frac{2\pi}{3a}, \frac{2\pi}{\sqrt{3}a} \right) \quad \text{and} \quad \mathbf{b}_2 = \left(\frac{2\pi}{3a}, -\frac{2\pi}{\sqrt{3}a} \right). \quad (3.22)$$

The coordinates of the high-symmetry points K and K' are given by

$$\mathbf{K} = \left(\frac{2\pi}{3a}, \frac{2\pi}{3\sqrt{3}a} \right) \quad \text{and} \quad \mathbf{K}' = \left(\frac{2\pi}{3a}, -\frac{2\pi}{3\sqrt{3}a} \right). \quad (3.23)$$

A minimal TB Hamiltonian for hBN, in second quantization¹³, reads

$$H = \sum_i E_A a_i^\dagger a_i + \sum_j E_B b_j^\dagger b_j - t \sum_{\langle ij \rangle} (a_i^\dagger b_j + b_j^\dagger a_i), \quad (3.24)$$

where the operator $a_i^\dagger/a_i$ creates/annihilates an electron in the A sublattice at the position $\mathbf{r}_i$, and $b_j^\dagger/b_j$ creates/annihilates an electron in the B sublattice at the position $\mathbf{r}_j$. The symbol $\langle ij \rangle$ under the sum symbol means that the sum is over nearest-neighbours. It can be rewritten in a more comprehensive form

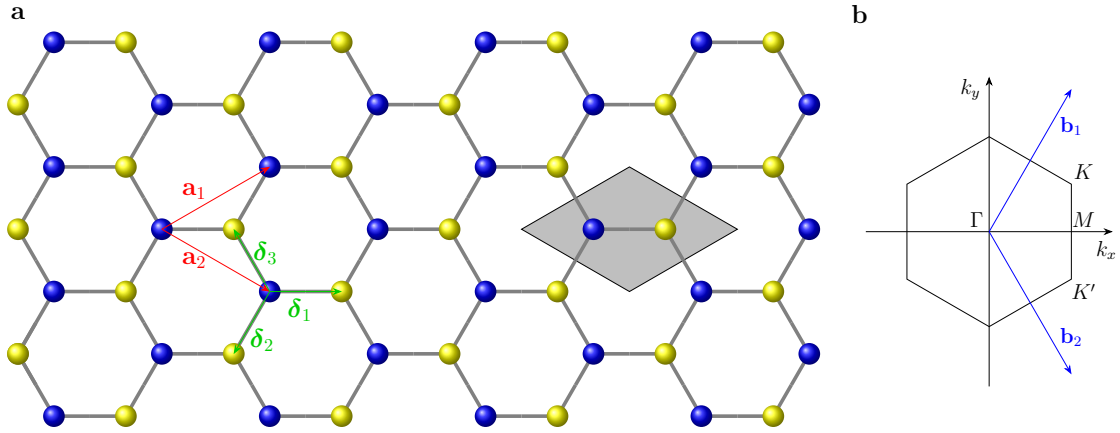


Figure 3.2: hBN lattice and Brillouin Zone. In panel **a**, blue and yellow atoms represent different site types in the hexagonal lattice. The blue atoms and yellow atoms correspond to boron and nitrogen, respectively, in hBN. The vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are possible primitive lattice vectors, and vectors $\boldsymbol{\delta}_{i=1,2,3}$ are vectors connecting the blue sites to their nearest-neighbours. The shaded gray area represents a possible choice for the primitive unit cell. In panel **b**, the BZ of the hexagonal lattice is represented together with the primitive reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$.

¹³In the second quantization formalism, we promote a ket state $|\alpha\rangle$ to an operator $\hat{a}_\alpha^\dagger$ that creates a particle in the state α . Then, a bra corresponds to an annihilation operator $\hat{a}_\alpha$, destroying a particle in the state α . These operators act on states living in the Fock space, and obey commutation/anti-commutation relations for bosons/fermions. In symbols, denoting $[\hat{O}_1, \hat{O}_2] = \hat{O}_1\hat{O}_2 - \hat{O}_2\hat{O}_1$ and $\{\hat{O}_1, \hat{O}_2\} = \hat{O}_1\hat{O}_2 + \hat{O}_2\hat{O}_1$ for the commutation and anticommutation operations, respectively, we have $[\hat{a}_\alpha, \hat{a}_{\alpha'}^\dagger] = \delta_{\alpha,\alpha'}$ and $[\hat{a}_\alpha, \hat{a}_{\alpha'}] = [\hat{a}_\alpha^\dagger, \hat{a}_{\alpha'}^\dagger] = 0$ for bosons, while the relations $\{\hat{a}_\alpha, \hat{a}_{\alpha'}^\dagger\} = \delta_{\alpha,\alpha'}$ and $\{\hat{a}_\alpha, \hat{a}_{\alpha'}\} = \{\hat{a}_\alpha^\dagger, \hat{a}_{\alpha'}^\dagger\} = 0$ are valid for fermions. The states that belong to the Fock space describe how many particles populate each ket state living in its corresponding single-particle Hilbert space. The second quantization formalism is also called the occupation number representation [91].

$$H = \sum_i E_A a_i^\dagger a_i + \sum_j E_B b_j^\dagger b_j - t \sum_{i \in A} \sum_{\boldsymbol{\delta}} \left(a_i^\dagger b_{i+\boldsymbol{\delta}} + b_{i+\boldsymbol{\delta}}^\dagger a_i \right). \quad (3.25)$$

Performing the change of basis

$$a_i^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} a_{\mathbf{k}}^\dagger, \quad \text{analogously for } b_j^\dagger, \quad (3.26)$$

and simplifying, we arrive at

$$H = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger H(\mathbf{k}) \Psi_{\mathbf{k}}, \quad (3.27)$$

where we can define the Bloch Hamiltonian $H(\mathbf{k})$, the vector $\Psi_{\mathbf{k}}$ and the function $\Delta_{\mathbf{k}}$ as

$$H(\mathbf{k}) = \begin{pmatrix} E_A & -t\Delta_{\mathbf{k}} \\ -t\Delta_{\mathbf{k}}^* & E_B \end{pmatrix}, \quad \Psi_{\mathbf{k}} = \begin{pmatrix} a_{\mathbf{k}} \\ b_{\mathbf{k}} \end{pmatrix}, \quad \Delta_{\mathbf{k}} = \sum_{\boldsymbol{\delta}} e^{-i\mathbf{k} \cdot \boldsymbol{\delta}}. \quad (3.28)$$

Recalling the vectors $\boldsymbol{\delta}$ from Eq. (3.21), we have

$$\Delta_{\mathbf{k}} = e^{-ik_x a} \left[1 + 2e^{i3k_x a/2} \cos\left(\frac{\sqrt{3}}{2} k_y a\right) \right]. \quad (3.29)$$

The dispersion relation for hBN is then given by [92]

$$\epsilon_{\pm}(\mathbf{k}) = E_0 \pm \frac{1}{2} \sqrt{E_g^2 + 4t^2 + 16t^2 \cos\left(\frac{3a}{2} k_x\right) \cos\left(\frac{\sqrt{3}a}{2} k_y\right) + 16t^2 \cos^2\left(\frac{\sqrt{3}a}{2} k_y\right)}, \quad (3.30)$$

where $E_0 = (E_A + E_B)/2$ is the energy reference (which is arbitrary) and $E_g = E_A - E_B$ is the bandgap, $+$ for the top, or conduction band, and $-$ for the bottom, or valence band. One can see clearly the opening of a gap (in the corners of the Brillouin zone K and K') in Figure 3.3. Even though identical to graphene in its crystal structure, contrary to it, monolayer hBN is an insulator as we have said before. The parameters of the TB model for hBN we use are $\hbar v_F = 5.06 \text{ eV \AA}$ and $E_g = 3.92 \text{ eV}$, and they are taken from Ref. [92]. This value for the bandgap comes from DFT calculations previously done in Ref. [92], within the generalized gradient approximation (GGA) with the Perdew–Burke–Ernzerhof functional [93]. It gives the right curvature, or effective mass, for the bands at the K and K' points in the BZ. But, it is known in the literature, that the bandgap using approximate schemes for the exchange–correlation effects is underestimated [94]. A better estimate for the bandgap from DFT calculations can be obtained by using functionals that better approximate the exchange–correlation effects, like hybrid ones. We will come back to this point later in this text, after better understanding the DFT formalism in subsection 3.2.2.2 and discussing these details in more depth.

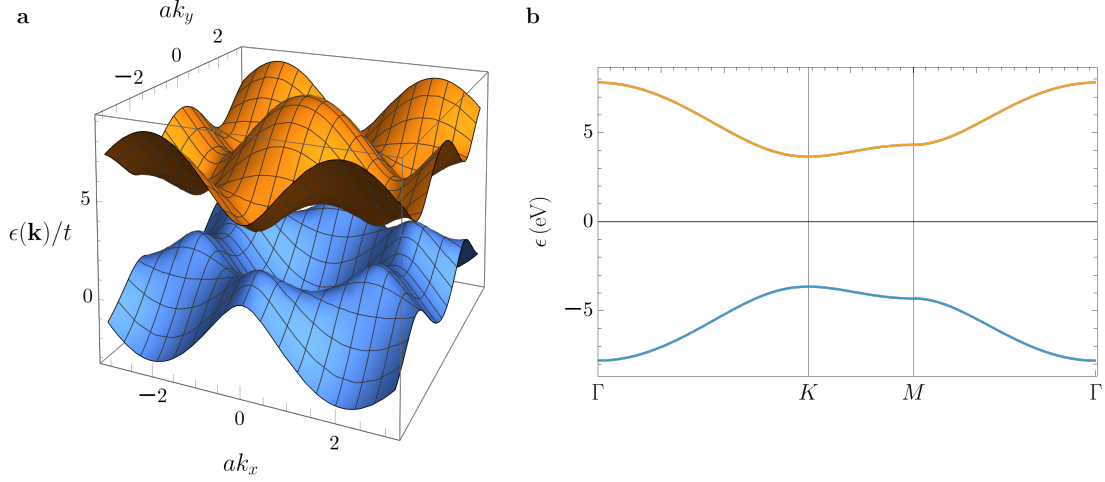


Figure 3.3: Band structure of monolayer hBN in the tight-binding approximation. In panel **a** we display a 3D representation of the bands, which have a parabolic dispersion near the corners of the BZ. In **b**, we plot the bands along the high-symmetry line $\Gamma - K - M - \Gamma$. We have used model parameters distinct from the ones from the literature, otherwise the band structure in the 3D plot would be harder to visualize.

Sometimes, a low-energy approximation suffices for describing the material at hand. Below, we do an analysis of the low-energy limit, or continuum limit, of monolayer hBN. Linearizing (3.29) with respect to momentum around the momentum $\mathbf{K} = (2\pi/3a, 2\pi/3\sqrt{3}a)$, and defining the relative momentum $\mathbf{q}$ by $\mathbf{k} = \mathbf{K} + \mathbf{q}$, with $|\mathbf{q}| \ll |\mathbf{K}|$, we have

$$\Delta_{\mathbf{K}+\mathbf{q}} \simeq -ie^{-i\frac{2\pi}{3}a} \frac{3a}{2} (q_x + iq_y) \rightarrow -\frac{3a}{2} (q_x + iq_y). \quad (3.31)$$

Replacing now in the Bloch Hamiltonian, we define the following low-energy Hamiltonian

$$\mathcal{H}_K \equiv H(\mathbf{K} + \mathbf{q}) \simeq \frac{3at}{2\hbar} (q_x \sigma_x - q_y \sigma_y) + M\sigma_z = \hbar v_F \bar{\mathbf{q}} \cdot \boldsymbol{\sigma} + M\sigma_z, \quad (3.32)$$

where M is half the bandgap, σ_x , σ_y and σ_z are the Pauli matrices, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$ is Pauli matrices vector, v_F is the Fermi velocity

$$v_F = \frac{3at}{2\hbar}, \quad (3.33)$$

and

$$\bar{\mathbf{q}} = (q_x, -q_y, 0). \quad (3.34)$$

Following the same procedure, but for the $\mathbf{K}'$ point, we obtain

$$\mathcal{H}_{K'} \equiv H(\mathbf{K}' + \mathbf{q}) = \hbar v_F \mathbf{q} \cdot \boldsymbol{\sigma} + M\sigma_z, \quad (3.35)$$

where now

$$\mathbf{q} = (q_x, q_y, 0). \quad (3.36)$$

So we can write

$$\Delta_{\tau\mathbf{K}+\mathbf{q}} \simeq -\frac{3a}{2}(q_x + i\tau q_y), \quad (3.37)$$

where $\tau = \pm 1$, with $\tau = 1$ for the K' point, also called K' valley, and $\tau = -1$ for the K point or valley. Thus, we say that τ represents the valley degree of freedom. In this way, the low-energy dispersion relation reads

$$E_{\pm}(\mathbf{q}) = \pm \sqrt{E_g^2 + \hbar^2 v_F^2 q^2}, \quad (3.38)$$

+ for the conduction band, and – for the valence band. One can see why the dispersion relation near the K and K' valleys look like parabolas for low energies; the dispersion is quadratic in the vicinity of those points, which can be seen if we further expand the previous expression up to leading order in $q = |\mathbf{q}|$. The energy spectrum has valley degeneracy, since the spin–orbit coupling (SOC), a contribution to the Hamiltonian which couples the spin $\vec{S}$ with the orbital angular momentum $\vec{L}$, in this system is negligible, since boron and nitrogen are very light elements. This is in contrast with other materials composed of heavier chemical elements, such as molybdenum disulfide, MoS_2 . We shall analyze this material in more detail in Chapter 6.

We can as well compute the effective masses of the electrons with momentum in the vicinity of $\mathbf{K}$ (or $\mathbf{K}'$), through

$$m^* = \left(\frac{1}{\hbar^2} \frac{d^2 E_{\pm}(q)}{dq^2} \Big|_{q=0} \right)^{-1} = \pm \frac{E_g}{2v_F^2}, \quad (3.39)$$

For later use, we compute the eigenvectors of the low-energy Hamiltonian. We will use now a different, but equivalent, configuration for the lattices (rotated by 30 degrees). The eigenvectors, also called spinors, are

$$\mathbf{C}^{v\mathbf{q}} = \begin{pmatrix} \tau e^{-i\tau\theta_q} \sin \frac{\xi_q}{2} \\ -\cos \frac{\xi_q}{2} \end{pmatrix} \quad \text{and} \quad \mathbf{C}^{c\mathbf{q}} = \begin{pmatrix} \cos \frac{\xi_q}{2} \\ \tau e^{-i\tau\theta_q} \sin \frac{\xi_q}{2} \end{pmatrix}, \quad (3.40)$$

for the valence band and for the conduction band, respectively, where

$$\theta_q = \arctan\left(\frac{q_y}{q_x}\right) \quad \text{and} \quad \xi_q = \arctan\left(\frac{2\hbar v_F q}{E_g}\right). \quad (3.41)$$

Even though the TB Hamiltonian for hBN provides a simple, yet insightful theoretical tool, we still need somehow to use realistic values for the parameters t and E_g . These can be obtained through fitting the TB band structure to a more realistic one, typically obtained through the first-principle methods, that we describe in the next section.

3.2.2.2 *Ab-initio* methods

If we wish to describe the electronic structure of a crystal, we are faced with the stationary many-body Schrödinger equation

$$H\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N), \quad (3.42)$$

where H is the many-body Hamiltonian in Eq. (3.5), and $\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$ is the many-body wavefunction. As previously stated, there is no hope in solving the previous equation for the many-body wavefunction. That was a motivation for introducing the nearly-free electron model and the TB model. But then, we are faced with the shortcomings of these models—they describe approximate opposing limits, and provide no way of determining the model parameters.

To overcome this issue, we have to take a different look into the many-body problem. Quite surprisingly, it is possible to describe a ground state of a crystal in terms of a function of a single variable, which is the electronic density $n(\mathbf{r})$, instead of dealing with 10^{23} variables in the many-body wavefunction. Physicists hypothesized that the ground-state energy could be written as a functional of the electronic density. This statement, which is a consequence of the first Hohenberg–Kohn theorem, was put in rigorous grounds by Hohenberg and Kohn [95], the founders of DFT. Together with the second Hohenberg–Kohn theorem, they constitute the pillars of DFT. The two theorems are very strong statements, that we proclaim below.

Theorem (Hohenberg–Kohn Theorem I) *The properties of any physical system, at zero temperature, are completely determined by its ground state electronic density. This statement is a consequence of the following claims:*

- (1) *The mapping from the potential $V_{e-\text{ion}}(\mathbf{r})$ to the electronic ground state density $n(\mathbf{r})$ is one-to-one;*
- (2) *The many-body wavefunction of the ground state of a system is solely determined by its many-body Hamiltonian, which in turn is determined solely by the potential $V_{e-\text{ion}}(\mathbf{r})$ (and the number of electrons N);*
- (3) *Per (1) and (2), the ground state density $n(\mathbf{r})$ determines uniquely the many-body wavefunction, and consequently, the ground and excited states of the system.*

As said before, this theorem implies that the ground-state energy is a functional of the electronic density, $E[n(\mathbf{r})]$. By application of the variational principle, we can state the second Hohenberg–Kohn theorem below.

Theorem (Hohenberg–Kohn Theorem II) *The ground state density of a system minimizes the energy functional $E[n(\mathbf{r})]$. The ground state can therefore be found by minimizing the ground-state energy functional with respect to the density.*

The two Hohenberg–Kohn theorems contextualize the problem of finding the electronic structure of any material using *ab-initio* methods. Nonetheless, even though the two theorems are powerful claims, they do not provide a practical method of finding the ground

state density, and therefore, the properties of the system. In order to do that, Kohn and Sham [96] proposed a method, called the Kohn–Sham scheme. Without going into much detail, the Kohn–Sham scheme is a self-consistent method, where the full interacting many-body problem is mapped into an auxiliary noninteracting one, but with the same ground state density as the initial problem.

The many-body effects are encoded in a contribution to the ground-state energy functional, called the exchange–correlation functional, denoted by $E_{xc}[n(\mathbf{r})]$. Even though this functional is universal, its exact analytical form (if it exists) is unknown. Therefore, different degrees of approximation are used, like the local-density approximation (LDA), or the GGA, the latter being a nonlocal approximation. It is well known that traditional DFT suffers from a fundamental problem, called the bandgap problem. In many materials, DFT calculations underestimate the bandgap, even drastically in some cases. In order to improve the outcome from DFT, there are some schemes that go beyond the Kohn–Sham one. One idea is to use hybrid functionals, consisting in mixing an approximated exchange–correlation functional, say LDA or GGA, with the exact Fock exchange. For a thorough treatment of these subjects, we refer the reader to Refs. [72, 97]. Later on in this work, we will mention the use of hybrid functionals in the first-principle calculations necessary for the framework developed for the screening in 2D materials, pertaining to Chapter 6. Our implementation for the screening is actually based on key concepts of the next section, where we present briefly the many-body perturbation theory formalism, which can also be used to predict electronic properties of materials.

The discussions in this subsection and in the next section, are in part adapted from lecture notes authored by Niels Walet for the course “Electrons in Solids”, offered at the 2023 edition of the “Physics by the Lake” summer school.

3.3 Many-Particle Description of Solids

Up until now, we have been discussing essentially single-particle properties of solids. Even though matter is composed of many particles, the single-particle problem serves as a good basis for the full many-body problem. In the single-particle picture, pairwise interactions are either discarded altogether, or treated at the mean-field level. Doing this can be a good starting point to study material properties, but it leaves out emergent phenomena coming from interactions. Just like in high-energy physics, where particles can be created or destroyed, in condensed matter some effects can only be described by introducing the concept of quasiparticle—a composite state coming from the coupling between different or identical particles in a solid, that can explain ground state or excited state properties we could not understand otherwise. For that, we need many-body perturbation theory formalism, which is the non-relativistic cousin of quantum field theory. As quasiparticles can be created or destroyed, the second quantization formalism, also called the canonical quantization formalism, is necessary. It is extremely useful in condensed matter, not only for single-particle descriptions of matter (looking at our example of a TB model in Section 3.2.2.1), but also

to incorporate interactions in the model Hamiltonian. Within many-body perturbation theory, the single-particle problem without interactions is assumed to be already solved (through the methods discussed in Section 3.2.2.2, most frequently), and the interactions are treated as a perturbation. Apart from a couple of simple models, a full *ab-initio* treatment of the full interacting many-body problem beyond perturbation theory is not possible. Nonetheless, the methods of many-body perturbation theory can explain exotic physical phenomena in condensed matter, such as a microscopic description for superconductivity as explained by the BCS theory, the Fermi liquid theory, quantum magnetism, among others.

In this section, we discuss the main concepts needed for future parts of this work, but in a briefly manner. A detailed discussion of this topic would be too technical for this thesis.

3.3.1 Many-body perturbation theory and the *GW* approximation

In physics, the concept of Green function is very important, with applications throughout many fields. In the context of condensed matter physics, its role in many-body perturbation theory is ubiquitous. The Green function, or propagator as in particle physics, is a quantity that helps providing a formal solution to the Schrödinger equation for the interacting problem.

In the context of many-body perturbation theory, working with units such that $\hbar = 1$, it can be defined as

$$\left[i\partial_t - \hat{H} \right] \mathcal{G}(\mathbf{r}\sigma t; \mathbf{r}'\sigma' t') = \delta(t - t')\delta(\mathbf{r} - \mathbf{r}')\delta_{\sigma\sigma'} , \quad (3.43)$$

if we represent the second-quantized Hamiltonian operator $\hat{H}$ in real space, and σ representing the spin degree of freedom. We keep the hat for operators in the canonical quantization formalism, and their respective first-quantized version without the hat. The Hamiltonian is assumed to have two parts, a noninteracting part $\hat{H}_0$, and an interaction part $\hat{V}$. In the real space basis representation, the total Hamiltonian reads

$$\hat{H} = \sum_{\sigma} \int d\mathbf{r} \, \hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}) H_0 \hat{\psi}_{\sigma}(\mathbf{r}) + \frac{1}{2} \sum_{\sigma, \sigma'} \int d\mathbf{r} \int d\mathbf{r}' \, \hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}) \hat{\psi}_{\sigma'}^{\dagger}(\mathbf{r}') V(\mathbf{r}, \mathbf{r}') \hat{\psi}_{\sigma'}(\mathbf{r}') \hat{\psi}_{\sigma}(\mathbf{r}) , \quad (3.44)$$

where the interaction potential $V(\mathbf{r}, \mathbf{r}')$ is the unscreened Coulomb interaction, if we assume that it is the only interaction in play. If the interaction allows spin flipping, then the potential should have two spin indices as well. The operators $\hat{\psi}_{\sigma}^{\dagger}(\mathbf{r})$ and $\hat{\psi}_{\sigma}(\mathbf{r})$ create and annihilate particles with spin σ at position $\mathbf{r}$, respectively. They are commonly called the field operators.

It is useful to also define the noninteracting Green function $\mathcal{G}^0$, which obeys the equation

$$\left[i\partial_t - \hat{H}_0 \right] \mathcal{G}^0(\mathbf{r}\sigma t; \mathbf{r}'\sigma' t') = \delta(t - t')\delta(\mathbf{r} - \mathbf{r}')\delta_{\sigma\sigma'} . \quad (3.45)$$

The noninteracting Green function can be written in the frequency domain as

$$\mathcal{G}^0 = \frac{1}{\omega - H_0} \equiv \text{---}\overleftarrow{\text{---}}, \quad (3.46)$$

if we use a compact notation by omitting the basis, and represent it as a single arrowed line using Feynman diagrams notation as represented in the right-hand side of the equation. In the Feynman diagrammatic jargon, it is also called the free fermion propagator. If we want to write something similar for the interacting Green function, we need to introduce the concept of self-energy Σ . Before that, we need to say that the introduction of the two-body interaction term in the Hamiltonian makes the interacting Green function $\mathcal{G}$ an infinite sum containing an infinite number of diagrams. Each diagram has two external free fermionic lines $\mathcal{G}^0$, connected by compositions of free propagator lines coming from $\mathcal{G}^0$, with interaction lines stemming from the screened potential W , which in diagrammatic representation is given by a double wavy line as

$$W \equiv \text{~~~~~} . \quad (3.47)$$

The self-energy, Σ , is defined as the sum of all irreducible diagrams appearing in the expansion of $\mathcal{G}$, without the external fermion lines. This results in writing for the interacting Green function

$$\mathcal{G} = \frac{1}{\omega - H_0 - \Sigma(\omega)} \equiv \text{====}\overleftarrow{\text{====}}, \quad (3.48)$$

a double arrowed line in diagram notation, and is called the full or interacting fermion propagator.

If in the previous expression we multiply both sides by the denominator from the left, we get $[\omega - H_0]\mathcal{G} - \Sigma(\omega)\mathcal{G} = (\mathcal{G}^0)^{-1}\mathcal{G} - \Sigma(\omega)\mathcal{G} = 1$. Multiplying by the Green function $\mathcal{G}^0$ from the left, simplifying, considering the $\mathbf{r}$ -space basis and going back to the time domain, we get a Dyson equation for the many-body Green function that reads

$$\boxed{\mathcal{G}(b, a) = \mathcal{G}^0(b, a) + \int d1 d2 \mathcal{G}^0(b, 1) \Sigma(1, 2) \mathcal{G}(2, a)}, \quad (3.49)$$

where a digit or letter is an abbreviation for all the degrees of freedom. With symbols, it means $i = (\mathbf{r}_i, \sigma_i, t_i)$, and $\int di = \sum_{\sigma_i} \int d\mathbf{r}_i \int dt_i$. This relation can also be written in diagrammatic notation as

$$\text{====}\overleftarrow{\text{====}}_{b \rightarrow a} = \text{---}\overleftarrow{\text{---}}_{b \rightarrow a} + \text{---}\overleftarrow{\text{---}}_{b \rightarrow 1} \text{---}\overrightarrow{\text{---}}_{2 \rightarrow a} \text{---}\overleftarrow{\text{---}}_{1 \rightarrow 2} \Sigma \text{---}\overrightarrow{\text{---}}_{2 \rightarrow a} \quad (3.50)$$

if we represent the self-energy¹⁴ by a gray filled circle. The self-energy derives from the

¹⁴The self-energy is relevant to introduce the necessary corrections to the particle (or quasiparticle) band structure obtained with DFT, mainly to correct the bandgap, as explained in Section 3.2.2.2. To see that, take a look at Eq. 3.48; the self-energy enters in the denominator summing with the noninteracting Hamiltonian, thus it can be seen as a renormalization of the Hamiltonian $\hat{H}_0$. We should say in fact it is the real part of Σ that corrects the band structure. The imaginary part has to do with the electron–electron scattering rates [74] and the quasiparticle life-times.

two-body interaction between the electrons, therefore, it is naturally related to the effective interaction between the electrons, W . The relation reads

$$\Sigma(1, 2) = i \int d3 d4 \mathcal{G}(1, 3^+) W(1, 4) \Lambda(3, 2; 4), \quad (3.51)$$

where 3^+ stands for the time instant t_3 plus an infinitely small imaginary shift τ as $t_{3+} = t_3 + i\tau$, with the purpose of performing contour integrals in the complex plane [74, 75], Λ is called the vertex function, and it is defined by

$$\Lambda(1, 2; 3) = \delta(1, 2)\delta(1, 3) + \int d4 d5 d6 d7 \frac{\delta \Sigma(1, 2)}{\delta \mathcal{G}(4, 5)} \mathcal{G}(4, 6) \mathcal{G}(7, 5) \Lambda(6, 7; 3). \quad (3.52)$$

It is possible to write for the self-energy as per (3.51) a diagrammatic expression by using a diagram for the vertex function introduced just now. Nonetheless, that is not needed, as we will soon introduce an approximation for the vertex function. Equations (3.49), (3.51) and (3.52) are not sufficient to describe the problem, as we have four unknowns; the interacting Green function $\mathcal{G}$, the self-energy Σ , the screened Coulomb interaction W and the vertex function Λ . We need as well the Dyson equation for the full interacting screened Coulomb potential W and the bare one v_c , which is given by

$$W(b, a) = v_c(b, a) + \int d1 d2 v_c(b, 1) \chi(1, 2) W(2, a), \quad (3.53)$$

where χ is the density-density response function in the language of linear response theory [74], also called polarizability in the context of many-body perturbation theory. It is easy to write this equation using Feynman diagrams, just as we did for the Green function. The Dyson equation for the Coulomb interaction in terms of diagrams reads

$$\begin{aligned} \text{wavy line} &= \text{bare wavy line} + \text{bare wavy line} \circlearrowleft \chi \text{ wavy line}, \\ \text{with } v_c &\equiv \text{bare wavy line} \quad \text{and} \quad \chi \equiv \text{loop with } \chi, \end{aligned} \quad (3.54)$$

where we introduced a new diagram for the polarizability χ , and the wavy single line for the bare Coulomb interaction. To close the problem, we need the equation relating the polarizability to the Green function through the vertex function, which reads

$$\chi(1, 2) = -i \int d3 d4 \mathcal{G}(1, 3) \Lambda(3, 4; 2) \mathcal{G}(4, 1^+). \quad (3.55)$$

Equations (3.49) to (3.55) constitute a set of five coupled equations, called Hedin's¹⁵ equations [98, 99]. The most widely-used approximation is to ignore the second term in the equation for the vertex, so that $\Gamma(1, 2; 3) \approx \delta(1, 2)\delta(1, 3)$. This approximation makes the

¹⁵As a historical curiosity, the Hedin's equations are named after the swedish physicist Lars Hedin, who was an emeritus professor at the university of Lund. He invented the *GW* approximation with its high impact article Ref. [98], igniting the Green function formalism of the many-body problem, used up until this day.

In the next subsection, we derive an expression for the RPA dielectric function from the RPA polarizability, which is routine in the literature [74, 99, 100], since the full interacting quantities are not possible to compute in practice. For that, let us take advantage of the many-body perturbation theory formalism introduced previously.

3.4.1 A derivation of the RPA dielectric function

We start by making a change of notation, and write for the polarizability χ^{irr} instead of just χ . This means that the polarizability introduced in the previous section is the irreducible polarizability, which is distinct from the full, or reducible, polarizability that we will soon introduce. Other changes of notation we make is to write V_c instead of simply v_c , and rewrite the convolutions from the previous section as mere multiplications, but that is just for the purpose of simplifying the writing. We would like to describe the screened potential W in terms of the bare one V_c , since the former is the one we want to find, and the latter the one we know. Let us then rewrite the Dyson equation for the Coulomb interaction (3.53) more compactly as

$$W = V_c + V_c \chi^{\text{irr}} W \equiv \varepsilon^{-1} V_c, \quad (3.59)$$

by defining the (microscopic) inverse dielectric function as

$$\varepsilon^{-1} = 1 + \chi^{\text{irr}} W. \quad (3.60)$$

This definition is not immediately useful, as it is still written in terms of the unknown W . Let us then manipulate the same Dyson equation once more, and cast the term in W from the right-hand side to the left-hand side, and solve for W to find

$$W = \frac{V_c}{1 - V_c \chi^{\text{irr}}}. \quad (3.61)$$

Now, it is just a matter of replacing the previous expression in the definition of the inverse dielectric function to get

$$\varepsilon^{-1} = 1 + \frac{\chi^{\text{irr}} V_c}{1 - V_c \chi^{\text{irr}}} = 1 + \chi V_c, \quad (3.62)$$

where we introduced the reducible polarizability χ

$$\textcircled{\chi} \equiv \chi = \frac{\chi^{\text{irr}}}{1 - V_c \chi^{\text{irr}}} = \chi^{\text{irr}} + \chi^{\text{irr}} V_c \chi, \quad (3.63)$$

where on the left-hand side we introduced a new diagram, now for the reducible polarizability χ , and the last equality was obtained by multiplying both sides of the first equality by the denominator in the right-hand side and solving for the polarizability χ . At this stage, no approximation for the polarizability was done, so χ^{irr} is the (irreducible) polarizability defined by Eq. (3.55), with its diagram introduced in Eq. (3.54). For completeness, we write the diagrammatic Dyson equation relating the two polarizabilities χ and χ^{irr} as



$$\chi = \chi + \chi \text{---} \chi. \quad (3.64)$$

Let us now manipulate the inverse dielectric function in Eq. (3.62) above, by reducing both terms in the right-hand side to the same denominator and simplifying to arrive at

$$\varepsilon^{-1} = \frac{1}{1 - V_c \chi^{\text{irr}}} \Rightarrow \varepsilon = 1 - V_c \chi^{\text{irr}}. \quad (3.65)$$

We have then found an expression for the microscopic dielectric function that, after recovering the notation from the previous section, reads

$$\varepsilon(b, a) = \delta(b - a) - \int d1 v_c(b, 1) \chi^{\text{irr}}(1, a). \quad (3.66)$$

Finally, to find the final expression for the microscopic dielectric function we will work with later on, we use the RPA irreducible polarizability, which reads $\chi^{\text{irr}} \approx \chi^{\text{irr,RPA}} = \chi^0$, where χ^0 is the noninteracting polarizability, that we shall discuss with detail in the next subsection. Within RPA, the irreducible polarizability written in the Hedin's equation (3.55) reduces to $\chi^{\text{irr,RPA}}(1, 2) = \chi^0(1, 2) = -i \int d3 \mathcal{G}^0(1, 3) \mathcal{G}^0(3, 2)$, whose diagram we have found before in Eq. (3.58). Henceforth, if we recover the explicit $(\mathbf{r}, t)$ -dependence, we can write

$$\varepsilon^{\text{RPA}}(\mathbf{r}, \mathbf{r}'; t, t') = \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') - \int d\mathbf{r}'' v_c(\mathbf{r} - \mathbf{r}'') \chi^0(\mathbf{r}'', \mathbf{r}'; t, t') d\mathbf{r}''.$$

(3.67)

After computing the polarizability χ^0 , not in the $\mathbf{r}$ -space but in the reciprocal $\mathbf{k}$ -space, we can *a posteriori* compute the dielectric function as per (3.67), but in the $\mathbf{k}$ -space as well. Only after computing the dielectric function, which will have a matrix form as we shall see soon, we can then invert it to obtain the inverse dielectric function.

As an alternative to computing the dielectric function and then inverting it, one could rewrite (3.53) within RPA as $W(b, a) = \int dc \varepsilon_{\text{RPA}}^{-1}(b, c) v_c(c, a)$, where [100]

$$\varepsilon_{\text{RPA}}^{-1}(a, b) = \delta(a - b) + \int dc v_c(a, c) \chi^{\text{RPA}}(c, b), \quad (3.68)$$

and χ^{RPA} is the interacting reducible polarizability at the RPA level, that, when resorting to the definition for the full polarizability (3.63) but within RPA, reads

$$\chi^{\text{RPA}}(1, 2) = \chi^0(1, 2) + \int d3 d4 \chi^0(1, 3) v_c(3, 4) \chi^{\text{RPA}}(4, 2). \quad (3.69)$$

We could have written the Dyson equation for the RPA reducible polarizabilities above in a diagrammatic notation. However, we do not work with this relationship in practice, and we would need an extra diagram for the RPA version of the reducible polarizability. At the end, it is only the RPA irreducible polarizability χ^0 that we work with, and all the other polarizabilities were introduced for self-contained discussion. But if we worked with it, the inverse dielectric function within RPA could then be obtained by replacing the reducible polarizability in Eq. (3.68) with its RPA form to write

$$\varepsilon_{\text{RPA}}^{-1}(a, b) = \delta(a - b) + \int dc v_c(a, c) \chi^{\text{RPA}}(c, b). \quad (3.70)$$

From now on, we omit the label RPA in the dielectric function or its inverse, and it will always be implicit that the dielectric function is computed within the RPA, and the polarizability is the noninteracting irreducible one χ^0 given by the Hartree diagram.

For a thorough discussion of these topics, we refer to Richard Martin's book in Ref. [100] and to the books by H. Bruus and K. Flensberg [74], and the one by Gerald D. Mahan [75]. The first reference studies the many-electron interacting problem in depth, providing also computational schemes to solve it in practice. The latter references address the interacting problem, with Ref. [74] working essentially with the jellium model for the electron gas (which has continuous translation symmetry). This allows for deep theoretical insights via analytical calculations. On the other hand, Ref. [75] is more general, and discusses at times the inhomogeneous electron gas.

3.4.2 The RPA dielectric function in reciprocal space

We recapitulate the RPA dielectric function in the $(\mathbf{r}, t)$ domain, which reads [74, 76]

$$\varepsilon(\mathbf{r}, \mathbf{r}'; t - t') = \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') - \int v_c(\mathbf{r} - \mathbf{r}'') \chi^0(\mathbf{r}'', \mathbf{r}'; t - t') d\mathbf{r}'', \quad (3.71)$$

where $\chi^0(\mathbf{r}, \mathbf{r}'; t - t')$ is the RPA irreducible polarizability introduced in the previous subsection, for which we have assumed time translation invariance¹⁶. The RPA irreducible polarizability, in the frequency domain, at finite temperature T and within a self-consistent field approach is given by [76]

$$\textcircled{\chi} \equiv \chi^0(\mathbf{r}, \mathbf{r}'; \omega) = \sum_{i,j} (f_i - f_j) \frac{\phi_i^*(\mathbf{r}) \phi_j(\mathbf{r}) \phi_j^*(\mathbf{r}') \phi_i(\mathbf{r}')}{\epsilon_i - \epsilon_j + \hbar\omega + i\eta}, \quad (3.72)$$

where η is a small infinitesimal parameter, $\phi_i(\mathbf{r})$ is the i -th state with eigenenergy ϵ_i , $f_i = (\exp\{\beta(\epsilon_i - \mu)\} + 1)^{-1}$ is the Fermi–Dirac distribution [102, 103] computed at the energy ϵ_i , with $\beta = 1/(k_B T)$ where k_B is the Boltzmann constant, T the absolute temperature (in Kelvin), and μ is the chemical potential. The states $\phi_i(\mathbf{r})$ are obtained within a mean-field approach, such as within HF theory or the Kohn–Sham framework of DFT. The many-body wavefunction of the associated independent particle system is then constructed through the Slater determinant of the orbitals $\phi_i(\mathbf{r})$, for which we do not specify the subjacent basis denoted by the indices i and j . The only thing left to define is the chemical potential. We will mention it in the next chapter as well, so we better define it now. Nonetheless, the

¹⁶Even though we can perceive the dielectric function as a response to an external potential perturbation of the form $\phi_0 \exp\{i(\mathbf{k} \cdot \mathbf{r} - \omega t)\}$, which is a time-periodic perturbation, we treat it within the linear regime. This is valid as long as the amplitude ϕ_0 is small enough. If it is too big, such that perturbative methods do not apply, then we have to account for the whole time dependence, as stated previously in footnote 3 of Chapter 2. If it still has periodicity, albeit discrete, we can work under the Floquet formalism of quantum mechanics, mentioned before in footnote 9. If the perturbation is not periodic, then we need a full non-equilibrium framework. That can be done, for instance, using the non-equilibrium Green function formalism, also called the Keldysh formalism [101].

precise technical definition is not very relevant for the moment. Hence, we define it in Box 3.4.1 below, which can be skipped in a first reading.

We notice that the expression in Eq. (3.71) is a two-time function with (continuous) time translation invariance. Whence, we may time-Fourier transform it to the frequency domain and write

$$\varepsilon(\mathbf{r}, \mathbf{r}'; \omega) = \delta(\mathbf{r} - \mathbf{r}') - \int v_c(\mathbf{r} - \mathbf{r}'') \chi^0(\mathbf{r}'', \mathbf{r}'; \omega) d\mathbf{r}'', \quad (3.73)$$

since the Fourier transform of the Dirac delta is the identity function, and the Fourier transform of a convolution translates to a multiplication. The microscopic dielectric function has *discrete* translation symmetry, so that

$$\varepsilon(\mathbf{r}, \mathbf{r}'; \omega) = \varepsilon(\mathbf{r} + \mathbf{R}, \mathbf{r}' + \mathbf{R}; \omega) \quad (3.74)$$

is true for any lattice vector $\mathbf{R}$, in a perfect crystal where discrete translation symmetry is preserved due to the absence of defects or boundaries. The same can be said about the polarizability χ^0 . This contrasts with the case of the previous chapter, where translation invariance was *continuous*¹⁷. We are still allowed to Fourier transform in space, but we need to keep track of two momenta, say $\mathbf{k}$ and $\mathbf{k}'$, instead of a single one, so that the polarizability is a two-point function in momentum when Fourier transformed from real to reciprocal space, $\mathcal{F}\{\chi^0(\mathbf{r}, \mathbf{r}')\}(\mathbf{k}, \mathbf{k}') \equiv \chi^0(\mathbf{k}, \mathbf{k}')$, if for moments we omit the ω -dependence. Luckily, as the polarizability still has discrete translation symmetry, one can show that $\chi^0(\mathbf{k}, \mathbf{k}')$ is non-zero only if $\mathbf{k}' = \mathbf{k} + \mathbf{g}$, where $\mathbf{g}$ is a reciprocal lattice vector. Furthermore, we can write any momentum $\mathbf{k}$ in terms of a vector $\mathbf{q}$ in the first BZ plus a vector of the reciprocal lattice $\mathbf{G}$, $\mathbf{k} = \mathbf{q} + \mathbf{G}$. The same can be said for the other momentum $\mathbf{k}' = \mathbf{q} + \mathbf{G}'$, with $\mathbf{G}' = \mathbf{G} + \mathbf{g}$ just being another vector of the reciprocal lattice. This allows us to write $\chi^0(\mathbf{k}, \mathbf{k}') = \chi^0(\mathbf{q} + \mathbf{G}, \mathbf{q}' + \mathbf{G}') \equiv \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$, where we can see the polarizability as an infinite matrix for each point $\mathbf{q} \in \text{BZ}$. For every momentum $\mathbf{q} \in \text{BZ}$, each pair of reciprocal lattice vectors $(\mathbf{G}, \mathbf{G}')$ corresponds to the matrix element $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$.

Using the property $\psi_{n,\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}} \psi_{n,\mathbf{k}}(\mathbf{r})$ for Bloch states, coming from the second statement (2) of Bloch theorem, we can show that the polarizability $\chi^0(\mathbf{r}, \mathbf{r}'; \omega)$ as per (3.72) has discrete translation symmetry under vectors of the lattice, if we particularize the basis states $\phi_i(\mathbf{r})$ to a basis of Bloch states $\psi_{n,\mathbf{k}}(\mathbf{r})$. The expression for the matrix elements $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$ can be obtained by Fourier transforming (3.72) represented in a Bloch state basis, and they read

$$\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega) = \frac{1}{\mathcal{V}} \sum_{n,n'} \sum_{\mathbf{k}\sigma} (f_{n,\mathbf{k}} - f_{n',\mathbf{k}+\mathbf{q}}) \frac{\langle n, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{r}} | n', \mathbf{k} + \mathbf{q} \rangle \langle n', \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{r}} | n, \mathbf{k} \rangle}{\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}+\mathbf{q}} + \hbar\omega + i\eta}, \quad (3.75)$$

where $\mathcal{V}$ is the volume of the crystal, n and n' are band indices, $\mathbf{G}$ and $\mathbf{G}'$ are reciprocal lattice vectors, σ represents spin and the sum over momentum $\mathbf{k}$ is over the BZ.

¹⁷We italicize the words continuous and discrete to stress that the distinction is important to make.

Box 3.4.1: Chemical potential

Usually introduced in the context of thermodynamics [102] and statistical physics [103], the chemical potential of a system, usually denoted by μ , can be defined in terms of the thermodynamical potential (or other potentials as well) $F = U - TS$, called the Helmholtz free energy, where U is the internal energy of the system, T its (absolute) temperature, and S its entropy. If the system is in thermal equilibrium and in contact with a reservoir (at the temperature T), with which it can exchange heat and particles, then the chemical potential can be defined as $\mu = F_{\mathcal{N}+1} - F_{\mathcal{N}}$, where $\mathcal{N}$ is the number of particles in the system. In words, μ is a measure of the energy cost to add or remove a particle from the system. Only in the thermodynamic limit where $\mathcal{N} \rightarrow \infty$, can we define the chemical potential in an equivalent way by the derivative $\partial F / \partial \mathcal{N}$ [102]. Otherwise, if the system is composed of few particles, then we are not allowed to take the derivative with respect to $\mathcal{N}$, henceforth the first definition applies.

In Fig. 3.4 below, we plot the Fermi–Dirac distribution for three different temperatures. At zero absolute temperature, the distribution is just a Heaviside step function $\Theta(\mu - \epsilon)$ centered at the chemical potential μ . This means that when the temperature is zero, the fermions are not thermally excited, therefore all states with energy $\epsilon > \mu$ are unpopulated, and all states with energy $\epsilon < \mu$ are occupied. Due to the Fermi–Dirac statistics and the Pauli exclusion principle [68], each state can only be occupied by one fermion. That explains how we have $f(\epsilon) \leq 1$ for any energy, chemical potential or temperature. If the fermions have a nonzero temperature, then they are thermally excited, meaning that states with energy $\epsilon > \mu$ start to have a nonzero occupation. As the temperature increases, from the continuous blue curve to the dashed blue one in the figure, then more fermions from lower energy levels are excited to even higher energy levels. In the limit $T \rightarrow \infty$, $f(\epsilon) \rightarrow 1/2$, because the thermal energy is so high that all fermions get uniformly distributed among all energy states, and the Fermi–Dirac distribution is a straight line at $f(\epsilon) = 1/2$.

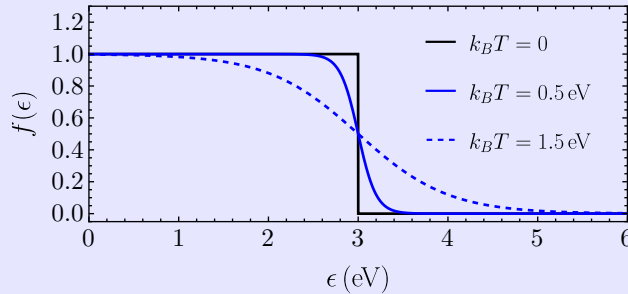


Figure 3.4: Fermi–Dirac distribution $f(\epsilon)$ as a function of the energy ϵ , for three different temperatures T ; absolute zero temperature $k_B T = 0$ in black, $k_B T = 0.5$ eV in continuous blue, and $k_B T = 1.5$ eV in dashed blue.

Defining the polarizability in reciprocal space as a matrix allows doing the same for the dielectric function, that Fourier transformed to reciprocal space reads

$$\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, \omega) = \delta_{\mathbf{G}\mathbf{G}'} - v_c(\mathbf{q} + \mathbf{G})\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega) . \quad (3.76)$$

This expression is usually introduced in condensed matter physics courses through the Lindhard function, which is the previous formula in the case of the jellium model, the free homogeneous electron gas. The matrix elements in Eq. (3.75) are equal to 1 when $n = n'$, the dispersion is quadratic and there is no crystalline structure, hence the polarizability and dielectric matrices are scalar functions of $\mathbf{q}$. The Lindhard function then reads [41, 57, 76]

$$\varepsilon(\mathbf{q}, \omega) = 1 - \frac{2e^2}{\varepsilon_0 q^2 \mathcal{V}} \sum_{\mathbf{k}} \frac{f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}}}{\epsilon_{\mathbf{k}+\mathbf{q}} - \epsilon_{\mathbf{k}} + \hbar\omega + i\eta} . \quad (3.77)$$

In the Lindhard function (where the prefactor of 2 comes from the spin degeneracy), we can notice its intricate dependence on both momentum and frequency. This was already a concern in the previous chapter, however we only analyzed the local and static limits separately. Taking the local limit $\mathbf{q} \rightarrow \mathbf{0}$ in the previous expression and computing the sum (converted into an integral), one arrives naturally at

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2} , \quad (3.78)$$

where ω_p is the electron plasma frequency. The previous expression is the Drude dielectric function discussed in the first chapter, found here through a microscopic approach. This provides a glimpse of the power of quantum mechanics, and reassures the claims made in the previous chapter on how models are used and introduced in classical physics, that in fact derive from physics that it cannot describe. The tools provided by quantum mechanics allowed obtaining the full dependence of the dielectric function on both momentum and frequency, something we were not able to do in the previous chapter (that can somewhat be achieved classically using nonlocal hydrodynamic models).

If the material in question is not a metal, but a semiconductor or insulator, then we can only aspire to compute the polarizability and the dielectric function numerically. The same can be stated about the screened Coulomb potential. In the jellium approximation, the screened Coulomb potential in reciprocal space is simply given by $W(\mathbf{q}, \omega) = v_c(\mathbf{q})/\varepsilon(\mathbf{q}, \omega)$, because there is continuous translation invariance, and as a consequence, the macroscopic dielectric function and the microscopic one coincide. However, assuming continuous translation invariance for insulators or semiconductors is questionable. In general, we need to account for the crystal structure, so that the screened Coulomb potential in momentum–frequency domain, obtained by Fourier transforming Eq. (3.59), reads

$$W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, \omega) = \varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}, \omega) v_c(\mathbf{q} + \mathbf{G}') , \quad (3.79)$$

for which one needs the inverse dielectric matrix, satisfying $\sum_{\mathbf{G}''} \varepsilon_{\mathbf{G}\mathbf{G}''}^{-1} \varepsilon_{\mathbf{G}''\mathbf{G}'} = \delta_{\mathbf{G}\mathbf{G}'}$ at every point $\mathbf{q}$ in the BZ and for any frequency ω . Only after having inverted the dielectric matrix, can we estimate the macroscopic dielectric function, $\varepsilon(\mathbf{q}, \omega)$, through the definition

$$\boxed{\varepsilon(\mathbf{q}, \omega) = \frac{1}{\varepsilon_{00}^{-1}(\mathbf{q}, \omega)} \neq \varepsilon_{00}(\mathbf{q}, \omega)} . \quad (3.80)$$

In the previous definition, the inequality is in general true, due to the local-field effects that come from the inhomogeneity of the crystal [104]. This will be a key aspect that we shall leave for later aspect, in Chapter 6. As we will understand in the next chapter, it is of upmost importance to capture accurately the screening in semiconductors and insulators, mainly 2D ones, in order to get realistic results for the excitonic states. Even if one is not doing excitons and simply working at the noninteracting level, incorporating screening effects in the calculations is necessary for accurate predictions. As a matter of fact, the absence of screening is one of the pitfalls of the HF method in extended systems. Now that we have described the ground state properties of materials, we are in position to introduce the exciton, a quasiparticle emerging from optical excitations of solids, that cannot be ignored when investigating the optical properties of gapped materials. The theory of excitons in 2D materials is thus the subject of the next chapter.

EXCITONIC EFFECTS IN TWO-DIMENSIONAL SEMICONDUCTORS

4.1 Introduction

In previous chapters, we have discussed how a classical description of electrodynamics in matter, together with a quantum description of its properties, we can describe different kinds of phenomena in solids. The tools of quantum mechanics provide means of calculating the dielectric function, and as we will see, also the conductivity. Given their equivalence established in the first chapter, knowing the dielectric function is sufficient to describe electrodynamic effects, such as dispersion, bulk plasmons in Drude metals, Ohmic losses and material absorption—the last two more directly through the conductivity. The fact that the classical description of plasmons, absorption, and reflection in metals via Maxwell’s equations and empiric dielectric functions is so successful, although it might seem surprising, can now be understood.

At the light of our quantum description of matter, we have seen that gapped materials can be identified with insulators and semiconductors, poor conducting materials. This is so because the valence bands in such materials are filled, requiring large amounts of energy to excite electrons to the conduction band such that there is a nonvanishing current [56, 70]. In contrast, metals are good conductors because the highest populated band is partially filled. This means that it only requires small amounts of energy to excite the electrons so that they can more easily participate in the charge flow. In other words, in gapped undoped materials and at low temperatures, the system can only be excited from the ground state via energetically costing interband transitions, whereas metals can be excited via energetically efficient intraband transitions. However, if the energies involved are high enough, a metal will experience interband transitions, which bring a contribution to the optical response that the Drude approach cannot capture. Such an example is gold [105], which displays a Drude-like optical conductivity until a certain frequency threshold, while above that frequency the optical measurements deviate from the classical model highlighting its shortcomings. Ad hoc corrections to the Drude model can be made [106], but a complete

picture is still missing. One rapidly sees that classical models cannot explain all phenomena in solids, in particular phenomena observed in optical spectroscopy experiments detecting quantum effects with no classical analogue.

Quantum mechanics sheds light on material properties stemming from interband transitions, usually included phenomenologically in classical models, but that are naturally incorporated in the microscopic dielectric function through the polarizability in Eq. (3.75), which in fact includes both interband and intraband transitions in a single expression. Therefore, we expect to be able to accurately describe the dielectric properties of insulators and semiconductors, which are necessary for describing many-body phenomena in these systems, as we will see. However, the expression for the dielectric function obtained in the previous chapter was written in terms of a noninteracting polarizability χ^0 , that accounts solely for single-particle transitions. An optical experiment, for instance a photoluminescence or optical absorption one, for frequencies above the bandgap would measure an optical response due to the excitation of electrons from the valence bands to the conduction bands when they absorb a photon. Then, in principle, the same experiment would not detect any response when tuning the frequency to sub-gap energies. However, that is not what happens. Photoluminescence experiments have measured an absorption resonance for monolayer hexagonal boron nitride (hBN) at an energy lying inside the electronic bandgap [107]. This observation can only be explained by taking into account the electron–electron interactions beyond the mean-field level, which is an effective single-particle description. The two-body interactions can promote a quantum state that we could not predict from the machinery introduced so far—that is called a many-body state.

Up until now, we have been stressing that the full interacting response of condensed matter systems cannot be computed in practice, which is true. If that was possible, then we could theoretically predict the whole response of a material at any range of frequencies; this dissertation would be unnecessary. That is why, in a first instance, we kept the discussion within the independent-particle picture which we can compute in practice, with the full interacting one being out of our Due to. Nonetheless, we do not need the full interacting picture to explain some experimental observations. To estimate the contribution from the most dominant resonances, it suffices to consider the Coulomb interaction between the electron that was excited to the conduction band, and the absence of the negative charge that it left behind in the valence band—the positively charged hole. This interaction between the electron and the hole, just like between the proton and the electron in the hydrogen atom, can be so strong that the electron–hole pair form a bound state—the exciton [108]. The situation is depicted in Figure 4.1 on the left, where the electron, depicted as the red filled circle, is in the conduction band, while the hole, depicted as the blue open circle, is in the valence band. Due to Coulomb attraction, extra states inside the bandgap can form, as represented in Figure 4.1 on the right. The material can be excited from its ground state to one of several excitonic states represented in the figure by a series of discrete energy levels, if the energy of the photon coincides with one of those energies. The minimum amount of energy required for that to happen is what defines the *optical*

bandgap, coinciding with the energy of the *excitonic* ground state. We can define the binding energy $E_{b,\nu}$ of a given exciton in a state ν , depicted in the schematic on the right in Fig 4.1, as the difference between the energy of the electronic bandgap E_g , a single-particle quantity, and the energy of the exciton E_ν . Just like the ionization energy in the hydrogen atom, it can be interpreted as the energy that we have to provide to separate the electron from the hole, resulting in their separation into an unpaired electron–hole pair, followed by the decay of the electron and its recombination with the hole.

These facts do not imply that all the techniques introduced until now in the last two chapters are not useful; actually the opposite. As we will see, getting the Coulomb attraction right in a solid is essential in order to compute the excitons correctly. We mean that we need to take screening into account by using an appropriate dielectric function. Screening effects at the noninteracting level can actually describe quite well interactions in a solid constituting the base of the calculations to determine the excitons. As we shall explain in a moment, this is crucial in two-dimensional (2D) materials. Before getting too technical, let us first try to draw some conclusions and gain intuition on the problem before we appropriately tackle it. We also have to make a disclaimer about Fig. 4.1, because the quasienergies of the bands are single-particle quantities, so in principle, it does not make sense to show the exciton states and the bands in the same plot. In panel **a**, we want to express that the exciton has an energy that lies inside the single-particle bandgap, and that it can be described by a coherent superposition of unbound electron–hole pairs, which are single-particle states. On the right side of the same figure, panel **b**, we want to express that there are eigenvalues of the Hamiltonian inside the bandgap. Even though exciton states are many-body states, the spectrum of the full many-body Hamiltonian contains all the allowed energy values—these form a continuum below and above the bandgap, but are discrete inside the bandgap.

As we shall see in this chapter, the excitonic problem that consists of finding the excitonic eigenenergies and eigenstates (or wavefunctions) can be quite intricate. Nonetheless, we have argued that determining the excitonic properties of a material is key to describe their optoelectronic properties, in particular those related to the sub-bandgap physics of gapped materials. At the optical frequencies considered in this work, no other properties come into play, such as phonon or magnetic excitations. Herein, we only model the electronic and optical properties of 2D materials, and how these relate with each other. There are several approaches to the exciton problem, and in this chapter we provide a couple of them, going up in difficulty and complexity.

Within a very crude approximation, the problem of bound electron–hole complexes in three-dimensional (3D) materials can be identified with the hydrogen-atom problem, if the energy surfaces for the electron and the hole are spherical and nondegenerate [70]. From a heuristic point of view, this implies that the binding energies are given by the modified Rydberg formula

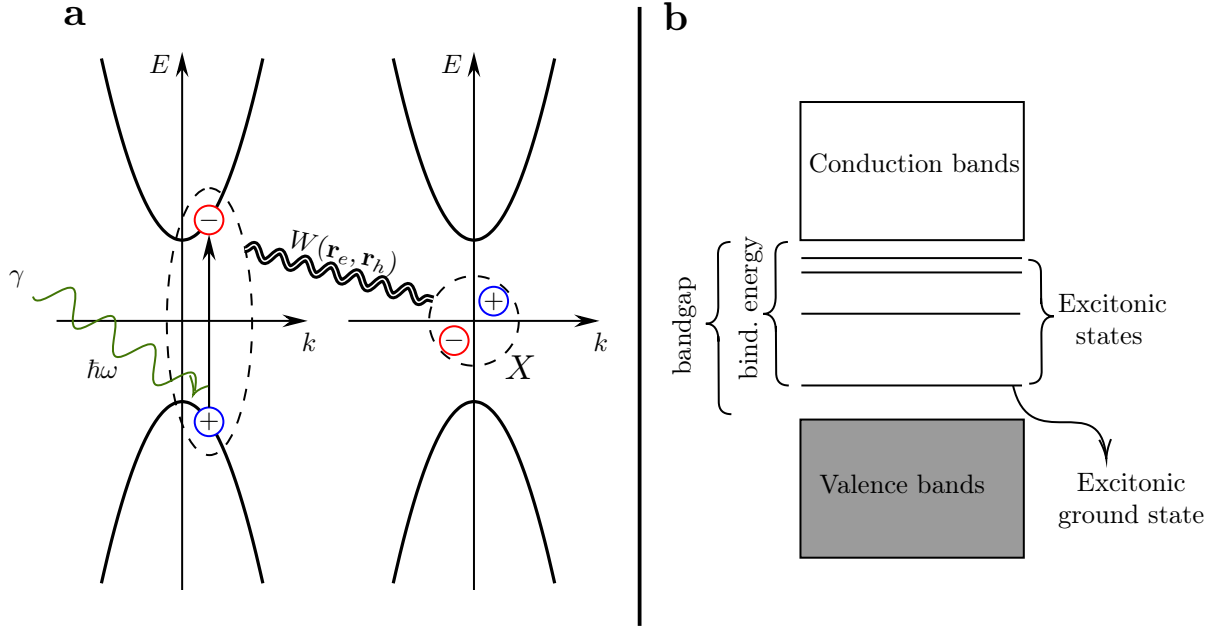


Figure 4.1: Formation of an exciton. **a** Schematic representation of the formation of an exciton X from an electron-hole pair. The electron (red circle) excited to the conduction band (thick black curve above the k axis) from absorbing a photon γ with frequency ω , and the hole (blue circle) left behind in the valence band (thick black curve below the k axis), attract each other due to the Coulomb interaction $W(\mathbf{r}_e, \mathbf{r}_h)$. The energy of the exciton E_X lies inside the bandgap. **b** Schematic representation of the several possible excitonic states, depicted as single straight horizontal lines located above the valence bands and below the conduction bands. The binding energy $E_{b,X}$ can be visualized as the bandgap subtracted by the energy of the exciton E_X . We remark that the figure on the right-hand side presents only the energies allowed in a semiconductor or insulator regardless of the momentum variable, including the sub-gap exciton states. It depicts the spectrum of the total Hamiltonian, including both those pertaining to single-particle and many-particle states, the latter at the excitonic level. We have in mind that the many-particle problem is different from the single-particle one and, in principle, these should not be mixed together in the same plot. We believe that the plot is reasonable, because we do not show the dependence of the energies on $\mathbf{k}$.

$$E_{b,n} = -\frac{R^*}{n^2}, \quad (4.1)$$

where

$$R^* = \frac{2\mu_{eh}^2 e^4}{\hbar^2 (8\pi\epsilon_r\epsilon_0)^2} \quad (4.2)$$

is the modified Rydberg constant which, instead of using the electron mass, it uses the reduced mass between the electron and hole effective masses

$$\frac{1}{\mu_{eh}} = \frac{1}{m_e} + \frac{1}{m_h}. \quad (4.3)$$

Here, ϵ_r is just the dielectric constant of the medium. This description is suited for weakly bound excitons, also-called Wannier excitons [109], originally proposed by Wannier [10]

and Mott [11–13]. The characteristic size of the exciton is given by the modified expression for the Bohr radius

$$a_B^* = \frac{4\pi\epsilon_r\epsilon_0\hbar^2}{\mu_{eh}^2 e^2}, \quad (4.4)$$

which can extend over a couple of atomic sites. If the exciton is tightly bound, then it is called a Frenkel exciton [110–112]. In this case, the excitation is localized at a single atom. Wannier excitons can form in inorganic semiconductors. Among all 3D materials, we can mention cuprous oxide, Cu_2O , gallium arsenide, GaAs, or krypton, Kr. For GaAs, the binding energy of the ground state exciton is 4.2 meV [113], whereas in solid krypton the ground-state excitonic binding energy is around 1.75 eV [114]. On the contrary, Frenkel excitons are more common in ionic crystals [115], such as CuCl or NaCl, or in organic molecular crystals [116, 117]. A recent theoretical work [118] predicted that when two materials, one hosting a Frenkel exciton and another hosting a Wannier exciton, are in close contact, the Frenkel exciton can hybridize with the Wannier exciton to form a Wannier–Mott–Frenkel exciton.

If we wish to apply the description above of Wannier excitons in 3D materials to excitons in 2D materials, the excitonic binding energies are given now by the modified Rydberg formula for the 2D hydrogen atom [119]

$$E_{b,n}^{2D} = -\frac{R^*}{\left(n - \frac{1}{2}\right)^2}. \quad (4.5)$$

Focusing on the ground state $n = 1$, we can see that $E_{b,n=1}^{2D} = 4E_{b,n=1}$. Furthermore, the 2D Bohr radius is half of the 3D one [119]. We can then predict that the exciton contribution to the optical spectrum is more prominent in 2D materials when compared to 3D ones. This is a consequence of the reduced dimensionality, deeming the screening of the electrostatic Coulomb interaction in 2D materials less effective than in 3D ones [26]. In 2D materials, excitons can form even at room temperatures [120]. Therefore, the hydrogen-atom picture for the excitons does not apply to 2D materials, as the binding energies do not follow the hydrogenic Rydberg series, and a modified Rydberg constant as per Eq. (4.2) with a dielectric constant is not valid. As we shall see later on, mainly in Chapter 6, the dielectric function in 2D gapped materials is strongly nonlocal. This implies that we cannot speak of a single dielectric constant ϵ_r as a relative permittivity, and writing for the screened potential $W(r) = v_c(r)/\epsilon_r$ is not sufficient, because screening in 2D materials is nonlocal. Actually, it has even been seen experimentally, at least for WS_2 , that the series of excitonic energies is nonhydrogenic [23], and this behavior is expected to be universal among 2D insulators. The 2D material WS_2 belongs to a class of semiconducting materials called transition metal dichalcogenides (TMDs) [121], which have by chemical formula MX_2 , where M is a transition metal, such as molybdenum (Mo) or wolfram (W), and X is a chalcogen, such as sulfur (S) or selenium (Se). In bulk, they are layered materials and present an indirect bandgap, while in their monolayer limit the bandgap is direct. Among

them, there is molybdenum disulfide [122], with chemical formula MoS_2 , a prototypical 2D material for studying excitons that we shall encounter later on in this work.

In order to explain the peculiarities of the excitons in two dimensions, we can as a first attempt adopt the hydrogen-atom model, but with a modified potential. The excitonic states and respective binding energies can be estimated by solving the Wannier–Mott equation [10], or simply Wannier equation, that reads

$$E_{b,\nu}\psi_\nu(\mathbf{r}) = \frac{p^2}{2\mu_{eh}}\psi_\nu(\mathbf{r}) - V(\mathbf{r})\psi_\nu(\mathbf{r}), \quad (4.6)$$

where μ_{eh} is the electron–hole reduced effective mass as before, and $V(\mathbf{r})$ is the attractive potential. One can notice similarities with the time-independent Schrödinger equation (TISE) for the hydrogen atom, but with an effective reduced mass (between two particles that have a comparable mass), and an attractive potential that includes now screening effects from the material itself. We have seen before, in the Section 2.2.1.1, that for 2D materials the dielectric function in the static limit has a nontrivial momentum-dependence. It is linear in q as $\varepsilon(q) = 1 + r_0q$, and we called this permittivity with the model parameter r_0 the Rytova–Keldysh dielectric function, giving origin to the Rytova–Keldysh potential [123].

The Wannier equation does not capture the full structure of the excitonic states (like nondegenerate p states [25, 124, 125] or the symmetries of the crystal), hence we need to resort to more accurate descriptions of the excitonic problem for complicate materials. For instance, the Wannier model breaks down if the electron–hole binding is strong, so that the exciton is delocalized in reciprocal space, introducing the need to look at the band structure beyond the effective mass approximation [76]. Nonetheless, the Wannier–Mott formalism offers good results when compared to experimental results [126], and provides theoretical insights even in the 2D case where an analytical solution cannot be found [25]. One can go beyond Wannier–Mott by using a reciprocal space description of the excitons through the so-called Bethe–Salpeter equation (BSE)—an integral equation where finer details of the band structure, the Coulomb interaction and the system Hamiltonian are included [15, 127].

In this work, we develop two versions of the BSE, one that we will apply to effective low-energy models, and another one using exact diagonalization methods. The *ab-initio* version of the BSE¹ is not addressed here. Besides being technically more involved, it is extremely demanding computationally, so it is necessary to come up with more efficient approaches. Nonetheless, it can be frequently encountered in the literature, in particular in works

¹For a reader fond of high-energy physics, we leave here a historical note. The BSE was first introduced in the literature by Y. Nambu in 1950 [128], and is named after the physicists Hans Bethe and Edwin Salpeter who derived it in 1951 [129], with posterior contributions by Gell-Mann and Low [130]. Its original formulation is in the context of quantum field theories, where composite two-fermion states can arise as well. One example is the electron–positron pair, positronium. Either in its covariant form or nonrelativistic form, the starting point to derive the BSE is defining a four-point, or two-particle, interacting Green function [100, 130]. Notwithstanding their differences, both formalisms can be conciliated into a relativistic many-body theory [131]. Just like in Section 3.3 for the many-particle description of solids, where we introduced the diagram notation, one can also write the BSE in a diagrammatic form (which of course has its own particle physics version).

using *ab-initio* methods for the optical response of materials [31, 33]. If the reader is curious about it, we refer to the original work by M. Rohlfing and S. G. Louie [127], or the PhD thesis by A. J. Uría-Álvarez [132] for a more recent text with a derivation adapted from [127]. Combined with the *GW* method for calculating quasiparticle band structures, the state-of-the-art for exciton and optical response calculations is called the *GW*+BSE method, constituting the theoretical basis of several computational packages [31–33, 133, 134], which has made its way to chemistry as well [135–139]. On par with *GW*+BSE, there is another formalism for predicting the optoelectronic properties of materials, the time-dependent density functional theory (TDDFT) formalism [104, 140], the extension of the density functional theory (DFT) formalism we mentioned in Section 3.2.2.2. The TDDFT method for linear response has also faced a recent extension to the relativistic case, which is called the quantum-electrodynamical density functional theory formalism [141].

In the following section, we derive the Bethe–Salpeter equation applied to effective models, followed by a derivation of the Wannier equation. Then in Section 4.3 we dwell once again on the BSE, this time using exact diagonalization methods as they are implemented in the code we contributed to in this thesis. To finalize the chapter, we derive expressions to study the linear optical response of materials, with and without excitonic effects.

4.2 The Bethe–Salpeter Equation: application to effective models

In this section, we derive the BSE as an integral equation written in momentum space, allowing a more accurate description of the material excitonic properties. A $\mathbf{k}$ -space description of the problem feels natural, given the concepts introduced in the previous chapter. We will use the tools therein acquired to our advantage. This derivation is adapted from the Master’s thesis of J. C. G. Henriques in Ref. [142], and the tutorial by M. F. C. Quintela in Ref. [15], where one can look into it with more detail. Here we explain as well how to arrive at the interaction Hamiltonian in momentum space, which is not as easy as in high-energy physics, because one has to account for the crystalline structure of the material. Instead of working in a plane-wave representation, we have to work with a basis of Bloch states, but we explain how in a moment.

We have seen that the ground state of insulators, or semiconductors at low enough temperatures, comprises all the valence bands filled and the conduction bands empty. In the context of many-particle physics, the ground state, that we can represent in ket notation as $|\text{GS}\rangle$, is many times called Fermi sea. That is why in some texts one often sees the ket $|\text{FS}\rangle$ for a systems ground state. The Fermi sea comes from populating the electronic states of the material starting from the lowest energy states, and filling up the states with increasing energy, until charge neutrality has been reached. If there is no electronic bandgap, the energy of the highest occupied state, at zero temperature, is called the Fermi energy or Fermi level, with symbol E_F . That is the case of metals, where E_F lies in the middle of the

partially filled highest valence band. In semiconductors and insulators, the Fermi energy lies in the midway between the energy of the highest occupied state and the energy of the lowest unoccupied state. More generally, it can be defined as the chemical potential μ (see Box 3.4.1 in Section 3.4.2) at zero temperature. This is explained in section 4.1 of the textbook in Ref. [56].

Since we introduced the exciton as a bound electron–hole complex, we will assume that the unbound electron–hole states serve as a good basis in which we can expand the excitonic states. To simplify the reasoning, we assume that only the top valence band and the lowest conduction band contribute to the exciton. In this way, and in the second quantization formalism, we can write the generic exciton state ν as the coherent superposition of electron–hole pairs

$$b_\nu^\dagger = \frac{1}{\sqrt{\mathcal{A}}} \sum_{\mathbf{k}} \phi_\nu(\mathbf{k}) c_{c\mathbf{k}}^\dagger c_{v\mathbf{k}}, \quad (4.7)$$

where $\sqrt{\mathcal{A}}$ is the extension area of the 2D system, $\phi_\nu(\mathbf{k})$ is the momentum-resolved excitonic wavefunction (that we wish to find), ν is a set of (good) exciton quantum numbers, and $c_{\lambda\mathbf{k}}^\dagger$ creates an electron with momentum $\mathbf{k}$ in the band λ ², and $c_{\lambda\mathbf{k}}$ annihilates an electron with momentum $\mathbf{k}$ in the valence band v , creating a hole in its place. For our two-band model, $\lambda = v, c$. In that case, $|\text{GS}\rangle$ consists of a completely filled valence band and an empty conduction band. This should be sufficient for a two-band model. Here, we assume only vertical transitions due to the small wavevector carried by the photon, meaning that the exciton center of mass carries no momentum. Often in the literature, one works with a fermionic operator for the hole as a quasiparticle, but that is not strictly necessary. We stick to electron operators and perceive the hole as the absence of an electron. Being quadratic on the fermion operators, the operators b and $b^\dagger$ obey commutation relations and are therefore bosonic (see footnote 13). The operator $b_\nu^\dagger$ can be seen as an exciton creation operator, that when acting on $|\text{GS}\rangle$ creates one exciton in the state $|X_\nu\rangle$; in the occupation number representation, we can write

$$b_\nu^\dagger |\text{GS}\rangle = |1_{v\mathbf{k}_1}, 1_{v\mathbf{k}_2}, \dots, 1_{v\mathbf{k}_f}, 0, 0, \dots, \underset{\downarrow X_\nu}{1}, 0, 0, \dots, 0_{c\mathbf{k}_1}, 0_{c\mathbf{k}_2}, \dots, 0_{c\mathbf{k}_f}\rangle. \quad (4.8)$$

Conversely, the operator b_ν annihilates an exciton in the state ν .

The total Hamiltonian of the system

$$\hat{H} = \hat{H}_0 + \hat{V} \quad (4.9)$$

has two parts, the kinetic or noninteracting one $\hat{H}_0$, and the interacting one $\hat{V}$. We assume that we know how to diagonalize $\hat{H}_0$, so that we can write it in the $|\lambda\mathbf{k}\rangle$ basis ignoring spin, as

²In the second quantization formalism, one usually uses c for fermions and b for bosons. The symbol a is commonly used for both fermions and bosons, depending on the context.

$$\hat{H}_0 = \sum_{\lambda \mathbf{k}} \epsilon_{\lambda \mathbf{k}} c_{\lambda \mathbf{k}}^\dagger c_{\lambda \mathbf{k}}, \quad (4.10)$$

where $\epsilon_{\lambda \mathbf{k}}$ is the energy of an electron with momentum $\mathbf{k}$ in the band λ . For the interacting part, we start from the real-space basis reading

$$\hat{V} = \frac{1}{2} \int d\mathbf{r} \int d\mathbf{r}' \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \hat{\psi}(\mathbf{r}') \hat{\psi}(\mathbf{r}), \quad (4.11)$$

if we ignore the spin degree of freedom once more, and $\hat{\psi}^\dagger(\mathbf{r})/\hat{\psi}(\mathbf{r})$ creates/destroys an electron at position $\mathbf{r}^3$. These operators are called field operators, and we keep the hat on them to stress that they are operators and to distinguish them from the wavefunction $\psi(\mathbf{r})$. In this work, we will not worry with spinful excitons, so there is no need to include spin indices in the field and fermionic operators. This is justified if we ignore spin–orbit coupling (SOC) in the Hamiltonian, and if we assume that the electron–electron interaction includes only the Coulomb potential, which induces no spin flipping. In this section, we tacitly model the Coulomb interaction by the Rytova–Keldysh potential, and that will be implicit until the end of this section. Therefore, to ease the notation we use the symbol V instead of W or V_{RK} . To write both terms in the Hamiltonian in momentum space, we have to convert the interaction term to the $\mathbf{k}$ -space. For that, we use the expansion of the field operators in terms of Bloch states

$$\hat{\psi}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\lambda, \mathbf{k}} \langle \mathbf{r} | \lambda \mathbf{k} \rangle c_{\lambda \mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\lambda \mathbf{k}} \psi_{\lambda, \mathbf{k}}(\mathbf{r}) c_{\lambda \mathbf{k}}, \quad (4.12)$$

where N is the number of unit cells (or $\mathbf{k}$ points), and write the potential in terms of its Fourier transform as (see Eq. (A.12) in Appendix A)

$$V(\mathbf{r} - \mathbf{r}') = \frac{1}{\mathcal{A}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')} V(\mathbf{q}). \quad (4.13)$$

Replacing everything in (4.11) gives

$$\begin{aligned} \hat{V} = & \frac{1}{2\mathcal{A}N^2} \sum_{\mathbf{q}} \sum_{\{\mathbf{k}_i, \lambda_i\}} V(\mathbf{q}) c_{\lambda_1 \mathbf{k}_1}^\dagger c_{\lambda_2 \mathbf{k}_2}^\dagger c_{\lambda_3 \mathbf{k}_3} c_{\lambda_4 \mathbf{k}_4} \times \\ & \times \left(\int_{\mathcal{A}} \psi_{\lambda_1, \mathbf{k}_1}^*(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_{\lambda_4, \mathbf{k}_4}(\mathbf{r}) d\mathbf{r} \right) \left(\int_{\mathcal{A}} \psi_{\lambda_2, \mathbf{k}_2}^*(\mathbf{r}') e^{-i\mathbf{q} \cdot \mathbf{r}'} \psi_{\lambda_3, \mathbf{k}_3}(\mathbf{r}') d\mathbf{r}' \right), \end{aligned} \quad (4.14)$$

where $\sum_{\{\mathbf{k}_i, \lambda_i\}} \equiv \sum_{\mathbf{k}_1 \mathbf{k}_2 \mathbf{k}_3 \mathbf{k}_4} \sum_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}$, and the integrals are over the plane of the material. If using bracket notation, then

³Naturally, these operators obey commutation relations for bosons and anticommutation relations for fermions. However, in the *continuous* real-space basis the Kronecker deltas become Dirac deltas like this: $[\hat{\psi}^\dagger(\mathbf{r}), \hat{\psi}(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}')$, if we are talking about bosons. For fermions it is similar.

$$\begin{aligned} \hat{V} = & \frac{1}{2\mathcal{A}N^2} \sum_{\mathbf{q}} \sum_{\{\mathbf{k}_i, \lambda_i\}} V(\mathbf{q}) c_{\lambda_1 \mathbf{k}_1}^\dagger c_{\lambda_2 \mathbf{k}_2}^\dagger c_{\lambda_3 \mathbf{k}_3} c_{\lambda_4 \mathbf{k}_4} \times \\ & \times \langle \lambda_1, \mathbf{k}_1 | e^{i\mathbf{q} \cdot \mathbf{r}} | \lambda_4, \mathbf{k}_4 \rangle \langle \lambda_2, \mathbf{k}_2 | e^{-i\mathbf{q} \cdot \mathbf{r}'} | \lambda_3, \mathbf{k}_3 \rangle. \end{aligned} \quad (4.15)$$

An auxiliary calculation for the matrix elements follows. The Bloch wavefunctions, in the atomic gauge that we used to write the TB Bloch Hamiltonian for hBN, have the form

$$\psi_{\lambda, \mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}, i\alpha} e^{i\mathbf{k} \cdot (\mathbf{R} + \mathbf{t}_i)} C_{i\alpha}^{\lambda \mathbf{k}} \phi_\alpha(\mathbf{r} - \mathbf{R} - \mathbf{t}_i). \quad (4.16)$$

Now we can go back and compute the matrix elements

$$\begin{aligned} \langle \lambda_1, \mathbf{k}_1 | e^{i\mathbf{q} \cdot \mathbf{r}} | \lambda_4, \mathbf{k}_4 \rangle &= \int_{\mathcal{A}} \psi_{\lambda_1, \mathbf{k}_1}^*(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_{\lambda_4, \mathbf{k}_4}(\mathbf{r}) d\mathbf{r} = \\ &= \int_{\mathcal{A}} u_{\lambda_1, \mathbf{k}_1}^*(\mathbf{r}) e^{i(\mathbf{q} - \mathbf{k}_1 + \mathbf{k}_4) \cdot \mathbf{r}} u_{\lambda_4, \mathbf{k}_4}(\mathbf{r}) d\mathbf{r} \approx \\ &\approx \langle u_{\lambda_1, \mathbf{k}_1}^*(\mathbf{r}) u_{\lambda_4, \mathbf{k}_4}(\mathbf{r}) \rangle_{\text{UC}} \underbrace{\int_{\mathcal{A}} e^{i(\mathbf{q} - \mathbf{k}_1 + \mathbf{k}_4) \cdot \mathbf{r}} d\mathbf{r}}_{\mathcal{A} \delta_{\mathbf{q}, \mathbf{k}_1 - \mathbf{k}_4}}. \end{aligned} \quad (4.17)$$

where we used the envelope function approximation. It consists in replacing the multiplying periodic parts of the Bloch wavefunctions (see Bloch Theorem 1) by their average over the unit cell. This approximation assumes that the periodic part of the Bloch function varies fast over the length scale of the unit cell, while the exponential part varies slowly. The average of the periodic part of the Bloch wavefunctions over a unit cell can be computed, also approximately using the point-like orbital approximation (see Eq. (4.56) in the next section), by

$$\begin{aligned}
 \langle u_{\lambda_1, \mathbf{k}_1}^*(\mathbf{r}) u_{\lambda_4, \mathbf{k}_4}(\mathbf{r}) \rangle_{\text{UC}} &= \frac{1}{\mathcal{A}_{\text{UC}}} \int_{\text{UC}} u_{\lambda_1, \mathbf{k}_1}^*(\mathbf{r}) u_{\lambda_4, \mathbf{k}_4}(\mathbf{r}) \, d\mathbf{r} = \\
 &= \frac{1}{N \mathcal{A}_{\text{UC}}} \int_{\text{UC}} \left[\sum_{\mathbf{R}, i\alpha} e^{-i\mathbf{k}_1 \cdot (\mathbf{R} + \mathbf{t}_i - \mathbf{r})} \left(C_{i\alpha}^{\lambda_1 \mathbf{k}_1} \right)^* \phi_\alpha(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) \right] \times \\
 &\quad \times \left[\sum_{\mathbf{R}', j\beta} e^{i\mathbf{k}_4 \cdot (\mathbf{R}' + \mathbf{t}_j - \mathbf{r})} C_{j\beta}^{\lambda_4 \mathbf{k}_4} \phi_\beta(\mathbf{r} - \mathbf{R}' - \mathbf{t}_j) \right] d\mathbf{r} = \\
 &= \frac{1}{N \mathcal{A}_{\text{UC}}} \sum_{\mathbf{R}, \mathbf{R}'} e^{-i\mathbf{k}_1 \cdot (\mathbf{R} + \mathbf{t}_i - \mathbf{r})} e^{i\mathbf{k}_4 \cdot (\mathbf{R}' + \mathbf{t}_j - \mathbf{r})} \sum_{i\alpha, j\beta} \left(C_{i\alpha}^{\lambda_1 \mathbf{k}_1} \right)^* C_{j\beta}^{\lambda_4 \mathbf{k}_4} \times \\
 &\quad \times \int_{\text{UC}} e^{i(\mathbf{k}_1 - \mathbf{k}_4) \cdot \mathbf{r}} \underbrace{\phi_\alpha(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) \phi_\beta(\mathbf{r} - \mathbf{R}' - \mathbf{t}_j)}_{\approx \delta_{ij} \delta_{\alpha\beta} \delta_{\mathbf{R}\mathbf{R}'} \delta(\mathbf{r} - \mathbf{R} - \mathbf{t}_i)} d\mathbf{r} \approx \\
 &\approx \frac{1}{N \mathcal{A}_{\text{UC}}} \sum_{\mathbf{R}} \cancel{e^{-i(\mathbf{k}_1 - \mathbf{k}_4) \cdot (\mathbf{R} + \mathbf{t}_i)}} \sum_{i\alpha, j\beta} \left(C_{i\alpha}^{\lambda_1 \mathbf{k}_1} \right)^* C_{j\beta}^{\lambda_4 \mathbf{k}_4} \cancel{e^{i(\mathbf{k}_1 - \mathbf{k}_4) \cdot (\mathbf{R} + \mathbf{t}_i)}} = \\
 &= \frac{1}{\mathcal{A}_{\text{UC}}} \sum_{i\alpha} \left(C_{i\alpha}^{\lambda_1 \mathbf{k}_1} \right)^* C_{i\alpha}^{\lambda_4 \mathbf{k}_4} = \tag{4.18} \\
 &= \frac{1}{\mathcal{A}_{\text{UC}}} \left(\mathbf{C}^{\lambda_1 \mathbf{k}_1} \right)^\dagger \mathbf{C}^{\lambda_4 \mathbf{k}_4}, \text{ since } \sum_{\mathbf{R}} 1 = N. \tag{4.19}
 \end{aligned}$$

Finally, the plane-wave matrix element reads

$$\begin{aligned}
 \langle \lambda_1, \mathbf{k}_1 | e^{i\mathbf{q} \cdot \mathbf{r}} | \lambda_4, \mathbf{k}_4 \rangle &= \frac{1}{\mathcal{A}_{\text{UC}}} \left(\mathbf{C}^{\lambda_1 \mathbf{k}_1} \right)^\dagger \mathbf{C}^{\lambda_4 \mathbf{k}_4} \mathcal{A} \delta_{\mathbf{q}, \mathbf{k}_1 - \mathbf{k}_4} = \\
 &= N \left(\mathbf{C}^{\lambda_1 \mathbf{k}_1} \right)^\dagger \mathbf{C}^{\lambda_4 \mathbf{k}_4} \delta_{\mathbf{q}, \mathbf{k}_1 - \mathbf{k}_4}, \tag{4.20}
 \end{aligned}$$

whereas for the other matrix element the same reasoning applies. In this way, the interaction term of the Hamiltonian in momentum space reads

$$\hat{V} = \frac{1}{2\mathcal{A}N^2} \sum_{\mathbf{q}} \sum_{\{\mathbf{k}_i, \lambda_i\}} V(\mathbf{q}) N^2 \left(\mathbf{C}^{\lambda_1 \mathbf{k}_1} \right)^\dagger \mathbf{C}^{\lambda_4 \mathbf{k}_4} \delta_{\mathbf{q}, \mathbf{k}_1 - \mathbf{k}_4} \left(\mathbf{C}^{\lambda_2 \mathbf{k}_2} \right)^\dagger \mathbf{C}^{\lambda_3 \mathbf{k}_3} \delta_{\mathbf{q}, \mathbf{k}_3 - \mathbf{k}_2} \times \tag{4.21}$$

$$\times c_{\lambda_1 \mathbf{k}_1}^\dagger c_{\lambda_2 \mathbf{k}_2}^\dagger c_{\lambda_3 \mathbf{k}_3} c_{\lambda_4 \mathbf{k}_4} = \tag{4.22}$$

$$= \frac{1}{2\mathcal{A}N^2} \sum_{\mathbf{q}, \mathbf{k}_3, \mathbf{k}_4} \sum_{\{\lambda_i\}} V(\mathbf{q}) N^2 \left(\mathbf{C}^{\lambda_1 \mathbf{k}_4 + \mathbf{q}} \right)^\dagger \mathbf{C}^{\lambda_4 \mathbf{k}_4} \left(\mathbf{C}^{\lambda_2 \mathbf{k}_3 - \mathbf{q}} \right)^\dagger \mathbf{C}^{\lambda_3 \mathbf{k}_3} \tag{4.23}$$

$$\times c_{\lambda_1 \mathbf{k}_4 + \mathbf{q}}^\dagger c_{\lambda_2 \mathbf{k}_3 - \mathbf{q}}^\dagger c_{\lambda_3 \mathbf{k}_3} c_{\lambda_4 \mathbf{k}_4} = \tag{4.24}$$

$$= \frac{1}{2\mathcal{A}} \sum_{\mathbf{q}, \mathbf{k}_3, \mathbf{k}_4} \sum_{\{\lambda_i\}} V(\mathbf{q}) F_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}(\mathbf{k}_3, \mathbf{k}_4, \mathbf{q}) c_{\lambda_1 \mathbf{k}_4 + \mathbf{q}}^\dagger c_{\lambda_2 \mathbf{k}_3 - \mathbf{q}}^\dagger c_{\lambda_3 \mathbf{k}_3} c_{\lambda_4 \mathbf{k}_4}, \tag{4.25}$$

where we defined the form factor $F_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}(\mathbf{k}, \mathbf{p}, \mathbf{q}) = \left(\mathbf{C}^{\lambda_1 \mathbf{p} + \mathbf{q}} \right)^\dagger \mathbf{C}^{\lambda_4 \mathbf{p}} \left(\mathbf{C}^{\lambda_2 \mathbf{k} - \mathbf{q}} \right)^\dagger \mathbf{C}^{\lambda_3 \mathbf{k}_3}$.

For simplicity, we can take spinors of the low-energy Bloch Hamiltonian for the vectors $\mathbf{C}^{\lambda \mathbf{k}}$, which is not a problem if the exciton is localized around a specific point in the Brillouin zone (BZ). In general, there may exist more than one of those points. For instance, in

the example of hBN, to which we will apply this formalism, there are the nonequivalent K and K' points in the BZ where the bandgap is located (and the points around which the exciton wavefunction $\phi_\nu(\mathbf{q})$ is localized). And in that case, the arrays $\mathbf{C}^{\lambda\mathbf{k}}$ assume the form of the spinors in Eq. (3.40). As both valleys share the same value for the bandgap, the exciton states enjoy valley degeneracy. This means that the BSE can be solved for only one of the valleys without loss of generality. If the exciton does not enjoy valley degeneracy, then the valley degree of freedom has to be accounted for.

We assume that the system can be well described by the bosonic operators $b_\nu^\dagger$ and b_ν , and that ν is a set of good quantum numbers, in the sense that the Hamiltonian is diagonal in this basis, which allows us to write

$$\hat{H}_X = \sum_\nu E_\nu b_\nu^\dagger b_\nu, \quad (4.26)$$

where we introduced the notation $\hat{H}_X$ for the Hamiltonian written in the basis of exciton states ν . Now that we have both terms in the Hamiltonian written in momentum space, the strategy is as follows: we take the commutator $[\hat{H}_X, b_\nu^\dagger]$ between $\hat{H}_X$ and the creation operator $b_\nu^\dagger$; next, we take the same commutator, but with the initial total Hamiltonian $\hat{H}$ instead (this step is the most involved one); finally, we equate both results, and the Bethe–Salpeter equation is then automatically extracted. The first commutator is straightforward

$$[\hat{H}_X, b_\nu^\dagger] = E_\nu b_\nu^\dagger = E_\nu \frac{1}{\sqrt{\mathcal{A}}} \sum_{\mathbf{k}} \phi_\nu(\mathbf{k}) c_{\mathbf{c}\mathbf{k}}^\dagger c_{v\mathbf{k}}, \quad (4.27)$$

where to write the second equality we explicitly wrote the bosonic creation operator in terms of electron–hole pairs as in (4.7). Now, we work with the total Hamiltonian $\hat{H}$ in terms of fermionic operators, and compute

$$[\hat{H}, b_\nu^\dagger] = [\hat{H}_0, b_\nu^\dagger] + [\hat{V}, b_\nu^\dagger], \quad (4.28)$$

but we better treat both contributions separately. First, the commutator with the kinetic component evaluates to

$$[\hat{H}_0, b_\nu^\dagger] = \sum_{\lambda\mathbf{k}} \epsilon_{\lambda\mathbf{k}} \sum_{\mathbf{q}} \frac{\phi_\nu(\mathbf{q})}{\sqrt{\mathcal{A}}} \left[c_{\lambda\mathbf{k}}^\dagger c_{\lambda\mathbf{k}}, c_{\mathbf{c}\mathbf{q}}^\dagger c_{v\mathbf{q}} \right] = \frac{1}{\sqrt{\mathcal{A}}} \sum_{\mathbf{k}} \phi_\nu(\mathbf{k}) (\epsilon_{\mathbf{c}\mathbf{k}} - \epsilon_{v\mathbf{k}}) c_{\mathbf{c}\mathbf{k}}^\dagger c_{v\mathbf{k}}, \quad (4.29)$$

after simplifying. At last, we have to do the commutator with the interacting term. This task is not hard, just very tedious. Therefore, we describe briefly the key steps and present the final result. For the commutator $[\hat{V}, b_\nu^\dagger]$, we have to compute the commutator between an operator quartic in c with one quadratic in c ; four contributions arise. Each contribution is an operator quartic in c . To make it comparable with the very first commutator in Eq. 4.27, we need to introduce an approximation that turns the quartic operators into quadratic ones—the mean-field approximation. It can also be called the Hartree–Fock decoupling [74]. Applying this approximation to one of the initial contributions in $[\hat{V}, b_\nu^\dagger]$ results in four

terms, and, in turn, each of these terms is a product between an operator quadratic in c and an equilibrium average of another operator quadratic in c . To better clarify, we write this decoupling using mathematical symbols. Assuming that the fermionic operators are written in a specific state basis $|i\rangle$, we write

$$c_{i_1}^\dagger c_{i_2}^\dagger c_{i_3} c_{i_4} \approx c_{i_2}^\dagger c_{i_3} \langle c_{i_1}^\dagger c_{i_4} \rangle + c_{i_1}^\dagger c_{i_4} \langle c_{i_2}^\dagger c_{i_3} \rangle - c_{i_2}^\dagger c_{i_4} \langle c_{i_1}^\dagger c_{i_3} \rangle - c_{i_1}^\dagger c_{i_3} \langle c_{i_2}^\dagger c_{i_4} \rangle. \quad (4.30)$$

This approximation comes from admitting that an operator of the form $c^\dagger c$ exhibits small fluctuations around its average value $\langle c^\dagger c \rangle$, which is an average computed at thermal equilibrium (without forgetting about the fermion statistics). One can consult Ref. [74] for more details. This approximation has to be done for each of the four contributions in $[\hat{V}, b_\nu^\dagger]$. Doing so results in a total of sixteen terms. In each of these sixteen terms, we have to evaluate an expectation value of the form $\langle c^\dagger c \rangle$. These expectation values are simply equilibrium thermal averages, and evaluated at zero temperature for simplicity⁴. Therefore, we can apply the rule

$$\langle c_{\lambda_1 \mathbf{k}}^\dagger c_{\lambda_2 \mathbf{k}'} \rangle = \begin{cases} 0 & \text{if } \lambda_1 = \lambda_2 = c, \\ \delta_{\mathbf{k}, \mathbf{k}'} & \text{if } \lambda_1 = \lambda_2 = v, \\ 0 & \text{if } \lambda_1 = c \text{ and } \lambda_2 = v, \\ 0 & \text{if } \lambda_1 = v \text{ and } \lambda_2 = c. \end{cases} \quad (4.31)$$

After that, we need to gather all the contributions, while recovering the sums in momentum to simplify the expression—equal terms simply add up or cancel. One can also smartly use the properties of the form factor, changing the order in which the $\mathbf{C}^{\lambda \mathbf{k}}$ arrays appear. To arrive at the final result, a couple of shifts in momentum might also be in order. At the end, we can set $[\hat{H}_X, b_\nu^\dagger] = [\hat{H}_0, b_\nu^\dagger] + [\hat{V}, b_\nu^\dagger]$, from which we can extract the Bethe–Salpeter equation for the exciton wavefunction $\phi_\nu(\mathbf{k})$ and its respective energy E_ν

$$\begin{aligned} E_\nu \phi_\nu(\mathbf{k}) = & \phi_\nu(\mathbf{k}) (\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}}) - \\ & \frac{1}{\mathcal{A}} \sum_{\mathbf{p}} \phi_\nu(\mathbf{p}) V(\mathbf{k} - \mathbf{p}) F_{cvvc}(\mathbf{k}, \mathbf{p}, \mathbf{k} - \mathbf{p}) + \\ & \frac{1}{\mathcal{A}} \phi_\nu(\mathbf{k}) \sum_{\mathbf{p}} V(\mathbf{p}) [F_{vvvv}(\mathbf{p} + \mathbf{k}, \mathbf{k}, \mathbf{p}) - F_{vcvc}(\mathbf{p} + \mathbf{k}, \mathbf{k}, \mathbf{p})]. \end{aligned} \quad (4.32)$$

The first term on the right-hand side is the gain in energy due to the transition of the system from the ground state to the excited state of an unbound electron–hole pair. The second term accounts for the attractive interaction between the electron and the hole, making it possible for the complex to form a bound state. The last term can be interpreted

⁴In the eventuality of the gapped material having a gap comparable with the thermal energy $k_B T$, then considering $T = 0$ K simplifies the analysis a lot, keeping the conduction bands unpopulated in the ground state.

as a renormalization of the bands due to electron–electron interactions, so it can be simply absorbed into the first term. It does not mean that it can be ignored, but we sweep it under the rug for now. Doing so, we arrive at the simplified form of the Bethe–Salpeter equation:

$$E_\nu \phi_\nu(\mathbf{k}) = (\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}}) \phi_\nu(\mathbf{k}) - \frac{1}{\mathcal{A}} \sum_{\mathbf{p}} \phi_\nu(\mathbf{p}) V(\mathbf{k} - \mathbf{p}) F_{cvvc}(\mathbf{k}, \mathbf{p}, \mathbf{k} - \mathbf{p}) . \quad (4.33)$$

The extension to multi-band systems is straightforward, but we choose to omit it here, referring to [15] instead. The exchange term in the Coulomb interaction is neglected, but this does not strongly affect the exciton binding energy [24]. The multi-band case is to be considered only in the next section, and with a different formalism altogether. We just make the remark that the negative sign going with the term in the potential is correct, and the electron–hole interaction is for sure attractive, since we use the convention of a positive interaction potential for $V(\mathbf{k})$.

We finish this section with a derivation of the Wannier equation as in (4.6) from the BSE in Eq. 4.33. For that, we need to introduce two approximations. First, we make the parabolic dispersion approximation, that is

$$\epsilon_{\lambda\mathbf{k}} \simeq \epsilon_{\lambda\mathbf{0}} + \frac{\hbar^2 k^2}{2m_\lambda}, \quad (4.34)$$

where m_λ is the effective mass of an electron/hole in the band λ , and $\mathbf{k}$ is measured from the point in the BZ where the exciton is localized. It follows that

$$\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}} \simeq E_g + \frac{\hbar^2 k^2}{2\mu_{eh}}, \quad (4.35)$$

where $E_g = \epsilon_{c\mathbf{0}} - \epsilon_{v\mathbf{0}}$ is the bandgap, and μ_{eh} is the electron–hole reduced mass. In the second term on the right-hand side of (4.33), we approximate $F \approx 1$, which is valid for large enough bandgaps [28]. In this way, the equation (4.33) leads to

$$E_{b,\nu} \phi_\nu(\mathbf{k}) = \frac{\hbar^2 k^2}{2\mu_{eh}} \phi_\nu(\mathbf{k}) - \frac{1}{\mathcal{A}} \sum_{\mathbf{p}} \phi_\nu(\mathbf{p}) V(\mathbf{k} - \mathbf{p}), \quad (4.36)$$

where $E_{b,\nu} = E_\nu - E_g$ is the binding energy of the excitonic state ν . Next, we apply a Fourier transform to the previous equation. The term on the left-hand side is direct, it becomes $E_{b,\nu} \phi_\nu(\mathbf{r})$. The first term on the right-hand side, after doing the inverse Fourier transform back to real-space representation, gives simply $p^2/2\mu_{eh}$ acting on the wavefunction $\phi_\nu(\mathbf{r})$. The second term on the right-hand side is a convolution in momentum space, so it becomes a product in real space. To see that, let us perform the inverse Fourier transform of just that term and write⁵

⁵All details about Fourier transforms can be found in App. A.

$$\begin{aligned}
 & \frac{1}{\mathcal{A}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} \frac{1}{\mathcal{A}} \sum_{\mathbf{p}} \phi_{\nu}(\mathbf{p}) V(\mathbf{k} - \mathbf{p}) = \frac{1}{\mathcal{A}^2} \sum_{\mathbf{k}} \sum_{\mathbf{p}} e^{i\mathbf{k}\cdot\mathbf{r}} e^{-i\mathbf{p}\cdot\mathbf{r}} e^{i\mathbf{p}\cdot\mathbf{r}} \phi_{\nu}(\mathbf{p}) V(\mathbf{k} - \mathbf{p}) = \\
 & = \frac{1}{\mathcal{A}^2} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}} \phi_{\nu}(\mathbf{p}) \sum_{\mathbf{k}} e^{i(\mathbf{k}-\mathbf{p})\cdot\mathbf{r}} V(\mathbf{k} - \mathbf{p}) \stackrel{\mathbf{k} \rightarrow \mathbf{k}+\mathbf{p}}{=} \frac{1}{\mathcal{A}} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}} \phi_{\nu}(\mathbf{p}) \frac{1}{\mathcal{A}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} V(\mathbf{k}) = \\
 & = \phi_{\nu}(\mathbf{r}) V(\mathbf{r}) .
 \end{aligned} \tag{4.37}$$

In this way, we recover the Wannier equation

$$E_{b,\nu} \psi_{\nu}(\mathbf{r}) = \frac{p^2}{2\mu_{eh}} \psi_{\nu}(\mathbf{r}) - V(\mathbf{r}) \psi_{\nu}(\mathbf{r}) . \tag{4.38}$$

As we can see, the (simplified) excitonic problem is a differential equation in real space, since in quantum mechanics $\mathbf{p} = -i\hbar\nabla_{\mathbf{r}}$, and an integral equation in momentum space. The real-space approach is more useful for analytic treatments (or semi-analytical if we use the Rytova–Keldysh potential). Unfortunately, it overlooks some features due to the approximations. As a consequence of approximating F to 1, excitonic states with symmetric angular momentum numbers are degenerate. On the contrary, the momentum space approach has no analytical solution, nor does it allow a simple semi-analytical approach, but has the advantage of providing more accurate results.

This formalism can be readily applied to isotropic systems, like hBN. Basically, one can separate the angular dependence of the wavefunction from the radial dependence, and in that way the problem can be cast into a one dimensional (1D) integral equation that, after discretizing the momentum, can be converted to an eigenvalue problem with only one momentum variable. How to do so in practice is explained in detail in Refs. [15] and [143], namely in their respective appendices. In contrast with isotropic systems, for anisotropic systems one cannot separate the angular and radial variables, and one has then to construct a big matrix with two momentum variables. But for such systems, the BSE formalism of the next section is more useful. Despite this observations, in Chapter 5 we approach the anisotropic excitonic problem for nanostructured hBN using the Wannier formalism—with this method, we obtain interesting results with very low computational effort. If one wishes to use a more detailed description for excitons, for instance, if a simple tight-binding (TB) description for the material is not possible (e.g. MoS₂, phosphorene, etc.), then the one presented in the next section is more appropriate. It constitutes as well a playground to test the results obtained in Chapter 6 for our numerical implementation of screening.

Before we report the research outcomes in this dissertation, we still need to introduce another picture for the excitonic problem, based on the numerical diagonalization of the BSE, but still within a TB approach. After that, we revisit the concept of dielectric function, but now in the local and dynamic limit. In that way, we shall reintroduce the concept of optical conductivity: the noninteracting we have already seen, and the interacting one at the excitonic level; finally, we will be able to interpret the model parameters of the conductivity in Eq. (2.27), herein derived in the light of quantum mechanics.

4.3 The Bethe–Salpeter Equation: exact diagonalization

In this section, we dwell on a second-principles-based approach to the calculation of the excitons, within an atomistic description where the TB approximation discussed previously is key. The discussion is adapted from [132] and [37]. This approach presents technical and computational advantages to first-principles-based methodologies for excitons pretty much established in the literature, such as the $GW+BSE$ method [31, 127, 144, 145]. Even though more numerically inefficient than solving the Wannier equation or the BSE applied to effective models, this method is still much more efficient than *ab-initio* ones. In spite of this, we can expect from it detailed insights into the excitonic problem that the ones introduced until now do not provide. It is very useful when one is interested in bigger systems, or when the symmetries of the excitonic states are relevant.

The starting point is to write the total electronic Hamiltonian⁶

$$\hat{H} = \hat{H}_0 + \hat{V} = \sum_{i,j} t_{ij} c_i^\dagger c_j + \frac{1}{2} \sum_{i,j,k,l} V_{ijkl} c_i^\dagger c_j^\dagger c_l c_k, \quad (4.39)$$

where the indices include the relevant degrees of freedom (orbital site and orbital type), the first term corresponds to the interaction between the valence electrons and the ions (with frozen degrees of freedom), and the second term corresponds to the electrostatic Coulomb interaction between the electrons. The single-particle physics is encoded in $\hat{H}_0$, to which all the techniques and results discussed in Section 3.2 apply (and the ones within the random-phase approximation (RPA) from Section 3.4 as well). In the Section 3.2, where we discussed single-particle properties of electrons in crystals, particle–particle interactions were not completely discarded, in the sense that the electronic eigenstates—the Bloch states—that describe the ground state of the material, were previously obtained through self-consistent calculations (Hartree–Fock (HF) or DFT), where interactions and correlations are treated at the mean-field level. The many-particle physics (e.g. excitons) is encoded in $\hat{V}$, without which excitons cannot be described, in concordance with the previous section. Instead of departing from the many-body Hamiltonian with the interacting part in terms of field operators, we depart from a Hamiltonian in real space already written in an atomistic basis, i.e., we adopt the TB description from the Section 3.2.2.1 of the previous chapter.

The sums in Eq. (4.39) run over indices including orbitals and position degrees of freedom (and spin, eventually). The parameters t_{ij} are the TB parameters, while V_{ijkl} are the interaction matrix elements⁷

⁶Commonly, when one looks at TB Hamiltonians in the literature, there is an extra term denoted by H.c. meaning Hermitian conjugate to ensure that the Hamiltonian is Hermitian. Here, we simply assume that the TB parameters t_{ij} per se render the Hamiltonian Hermitian. Something similar can be said about the interaction term, so the reader does not have to worry about Hermiticity of the Hamiltonians we consider.

⁷Note that by matrix elements, we mean the symbols V_{ijkl} , which are matrix elements of the first quantized interaction term V between single-particle states belonging to the one-particle Hilbert space, and not matrix elements of the second quantized operator $\hat{V}$ between states living in the Fock space.

$$V_{ijkl} = \langle i, j | V | k, l \rangle = \int d\mathbf{r} \int d\mathbf{r}' \phi_i^*(\mathbf{r}) \phi_j^*(\mathbf{r}') V(\mathbf{r}, \mathbf{r}') \phi_k(\mathbf{r}) \phi_l(\mathbf{r}'). \quad (4.40)$$

At this point, we do not particularize how we model the screening, postponing that for later on.

4.3.1 The Bethe–Salpeter equation rederived

Now, with the purpose of describing excitons in the more general multi-band case, we need to start from a ground state $|\text{GS}\rangle$ taking all bands into account. For undoped 2D semiconductors or insulators, whose ground state consists of all the valence bands filled and empty conduction bands, we can write mathematically

$$|\text{GS}\rangle = \prod_{n, \mathbf{k}}^{\epsilon_{n\mathbf{k}} \leq E_F} c_{n\mathbf{k}}^\dagger |0\rangle, \quad (4.41)$$

where $|0\rangle$ denotes the “vacuum” state (the crystal in the absence of valence electrons) and E_F is the Fermi energy. Exciting an electron from the valence band v with momentum $\mathbf{k}$ to the conduction band c with momentum $\mathbf{k} + \mathbf{Q}$, we are before an electron–hole pair, whose state can be formally written as

$$|v, c, \mathbf{k}, \mathbf{Q}\rangle \equiv c_{c\mathbf{k}+\mathbf{Q}}^\dagger c_{v\mathbf{k}} |\text{GS}\rangle. \quad (4.42)$$

As a passing note, there is no technical reason why we should adopt a quasiparticle hole description for the holes (i.e., define a hole creation/annihilation operator h as $h = c/c^\dagger$). Just like in the previous section, we only use fermion operators.

If the exciton has a center of mass momentum $\mathbf{Q}$, and assuming that a basis of electron–hole pairs is appropriate for the excitonic states, then we can write as an ansatz for the exciton state $|X_{\mathbf{Q}}\rangle$

$$|X_{\mathbf{Q}}\rangle = \sum_{vc, \mathbf{k}} A_{vc}^{\mathbf{Q}}(\mathbf{k}) |v, c, \mathbf{k}, \mathbf{Q}\rangle = \sum_{vc, \mathbf{k}} A_{vc}^{\mathbf{Q}}(\mathbf{k}) c_{c\mathbf{k}+\mathbf{Q}}^\dagger c_{v\mathbf{k}} |\text{GS}\rangle. \quad (4.43)$$

In this way, the exciton state is written as a linear combination of electron–hole pair states summing over valence bands, conduction bands, and momentum. We can consider the quantum number $\mathbf{Q}$ a good one since the interaction conserves momentum, as all electron–hole pairs share the same center of mass momentum $\mathbf{Q}$. Up until now, we have just reintroduced the problem as in the beginning of Section 4.2, albeit with a different notation. The main difference resides in the approach for determining the coefficients $A_{vc}^{\mathbf{Q}}(\mathbf{k})$ (previously labeled by $\phi_\nu(\mathbf{k})$, that we identified as the exciton wavefunction). Herein, we follow a variational approach (see the beginning of Section 3.1.3). Hence, the expansion coefficients $A_{vc}^{\mathbf{Q}}(\mathbf{k})$ are determined by minimization of the expectation value $\langle X | \hat{H} | X \rangle$ (omitting the $\mathbf{Q}$ subscript momentarily) with respect to $|X\rangle$

$$\frac{\delta E[X]}{\delta X} = \frac{\delta}{\delta X} \left[\frac{\langle X | \hat{H} | X \rangle}{\langle X | X \rangle} \right] = \frac{\delta \langle X | \hat{H} | X \rangle}{\delta X} \underbrace{\langle X | X \rangle}_{E_X} - \frac{\langle X | \hat{H} | X \rangle}{\langle X | X \rangle} \frac{\delta \langle X | X \rangle}{\delta X} = 0. \quad (4.44)$$

Upon performing the variational derivatives and using the notation $H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q})$ for the matrix elements of the Hamiltonian $\langle v, c, \mathbf{k}, \mathbf{Q} | \hat{H} | v', c', \mathbf{k}', \mathbf{Q} \rangle \equiv H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q})$ between two electron–hole pair states, we arrive at the equation

$$\sum_{v'c', \mathbf{k}'} H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) A_{v'c'}^{\mathbf{Q}}(\mathbf{k}') = E_X A_{vc}^{\mathbf{Q}}(\mathbf{k}), \quad (4.45)$$

which is a different way of writing the BSE, and the symbols $H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q})$ are the matrix elements of what we call the Bethe–Salpeter Hamiltonian.

The expansion of the excitonic state in a basis of electron–hole pairs is an ansatz under the assumption that the basis is adequate (or complete, technically speaking), so that we can write the true excitonic states in that basis. Instead of considering all the possible combinations of electron–hole excitations, where several electrons would populate the conduction bands leaving as many holes behind in the valence bands, we do the calculations in a reduced Fock space where the only excited states are unbound electron–hole pairs, presuming that this is a good enough approximation. For a more rigorous discussion of this aspect, we refer to Refs. [37, 132].

We need to compute the matrix elements $H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q})$ of the Bethe–Salpeter Hamiltonian. Naturally, this has to be done numerically in practice, which also means that we have to truncate the Bethe–Salpeter Hamiltonian and work with an approximate finite-matrix representation. Physically speaking, this approximation means that we are considering only a finite number of valence and conduction bands, and a finite number of momentum points in the BZ (or unit cells, equivalently). Mathematically, it means that we are working with a truncated basis for $\hat{H}_0$, which in practice is not complete. Numerically, we have to compute the matrix elements for each pair of momenta $(\mathbf{k}, \mathbf{k}')$ in the BZ mesh, and for each pair $(vc, v'c')$ of pairs of valence bands v, v' and of conduction bands c, c' and ensure numerical convergence. We mean that the numbers of $\mathbf{k}$ points, valence bands and conduction bands included have to be enough to obtain the expected degeneracies and symmetries. Naturally, we select only the valence and conduction bands more relevant for the excitons, those closer to the Fermi level.

Before we explain how to compute the matrix elements for the BSE, let us see how far we can go with analytical expressions. The Bethe–Salpeter Hamiltonian matrix elements are given by $H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \langle v, c, \mathbf{k}, \mathbf{Q} | \hat{H}_0 | v', c', \mathbf{k}', \mathbf{Q} \rangle + \langle v, c, \mathbf{k}, \mathbf{Q} | \hat{V} | v', c', \mathbf{k}', \mathbf{Q} \rangle$. The first term corresponds to the matrix elements of the Hamiltonian at the mean-field level between two electron–hole pair states, while the second term corresponds to the interaction matrix elements. We work out first the matrix elements of $\hat{H}_0$, as they are easier to deal with. For that, let us assume that the operator $\hat{H}_0$ can be written in a spectral representation of

Bloch states labeled by $|n, \mathbf{k}\rangle$ ($|\lambda, \mathbf{k}\rangle$ in the notation of the previous section). Then, in its second-quantized form, the single-particle Hamiltonian reads

$$\hat{H}_0 = \sum_{n, \mathbf{k}} \epsilon_{n\mathbf{k}} c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}}, \quad (4.46)$$

assuming that the diagonalization has been done. Replacing the expression above in the matrix element of $\hat{H}_0$, it is not hard to show that

$$\langle v, c, \mathbf{k}, \mathbf{Q} | \hat{H}_0 | v', c', \mathbf{k}', \mathbf{Q} \rangle = (\epsilon_{c\mathbf{k}+\mathbf{Q}} - \epsilon_{v\mathbf{k}}) \delta_{v,v'} \delta_{\mathbf{k},\mathbf{k}'} \delta_{c,c'} \quad \text{up to a shift of } E_F. \quad (4.47)$$

The interaction matrix elements are tricky to compute, as the interaction operator $\hat{V}$ is a two-body operator, quartic in the creation and annihilation operators. Nonetheless, in this section we refrain from performing any kind of mean-field approximation, and compute the matrix elements exactly. Doing so is not hard, it is just tedious. We do not perform all the steps, and instead explain briefly how to proceed. For convenience, we make a brief change and abuse of notation, and relabel $(v', \mathbf{k}') \rightarrow v'$, $(c', \mathbf{k}' + \mathbf{Q}) \rightarrow c'$, $(v, \mathbf{k}) \rightarrow v$ and $(c, \mathbf{k} + \mathbf{Q}) \rightarrow c$, and keep the notation for $\hat{V}$ used above. Then, we have

$$\begin{aligned} \langle v, c, \mathbf{k}, \mathbf{Q} | \hat{V} | v', c', \mathbf{k}', \mathbf{Q} \rangle &= \sum_{i,j,k,l} V_{ijkl} \langle v, c, \mathbf{k}, \mathbf{Q} | c_i^\dagger c_j^\dagger c_l c_k | v', c', \mathbf{k}', \mathbf{Q} \rangle = \\ &= \sum_{i,j,k,l} V_{ijkl} \langle \text{GS} | c_v^\dagger c_c c_i^\dagger c_j^\dagger c_l c_k c_{v'}^\dagger c_{c'} | \text{GS} \rangle. \end{aligned}$$

For instance, we pick the pair $c_c c_i^\dagger$ and use the anti-commutation relations to move the creation operator to the left, and the annihilation operator to the right, giving rise to two new matrix elements. Then, in each term, we select a new pair of creation and annihilation operators $c^\dagger c$ and repeat the process, making more terms appear. Following this procedure recursively, until we exhaust all our options of pairs of creation and annihilation operators, and end up only with a matrix element of a single $c^\dagger c$ pair to evaluate in each term of the final expression, we arrive at

$$\begin{aligned} \langle v, c, \mathbf{k}, \mathbf{Q} | \hat{V} | v', c', \mathbf{k}', \mathbf{Q} \rangle &= V_{c'vv'c} - V_{c'vcv'} + \delta_{vv'} \sum_j (V_{c'jcj} - V_{c'jjc}) + \\ &\delta_{cc'} \sum_j (V_{vjv'j} - V_{vjv'j}) + \delta_{vv'} \delta_{cc'} \frac{1}{2} \sum_{ij} (V_{ijij} - V_{ijji}). \end{aligned} \quad (4.48)$$

Summing the two contributions (4.47) and (4.48), we have for the Bethe–Salpeter Hamiltonian matrix elements

$$\begin{aligned} H_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) &= \delta_{\mathbf{k},\mathbf{k}'} \delta_{v,v'} [\epsilon_{c\mathbf{k}+\mathbf{Q}} \delta_{c,c'} + \Sigma_{cc'}(\mathbf{k} + \mathbf{Q}, \mathbf{k}' + \mathbf{Q})] - \\ &\delta_{\mathbf{k},\mathbf{k}'} \delta_{c,c'} [\epsilon_{v\mathbf{k}} \delta_{v,v'} + \Sigma_{v'v}(\mathbf{k} + \mathbf{Q}, \mathbf{k}' + \mathbf{Q})] - \\ &(D - X)_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}), \end{aligned} \quad (4.49)$$

where we have defined the direct and exchange interaction matrix elements as

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = V_{c\mathbf{k}+\mathbf{Q},v'\mathbf{k}',c'\mathbf{k}'+\mathbf{Q},v\mathbf{k}} \quad \text{and} \quad (4.50a)$$

$$X_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = V_{c\mathbf{k}+\mathbf{Q},v'\mathbf{k}',v\mathbf{k},c'\mathbf{k}'+\mathbf{Q}}, \quad (4.50b)$$

respectively, and the self-energy accounting for the interaction between the electron (or hole) with the Fermi sea as

$$\Sigma_{nm}(\mathbf{k}, \mathbf{k}') = \sum_{j, \mathbf{k}''}^{\text{occ}} (V_{n\mathbf{k},j\mathbf{k}'',m\mathbf{k}',j\mathbf{k}''} - V_{n\mathbf{k},j\mathbf{k}'',j\mathbf{k}'',m\mathbf{k}'}), \quad (4.51)$$

where the sum is only over occupied bands. About the self-energy, we have to make a technical remark. In principle, we could diagonalize the Bethe–Salpeter Hamiltonian with matrix elements given by Eq. (4.49) as is. But that is not what we do. The Hamiltonian that we depart from in Eq. (4.39), is such that the single-particle part has TB parameters t_{ij} that come from diagonalizing the Hamiltonian (3.8) including interactions, but at the mean-field level via HF theory or DFT [146]. In that way, we can then see the self-energy corrections as already incorporated in the quasiparticle energies $\epsilon_{n\mathbf{k}}$. On the other hand, if we were to solve the BSE as per (4.49), the Hamiltonian to diagonalize in order to obtain the single-particle properties would have to be the one in (3.6), the noninteracting limit of the many-body crystal Hamiltonian in the Born–Oppenheimer approximation. Then, we would have a distinct set of TB parameters t'_{ij} , and another initial band structure $\epsilon'_{n\mathbf{k}}$. The caveat is that the cancellation might not be exact. In math notation, it means that $\epsilon_{n\mathbf{k}} = \epsilon'_{n\mathbf{k}} + \Sigma_{nn}(\mathbf{k} + \mathbf{Q}, \mathbf{k} + \mathbf{Q})$ might be false. But if we assume that it is true, then the BSE in Eq. (4.49) reduces to

$$(\epsilon_{c\mathbf{k}+\mathbf{Q}} - \epsilon_{v\mathbf{k}})A_{vc}^{\mathbf{Q}}(\mathbf{k}) + \sum_{v'c', \mathbf{k}'} K_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q})A_{v'c'}^{\mathbf{Q}}(\mathbf{k}') = E_X A_{vc}^{\mathbf{Q}}(\mathbf{k}), \quad (4.52)$$

where the energies $\{\epsilon_{n\mathbf{k}}\}$ are now mean-field quasiparticle energies, and $K = -(D - X)$ is the interaction kernel. This BSE is similar to the one obtained with many-body perturbation theory [144, 145]. However, our version is an already approximated version when compared to the many-body perturbation theory one, which includes terms that render the Bethe–Salpeter Hamiltonian non-Hermitian. Ignoring those terms is the Tamm–Dancoff approximation [147, 148], and it is routine in the literature. The BSE obtained in that way resembles ours. Nonetheless, there are two striking differences between the BSE we solve and the one obtained within many-body perturbation theory [144, 145]. In the latter, the problem has a frequency dependence that we do not find here. The screening is dynamic, and that entails performing an integral in the frequency to compute the interaction matrix elements. Here we model the screening using a static dielectric function, so we work always in the static limit $\omega \rightarrow 0$. On one hand, this approximation is also common in *ab-initio* calculations in practice, although in that context it has to be justified. On the other hand,

the general case $\omega \neq 0$ hinders heavily the numerical calculations. The other difference we can point out is that within many-body perturbation theory, the exchange term X is a matrix element of the **bare** Coulomb potential, while only the direct term is a matrix element of the **screened** Coulomb potential. In our case, both direct and exchange terms are matrix elements of the **screened** Coulomb potential, and there is no reason for the exchange term to involve the bare Coulomb potential (either way, it is usually discarded).

We have not hitherto specified the form of the Coulomb potential, even though screening was an important topic in previous chapters. We shall discuss that right in the next section, where we explain how we can compute the interaction matrix elements in the Bethe–Salpeter equation taking screening into account.

4.3.2 The interaction matrix elements and screening

Before working out the expression for the interaction matrix elements D and X , let us first remind the reader that we use Bloch wavefunctions, that in the lattice gauge, read

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \phi_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i), \quad (4.53)$$

where the TB coefficients are determined by diagonalizing the Bloch Hamiltonian (as explained in Sec. 3.2.2.1), and the atomic orbitals denoted by ϕ_{α} are orthogonal, as we work under the TB formalism⁸. We focus first on the direct interaction term as per (4.50a), that in terms of Bloch states is

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \int d\mathbf{r} \int d\mathbf{r}' \psi_{c,\mathbf{k}+\mathbf{Q}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') W(\mathbf{r}, \mathbf{r}') \psi_{c',\mathbf{k}'+\mathbf{Q}}(\mathbf{r}) \psi_{v\mathbf{k}}(\mathbf{r}'), \quad (4.54)$$

where we have denoted now the screened potential by W instead of V . We have also omitted the region of integration, but it is implied that it is over the extension of the crystal (over its volume $\mathcal{V}$ if it is 3D, or over its area $\mathcal{A}$ if it is 2D). If the integration region is not explicit, then the previous statement is assumed⁹. Upon substitution of the Bloch states by their expression in Eq. (4.53), at some point we need to evaluate four-body integrals of the form

$$\int d\mathbf{r} \int d\mathbf{r}' \phi_{\alpha}^*(\mathbf{r}) \phi_{\beta}^*(\mathbf{r}') W(\mathbf{r}, \mathbf{r}') \phi_{\alpha'}(\mathbf{r}) \phi_{\beta'}(\mathbf{r}'), \quad (4.55)$$

⁸In general, they are not orthogonal. If the model Hamiltonian has been obtained by a Wannier interpolation and is written in a basis of Wannier functions, then they are orthogonal by construction. However, if the Hamiltonian comes from calculations on a basis of localized functions other than Wannier functions, they may not be orthogonal. This is not a problem at the end, because one can always follow an orthogonalization procedure. The code we work with does that. We shall remind the reader again, once it comes the time of discussing the code developed in Chapter 6.

⁹We justify this choice for the notation with the fact that the integrals for D and X will have to be computed numerically. In practice, the system has to be truncated (with periodic boundary conditions), and has as many $\mathbf{k}$ points in the BZ as unit cells, N . The results then have to be converged with N , which is finite.

where the position vectors in the argument of the orbitals may have a shift in position of the form $\mathbf{r} \rightarrow \mathbf{r} - \mathbf{R} - \mathbf{t}_i$ that we did not write explicitly. In *ab-initio* calculations, these integrals are typically evaluated by Fourier transforming the screened Coulomb potential, so that the integrals in $\mathbf{r}$ and $\mathbf{r}'$ factor out, and each integral ends up being a Fourier transform of the product between two orbitals. Those Fourier transforms are computed numerically via fast Fourier transforms algorithms, as there are libraries providing them [149]. However, in this work we can reduce the degree of complexity by making the point-like orbital approximation, that we have already introduced in Eq. (4.18) from Section 4.2. It consists in overlooking the spatial structure of the orbitals and assuming orthogonality whenever a product between orbitals appears. We can make the substitution

$$\phi_\alpha(\mathbf{r} - \mathbf{R} - \mathbf{t}_i)\phi_\beta(\mathbf{r} - \mathbf{R}' - \mathbf{t}_j) \approx \delta_{\alpha\beta}\delta_{\mathbf{R}\mathbf{R}'}\delta_{ij}\delta(\mathbf{r} - \mathbf{R} - \mathbf{t}_i). \quad (4.56)$$

This allows for performing the space integrals directly and analytically, so that we obtain a relatively simple expression for the direct term:

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{N} \sum_{i\alpha} \sum_{j\beta} \left(C_{i\alpha}^{c\mathbf{k}+\mathbf{Q}} \right)^* C_{i\alpha}^{c'\mathbf{k}'+\mathbf{Q}} \left(C_{j\beta}^{v'\mathbf{k}'} \right)^* C_{j\beta}^{v\mathbf{k}} V_{ij}(\mathbf{k}' - \mathbf{k}), \quad (4.57)$$

where

$$V_{ij}(\mathbf{k}' - \mathbf{k}) = \sum_{\mathbf{R}} e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}} W(\mathbf{R} + \mathbf{t}_i; \mathbf{t}_j), \quad (4.58)$$

which **is not** a Fourier transform. If we model the screening using the Rytova–Keldysh potential, then we can further write $W(\mathbf{R} + \mathbf{t}_i; \mathbf{t}_j) = V_{\text{RK}}(\mathbf{R} + \mathbf{t}_i - \mathbf{t}_j)$. The Rytova–Keldysh model is classical, as the dielectric function depends on a single momentum variable, an implication of assuming the material enjoys continuous translation invariance, which is clearly an approximation. In the spirit of *ab-initio* approaches, where the matrix elements are evaluated in $\mathbf{k}$ -space, we can go beyond classical models for screening and take into account the inhomogeneity of the crystal. For that, in the expression (4.54) one performs the inverse Fourier transform of the screened Coulomb potential. According to Eq. (A.16) in Appendix A, and assuming that the system is 3D with volume $\mathcal{V}$, we can write

$$\begin{aligned} D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) &= \int d\mathbf{r} \int d\mathbf{r}' \psi_{c,\mathbf{k}+\mathbf{Q}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') W(\mathbf{r}, \mathbf{r}') \psi_{c',\mathbf{k}'+\mathbf{Q}}(\mathbf{r}) \psi_{v,\mathbf{k}}(\mathbf{r}') = \\ &= \frac{1}{\mathcal{V}} \sum_{\mathbf{q} \in \text{BZ}} \delta_{\mathbf{q}, \mathbf{k} - \mathbf{k}'} \sum_{\mathbf{G}, \mathbf{G}'} \langle c, \mathbf{k} + \mathbf{Q} | e^{i(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r}} | c', \mathbf{k}' + \mathbf{Q} \rangle W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) \times \\ &\quad \langle v', \mathbf{k}' | e^{-i(\mathbf{q} + \mathbf{G}') \cdot \mathbf{r}} | v, \mathbf{k} \rangle = \\ &= \frac{1}{\mathcal{V}} \sum_{\mathbf{G}, \mathbf{G}'} (I_{c'\mathbf{k}'+\mathbf{Q}, c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}})^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{k} - \mathbf{k}') I_{v'\mathbf{k}', v\mathbf{k}}^{\mathbf{G}'}, \end{aligned} \quad (4.59)$$

where the second equality was obtained by replacing the $\mathbf{r}$ -resolved potential with its inverse Fourier transform, alongside the bracket notation for the real-space integrals, and in the last equality we introduced the definition

$$I_{n\mathbf{k},n'\mathbf{k}'}^{\mathbf{G}} \equiv \langle n, \mathbf{k} | e^{-i(\mathbf{k}-\mathbf{k}'+\mathbf{G})\cdot\mathbf{r}} | n', \mathbf{k}' \rangle = \int d\mathbf{r} \psi_{n,\mathbf{k}}^*(\mathbf{r}) e^{-i(\mathbf{k}-\mathbf{k}'+\mathbf{G})\cdot\mathbf{r}} \psi_{n',\mathbf{k}'}(\mathbf{r}). \quad (4.60)$$

In this definition, if the system is 3D, then the integration is over the volume $\mathcal{V}$, while if the system is 2D, then the integration is over the area $\mathcal{A}$. In this way, we are ignoring the z coordinate if the material is 2D. We did so when computing the plane-wave matrix elements of the form $\langle \lambda \mathbf{k} | \exp\{i\mathbf{q} \cdot \mathbf{r}\} | \lambda' \mathbf{k}' \rangle$ in Section 4.2. This detail will be a concern again only when we reach to Chapter 6.

The final expression in equation (4.59) is the one usually encountered in the literature pertaining to first-principle methods [31, 33]. Those methods allow also computing the dielectric matrix and its inverse *a priori*, so that one can calculate the matrix elements of the screened Coulomb potential through $W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) = \varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}) v_c(\mathbf{q} + \mathbf{G}')$, the static limit of Eq. (3.79) (if dynamic screening is not implemented in the package or library).

Regarding the exchange term, even though one typically discards it, for completion we present its expression

$$\begin{aligned} X_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) &= \int d\mathbf{r} \int d\mathbf{r}' \psi_{c,\mathbf{k}+\mathbf{Q}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') W(\mathbf{r}, \mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}) \psi_{c',\mathbf{k}'+\mathbf{Q}}(\mathbf{r}') = \\ &= \frac{1}{N} \sum_{i\alpha} \sum_{j\beta} \left(C_{i\alpha}^{c\mathbf{k}+\mathbf{Q}} \right)^* C_{i\alpha}^{v\mathbf{k}} C_{j\beta}^{c'\mathbf{k}'+\mathbf{Q}} \left(C_{j\beta}^{v'\mathbf{k}'} \right)^* V_{ij}(\mathbf{Q}) = \\ &= \frac{1}{\mathcal{V}} \sum_{\mathbf{G}, \mathbf{G}'} \left(I_{v\mathbf{k},c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}} \right)^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{Q}) I_{v'\mathbf{k}',c'\mathbf{k}'+\mathbf{Q}}^{\mathbf{G}'}. \end{aligned} \quad (4.61)$$

If we use for the screened Coulomb potential the Rytova–Keldysh model, which applies only to 2D materials, and work in reciprocal space, then the expressions for the direct (D) and exchange (X) terms are given by

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{\mathcal{A}} \sum_{\mathbf{G}} \left(I_{c'\mathbf{k}'+\mathbf{Q},c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}} \right)^* V_{\text{RK}}(\mathbf{k} - \mathbf{k}' + \mathbf{G}) I_{v'\mathbf{k}',v\mathbf{k}}^{\mathbf{G}'} \quad \text{and} \quad (4.62a)$$

$$X_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{\mathcal{A}} \sum_{\mathbf{G}} \left(I_{v\mathbf{k},c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}} \right)^* V_{\text{RK}}(\mathbf{Q} + \mathbf{G}) I_{v'\mathbf{k}',c'\mathbf{k}'+\mathbf{Q}}^{\mathbf{G}'}, \quad (4.62b)$$

where $V_{\text{RK}}(\mathbf{q}) = \frac{e}{2\varepsilon_0 q(1+r_0 q)}$ is the momentum-resolved Rytova–Keldysh potential, encountered in this work for the first time in the Section 2.2.1.1. We also had to replace $\mathcal{V} \rightarrow \mathcal{A}$. A striking difference between the expressions (4.61) and (4.59) and the expressions (4.62a) and (4.62b), is that in the first two there is a double sum over reciprocal lattice vectors $\mathbf{G}$ and $\mathbf{G}'$, while in the last two there is only a single sum over reciprocal lattice vectors, without any possibility to account for nonlocal-field effects since they use a model potential.

The double sum introduces nonlocal-field effects in the interaction matrix elements, while in the single sum, the inclusion of the reciprocal lattice vectors has the purpose of preserving the symmetry in the interaction between electrons [150]. We will touch on this detail again later on in this work, in subsection 6.2.5 about excitons in hBN.

Thus far, we have learned how to describe the emergence of excitons; many-body states arising from electronic interactions and correlations in a gapped material. We have also learned that the electronic, single-particle, bandgap does not coincide with the optical gap in these type of systems—the formation of excitons allows for photon absorption at energies that, according to the single-particle picture, would not be allowed, and it is the excitonic ground state defining the optical gap. In the next section, we focus on how the system responds to incident radiation, and how to describe the response depending on the range of frequencies used to probe the system.

4.4 Optical effects in two-dimensional semiconductors

In this section, we provide the theory for the optical conductivity at the noninteracting level, and at the interacting level due to excitonic effects. From the Wannier formalism or, in alternative, from the BSE formalism, we should be able to compute the excitonic optical conductivity from knowing the excitonic energies and eigenstates. As we will see later on in this work, probing a gapped material with light is a way of measuring the imprint of many-body effects in its absorption spectrum. First, let us look at the optical response in the noninteracting limit in the next section. The excitonic response will follow in Section 4.4.2.

The expressions derived are designed for 3D materials, as they are based on the dielectric function in Eq. (3.76), with the polarizability given by Eq. (3.75) which applies to the 3D case. Nonetheless, they can be readily adapted to 2D materials by analogy (by rigorous methods, one can consult Appendix D that the permittivity–conductivity equivalence is different in the 2D case). Therefore, this section serves as a technical base for what will be discussed in Chapters 5 and 6, when we shall put in practice the tools developed until now.

4.4.1 Noninteracting optical conductivity

Herein, we derive the expression for the dynamical optical conductivity in the local limit, $\sigma(\omega)$. To do so, we could adopt a more technical approach as in Refs [74, 75], but since we have at our disposal the dielectric function, we promptly use it to reach our goal. We start from the dynamical longitudinal dielectric function as per (3.76), that we recall it is given by

$$\begin{aligned}
 \varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, \omega) &= \delta_{\mathbf{G}\mathbf{G}'} - v_c(\mathbf{q} + \mathbf{G})\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega) = \\
 &= \delta_{\mathbf{G}\mathbf{G}'} - \frac{e^2}{\varepsilon_0|\mathbf{q} + \mathbf{G}|^2\mathcal{V}} \sum_{nn', \mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}+\mathbf{q}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}+\mathbf{q}} - \epsilon_{n\mathbf{k}}) + i\eta} \times \\
 &\quad \langle n, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | n', \mathbf{k} + \mathbf{q} \rangle \langle n', \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | n, \mathbf{k} \rangle,
 \end{aligned} \tag{4.63}$$

where we used the 3D Fourier transform of the Coulomb potential, and recovered the square in the electron charge for convenience, whereas $\mathcal{V}$ is the volume of the system assumed to be 3D. Taking the local limit $\mathbf{q} \rightarrow \mathbf{0}$ in the $\mathbf{G} = \mathbf{G}' = \mathbf{0}$ matrix element, we get

$$\begin{aligned}
 \varepsilon_{00}(\mathbf{q} \rightarrow \mathbf{0}, \omega) &= 1 - \frac{e^2}{\varepsilon_0 q^2 \mathcal{V}} \sum_{nn', \mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} \langle n, \mathbf{k} | -i\mathbf{q} \cdot \mathbf{r} | n', \mathbf{k} \rangle \langle n', \mathbf{k} | i\mathbf{q} \cdot \mathbf{r} | n, \mathbf{k} \rangle = \\
 &= 1 - \frac{e^2}{\varepsilon_0 \mathcal{V}} \sum_{nn', \mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} |\hat{\mathbf{e}}_{\mathbf{q}} \cdot \langle n, \mathbf{k} | \mathbf{r} | n', \mathbf{k} \rangle|^2,
 \end{aligned} \tag{4.64}$$

where to arrive at the last expression, we expanded the exponential to first order in $\mathbf{q}$, and defined the unit vector with the direction of $\mathbf{q}$ as $\hat{\mathbf{e}}_{\mathbf{q}} = \mathbf{q}/q$. The expansion of the exponential only up to first order is called the dipole approximation¹⁰. It is valid as long as the wavelength is larger than typical length scales of the system, for instance, the unit cell, if the context is solid state physics. This approximation simplifies calculations a lot in other contexts, such as optics [54, 151, 152] or atomic physics [69, 153, 154] (which, for a matter of fact, are not completely unrelated). Using the famous relation $\varepsilon = 1 + i\frac{\sigma}{\omega\varepsilon_0}$, mentioned before for the first time in Eq. (2.14) from Section 2.2, we find

$$\begin{aligned}
 \sigma(\omega) &= -i\omega\varepsilon_0[\varepsilon(\omega) - 1] = i\frac{\omega e^2}{\mathcal{V}} \sum_{nn', \mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} |\hat{\mathbf{e}}_{\mathbf{q}} \cdot \langle n, \mathbf{k} | \mathbf{r} | n', \mathbf{k} \rangle|^2 \stackrel{!1}{=} \\
 &= i\frac{\omega e^2}{\mathcal{V}} \sum_{nn', \mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} \frac{|\hat{\mathbf{e}}_{\mathbf{q}} \cdot \langle n, \mathbf{k} | [H_0, \mathbf{r}] | n', \mathbf{k} \rangle|^2}{(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}})^2} \stackrel{\checkmark}{=} \\
 &= i\frac{\omega e^2 \hbar^2}{\mathcal{V}} \sum_{nn', \mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} \frac{|\hat{\mathbf{e}}_{\mathbf{q}} \cdot \langle n, \mathbf{k} | \mathbf{v} | n', \mathbf{k} \rangle|^2}{(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}})^2} \stackrel{\checkmark}{=} \\
 &= \frac{i\hbar}{\mathcal{V}} \sum_{nn', \mathbf{k}} \hbar\omega \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} \frac{|\hat{\mathbf{e}}_{\mathbf{q}} \cdot \langle n, \mathbf{k} | \mathbf{j} | n', \mathbf{k} \rangle|^2}{(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}})^2} \stackrel{!2}{=} \\
 &= \frac{ie^2\hbar}{\mathcal{V}} \sum_{nn', \mathbf{k}} \hbar\omega \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} \frac{|\hat{\mathbf{e}}_{\mathbf{q}} \cdot \langle n, \mathbf{k} | \frac{\mathbf{p}}{m} | n', \mathbf{k} \rangle|^2}{(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}})^2},
 \end{aligned} \tag{4.65}$$

where we have used the identities $\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{i}{\hbar} [H_0, \mathbf{r}]$, $\mathbf{j} = -e\mathbf{v}$, and also $\mathbf{p} = m\mathbf{v}$, with m being the electron mass. The Hamiltonian at hand H_0 is the Hamiltonian in the noninteracting limit (or interacting, but at the mean-field level), that when commutated

¹⁰The optical transition is induced by the dipole operator $-e\mathbf{r}$, hence the name of the approximation.

with another operator $\hat{O}$ gives that operator's time derivative as $\frac{d\hat{O}}{dt} = i/\hbar[H_0, \hat{O}]$ ¹¹. We advert here the reader for important technical details commonly overlooked in the literature. In the equation above, only the equalities marked with $\checkmark$ are exact. The others marked with exclamation marks are not, because in general, the position $\mathbf{r}$, velocity $\mathbf{v}$ and momentum $\mathbf{p}$ operators are not interchangeable when doing calculations for extended systems. For the details regarding this issue, the reader can consult the box below, but they can be skipped in a first reading.

The framework consisting of using only position matrix elements is called the length gauge. It is called so because the interaction Hamiltonian between the material and the impinging electromagnetic field is written in terms of the dipole operator, $-e\mathbf{r}$. This is valid in the dipole approximation, that we mentioned above. On the contrary, one can work with momentum matrix elements, if we determine the light-matter interaction Hamiltonian by using the minimal substitution in the canonical momentum as $\mathbf{p} \rightarrow \mathbf{q} + e/c\mathbf{A}$ [66], where $\mathbf{A}$ is the vector potential [38]. This is called the velocity gauge. Nonetheless, it is possible to go from one gauge to the other [156, 157]. For a discussion on both gauges we recommend Ref. [158], while for a more thorough discussion of all these technical details we refer to the PhD thesis by J. J. Esteve-Paredes in Ref. [159].

In the manipulation of the expression for the optical conductivity in Eq (4.65), we took the chance to write it in terms of the current operator as well. We will use that expression to make a bridge between the RPA formalism adopted here, and the Kubo or linear response formalism [160]. We start by introducing the conductivity tensor $\overset{\leftrightarrow}{\sigma}$ with components $\sigma_{\alpha\beta}$ given by

$$\sigma_{\alpha\beta}(\omega) = \frac{i\hbar}{\mathcal{V}} \sum_{nn',\mathbf{k}} \hbar\omega \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}}) + i\eta} \frac{\langle n, \mathbf{k} | j_\alpha | n', \mathbf{k} \rangle \langle n', \mathbf{k} | j_\beta | n, \mathbf{k} \rangle}{(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}})^2}. \quad (4.66)$$

We just generalized the matrix element of the current operator in the fourth line of Eq. (4.65) in absolute value squared, to a product between the matrix elements of two current operator components α and β . The initial conductivity can be obtained by simply extracting the diagonal component of the conductivity tensor defined just now. Using symbols, $\sigma(\omega) = \hat{\mathbf{e}}_{\mathbf{q}} \cdot \overset{\leftrightarrow}{\sigma} \cdot \hat{\mathbf{e}}_{\mathbf{q}}$.

¹¹If the reader is not very familiar with concepts of quantum mechanics, this identity might require justification. There is in quantum mechanics the Heisenberg picture, where the states are static, but the operators evolve in time. Given an operator $\hat{O}_S$ in the picture we have been working with so far, the Schrödinger picture, the same operator in the Heisenberg picture reads $\hat{O}_H = \hat{U}^\dagger(t)\hat{O}_S(t)\hat{U}(t)$, where $\hat{U}(t)$ is the time-evolution operator. By using the time-dependent Schrödinger equation (TDSE), one can show that $\frac{d\hat{O}_H}{dt} = i/\hbar[H_H, \hat{O}_H] + \left(\frac{\partial \hat{O}_S}{\partial t}\right)_H$. If the Hamiltonian has no explicit time-dependence, which could be that of a crystal, then one can show that if $\hat{O}_S$ is also time-independent, then (by virtue of the time-evolution operator as in Eq. (3.2)) $H_H = H_S \equiv H$ and $\frac{d\hat{O}_H}{dt} = i/\hbar[H_H, \hat{O}_H]$. This is the quantum version of the identity $\frac{dA}{dt} = \{A, H\}$ in classical mechanics, where A is an observable with no explicit t -dependence, H the Hamiltonian function and $\{\cdot, \cdot\}$ are the Poisson brackets [66, 155].

Box 4.4.1: Position, velocity, and momentum matrix elements

In the context of atomic physics, there is no issue at all and the identity $\mathbf{p} = im/\hbar[H_0, \mathbf{r}]$ is exact. The same cannot be said about infinite crystalline solids. For instance, the relation $\mathbf{p} = m\mathbf{v}$ is only true if the potential entering the Hamiltonian is local ^a, since in such a case $-i\hbar\mathbf{v} = [H_0, \mathbf{r}] = [p^2/2m, \mathbf{r}] + [V(\mathbf{r}), \mathbf{r}] = -\frac{i\hbar}{m}\mathbf{p} + 0 = -\frac{i\hbar}{m}\mathbf{p}$, by virtue of the identity $[AB, C] = A[B, C] + [A, C]B$ and the canonical commutation relation $[x_i, p_j] = i\hbar\delta_{ij}$ [40]. If the potential is nonlocal, an issue that may arise in *ab-initio* calculations using the pseudopotentials [72], then we have undesired terms in the commutator $[V(\mathbf{r}, \mathbf{r}'), \mathbf{r}] \neq 0$, making the passage marked with !2 not exact. Moreover, we highlighted the equality from the second line to the third line with !1 because it is not exact as well, if the matrix elements are computed for a finite volume (with periodic boundary conditions). If the matrix element is computed in a finite volume, according to B. Gu and collaborators [161], there is an extra term $\mathcal{C}$ in the relation between the position and velocity matrix elements, reading

$$\langle n, \mathbf{k} | \mathbf{v} | n', \mathbf{k}' \rangle_{\mathcal{V}} = \frac{i}{\hbar}(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}'})\langle n, \mathbf{k} | \mathbf{r} | n', \mathbf{k}' \rangle_{\mathcal{V}} + \mathcal{C}_{n\mathbf{k}, n'\mathbf{k}'}, \quad (4.67)$$

with the term $\mathcal{C}_{n\mathbf{k}, n'\mathbf{k}'}$ coming from an integral over the boundary of the solid. Notice that we have used the subscript $\mathcal{V}$ to represent a bounded integration region. Retaining only the first term in the right-hand side is incorrect, because the integral of the position operator over a finite volume depends on the choice of the origin, while the left-hand side clearly does not. The second term on the right-hand side compensates for this. For the position matrix elements to not depend on the placement of the origin, we could extend the integral to infinity. However, the Bloch states $\psi_{n,\mathbf{k}}(\mathbf{r})$ do not decay to zero at infinity, henceforth the integral is ill-defined. To circumvent this technical difficulty, Blount in Ref. [162] obtained an expression for the position matrix elements, where only the diagonal ones depend on the origin, by choosing a distribution normalization for the Bloch states, instead of normalizing them to a finite volume. That is, instead of the finite case with $\langle n, \mathbf{k} | n', \mathbf{k}' \rangle_{\mathcal{V}} = \delta_{n,n'}\delta_{\mathbf{k},\mathbf{k}'}$, one has $\langle n, \mathbf{k} | n', \mathbf{k}' \rangle_{\infty} = \delta_{nn'}\delta(\mathbf{k} - \mathbf{k}')$, with the ∞ in subscript to denote an integral over all space. The work by J. J. Esteve-Paredes and J. J. Palacios in Ref. [163] showed that the relation

$$\langle n, \mathbf{k} | \mathbf{v} | n', \mathbf{k}' \rangle_{\infty} = \frac{i}{\hbar}(\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}'})\langle n, \mathbf{k} | \mathbf{r} | n', \mathbf{k}' \rangle_{\infty} \quad (4.68)$$

is true when the integrals are computed for an infinite system with open boundary conditions. Only in this way the relation above matches with the one from atomic physics. On a closing note, it is also possible to compute the momentum matrix elements in practice using the representation $\mathbf{p} = -i/\hbar\nabla_{\mathbf{r}}$. However, that involves taking spatial derivatives of atomic orbitals, if we work with a Bloch state basis within linear combination of atomic orbitals, which is not favorable. We decide then

to keep going with velocity or position matrix elements, which we will calculate in practice later on.

^aAnd also if SOC is not included [159].

We manipulate now the components of the conductivity tensor. By introducing the Sokhotski–Plemelj identity [164]

$$\frac{1}{(x-a) \pm i\eta} = \text{P.V.} \left(\frac{1}{x-a} \right) \mp i\pi\delta(x-a), \quad \eta = 0^+, \quad (4.69)$$

where P.V. denotes Cauchy principal value, in the first fraction inside the sum in Eq. (4.66), we obtain the so-called Kubo–Greenwood formula for the real part of the conductivity

$$\text{Re}\{\sigma_{\alpha\beta}(\omega)\} = \frac{\pi\hbar}{\mathcal{V}} \sum_{nn',\mathbf{k}} \frac{f(\epsilon_{n'\mathbf{k}}) - f(\epsilon_{n\mathbf{k}})}{\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}}} j_{nn'}^\alpha(\mathbf{k}) j_{n'n}^\beta(\mathbf{k}) \delta[\hbar\omega - (\epsilon_{n'\mathbf{k}} - \epsilon_{n\mathbf{k}})], \quad (4.70)$$

if written in terms of the current operator matrix elements $j_{nn'}^\alpha(\mathbf{k}) \equiv \langle n, \mathbf{k} | j_\alpha | n', \mathbf{k} \rangle$, where we also did the trick of replacing $\hbar\omega \rightarrow \epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}}$ in the first factor inside the sum in Eq. (4.66) to cancel the frequency in the numerator, which is allowed due to the presence of the Dirac delta function. That is why the difference $\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}}$ in the denominator appears with power one, and not two. Being in practice impossible to calculate at the full interacting level, normally some sort of approximation is needed. The Kubo–Greenwood formula is a particular case of the more general Kubo formula [75, 165], accounting only for transitions between single-particle states—this is called the independent-particle approximation (IPA) [166]. Particularizing for a semiconductor at zero temperature, such that the Fermi–Dirac distribution becomes a Heaviside step function (see Box 3.4.1), and simplifying, we can write the real part of the conductivity tensor, now in terms of velocity operator matrix elements, as

$$\text{Re}\{\sigma_{\alpha\beta}(\omega)\} = \frac{\pi e^2 \hbar}{\mathcal{V}} \sum_{vc,\mathbf{k}} \frac{v_{cv}^\alpha(\mathbf{k}) v_{vc}^\beta(\mathbf{k})}{\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}}} \delta[\hbar\omega - (\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}})], \quad (4.71)$$

where we used for the velocity matrix elements the notation $v_{cv}^\alpha(\mathbf{k}) \equiv \langle c, \mathbf{k} | v_\alpha | v, \mathbf{k} \rangle$. At this point we will not explain how to compute the position or velocity matrix elements in practice. They will be addressed in two different ways in chapters 5 and 6, so we postpone the explanation to those two chapters.

4.4.2 Excitonic optical conductivity

In this section we are concerned with the contribution of the excitons to the optical conductivity. Only going beyond the IPA can we compare the theoretical predictions with the optical measurements for frequencies lying inside the bandgap, as we discussed in the Section 4.1 of this chapter.

If the material is subject to a drive from a spatially uniform time-dependent electric field composed of a single harmonic ω , the linear optical conductivity, according to the work of T. Pedersen in [167], is given by

$$\sigma(\omega) = \frac{e^2}{8\pi^3 i \hbar} \sum_{\nu} \left[E_{\nu} \frac{|\Omega_{\nu}|^2}{E_{\nu} - \hbar\omega + i\hbar\Gamma} + E_{\nu} \frac{|\Omega_{\nu}|^2}{E_{\nu} + \hbar\omega - i\hbar\Gamma} \right], \quad (4.72)$$

where the sum is over the excitonic states, E_{ν} being the energy of the ν -th state, and the oscillator strength $|\Omega_{\nu}|^2$ is given in terms of the quantity Ω_{ν} , which reads

$$\Omega_{\nu} = \sum_{cv, \mathbf{k}} \psi_{cv}^{\nu}(\mathbf{k}) \langle v, \mathbf{k} | \mathbf{r} \cdot \hat{\mathbf{e}} | c, \mathbf{k} \rangle, \quad (4.73)$$

where $\mathbf{r}$ is the position operator (or dipole operator, up to a factor of $-e$), $\hat{\mathbf{e}}$ is the polarization unit vector, and $\psi_{cv}^{\nu}(\mathbf{k})$ is the excitonic wavefunction for the pair of bands (c, v) . This quantity is connected to the dipole matrix element $\langle v, \mathbf{k} | \mathbf{r} | c, \mathbf{k} \rangle$, which can be identified with the Berry connection $\vec{\Omega}^{nm}(\mathbf{k}) = \langle u_{n\mathbf{k}} | i \nabla_{\mathbf{k}} u_{m\mathbf{k}} \rangle_{\text{UC}}$ ¹². If, once more, we wish to define the optical conductivity as a tensor, then we can deduce an expression similar to (4.71), we just have to replace the transition energy $\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}}$ by the exciton energy E_{ν} , and the velocity matrix elements by their interacting version given by a weighted sum over momentum of the noninteracting matrix elements, with the weight coinciding with the excitons wavefunction, similarly to what is written in (4.73). We can therefore write

$$\text{Re}\{\sigma_{\alpha\beta}(\omega)\} = \frac{\pi e^2 \hbar}{\mathcal{V}} \sum_{\nu} \frac{1}{E_{\nu}} V_{\nu}^{\alpha} (V_{\nu}^{\beta})^* \delta(\hbar\omega - E_{\nu}), \quad (4.74)$$

if we define the interacting velocity matrix elements as $V_{\nu}^{\alpha} = \sum_{vc, \mathbf{k}} A_{vc}^{\mathbf{Q}}(\mathbf{k}) v_{vc}^{\alpha}(\mathbf{k})$, where the coefficients $A_{vc}^{\mathbf{Q}}(\mathbf{k})$ are extracted from diagonalizing the Bethe–Salpeter Hamiltonian.

Everything is set for the main contribution developed during the PhD. We devote the next two chapters to each of the two first-author publications, consisting most of the novel contributions to the literature.

¹²In general, the Berry connection $\vec{\Omega}$ of a given quantum system when it goes around a closed loop in a parameter space, where we can denote the set of parameters by $\boldsymbol{\lambda}$ is given by $\vec{\Omega}(\boldsymbol{\lambda}) = \langle u_{\boldsymbol{\lambda}} | i \nabla_{\boldsymbol{\lambda}} u_{\boldsymbol{\lambda}} \rangle$, and the Berry phase is given by $\phi = \oint \vec{\Omega}(\boldsymbol{\lambda}) \cdot d\boldsymbol{\lambda}$. In 3D through Stokes theorem and defining the Berry curvature $\boldsymbol{\Omega} = \nabla_{\boldsymbol{\lambda}} \times \vec{\Omega}$, one can write for the Berry phase, also called geometrical phase, an equivalent expression in terms of a surface integral. The concepts of Berry connection, Berry curvature and Berry phase have applications in different disciplines in physics. One particular case are Chern insulators, where a nonvanishing surface integral of the Berry curvature over a closed manifold—the Chern number—related with the quantum anomalous Hall effect, the appearance of a nonvanishing Hall current in a nonmagnetized medium [168]. In a 2D periodic material, a crystal, we can take for the parameters $\boldsymbol{\lambda} = \mathbf{k} = (k_x, k_y)$, and one can then define a more general Berry connection $\vec{\Omega}^{nm}(\mathbf{k}) = \langle u_{n\mathbf{k}} | i \nabla_{\mathbf{k}} u_{m\mathbf{k}} \rangle_{\text{UC}}$, where $|u_{n\mathbf{k}}\rangle$ is the periodic part of the Bloch state, and the bracket is taken over the unit cell. These concepts are relevant in photonic crystals [169, 170] and magnetic materials.

OPTICAL PROPERTIES OF NANOSTRUCTURED hBN

In previous chapters, we studied the phenomenological and theoretical aspects of exciton formation and their contribution to the optical response. We have seen also at the end of Chapter 2 how light can couple to material polar excitations to form polaritons. In this chapter, we analyze the exciton–polariton resultant from the hybridization of excitons with the electromagnetic (EM) field.

We revisit hexagonal boron nitride (hBN), whose tight-binding model has been presented in Section 3.2.2.1, a prototypical two-dimensional (2D) material to study excitons. We wish to engineer the optical properties of hBN by applying a one dimensional (1D) electrostatic potential, resulting in an emergent 1D superlattice. Choosing an appropriate form for the potential, it is possible to obtain an analytical expression for the renormalized Hamiltonian for any potential strength, without resorting to perturbative methods. Since we have an analytical expression for the low-energy Hamiltonian, we can extract the effective masses analytically, which speeds up the process of obtaining the binding energies of the excitons. Those are obtained within the Wannier–Mott formalism discussed in the previous chapter, where the Wannier equation is solved using the variational method. However, due to the potential being 1D, we introduce a degree of anisotropy on the dispersion which depends on the periodicity and amplitude of the potential. The fact that the method of the Bethe–Salpeter equation (BSE) introduced before for effective models is only useful for isotropic systems, justifies the choice for the methods adopted. The BSE method through exact diagonalization of the Bethe–Salpeter Hamiltonian is unfortunately not applicable here—the introduction of the electrostatic potential may spoil the translational invariance, and the code developed in this work, to be discussed in the next chapter, is only relevant for periodic 2D materials.

The approach adopted in this chapter allows for considering any amplitude or periodicity for the potential, either commensurate or incommensurate with the underlying lattice. Eventual problems posed by an incommensurate potential are not critical, as we work with the low-energy limit of the tight-binding Hamiltonian for hBN. The variational method

offers also the advantage of obtaining an analytical expression for the optical conductivity, apart from a final angular integral that can only be performed analytically in the anisotropic case. With the optical conductivity, we can finally compute the dispersion relation of the exciton–polariton. The results that we provide in this chapter have been reported in Ref. [171]. Herein, we adapt the discussion to the structure of this thesis, giving at certain times insights that were not given in the original publication.

We divide this chapter into five sections. The first four consist of the theory that we apply in order to get our results, to which we reserve the last section. In the first Section 5.1, we introduce the system under study, while properly contextualizing and motivating this work. In particular, the peculiar properties of hBN on different application fronts shall be stressed once again, as we also did in Section 3.2.2.1 when we introduced the tight-binding model for hBN. After motivating this work, its technical aspects follow in the next sections. In the Section 5.2, we derive the expression for the renormalized low-energy Hamiltonian and discuss the new emergent electronic structure. Next, in Section 5.3 we focus on solving the excitonic problem, that is, finding the excitonic binding energies and wavefunctions for the “new” engineered hBN. Afterward, in Section 5.4 we focus on the optical conductivity that we later use to compute the dispersion relation for the exciton–polariton. Finally, in Section 5.5 we present and discuss our results, previously reported in Ref. [171]. Some techniques introduced in the previous chapters will be put in practice.

5.1 Nanostructured hBN: a One-dimensional Superlattice

The material we study in this chapter, hexagonal boron nitride (hBN), is a 2D material, one among the big family of all the 2D materials that have been proposed or discovered since the isolation of a single layer of graphene in 2004 [6]. Many materials exhibiting physical phenomena once classified as exotic by the scientific community, like superconductors [172], strange metals [173], topological insulators [174], and ferroelectric materials [175] are now quite familiar to physicists working in 2D materials. Using natural 2D materials, which already have alluring properties per se, both experimental and theoretical physicists envision stacking monolayers of the same or different materials, just like pieces of Lego [176], as a way of producing new materials. If these new structures, called van der Waals (vdW) materials, consist of monolayers of distinct compounds, then they are called vdW heterostructures. Otherwise, if they consist of monolayers of the same kind, they are called vdW homostructures. The several stacked monolayers present strong chemical bonds along their in-plane extension, and bond with neighboring layers through vdW forces, which are much weaker than covalent or ionic bonds. A typical naturally occurring example is graphite, that some scientists like to call multilayer graphene, which is known to conduct electricity.

The material we study, hBN, unlike the semi-metal graphene, presents a wide bandgap as explained before in Section 3.2.2.1. These two materials have been combined before by using two hBN layers to encapsulate one layer of graphene [177] to achieve highly

confined plasmonic modes in graphene with minimum losses. It has also been observed experimentally that graphene displays ballistic transport when encapsulated with a thin film of hBN [178]. The choice of hBN as an encapsulating material is justified by its extremely flat surface and low defect-density [179]. Its low density of defects is actually one reason why this material is also promising for applications in quantum optics, as the defects can act as single-photon emitters [180]. In its bulk form, hBN is also promising for applications in nanophotonics, as we have already discussed in Section 3.2.2.1, due to its intriguing optical properties in the mid-infrared (IR), in which hBN can sustain phonon-polaritons [181].

The observation of superconductivity in twisted bilayer graphene [182] gave birth to a new field in condensed matter physics, called twistrionics [183, 184], where bilayer systems with a relative twist angle display emergent phenomena that its isolated counterparts do not. The exotic phenomena mentioned above that physicists have been so hard trying to seek, are now within reach by tuning the twist angle between stacked monolayers [185–187]. By tuning the twist angle, one can produce different patterns on the atomic arrangement, called moiré patterns or superlattices, and in that way invent completely new materials with different electronic, optical, and magnetic properties, among others. Nowadays, the field of twistrionics is a hot playground for theoretical physicists, where they predict exotic effects stemming from strong correlations in moiré materials [188–190]. The effect the moiré superlattice has on the electronic properties of a material can also be emulated by applying an external potential on a 2D material of interest [191, 192].

In this work, we emulate a 1D superlattice by externally modulating a hBN monolayer with an electrostatic potential induced by a patterned grating, as shown in Fig. 5.1. In this figure, the grating is composed of long metallic rods with an alternated bias, that lie under a quartz substrate with permittivity ε_2 that acts as a substrate for the hBN monolayer. The period of the potential L is also marked in the figure. This display serves as a suggestion for a possible experimental setup to realize a 1D superlattice.

Admittedly interesting in the ultra violet (UV) range of the radiation spectrum due to its strong excitonic optical response [193], we can deem hBN promising for polaritonics in the deep UV down to the mid IR by tuning the electrostatic modulation. Quite relevant to this work is the experiment in Ref. [194] reported after our publication [171], where they observed exciton-polaritons in a Fourier-tailored hBN flake in a vdW heterostructure with a nonzero Rabi splitting, which indicates that the strong-coupling regime was reached [194]. However, in the experimental work the Fourier surface has more than one Fourier component, whereas in our theoretical work we consider a single spatial harmonic defined by the period L . Notwithstanding the use of a more simple potential, we believe that qualitative comparisons are fair.

In Figure 5.1 one can see a schematic representation of a plane wave with frequency ω impinging on the hBN flake that responds to this external perturbation. We quantify that response when computing the optical conductivity detailed in Section 5.4.

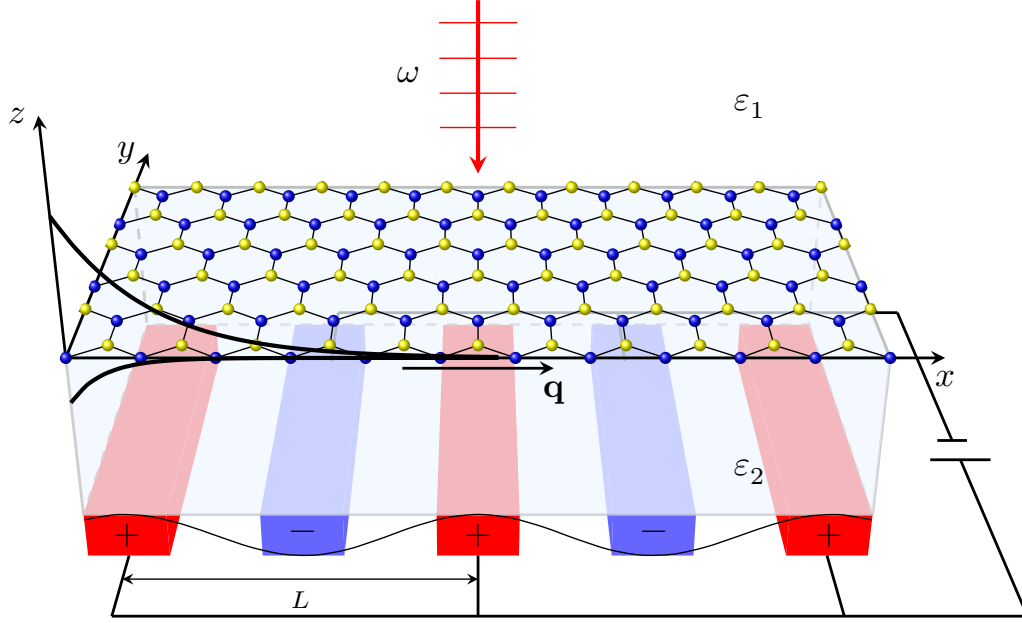


Figure 5.1: A single hBN layer electrostatically modulated. A hBN monolayer over a substrate with permittivity ε_2 , subjected to an external periodic potential with period L along the x -direction, under a time-periodic drive of a linearly polarized optical field. The environment has permittivity ε_1 .

5.2 Electronic Band Structure of Modulated hBN

The idea of subjecting a 2D material to an external 1D periodic potential in order to change its electronic properties is not exactly new. For instance, the authors in Refs. [195, 196] investigated a graphene monolayer subjected to a 1D potential to find that the dispersion, initially isotropic, renormalizes along one of the directions while leaving the other intact; new Dirac quasiparticles with an anisotropic low-energy dispersion emerge. In Ref. [197], an identical idea for a graphene 1D superlattice was put in practice, this time with a periodically modulated magnetic field, to see that the carrier velocity renormalizes isotropically, contrary to what was expected. In another theoretical work with graphene [198], it was seen that the effect of introducing an external potential affects the plasmonic effects. In particular, it affects the plasmonic modes along the direction of the potential variation, while the ones perpendicular to it are left unchanged. We shall see that the introduction of a nonvanishing mass term in the Hamiltonian, when going from graphene to hBN, leads to an anisotropic effective mass tensor, where both components in the diagonal renormalize; the enhancement of the anisotropy in the electron dispersion is more pronounced than it is in the massless case. Only after determining how exactly this anisotropy affects the dispersion of hBN, will we see how the excitonic properties are affected. First and foremost, we need to determine the renormalized Hamiltonian, in the light of what was done for the gapless case of graphene in Ref. [196].

We start from the low-energy Hamiltonian for hBN around the point K' of the Brillouin zone (BZ) given by Eq. (3.35), and convert the momentum coordinates into spatial derivatives as

$$\mathcal{H}_{K'} = \hbar v_F \mathbf{k} \cdot \boldsymbol{\sigma} + M\sigma_z = -i\hbar v_F \left(\frac{\partial}{\partial x} \sigma_x + \frac{\partial}{\partial y} \sigma_y \right) + M\sigma_z. \quad (5.1)$$

Now, we include a periodic potential $V(x) = V(x + L)$, with the period L being much larger than the atomic lattice spacing, in the Hamiltonian

$$H = \mathcal{H}_{K'} + V(x), \quad (5.2)$$

with the two-by-two identity matrix implicit in the potential term. It is not useful to have the initial Hamiltonian in reciprocal space with the external potential in real space. The potential term can be removed by introducing the unitary transformation

$$\mathbb{U}_\alpha = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\alpha(x)/2} & -e^{i\alpha(x)/2} \\ e^{-i\alpha(x)/2} & e^{i\alpha(x)/2} \end{pmatrix}, \quad (5.3)$$

where

$$\alpha(x) = \frac{2}{\hbar v_F} \int_0^x V(x') dx'. \quad (5.4)$$

Applying this transformation to the Hamiltonian in Eq. (5.2), we obtain

$$\overline{H} = \mathbb{U}_\alpha^\dagger [\mathcal{H}_{K'} + V(x)] \mathbb{U}_\alpha = \mathbb{U}_\alpha^\dagger \mathcal{H}_{K'} \mathbb{U}_\alpha + V(x). \quad (5.5)$$

Now, we need to massage the term in $\mathcal{H}_{K'}$, reading

$$\mathbb{U}_\alpha^\dagger \mathcal{H}_{K'} \mathbb{U}_\alpha = \hbar v_F \left[\mathbb{U}_\alpha^\dagger \sigma_x \frac{\partial}{\partial x} \mathbb{U}_\alpha + \mathbb{U}_\alpha^\dagger \sigma_y \mathbb{U}_\alpha \frac{\partial}{\partial y} \right] + M \mathbb{U}_\alpha^\dagger \sigma_z \mathbb{U}_\alpha. \quad (5.6)$$

We point out that, since we are with dealing operators, the partial derivative with respect to x does not act only on the matrix $\mathbb{U}_\alpha$. When applying the operator above on a test function $f(\mathbf{r})$, the derivative acts on the product $\mathbb{U}_\alpha f(\mathbf{r})$. This means that $\partial_x \mathbb{U}_\alpha$ should be read as $\partial_x (\mathbb{U}_\alpha f(\mathbf{r}))$, and the product rule for differentiation applies. Before we work out the terms with the spatial derivatives, let us first simplify the term in σ_z , since it is the simplest. It evaluates to

$$M \mathbb{U}_\alpha^\dagger \sigma_z \mathbb{U}_\alpha = \begin{pmatrix} 0 & -M e^{i\alpha} \\ -M e^{-i\alpha} & 0 \end{pmatrix}, \quad \text{with} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (5.7)$$

To work out the term involving the y -derivative, we evaluate the matrix element of that term between two arbitrary functions resolved in real space, $f(\mathbf{r})$ and $g(\mathbf{r})$, where $\mathbf{r} = (x, y)$. Later on, we particularize those wavefunctions to wavefunctions of electronic states. With $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, we can write

$$\begin{aligned}
 H_y[f, g] &= \hbar v_F \int f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_y \mathbb{U}_\alpha \frac{\partial g(\mathbf{r})}{\partial y} d\mathbf{r} = \\
 &= \frac{\hbar v_F}{2} \int f^*(\mathbf{r}) \begin{pmatrix} 0 & -e^{i\alpha(x)} \\ e^{-i\alpha(x)} & 0 \end{pmatrix} \frac{\partial g(\mathbf{r})}{\partial y} d\mathbf{r} - \\
 &\quad \frac{\hbar v_F}{2} \int \frac{\partial f^*(\mathbf{r})}{\partial y} \begin{pmatrix} 0 & -e^{i\alpha(x)} \\ e^{-i\alpha(x)} & 0 \end{pmatrix} g(\mathbf{r}) d\mathbf{r}, \tag{5.8}
 \end{aligned}$$

where some clarifications are needed. First, the integrals are over the region $[0, L] \times [0, L_y]$ in the xy plane, with L_y being the length of the material in the y direction, and $\times$ denotes Cartesian product. In the previous equation, the subscript y in $H_y[f, g]$ stands for the term with σ_y (and with $\frac{\partial}{\partial y}$) in equation (5.6). We also split the initial integral into two equal terms with a prefactor of $1/2$, and performed an integration by parts in y on one of them.

The term in (5.6) that goes with σ_x is more subtle, since both the unitary transformation and the wavefunctions depend on x . We compute the matrix element between two general functions for that term. We write

$$\begin{aligned}
 H_x[f, g] &= \hbar v_F \int f(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_x \frac{\partial}{\partial x} [\mathbb{U}_\alpha g(\mathbf{r})] d\mathbf{r} = \\
 &= \frac{\hbar v_F}{2} \int f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_x \frac{\partial}{\partial x} [\mathbb{U}_\alpha g(\mathbf{r})] d\mathbf{r} - \frac{\hbar v_F}{2} \int \frac{\partial}{\partial x} [f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger] \sigma_x \mathbb{U}_\alpha g(\mathbf{r}) d\mathbf{r} = \\
 &= \frac{\hbar v_F}{2} \int \left[f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_x \frac{d\mathbb{U}_\alpha}{dx} g(\mathbf{r}) + f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_x \mathbb{U}_\alpha \frac{\partial g(\mathbf{r})}{\partial x} \right] d\mathbf{r} - \\
 &\quad \frac{\hbar v_F}{2} \int \left[f^*(\mathbf{r}) \frac{d\mathbb{U}_\alpha^\dagger}{dx} \sigma_x \mathbb{U}_\alpha g(\mathbf{r}) + \frac{\partial f^*(\mathbf{r})}{\partial x} \mathbb{U}_\alpha^\dagger \sigma_x \mathbb{U}_\alpha g(\mathbf{r}) \right] d\mathbf{r}, \tag{5.9}
 \end{aligned}$$

where an integration by parts was done from the first line to the second line, just like we did for H_y . We associate the terms involving the derivative of the unitary operator and obtain

$$\frac{\hbar v_F}{2} \int \left[f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_x \frac{d\mathbb{U}_\alpha}{dx} g(\mathbf{r}) - f^*(\mathbf{r}) \frac{d\mathbb{U}_\alpha^\dagger}{dx} \sigma_x \mathbb{U}_\alpha g(\mathbf{r}) \right] d\mathbf{r} = -V(x) \mathbb{1}, \tag{5.10}$$

where $\mathbb{1}$ is the two-by-two identity matrix, after doing all simplifications. The previous term cancels with the last term in equation (5.5), which leaves only the terms

$$\overline{H}[f, g] = \sum_{\iota=x,y,z} H_\iota[f, g], \tag{5.11}$$

where we have to redefine the term $H_x[f, g]$ as

$$H_x[f, g] = \frac{\hbar v_F}{2} \int \left[f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_x \mathbb{U}_\alpha \frac{\partial g(\mathbf{r})}{\partial x} - \frac{\partial f^*(\mathbf{r})}{\partial x} \mathbb{U}_\alpha^\dagger \sigma_x \mathbb{U}_\alpha g(\mathbf{r}) \right] d\mathbf{r}. \tag{5.12}$$

The term H_y has already been defined above, and

$$H_z[f, g] = M \int f^*(\mathbf{r}) \mathbb{U}_\alpha^\dagger \sigma_z \mathbb{U}_\alpha g(\mathbf{r}) d\mathbf{r}. \quad (5.13)$$

Later on, we will particularize f and g to physical states. Before that, we expand $\exp\{i\alpha(x)\}$ in Fourier series as

$$e^{i\alpha(x)} = \sum_{\ell=-\infty}^{\infty} f_\ell e^{i\ell G_0 x}, \quad (5.14)$$

where $G_0 = \frac{2\pi}{L}$ is defined by the periodicity of the potential, and the Fourier coefficients read

$$f_\ell = \frac{1}{L} \int_0^L e^{i[\alpha(x) - \ell G_0 x]} dx. \quad (5.15)$$

We assume an even potential of the form

$$V(x) = V_0 \cos(G_0 x), \quad (5.16)$$

which implies

$$\alpha(x) = \frac{2V_0}{\hbar v_F G_0} \sin(G_0 x). \quad (5.17)$$

Computing f_ℓ for this potential, we obtain

$$\begin{aligned} f_\ell &= \frac{1}{L} \int_0^L e^{i[\alpha(x) - \ell G_0 x]} dx = \\ &= \frac{1}{L} \int_0^L e^{i\left[\frac{2V_0 L}{\hbar v_F 2\pi} \sin\left(\frac{2\pi}{L}x\right) - \ell \frac{2\pi}{L}x\right]} dx = \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(\beta \sin s - \ell s)} ds = \frac{1}{2\pi} \int_{\pi}^{-\pi} e^{i(\ell s - \beta \sin s)} (-ds) = \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(\ell s - \beta \sin s)} ds = J_\ell(\beta), \end{aligned} \quad (5.18)$$

where we defined the dimensionless parameter

$$\beta = \frac{V_0 L}{\hbar v_F \pi}. \quad (5.19)$$

Since the potential is even, in the Fourier series only the terms with even ℓ appear. As $J_{-\ell}(x) = (-1)^\ell J_\ell(x)$ and ℓ is even, we have $J_{-\ell}(x) = J_\ell(x)$. We use the following basis states

$$\phi_+^m(\mathbf{r}) = \frac{1}{\sqrt{LL_y}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{i(\mathbf{k} + \frac{1}{2}\mathbf{G}_m) \cdot \mathbf{r}} \quad \text{and} \quad \phi_-^m(\mathbf{r}) = \frac{1}{\sqrt{LL_y}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i(\mathbf{k} - \frac{1}{2}\mathbf{G}_m) \cdot \mathbf{r}}, \quad (5.20)$$

denoted previously by f and g , that have a plane-wave-like spatial part $\varphi_\pm^m(\mathbf{r}) = e^{i(\mathbf{k} \pm \mathbf{G}_m/2) \cdot \mathbf{r}}$ and a spinorial part, where $\mathbf{G}_m$ are the vectors of the reciprocal superlattice given by

$\mathbf{G}_m = m\mathbf{G} = mG_0\hat{\mathbf{x}}$, with $|\mathbf{k}| \ll |\mathbf{G}|$. We build the new Hamiltonian using these base states as

$$\bar{H}_{\pm,\pm} = \sum_{\iota=x,y,z} H_{\iota}[\phi_{\pm}^m, \phi_{\pm}^m] = \int (\phi_{\pm}^m(\mathbf{r}))^{\dagger} \bar{H} \phi_{\pm}^m(\mathbf{r}) d\mathbf{r}. \quad (5.21)$$

For the calculation of the matrix elements for the Hamiltonian, we need to evaluate the following quantities

$$\langle \varphi_{\pm}^m | e^{i\alpha(x)} | \varphi_{\pm}^m \rangle = \int (\varphi_{\pm}^m(\mathbf{r}))^* e^{i\alpha(x)} \varphi_{\pm}^m(\mathbf{r}) d\mathbf{r} = \sum_{\ell=-\infty}^{\infty} f_{\ell} \int (\varphi_{\pm}^m(\mathbf{r}))^* e^{i\ell G_0 x} \varphi_{\pm}^m(\mathbf{r}) d\mathbf{r}. \quad (5.22)$$

Using the spatial parts $\varphi_{\pm}^m(\mathbf{r})$ of the states $\phi_{\pm}^m(\mathbf{r})$, we have

$$\begin{aligned} \langle \varphi_{\pm}^m | e^{\pm i\alpha(x)} | \varphi_{\pm}^m \rangle &= \sum_{\ell=-\infty}^{\infty} f_{\ell} \frac{1}{LL_y} \int e^{-i(\mathbf{k} \pm \frac{1}{2}\mathbf{G}_m) \cdot \mathbf{r}} e^{\pm i\ell G_0 x} e^{i(\mathbf{k} \pm \frac{1}{2}\mathbf{G}_m) \cdot \mathbf{r}} d\mathbf{r} = \\ &= \begin{cases} \langle \varphi_+^m | e^{\pm i\alpha(x)} | \varphi_+^m \rangle = \sum_{\ell=-\infty}^{\infty} f_{\ell} \frac{1}{L} \int_0^L e^{\pm i\ell G_0 x} dx = f_0, \\ \langle \varphi_+^m | e^{\pm i\alpha(x)} | \varphi_-^m \rangle = \sum_{\ell=-\infty}^{\infty} f_{\ell} \frac{1}{L} \int_0^L e^{i(\pm\ell-m)G_0 x} dx = f_{\pm m}, \\ \langle \varphi_-^m | e^{\pm i\alpha(x)} | \varphi_+^m \rangle = \sum_{\ell=-\infty}^{\infty} f_{\ell} \frac{1}{L} \int_0^L e^{i(\pm\ell+m)G_0 x} dx = f_{\mp m}, \\ \langle \varphi_-^m | e^{\pm i\alpha(x)} | \varphi_-^m \rangle = \sum_{\ell=-\infty}^{\infty} f_{\ell} \frac{1}{L} \int_0^L e^{\pm i\ell G_0 x} dx = f_0. \end{cases} \end{aligned} \quad (5.23)$$

We also note that the spatial derivatives are simply given by

$$\begin{cases} \frac{\partial \phi_{\pm}^m}{\partial x} = i[k_x \pm m\frac{G_0}{2}] \phi_{\pm}^m, \text{ and} \\ \frac{\partial \phi_{\pm}^m}{\partial y} = ik_y \phi_{\pm}^m. \end{cases} \quad (5.24)$$

For the new Hamiltonian, we need all the following matrix elements (omitting the superscript m in the states for clarity)

$$\begin{aligned} H_x[\phi_+, \phi_+] &= \hbar v_F \left(k_x + m\frac{G_0}{2} \right), \\ H_y[\phi_+, \phi_+] &= 0, \\ H_z[\phi_+, \phi_+] &= 0, \\ H_x[\phi_+, \phi_-] &= H_x[\phi_-, \phi_+] = 0, \\ H_y[\phi_+, \phi_-] &= i\hbar v_F k_y J_m(\beta) = H_y^*[\phi_+, \phi_-], \\ H_z[\phi_+, \phi_-] &= H_z[\phi_-, \phi_+] = -MJ_m(\beta), \\ H_x[\phi_-, \phi_-] &= \hbar v_F \left(-k_x + m\frac{G_0}{2} \right), \\ H_y[\phi_-, \phi_-] &= 0, \\ H_z[\phi_-, \phi_-] &= 0, \end{aligned} \quad (5.25)$$

where to compute the matrix elements of H_x we used the first Pauli matrix $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Inserting the previous expressions in equation (5.21), we obtain the Hamiltonian

$$\bar{H} = \begin{pmatrix} \bar{H}_{+,+} & \bar{H}_{+,-} \\ \bar{H}_{-,+} & \bar{H}_{-,-} \end{pmatrix} = \begin{pmatrix} \hbar v_F (k_x + m \frac{G_0}{2}) & i \hbar v_F k_y J_m(\beta) \\ -i \hbar v_F k_y J_m(\beta) & \hbar v_F (-k_x + m \frac{G_0}{2}) \end{pmatrix}. \quad (5.26)$$

For convenience, we symmetrize and antisymmetrize the basis states as

$$\begin{aligned} \phi_s(\mathbf{r}) &= \frac{1}{\sqrt{2}}[\phi_+(\mathbf{r}) + \phi_-(\mathbf{r})], \\ \phi_a(\mathbf{r}) &= \frac{1}{\sqrt{2}}[\phi_+(\mathbf{r}) - \phi_-(\mathbf{r})]. \end{aligned} \quad (5.27)$$

In this new basis, the Hamiltonian reads

$$\bar{H} = \frac{1}{2} \begin{pmatrix} \bar{H}_{+,+} + \bar{H}_{-,-} + \bar{H}_{+,-} + \bar{H}_{-,+} & \bar{H}_{+,+} - \bar{H}_{-,-} - \bar{H}_{+,-} + \bar{H}_{-,+} \\ \bar{H}_{+,+} - \bar{H}_{-,-} + \bar{H}_{+,-} - \bar{H}_{-,+} & \bar{H}_{+,+} + \bar{H}_{-,-} - \bar{H}_{+,-} - \bar{H}_{-,+} \end{pmatrix}, \quad (5.28)$$

which, after computing all the elements, yields

$$\begin{aligned} \bar{H}_m &= \begin{pmatrix} \frac{1}{2} G_0 m \hbar v_F - |J_m(\beta)| M & \hbar v_F [k_x - i J_m(\beta) k_y] \\ \hbar v_F [k_x + i J_m(\beta) k_y] & \frac{1}{2} G_0 m \hbar v_F + |J_m(\beta)| M \end{pmatrix} \\ &= \mathbb{1} \frac{m}{2} G_0 \hbar v_F + \hbar v_F [\sigma_x k_x + \sigma_y J_m(\beta) k_y] - \sigma_z M J_m(\beta). \end{aligned} \quad (5.29)$$

This Hamiltonian is massive-Dirac-like with renormalized parameters, where we recovered the m label as subscript in the Hamiltonian. We see that there is a new term which shifts the Hamiltonian in energy. If we focus on the Hamiltonian matching with the original, which corresponds to the case $m = 0$, the y -component of the Fermi velocity changes as $v_F \rightarrow J_0(\beta) v_F$. Since $|J_0(\beta)| < 1$, an electron traveling along the y -direction is slowed down compared to its speed if traveling along the x -direction. This leads to an anisotropic low-energy dispersion. Also, the energy gap $E_g = 2M$ renormalizes to $E_g \rightarrow 2M|J_0(\beta)| \equiv \Delta_g(\beta)$, if we denote the renormalized parameter E_g as a function of β by $\Delta_g(\beta)$. We recall that the energy gap E_g was identified previously, in the paragraph before Fig. 3.3, as the bandgap obtained by density functional theory (DFT) calculations within the generalized gradient approximation (GGA). More precisely, it is extracted from fitting the bands of the minimal tight-binding model from subsection 3.2.2.1 to the DFT band structure. As already discussed in that paragraph, the value $E_g = 3.92$ eV for this bandgap, that we took from Ref. [92], is smaller than the true bandgap of hBN, which is approximately 6.0 eV [107, 199]. The DFT bandgap within GGA can be corrected by summing an exchange correction, Δ_{xc} , coming from the third term on the right-hand side of Eq. (4.32) that we discarded at the time. We will see in Section 5.5 that this contribution to the bandgap renormalizes as well. Choosing an appropriate value for either V_0 or L (actually, the product $V_0 L$), it is possible to reduce the gap of hBN. With this

mechanism, we can control both the gap and the excitonic states, which now become those of an anisotropic material (similarly to phosphorene).

At first glance, it is hard to realize how the previous Hamiltonian falls on the original one in the case where there is no potential. It is actually possible to conciliate the renormalized Hamiltonian with the original one, which is done in Appendix B. In the same appendix, we also recover the original Hamiltonian when the period L tends to infinity.

The group velocity components and the two effective masses follow directly from the spectrum of the previous Hamiltonian. These are given by

$$\begin{aligned} v_x &= \frac{1}{\hbar} \frac{\partial \epsilon_c}{\partial k_x} = \frac{\hbar v_F^2 k_x}{\epsilon_c(\mathbf{k}, \beta)} \quad \text{and} \\ v_y &= \frac{1}{\hbar} \frac{\partial \epsilon_c}{\partial k_y} = \frac{\hbar v_F^2 J_0^2(\beta) k_y}{\epsilon_c(\mathbf{k}, \beta)}, \end{aligned} \quad (5.30)$$

where

$$\epsilon_c(\mathbf{k}, \beta) = \sqrt{\hbar^2 v_F^2 [k_x^2 + J_0^2(\beta) k_y^2] + J_0^2(\beta) M^2} \quad (5.31)$$

is the renormalized low-energy dispersion of the conduction band. The diagonal components of the effective mass tensor computed at momentum $\mathbf{k}$ are given by

$$m_x(k_x, k_y) = \frac{[J_0^2(\beta) M^2 + \hbar^2 v_F^2 (k_x^2 + J_0^2(\beta) k_y^2)]^{3/2}}{J_0^2(\beta) v_F^2 (M^2 + \hbar^2 v_F^2 k_y^2)} \quad \text{and} \quad (5.32a)$$

$$m_y(k_x, k_y) = \frac{[J_0^2(\beta) M^2 + \hbar^2 v_F^2 (k_x^2 + J_0^2(\beta) k_y^2)]^{3/2}}{J_0^2(\beta) v_F^2 (J_0^2(\beta) M^2 + \hbar^2 v_F^2 k_x^2)}, \quad (5.32b)$$

that at zero momentum evaluate to

$$m_x = \frac{|J_0(\beta)| M}{v_F^2} \quad \text{and} \quad (5.33a)$$

$$m_y = \frac{M}{|J_0(\beta)| v_F^2}. \quad (5.33b)$$

The spinors of the new low-energy Hamiltonian are also needed to compute the optical matrix element for the conductivity, as discussed in sections 4.4.1 and 4.4.2. We change the notation slightly, replacing M with $E_g/2$. Besides that, we define $k_x + iJ_0(\beta)k_y = k(\beta)e^{i\theta_k(\beta)}$, with $k(\beta) = \sqrt{k_x^2 + J_0^2(\beta)k_y^2}$, and define

$$\theta_k(\beta) = \arctan\left(\frac{J_0(\beta)k_y}{k_x}\right) \quad \text{and} \quad \xi_k = \arctan\left(\frac{2\hbar v_F k(\beta)}{J_0(\beta)E_g}\right), \quad (5.34)$$

which are the anisotropic analogues of their respective isotropic counterparts in Eq. (3.41) (apart from changing the notation for momentum from k and $\mathbf{k}$ to q and $\mathbf{q}$, respectively). In this notation, and after some straightforward algebra, the spinor for the conduction band reads

$$\mathbf{C}^{\text{ck}} = \begin{pmatrix} e^{-i\theta_k(\beta)} \frac{\sin \frac{\xi_k}{2}}{\cos \frac{\xi_k}{2}} \\ 1 \end{pmatrix}, \quad (5.35)$$

that, when multiplied by $\cos \frac{\xi_k}{2}$, yields the spinor in the final form that we will use

$$\mathbf{C}^{\text{ck}} = \begin{pmatrix} e^{-i\theta_k(\beta)} \sin \frac{\xi_k}{2} \\ \cos \frac{\xi_k}{2} \end{pmatrix}. \quad (5.36)$$

Analogously, the spinor for the valence band can be written as

$$\mathbf{C}^{vk} = \begin{pmatrix} \cos \frac{\xi_k}{2} \\ -e^{i\theta_k(\beta)} \sin \frac{\xi_k}{2} \end{pmatrix}. \quad (5.37)$$

These last two equations provide the spinors of the renormalized low-energy Hamiltonian, similar to their isotropic versions found before in Eq. (3.40). They do not exactly coincide when $\beta = 0$, because the mass term in the renormalized low-energy Hamiltonian carries a negative sign, as it can be seen in Eq. (5.29) when $m = 0$.

5.3 Anisotropic Excitonic States

In this section, we provide the theory necessary for the variational calculation of the excitonic states and energies, within a Wannier–Mott description for the excitons. The variational method was first mentioned in Section 3.1.3, and it is here applied to solve the Wannier equation. We wish to compare with the excitons of pristine monolayer hBN. Therefore, we focus on the particular case $m = 0$ of Eq. (5.29), for which the effective low-energy Hamiltonian reads

$$\overline{H}_0 = \hbar v_F [k_x \sigma_x + J_0(\beta) k_y \sigma_y] - \frac{E_g}{2} J_0(\beta) \sigma_z. \quad (5.38)$$

The variational wavefunctions need to capture the anisotropy of the dispersion, and the matrix elements of the Hamiltonian for those wavefunctions are not easy to work out. We separate the kinetic part from the potential part and treat each state separately in different subsections. The kinetic energy contribution to the Wannier Hamiltonian is given by

$$\mathcal{K} = \frac{p_x^2}{2\mu_x} + \frac{p_y^2}{2\mu_y}, \quad (5.39)$$

where the effective masses μ_x and μ_y are now in general different. We do the calculations for the Rytova–Keldysh potential

$$V_{\text{RK}}(r) = \frac{e}{8\varepsilon_0 r_0} \left[\mathbf{H}_0 \left(\frac{\kappa r}{r_0} \right) - Y_0 \left(\frac{\kappa r}{r_0} \right) \right], \quad (5.40)$$

which we have already encountered in Section 2.2.1.1, where we have a small modification in the argument of the special functions compared to Eq. (2.21). We introduced the parameter

$\kappa = (\varepsilon_1 + \varepsilon_2)/2$, which is the average between the permittivity ε_1 of the material on top and the permittivity ε_2 of the material supporting the hBN layer. These two permittivities are the static ones and account for electrostatic screening in the materials surrounding the hBN monolayer [28]. The parameter r_0 is only material dependent, and for monolayer hBN we use $r_0 = 10$ Å [28, 30, 200]. Joining the kinetic and potential contributions, we have for the Wannier Hamiltonian

$$H = \frac{p_x^2}{2\mu_x} + \frac{p_y^2}{2\mu_y} + V_{\text{RK}}(r), \quad (5.41)$$

so that the excitonic problem can be described by the Wannier equation

$$H\Psi_\nu = E_{b,\nu}\Psi_\nu. \quad (5.42)$$

As trial wavefunctions for the ground state $1s$ and the first 3 excited states, $2x$, $2y$ and $2s$, we pick the expressions

$$\begin{aligned} \Psi_{1s}(x, y) &= C_{1s} e^{-\sqrt{a_{1s}x^2 + b_{1s}y^2}}, \\ \Psi_{2x}(x, y) &= C_{2x} x e^{-\sqrt{a_{2x}x^2 + b_{2x}y^2}}, \\ \Psi_{2y}(x, y) &= C_{2y} y e^{-\sqrt{a_{2y}x^2 + b_{2y}y^2}}, \quad \text{and} \\ \Psi_{2s}(x, y) &= C_{2s} \left(1 - d\sqrt{a_{2s}x^2 + b_{2s}y^2}\right) e^{-\sqrt{a_{2s}x^2 + b_{2s}y^2}}. \end{aligned}$$

The parameter d in the $2s$ wavefunction is determined by the orthogonality condition $\langle \Psi_{1s} | \Psi_{2s} \rangle = 0$, and the constants C_ν are normalization factors. These wavefunctions are an approximation of the true ones, but we assume that they describe the problem well enough. They coincide with the wavefunctions used by J. N. S. Gomes and co-authors in Ref. [29] for phosphorene. Therefore, the expressions that we use to calculate the binding energies are the same ones as in Ref. [29]. We just complement the information therein provided with abbreviated derivations. Each wavefunction has two variational parameters that we denote by a_ν and b_ν , with $\alpha = 1s, 2x, 2y, 2s$. In this way the anisotropy of the dispersion is accounted for. In the isotropic case, where $a_\nu = b_\nu$, we recover the wavefunctions that are the analytical solutions of the 2D hydrogen atom [119]. For each of these states, we compute the expectation value of the Hamiltonian and minimize it with respect to both parameters. By solving the minimization condition, we find the values of a_ν and b_ν . Then, we go back to the expression for the wavefunction and for the expectation value to obtain the (approximate) eigenstate and energy, respectively. We repeat the procedure separately for each of the four states. Naturally, we start with the easiest case, the ground state. In the discussions that follow, for ease of language and notation, we may tacitly refer to the binding energies as just energies, and use the symbol E_ν instead of $E_{b,\nu}$.

5.3.1 Ground State $1s$

As a trial wavefunction for the ground state we use

$$\Psi_{1s}(x, y) = C_{1s} e^{-\sqrt{a_{1s}x^2 + b_{1s}y^2}}, \quad (5.43)$$

where

$$C_{1s} = \sqrt{\frac{2}{\pi}} (a_{1s} b_{1s})^{1/4}. \quad (5.44)$$

To compute the expression for the ground-state energy E_{1s} in terms of the variational parameters, we need to evaluate

$$E_{1s}(a_{1s}, b_{1s}) = \langle \Psi_{1s} | H | \Psi_{1s} \rangle = \langle \Psi_{1s} | \mathcal{K} | \Psi_{1s} \rangle + \langle \Psi_{1s} | V_{\text{RK}} | \Psi_{1s} \rangle. \quad (5.45)$$

We compute first the term involving the kinetic energy

$$\langle \Psi_{1s} | \mathcal{K} | \Psi_{1s} \rangle = \int \Psi_{1s}^*(x, y) \left[\frac{p_x^2}{2\mu_x} + \frac{p_y^2}{2\mu_y} \right] \Psi_{1s}(x, y) d\mathbf{r}. \quad (5.46)$$

As a first step, we perform a change of variables proposed by Rodin et al. in [201],

$$X = s_x x, Y = s_y y, \quad (5.47)$$

where

$$s_{x/y} = \sqrt{\frac{\mu_{x/y}}{2\bar{\mu}}} \quad \text{with} \quad \bar{\mu} = \frac{\mu_x \mu_y}{\mu_x + \mu_y}. \quad (5.48)$$

In this way, the kinetic energy that now reads

$$\mathcal{K} = -\frac{\hbar^2}{4\bar{\mu}} \nabla^2 \quad (5.49)$$

has the usual form of isotropic systems but with a different mass, and the Laplacian is with respect to the new coordinates (X, Y) . The infinitesimal element of area in the old coordinates transforms into

$$dx dy = \frac{dX dY}{s_x s_y}, \quad (5.50)$$

so the integral in Eq. (5.46) reads

$$-\frac{\hbar^2}{4\bar{\mu}} \frac{1}{s_x s_y} \int \Psi_{1s}^*(X, Y) \nabla^2 \Psi_{1s}(X, Y) dX dY. \quad (5.51)$$

The wavefunction as a function of the new variables is

$$\Psi_{1s}(X, Y) = C_{1s} e^{-\sqrt{\delta_x X^2 + \delta_y Y^2}}, \quad (5.52)$$

where we introduced new parameters $\delta_x = a_{1s} s_x^{-2}$ and $\delta_y = b_{1s} s_y^{-2}$. Next, we compute the Laplacian of the wavefunction in the (X, Y) coordinates and then perform another change of variables to elliptical coordinates at the end, given by

$$X = \frac{\rho}{\sqrt{\delta_x}} \cos \theta, Y = \frac{\rho}{\sqrt{\delta_y}} \sin \theta. \quad (5.53)$$

For the integral, we need the absolute value of the Jacobian, which is $\rho/\sqrt{\delta_x\delta_y}$. Then, the integral reads

$$\begin{aligned} & -\frac{\hbar^2}{4\bar{\mu}} \frac{1}{s_x s_y} \int \Psi_{1s}^*(X, Y) \nabla^2 \Psi_{1s}(X, Y) dX dY = \\ & = -\frac{\hbar^2}{4\bar{\mu}} \frac{C^2}{s_x s_y \sqrt{\delta_x \delta_y}} \int_0^{2\pi} \int_0^\infty \rho e^{-2\rho} \left[\delta_x \left(\cos^2 \theta - \frac{\sin^2 \theta}{\rho} \right) + \delta_y \left(\sin^2 \theta - \frac{\cos^2 \theta}{\rho} \right) \right] d\rho d\theta = \\ & = -\frac{\hbar^2}{4\bar{\mu}} \frac{C^2}{s_x s_y \sqrt{\delta_x \delta_y}} \frac{-1}{4} \pi (\delta_x + \delta_y) = \frac{\hbar^2}{4} \left(\frac{a_{1s}}{\mu_x} + \frac{b_{1s}}{\mu_y} \right), \end{aligned} \quad (5.54)$$

after making all substitutions. Now, are left with the calculation of the term in the matrix element derived from the potential to calculate. For this term, we change from Cartesian to polar coordinates, and define the function $\alpha(\theta) = 2\sqrt{a_{1s} + (b_{1s} - a_{1s}) \sin^2 \theta}$, to obtain

$$\langle \Psi_{1s} | V_{\text{RK}} | \Psi_{1s} \rangle = \sqrt{a_{1s} b_{1s}} \frac{e}{4\pi\epsilon_0 r_0} \frac{\pi}{2} \int_0^{2\pi} \int_0^\infty \mathbb{K} \left(\frac{\kappa r}{r_0} \right) r e^{-\alpha(\theta)r} dr d\theta, \quad (5.55)$$

where we also defined $\mathbb{K}(z) \equiv \mathbf{H}_0(z) - Y_0(z)$. Fortunately, the radial integral has an analytical expression. We calculate analytically the auxiliary integral

$$\begin{aligned} I_0(\alpha, \kappa_0) &= \int_0^\infty \mathbb{K}(\kappa_0 r) e^{-\alpha r} dr = \\ &= \frac{2}{\pi \sqrt{\alpha^2 + \kappa_0^2}} \left[\log \left(\frac{\kappa_0 + \sqrt{\alpha^2 + \kappa_0^2}}{\alpha} \right) + \log \left(\frac{\alpha + \sqrt{\alpha^2 + \kappa_0^2}}{\kappa_0} \right) \right], \end{aligned} \quad (5.56)$$

which depends on θ through the function $\alpha(\theta)$, and where we introduced as well $\kappa_0 = \kappa/r_0$. The radial integral of interest in Eq. (5.55), that we will denote by I , relates to the previous one through

$$I(\alpha, \kappa_0) \equiv \int_0^\infty \mathbb{K}(\kappa_0 r) r e^{-\alpha r} dr = -\frac{\partial}{\partial \alpha} I_0(\alpha, \kappa_0). \quad (5.57)$$

The final expression for the energy of the excitonic ground state can only be written in terms of this last integral, and it reads

$$E_{1s} = \frac{\hbar^2}{4} \left(\frac{a_{1s}}{\mu_x} + \frac{b_{1s}}{\mu_y} \right) - \sqrt{a_{1s} b_{1s}} \frac{e}{4\pi\epsilon_0 r_0} \frac{\pi}{2} \int_0^{2\pi} I(\alpha, \kappa_0) d\theta. \quad (5.58)$$

Requiring an absolute minimum for this energy consists in solving $\partial E_{1s}/\partial a_{1s} = \partial E_{1s}/\partial b_{1s} = 0$ for a_{1s} and b_{1s} . As an example, for black phosphorus encapsulated in hBN [29], we have the parameters $\mu_x = 0.99m_e$ and $\mu_y = 0.12m_e$ where m_e is the electron mass, $\kappa = 4.5$ and $r_0 = 2.5$ nm. For this set of parameters, the binding energy evaluates

to $E_{1s} = -257$ meV, and the variational parameters evaluate to $a_{1s} = 2.45$ nm⁻² and $b_{1s} = 0.533$ nm⁻². Computing the derivatives is straightforward, so we present here only the final expressions

$$\frac{\partial E_{1s}}{\partial a_{1s}} = \frac{\hbar^2}{4\mu_x} - \frac{e}{8\pi\epsilon_0 r_0} \sqrt{\frac{b_{1s}}{a_{1s}}} \int_0^{2\pi} I(\alpha, \kappa_0) + 4 \frac{a_{1s} \cos^2 \theta}{\alpha} \partial_\alpha I \, d\theta, \quad (5.59a)$$

$$\frac{\partial E_{1s}}{\partial b_{1s}} = \frac{\hbar^2}{4\mu_y} - \frac{e}{8\pi\epsilon_0 r_0} \sqrt{\frac{a_{1s}}{b_{1s}}} \int_0^{2\pi} I(\alpha, \kappa_0) + 4 \frac{b_{1s} \sin^2 \theta}{\alpha} \partial_\alpha I \, d\theta. \quad (5.59b)$$

The excited states follow, starting with the states $2x$ and $2y$.

5.3.2 Excited States $2x$ and $2y$

We discuss first the calculations for the $2x$ state. We recall the expression for the trial wavefunction

$$\Psi_{2x}(x, y) = C_{2x} x e^{-\sqrt{a_{2x}x^2 + b_{2x}y^2}}, \quad (5.60)$$

where the normalisation constant can be found using the same change of variables as in Eq. (5.53) for the integral $\int |\Psi_{2x}(\mathbf{r})|^2 \, d\mathbf{r}$, and it is given by

$$|C_{2x}|^2 = \frac{8a_{2x}\sqrt{a_{2x}b_{2x}}}{3\pi}. \quad (5.61)$$

To obtain the kinetic contribution to the energy, we followed the same strategy as in the $1s$ case. We change from (x, y) to (X, Y) coordinates and compute the Laplacian in these new coordinates, followed by a change of variables similar to (5.53), performing the radial integral and finally the angular integral. After all substitutions, we obtain

$$\langle \Psi_{2x} | \mathcal{K} | \Psi_{2x} \rangle = \frac{\hbar^2}{8} \left(\frac{3a_{2x}}{\mu_x} + \frac{b_{2x}}{\mu_y} \right). \quad (5.62)$$

The potential term reads

$$\begin{aligned} \langle \Psi_{2x} | V_{\text{RK}} | \Psi_{2x} \rangle &= \frac{8a_{2x}\sqrt{a_{2x}b_{2x}}}{3\pi} \frac{e}{4\pi\epsilon_0 r_0} \frac{\pi}{2} \int_0^{2\pi} \cos^2 \theta \frac{\partial^2}{\partial \alpha^2} I(\alpha, \kappa_0) \, d\theta = \\ &= \frac{8a_{2x}\sqrt{a_{2x}b_{2x}}}{3\pi} \frac{-e}{4\pi\epsilon_0 r_0} \frac{\pi}{2} \int_0^{2\pi} \cos^2 \theta \frac{\partial^3}{\partial \alpha^3} I_0(\alpha, \kappa_0) \, d\theta, \end{aligned} \quad (5.63)$$

where we made use of the auxiliary function $I_0(\alpha, \kappa_0)$ defined in Eq. (5.56), having thus for the binding energy

$$E_{2x} = \frac{\hbar^2}{8} \left(\frac{3a_{2x}}{\mu_x} + \frac{b_{2x}}{\mu_y} \right) - \frac{8a_{2x}\sqrt{a_{2x}b_{2x}}}{3\pi} \frac{e}{4\pi\epsilon_0 r_0} \frac{\pi}{2} \int_0^{2\pi} \cos^2 \theta \frac{\partial^2}{\partial \alpha^2} I(\alpha, \kappa_0) \, d\theta. \quad (5.64)$$

The derivatives with respect to a_{2x} and b_{2x} , respectively, read

$$\frac{\partial E_{2x}}{\partial a_{2x}} = \frac{3\hbar^2}{8\mu_x} - \quad (5.65a)$$

$$\frac{e}{8\varepsilon_0 r_0} \left[\frac{4\sqrt{a_{2x}b_{2x}}}{\pi} \int_0^{2\pi} \cos^2 \theta \frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} d\theta + \frac{8\sqrt{a_{2x}^3 b_{2x}}}{3\pi} \int_0^{2\pi} \cos^2 \theta \frac{\partial}{\partial a_{2x}} \left(\frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} \right) d\theta \right],$$

$$\frac{\partial E_{2x}}{\partial b_{2x}} = \frac{\hbar^2}{8\mu_y} - \quad (5.65b)$$

$$\frac{e}{8\varepsilon_0 r_0} \left[\frac{4\sqrt{a_{2x}^3/b_{2x}}}{3\pi} \int_0^{2\pi} \cos^2 \theta \frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} d\theta + \frac{8\sqrt{a_{2x}^3 b_{2x}}}{3\pi} \int_0^{2\pi} \cos^2 \theta \frac{\partial}{\partial b_{2x}} \left(\frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} \right) d\theta \right].$$

For simplicity, we omitted the explicit forms of the derivatives $\frac{\partial}{\partial a_{2x}} \left(\frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} \right)$ and $\frac{\partial}{\partial b_{2x}} \left(\frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} \right)$.

Since the calculations for the state $2y$ are analogous, we show only the key results. For the binding energy, we have

$$E_{2y} = \frac{\hbar^2}{8} \left(\frac{a_{2y}}{\mu_x} + \frac{3b_{2y}}{\mu_y} \right) - \frac{8b_{2y}\sqrt{a_{2y}b_{2y}}}{3\pi} \frac{e}{4\pi\varepsilon_0 r_0} \frac{\pi}{2} \int_0^{2\pi} \sin^2 \theta \frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} d\theta, \quad (5.66)$$

and for the derivatives of the binding energy we have

$$\frac{\partial E_{2y}}{\partial a_{2y}} = \frac{\hbar^2}{8\mu_x} - \quad (5.67a)$$

$$\frac{e}{8\varepsilon_0 r_0} \left[\frac{4\sqrt{b_{2y}^3/a_{2y}}}{3\pi} \int_0^{2\pi} \cos^2 \theta \frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} d\theta + \frac{8\sqrt{a_{2y}b_{2y}^3}}{3\pi} \int_0^{2\pi} \sin^2 \theta \frac{\partial}{\partial b_{2y}} \left(\frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} \right) d\theta \right],$$

$$\frac{\partial E_{2y}}{\partial b_{2y}} = \frac{3\hbar^2}{8\mu_y} - \quad (5.67b)$$

$$\frac{e}{8\varepsilon_0 r_0} \left[\frac{4\sqrt{a_{2y}b_{2y}}}{\pi} \int_0^{2\pi} \sin^2 \theta \frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} d\theta + \frac{8\sqrt{a_{2y}b_{2y}^3}}{3\pi} \int_0^{2\pi} \sin^2 \theta \frac{\partial}{\partial a_{2y}} \left(\frac{\partial^2 I(\alpha, \kappa_0)}{\partial \alpha^2} \right) d\theta \right].$$

5.3.3 Excited State $2s$

We devote this entire section to the calculations for the $2s$ state, as for this state there are more intermediate steps. First, the normalisation condition leads to

$$|C_{2s}|^2 = \frac{4\sqrt{a_{2s}b_{2s}}}{\pi(2 - 4d + 3d^2)}, \quad (5.68)$$

where the value of d is obtained by the orthogonality condition $\langle \Psi_{1s} | \Psi_{2s} \rangle = 0$. To ease the notation, from now on we will write a and b instead of a_{2s} and b_{2s} . Since the system is anisotropic, d is not a simple function of the parameters a_{1s} and b_{1s} . To see this, let us write

$$\begin{aligned}\langle \Psi_{1s} | \Psi_{2s} \rangle &= C_{1s} C_{2s} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(1 - d\sqrt{ax^2 + by^2}\right) e^{-\sqrt{ax^2 + by^2}} e^{-\sqrt{a_{1s}x^2 + b_{1s}y^2}} dx dy = 0 \rightarrow \\ &\rightarrow d = \frac{\mathcal{I}_1}{\mathcal{I}_2},\end{aligned}\quad (5.69)$$

where we defined

$$\begin{aligned}\mathcal{I}_1 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\sqrt{ax^2 + by^2}} e^{-\sqrt{a_{1s}x^2 + b_{1s}y^2}} dx dy \quad \text{and} \\ \mathcal{I}_2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sqrt{ax^2 + by^2} e^{-\sqrt{ax^2 + by^2}} e^{-\sqrt{a_{1s}x^2 + b_{1s}y^2}} dx dy.\end{aligned}\quad (5.70)$$

For both integrals, we perform a change of variables like in (5.53). The first integral reads

$$\begin{aligned}\mathcal{I}_1 &= \int_0^{2\pi} \int_0^{\infty} \rho e^{-\rho} e^{-\rho \sqrt{\frac{a_{1s}}{a} \cos^2 \theta + \frac{b_{1s}}{b} \sin^2 \theta}} \frac{d\rho d\theta}{\sqrt{ab}} = \int_0^{2\pi} \int_0^{\infty} \rho e^{-t(\theta)\rho} \frac{d\rho d\theta}{\sqrt{ab}} = \\ &= \int_0^{2\pi} t^{-2}(\theta) \frac{d\theta}{\sqrt{ab}},\end{aligned}\quad (5.71)$$

if we also define $t(\theta) \equiv 1 + \sqrt{\frac{a_{1s}}{a} \cos^2 \theta + \frac{b_{1s}}{b} \sin^2 \theta}$. For the second integral, we have

$$\begin{aligned}\mathcal{I}_2 &= \int_0^{2\pi} \int_0^{\infty} \rho^2 e^{-\rho} e^{-\rho \sqrt{\frac{a_{1s}}{a} \cos^2 \theta + \frac{b_{1s}}{b} \sin^2 \theta}} \frac{d\rho d\theta}{\sqrt{ab}} = \int_0^{2\pi} \int_0^{\infty} \rho^2 e^{-t(\theta)\rho} \frac{d\rho d\theta}{\sqrt{ab}} = \\ &= \int_0^{2\pi} \frac{2}{t^3(\theta)} \frac{d\theta}{\sqrt{ab}}.\end{aligned}\quad (5.72)$$

For convenience, we define the auxiliary functions

$$f_1(a, b) = \int_0^{2\pi} t^{-2}(\theta) d\theta \quad \text{and} \quad f_2(a, b) = \int_0^{2\pi} t^{-3}(\theta) d\theta, \quad (5.73)$$

so that the orthogonality condition reads

$$d(a, b) = \frac{f_1(a, b)}{2f_2(a, b)}. \quad (5.74)$$

Let us now compute the energy of the $2s$ state. The kinetic term is obtained similarly to the previous cases, and equals

$$\langle \Psi_{2s} | \mathcal{K} | \Psi_{2s} \rangle = \frac{\hbar^2}{4} \left(\frac{a}{\mu_x} + \frac{b}{\mu_y} \right) \frac{2 + d^2}{2 - 4d + 3d^2} = \frac{\hbar^2}{4} F(a, b), \quad (5.75)$$

where we defined

$$F(a, b) = \left(\frac{a}{\mu_x} + \frac{b}{\mu_y} \right) \frac{2 + d^2}{2 - 4d + 3d^2}. \quad (5.76)$$

The potential term is computed through the integral

$$\begin{aligned}
 \langle \Psi_{2s} | V_{\text{RK}} | \Psi_{2s} \rangle &= C_{2s}^2 \int \left(1 - d\sqrt{ax^2 + by^2} \right)^2 e^{-2\sqrt{ax^2 + by^2}} V_{\text{RK}}(r) \, d\mathbf{r} = \\
 &= 2 \frac{\sqrt{ab}}{(2 - 4d + 3d^2)} \frac{e}{4\pi\epsilon_0 r_0} \int_0^{2\pi} \left[1 + \alpha(\theta) \frac{d}{2} \partial_\alpha \right]^2 I(\alpha, \kappa_0) \, d\theta.
 \end{aligned} \tag{5.77}$$

We define now the function

$$G(a, b) = \frac{1}{(2 - 4d + 3d^2)} \int_0^{2\pi} \int_0^\infty \mathbb{K}(\kappa_0 \alpha(\theta) \rho) (1 - d\rho)^2 \rho e^{-2\rho} \, d\rho \, d\theta, \tag{5.78}$$

where, for the integral, we performed a change of variables as in (5.53), so that we can write the potential term in the alternative way

$$\langle \Psi_{2s} | V_{\text{RK}} | \Psi_{2s} \rangle = 2 \frac{e}{4\pi\epsilon_0 r_0} G(a, b). \tag{5.79}$$

In this way, the expression for the binding energy of the $2s$ state can be more compactly written as

$$E_{2s} = \frac{\hbar^2}{4} F(a, b) - 2 \frac{e}{4\pi\epsilon_0 r_0} G(a, b), \tag{5.80}$$

which is the expression that we need to minimize. For that, we need the derivatives of the function $d(a, b)$, and they read

$$\partial_a d = \frac{\partial_a f_1 f_2 - f_1 \partial_a f_2}{2f_2^2} \quad \text{and} \quad \partial_b d = \frac{\partial_b f_1 f_2 - f_1 \partial_b f_2}{2f_2^2}, \tag{5.81}$$

and also the derivatives of f_i with $i = 1, 2$, which are given by

$$\partial_a f_i = \frac{a_{1s}(1+i)}{2a^2} \int_0^{2\pi} \frac{\cos^2 \theta \, t^{-(2+i)}(\theta)}{t(\theta) - 1} \, d\theta \quad \text{and} \quad \partial_b f_i = \frac{b_{1s}(1+i)}{2b^2} \int_0^{2\pi} \frac{\sin^2 \theta \, t^{-(2+i)}(\theta)}{t(\theta) - 1} \, d\theta. \tag{5.82}$$

The derivatives of the function F are straightforward to calculate

$$\begin{aligned}
 \partial_a F &= \frac{2 + d^2}{2 - 4d + 3d^2} \frac{1}{\mu_x} + \left(\frac{a}{\mu_x} + \frac{b}{\mu_y} \right) \frac{8 - 8d - 4d^2}{2 - 4d + 3d^2} \partial_a d, \\
 \partial_b F &= \frac{2 + d^2}{2 - 4d + 3d^2} \frac{1}{\mu_y} + \left(\frac{a}{\mu_x} + \frac{b}{\mu_y} \right) \frac{8 - 8d - 4d^2}{2 - 4d + 3d^2} \partial_b d.
 \end{aligned} \tag{5.83}$$

Regarding the derivatives of G , we define two additional auxiliary integrals,

$$\begin{aligned}
 I_1(\theta) &= \int_0^\infty \mathbb{K}(\kappa_0 \alpha(\theta) \rho) (1 - d\rho) \rho^2 e^{-2\rho} \, d\rho \quad \text{and} \\
 I_2(\theta) &= \int_0^\infty \mathbb{K}'(\kappa_0 \alpha(\theta) \rho) (1 - d\rho)^2 \rho^2 e^{-2\rho} \, d\rho,
 \end{aligned} \tag{5.84}$$

where $\mathbb{K}'(\kappa_0 \alpha(\theta) \rho)$ denotes the derivative of $\mathbb{K}(z)$ computed at the point $z = \kappa_0 \alpha(\theta) \rho$. In this way, the derivatives of G read

$$\begin{aligned}
 \partial_a G = & -\frac{1}{2-4d+3d^2} \int_0^{2\pi} \frac{\kappa}{2r_0 a^2} \frac{\cos^2 \theta}{\sqrt{a^{-1} \cos^2 \theta + b^{-1} \sin^2 \theta}} I_2(\theta) d\theta + \\
 & \frac{\partial_a d}{2-4d+3d^2} \left[(4-6d)G - 2 \int_0^{2\pi} I_1(\theta) d\theta \right] \quad \text{and} \\
 \partial_b G = & -\frac{1}{2-4d+3d^2} \int_0^{2\pi} \frac{\kappa}{2r_0 b^2} \frac{\sin^2 \theta}{\sqrt{a^{-1} \cos^2 \theta + b^{-1} \sin^2 \theta}} I_2(\theta) d\theta + \\
 & \frac{\partial_b d}{2-4d+3d^2} \left[(4-6d)G - 2 \int_0^{2\pi} I_1(\theta) d\theta \right].
 \end{aligned} \tag{5.85}$$

Finally, the derivatives of the energy of the $2s$ state can be compactly written as

$$\frac{\partial E_{2s}}{\partial a} = \frac{\hbar^2}{4} \partial_a F(a, b, d(a, b)) - 2 \frac{e}{4\pi\epsilon_0 r_0} \partial_a G(a, b, d(a, b)), \tag{5.86a}$$

$$\frac{\partial E_{2s}}{\partial b} = \frac{\hbar^2}{4} \partial_b F(a, b, d(a, b)) - 2 \frac{e}{4\pi\epsilon_0 r_0} \partial_b G(a, b, d(a, b)). \tag{5.86b}$$

We stress that, to compute the excited state $2s$, the ground state $1s$ has to be computed first, since the orthogonalization parameter d depends on the variational parameters of the ground state. In the next section, we see how we can compute the excitonic optical conductivity using the exciton energies and wavefunctions.

5.4 Excitonic Optical Response

If we subject a material to a drive from a monochromatic field with frequency ω , the expression for the excitonic optical conductivity in the linear regime, and in the dipole approximation, is given by the following sum over the excitonic states ν [167]

$$\sigma(\omega) = \frac{e^2}{8\pi^3 i \hbar} \sum_{\nu} \left[E_{\nu} \frac{|\Omega_{\nu}|^2}{E_{\nu} - (\hbar\omega + i\Gamma)} + E_{\nu} \frac{|\Omega_{\nu}|^2}{E_{\nu} + (\hbar\omega + i\Gamma)} \right], \tag{5.87}$$

where the quantity Ω_{ν} depends on the exciton wavefunction $\Psi_{\nu}(\mathbf{k})$, and recalling its expression (4.73), it is computed by

$$\Omega_{\nu} = \sum_{\mathbf{k}} \Psi_{\nu}(\mathbf{k}) \hat{\mathbf{e}} \cdot \langle v, \mathbf{k} | \mathbf{r} | c, \mathbf{k} \rangle \tag{5.88}$$

with no sum over bands, since we consider the most simple two-band model for hBN, $\hat{\mathbf{e}}$ is the polarization vector, $\Psi_{\nu}(\mathbf{k})$ is the momentum-space wavefunction of the exciton in the state ν , and the bracket is the optical matrix element we have encountered before in Section 4.4.1. However, we still need to explain how we compute it in practice. Let us then call Ω_{ν} the interacting optical matrix element, or excitonic optical matrix element.

In this work, we consider light polarized along the x -direction and along the y -direction separately, so that the polarization vector will write $\hat{\mathbf{e}} = \hat{\mathbf{x}}$ or $\hat{\mathbf{e}} = \hat{\mathbf{y}}$ accordingly. After computing the optical matrix element, we shall compute its dot product with $\hat{\mathbf{e}}$. We will then

have the two distinct conductivities σ_{xx} and σ_{yy} in the main diagonal of the conductivity tensor. But before that, we need to list the exciton wavefunctions in momentum space to compute the excitonic optical matrix element.

5.4.1 Exciton wavefunctions in momentum space

Now, we derive the wavefunction of the excitonic ground state in momentum space, and the excited states will follow. For that, we just need to Fourier transform $\Psi_{1s}(\mathbf{r})$. In the isotropic case, the integral is calculated by fixing the orientation of the wavenumber $\mathbf{k}$, and then write $\mathbf{k} \cdot \mathbf{r} = kr \cos \theta$, where θ is the angle between $\mathbf{k}$ and $\mathbf{r}$, alongside with a change of variables from Cartesian to polar coordinates. Now we need to do different, because we are evaluating the anisotropic case. We perform instead the change of variables

$$\begin{cases} x = \frac{\rho}{\sqrt{a_{1s}}} \cos \theta, \\ y = \frac{\rho}{\sqrt{b_{1s}}} \sin \theta, \end{cases} \quad (5.89)$$

which is in fact a change from Cartesian to elliptic coordinates. Then, the Fourier transform evaluates to

$$\begin{aligned} \Psi_{1s}(\mathbf{k}) &= \int e^{-i\mathbf{k} \cdot \mathbf{r}} \Psi_{1s}(\mathbf{r}) d\mathbf{r} = \\ &= C_{1s} \int_0^{2\pi} \int_0^\infty e^{-i\rho \left(\frac{k_x}{\sqrt{a_{1s}}} \cos \theta + \frac{k_y}{\sqrt{b_{1s}}} \sin \theta \right)} \frac{\rho e^{-\rho}}{\sqrt{a_{1s}b_{1s}}} d\rho d\theta = \\ &= C_{1s} \int_0^{2\pi} \frac{1}{\sqrt{a_{1s}b_{1s}}} \frac{1}{\left(1 + i \frac{k_x}{\sqrt{a_{1s}}} \cos \theta + i \frac{k_y}{\sqrt{b_{1s}}} \sin \theta \right)^2} d\theta = \\ &= \frac{2\sqrt{2\pi}}{(a_{1s}b_{1s})^{1/4}} \frac{1}{\left(1 + \frac{k_x^2}{a_{1s}} + \frac{k_y^2}{b_{1s}} \right)^{3/2}}, \end{aligned} \quad (5.90)$$

which is an analytical expression after all. Fortunately, that will be the case for all states. We note that when $a_{1s} = b_{1s}$, the previous expression depends only on $k = |\mathbf{k}|$, and we recover the isotropic case. For all the excited states, we follow a similar reasoning. For the excited states $2x$ and $2y$, respectively, we have

$$\begin{aligned} \Psi_{2x}(\mathbf{k}) &= -C_{2x} \frac{3i\pi k_x / a_{2x}}{\left(1 + \frac{k_x^2}{a_{2x}} + \frac{k_y^2}{b_{2x}} \right)^{5/2}} \quad \text{and} \\ \Psi_{2y}(\mathbf{k}) &= -C_{2y} \frac{3i\pi k_y / b_{2y}}{\left(1 + \frac{k_x^2}{a_{2y}} + \frac{k_y^2}{b_{2y}} \right)^{5/2}}. \end{aligned} \quad (5.91)$$

The excited state $2s$, even though more involved, has also an analytical expression for its Fourier transform, which reads

$$\Psi_{2s}(\mathbf{k}) = \sqrt{\frac{16\pi}{(2-4d+3d^2)}} (a_{2s}b_{2s})^{-1/4} \frac{1-2d+(1+d)\left(\frac{k_x^2}{a_{2s}} + \frac{k_y^2}{b_{2s}}\right)}{\left(1 + \frac{k_x^2}{a_{2s}} + \frac{k_y^2}{b_{2s}}\right)^{5/2}}. \quad (5.92)$$

5.4.2 Optical matrix element: the anisotropic case

The optical matrix element can be computed by using the identity

$$\langle v, \mathbf{k} | \mathbf{r} | c, \mathbf{k} \rangle = \frac{\langle v, \mathbf{k} | [H_0, \mathbf{r}] | c, \mathbf{k} \rangle}{\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}}}, \quad (5.93)$$

where the Hamiltonian is the noninteracting one, the ket and the bra correspond to Bloch states of an electron in the conduction and valence bands, respectively, and $\epsilon_{v\mathbf{k}}$ and $\epsilon_{c\mathbf{k}}$ are the energies of an electron in the corresponding Bloch states. By coincidence, the subscript 0 we use in the Hamiltonian $\bar{H}_0$, which stands for the particular case of $\bar{H}_m$ when $m = 0$, coincides with the same symbol used in Section 4.4 to denote the noninteracting Hamiltonian H_0 . It happens that $\bar{H}_0$ is our noninteracting Hamiltonian at the moment, so there is no problem in repeating the symbol, we just advert for the different meanings it may have. In practice, the numerator can be evaluated in terms of the commutator $[\bar{H}_0, \mathbf{r}]$, if in the low-energy Hamiltonian we perceive the momentum components as operators, and take the commutator with $\mathbf{r}$ as an operator by using the canonical commutation relations $[x_i, p_j] = i\hbar\delta_{ij}$. After getting the commutator, we compute the matrix element replacing the bra and the ket with their respective spinors of the low-energy Hamiltonian, denoted by $\mathbf{C}^{v\mathbf{k}}$ and $\mathbf{C}^{c\mathbf{k}}$. However, it may not seem obvious that the optical matrix element can be computed in this way, without referring explicitly to the Bloch wavefunctions. We are not aware of any justification for this in the literature, even though it has been used before and with good final results [15, 28, 143, 202]. Therefore, we give an instructive explanation, even though it may feel ad hoc.

To evaluate the optical matrix element, one considers first the term $H_0\mathbf{r}$ in the commutator. Then, we can insert a resolution of the identity $\sum_{n,\mathbf{k}'} |n, \mathbf{k}'\rangle \langle n, \mathbf{k}'|$ between H_0 and $\mathbf{r}$, assuming it is complete. The matrix elements of H_0 between the Bloch states $\langle v, \mathbf{k} |$ and $|n, \mathbf{k}'\rangle$ can be written as real-space integrals. If we represent the Bloch wavefunctions in the tight-binding approximation per Eq. (3.11) as a linear combination of the atomic orbitals, then the matrix element transforms into a linear combination of integrals, which are the matrix elements of the Hamiltonian between the atomic orbitals, eventually located at different positions in the lattice—these are nothing more than the transfer integrals, which define the parameters of the tight-binding model. Since in the wavefunction we have a sum over unit cells $\mathbf{R}$ with the imaginary exponential $e^{-i\mathbf{k}\cdot\mathbf{R}}$, then, at the end, we would just have to calculate $(\mathbf{C}^{v\mathbf{k}})^\dagger H(\mathbf{k}) \mathbf{C}^{n\mathbf{k}'}$, where $H(\mathbf{k})$ is the Bloch Hamiltonian and the arrays $\mathbf{C}^{n\mathbf{k}}$ are its eigenvectors, that we derived earlier and are given by Eqs. (5.37) and (5.36) for the valence and conduction bands, respectively. For the position operator, we have integrals of the form $\int \phi_\beta(\mathbf{r} - \mathbf{R} - \mathbf{t}_j) \mathbf{r} \phi_{\beta'}(\mathbf{r} - \mathbf{R}' - \mathbf{t}_{j'}) d\mathbf{r}$. In the tight-binding approximation,

if the orbitals are point-like, the integrals amount to matrix elements $\mathbf{r}_{j\beta,j'\beta'}\delta_{\mathbf{R}\mathbf{R}'}$ of a position operator with a matrix representation that is diagonal in the $\mathbf{R}$ variable, that we can denote by $\bar{\mathbf{r}}$. The Kronecker delta $\delta_{\mathbf{R}\mathbf{R}'}$ simplifies the calculations. Then, we have to use the fact that $|n, \mathbf{k}'\rangle$ is an eigenstate of H_0 , so in the sum in the resolution of the identity, only the term $|n, \mathbf{k}'\rangle = |v, \mathbf{k}\rangle$ survives, simplifying even more the calculation. Then at the end, if we choose the lattice gauge without loss of generality, we would have at the end an expression of the form

$$\begin{aligned} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \sum_{i\alpha,i'\alpha'} (C_{i\alpha}^{v\mathbf{k}})^* H_{i\alpha,i'\alpha'}(\mathbf{R}) C_{i'\alpha'}^{v\mathbf{k}} \sum_{j'\beta',j\beta} (C_{j'\beta'}^{v\mathbf{k}})^* \bar{\mathbf{r}}_{j'\beta',j\beta} C_{j\beta}^{c\mathbf{k}} = \\ = (\mathbf{C}^{v\mathbf{k}})^\dagger H(\mathbf{k}) \mathbf{C}^{v\mathbf{k}} (\mathbf{C}^{v\mathbf{k}})^\dagger \bar{\mathbf{r}} \mathbf{C}^{c\mathbf{k}} = \\ = (\mathbf{C}^{v\mathbf{k}})^\dagger H(\mathbf{k}) \bar{\mathbf{r}} \mathbf{C}^{c\mathbf{k}}, \end{aligned}$$

where $H(\mathbf{k})$ is simply the Bloch Hamiltonian that we have defined before in Eq. (3.19). Upon doing something similar for the other term in the commutator $\mathbf{r}H$ and taking the difference with the term above, we arrive at the intermediate formula

$$\langle v, \mathbf{k} | [H_0, \mathbf{r}] | c, \mathbf{k} \rangle = (\mathbf{C}^{v\mathbf{k}})^\dagger [H(\mathbf{k}), \bar{\mathbf{r}}] \mathbf{C}^{c\mathbf{k}}. \quad (5.94)$$

Now, we expand the Bloch Hamiltonian $H(\mathbf{k})$ around the point where the exciton is localized, namely the points $\mathbf{K}$ or $\mathbf{K}'$ in the case of hBN, so that we can replace $H(\mathbf{k}) \rightarrow \bar{H}_0$, where the low-energy Hamiltonian is now the renormalized one, and we allow for the momentum variable $\mathbf{k}$ to become the momentum operator, and $\bar{\mathbf{r}}$ the position operator. Then, we can replace the commutator $[H_0, \mathbf{r}]$ with the commutator

$$[\bar{H}_0, \mathbf{r}] = -i\hbar v_F [\sigma_x \hat{\mathbf{x}} + J_0(\beta) \sigma_y \hat{\mathbf{y}}], \quad (5.95)$$

which was obtained by using the canonical commutation relations. Then, the initial matrix element between the Bloch states of the commutator between the Hamiltonian and the position operator evaluates to

$$\langle v, \mathbf{k} | [H, \mathbf{r}] | c, \mathbf{k} \rangle \approx -i\hbar v_F (\mathbf{C}^{v\mathbf{k}})^\dagger [\sigma_x \hat{\mathbf{x}} + J_0(\beta) \sigma_y \hat{\mathbf{y}}] \mathbf{C}^{c\mathbf{k}}, \quad (5.96)$$

where the arrays $\mathbf{C}^{v\mathbf{k}}$ and $\mathbf{C}^{c\mathbf{k}}$ are the spinors of the low-energy Hamiltonian that we have defined before in Eqs. (5.37) and (5.36) for the valence and conduction bands, respectively. Inserting them in (5.96) and going back to the optical matrix element (5.93), we find

$$\langle v, \mathbf{k} | \mathbf{r} | c, \mathbf{k} \rangle \approx \frac{i\hbar v_F}{\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}}} \left\{ \cos^2\left(\frac{\xi_q}{2}\right) [\hat{\mathbf{x}} - J_0(\beta) \hat{\mathbf{y}}] - e^{-2i\theta_q(\beta)} \sin^2\left(\frac{\xi_q}{2}\right) [\hat{\mathbf{x}} + J_0(\beta) \hat{\mathbf{y}}] \right\}. \quad (5.97)$$

Finally, we can compute Ω_ν for each excitonic state. We split the discussion into the ground state and the excited states.

5.4.2.1 Ground state $1s$

We recall that the variational wavefunction of the $1s$ state in momentum space, easing the notation and dropping the subscripts accompanying the variational parameters, is given by

$$\Psi_{1s}(\mathbf{k}) = \frac{2\sqrt{2\pi}}{(ab)^{1/4}} \frac{1}{\left(1 + \frac{k_x^2}{a} + \frac{k_y^2}{b}\right)^{3/2}}. \quad (5.98)$$

To compute the interacting optical matrix element (5.88), we convert the sum over momentum into an integral, for which the following functions will be useful

$$\begin{aligned} \gamma(\theta) &= \sqrt{\cos^2 \theta + J_0^2(\beta) \sin^2 \theta}, \\ \sigma(\theta) &= \sqrt{\frac{\cos^2 \theta}{a} + \frac{\sin^2 \theta}{b}}. \end{aligned} \quad (5.99)$$

With these functions, we can write the quantities

$$\begin{aligned} \epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}} &= \sqrt{E_g^2 J_0^2(\beta) + 4\hbar^2 v_F^2 [k_x^2 + J_0^2(\beta) k_y^2]} = \sqrt{E_g^2 J_0^2(\beta) + 4\hbar^2 v_F^2 k^2 \gamma^2(\theta)}, \\ \cos^2\left(\frac{\xi_q}{2}\right) &= \frac{1}{2} + \frac{1}{2\sqrt{1 + \left(\frac{2\hbar v_F k \gamma(\theta)}{J_0(\beta) E_g}\right)^2}}, \quad \text{and} \\ \sin^2\left(\frac{\xi_q}{2}\right) &= \frac{1}{2} - \frac{1}{2\sqrt{1 + \left(\frac{2\hbar v_F k \gamma(\theta)}{J_0(\beta) E_g}\right)^2}} \end{aligned} \quad (5.100)$$

that appear in the optical matrix element, and for the wavefunction

$$\Psi_{1s}(\mathbf{k}) = \frac{2\sqrt{2\pi}}{(ab)^{1/4}} \frac{1}{[1 + k^2 \sigma^2(\theta)]^{3/2}}. \quad (5.101)$$

In the excitonic optical matrix element, we have two terms; one that comes from the cosine squared in the optical matrix element in Eq. (5.97), and another that comes from the sine squared in the same equation. So we write $\Omega = \Omega_{1s}^{\cos} + \Omega_{1s}^{\sin}$, and treat both terms separately. We first consider light polarized along the x -axis. The term that comes from the cosine reads

$$\Omega_{1s}^{\cos} = \int_0^{2\pi} d\theta \int_0^\infty dk \frac{2\sqrt{2\pi} i \hbar v_F k \left\{ \frac{1}{2} + \left[2\sqrt{1 + \left(\frac{2\hbar v_F k \gamma(\theta)}{J_0(\beta) E_g}\right)^2} \right]^{-1} \right\}}{(ab)^{1/4} [1 + k^2 \sigma^2(\theta)]^{3/2} \sqrt{E_g^2 J_0^2(\beta) + 4\hbar^2 v_F^2 k^2 \gamma^2(\theta)}}, \quad (5.102)$$

where we have already performed the usual change of variables to polar coordinates. In the isotropic case, the integrand function does not depend on θ , so from the angular integral it only comes a factor of 2π , and the radial integral has an analytical form. However, the

angular integral is not trivial anymore, but the radial integral is still analytical, so we perform it first. We obtain

$$\Omega_{1s}^{\cos} = i\hbar v_F \frac{-2\sqrt{2\pi}}{(ab)^{1/4}} \int_0^{2\pi} \left\{ \frac{\hbar v_F \gamma(\theta) - E_g |J_0(\beta)| \sigma(\theta)}{E_g^2 J_0^2(\beta) \sigma^3(\theta) - 4\hbar^2 v_F^2 \sigma(\theta) \gamma^2(\theta)} - \frac{E_g \hbar v_F |J_0(\beta)| \gamma(\theta) \operatorname{arccsc}\left(\frac{E_g |J_0(\beta)| \sigma(\theta)}{2\hbar v_F \gamma(\theta)}\right)}{[E_g^2 J_0^2(\beta) \sigma^2(\theta) - 4\hbar^2 v_F^2 \gamma^2(\theta)]^{3/2}} \right\} d\theta, \quad (5.103)$$

leaving just the angular integral that can only be evaluated numerically. The term that goes with the sine reads

$$\Omega_{1s}^{\sin} = \int_0^{2\pi} d\theta \int_0^\infty dk \frac{2\sqrt{2\pi} i\hbar v_F k e^{-2i\theta_k(\beta)} \left\{ \frac{1}{2} - \left[2\sqrt{1 + \left(\frac{2\hbar v_F k \gamma(\theta)}{J_0(\beta) E_g} \right)^2} \right]^{-1} \right\}}{(ab)^{1/4} [1 + k^2 \sigma^2(\theta)]^{3/2} \sqrt{E_g^2 J_0^2(\beta) + 4\hbar^2 v_F^2 k^2 \gamma^2(\theta)}}, \quad (5.104)$$

and performing the radial integral we obtain

$$\Omega_{1s}^{\sin} = i\hbar v_F \frac{-2\sqrt{2\pi}}{(ab)^{1/4}} \int_0^{2\pi} \left\{ \frac{-\hbar v_F \gamma(\theta)}{E_g^2 J_0^2(\beta) \sigma^3(\theta) - 4\hbar^2 v_F^2 \sigma(\theta) \gamma^2(\theta)} + \frac{E_g \hbar v_F |J_0(\beta)| \gamma(\theta) \operatorname{arccsc}\left(\frac{E_g |J_0(\beta)| \sigma(\theta)}{2\hbar v_F \gamma(\theta)}\right)}{[E_g^2 J_0^2(\beta) \sigma^2(\theta) - 4\hbar^2 v_F^2 \gamma^2(\theta)]^{3/2}} \right\} e^{-2i\theta_k(\beta)} d\theta, \quad (5.105)$$

to be evaluated numerically as well. The excitonic optical matrix element, when taking the polarization along the x -axis, is then simply $\Omega_{1s}^x = \Omega_{1s}^{\cos} + \Omega_{1s}^{\sin}$, where we put an x in superscript to stress that the calculation was for light polarized along the x -direction. For light polarized along the y -direction, we can identify from the optical matrix element (5.93) an extra factor of $J_0(\beta)$, and also a relative sign between the cosine and sine terms, so that $\Omega_{1s}^y = J_0(\beta)(\Omega_{1s}^{\cos} - \Omega_{1s}^{\sin})$.

5.4.2.2 Excited states $2x$, $2y$ and $2s$

Regarding the excited states $2x$ and $2y$, we need to argue why they do not contribute to the optical conductivity, since it is not straightforward to see why is that (even in the anisotropic case). We consider the wavefunction of the $2x$ state given in equation (5.91), and note that it has a factor of $\cos \theta$ if we change to polar coordinates. The angular integral in the interacting optical matrix element that comes from the cosine in the optical matrix element is of the form

$$\int_0^{2\pi} f(\cos^2 \theta) \cos \theta d\theta, \quad (5.106)$$

where the factor $\cos \theta$ comes from the wavefunction, and the function f encodes all the other factors. We can perform a change of variables $\vartheta = \pi/2 - \theta$ and write

$$\int_0^{2\pi} f(\cos^2 \theta) \cos \theta \, d\theta = - \int_{\pi/2}^{-3\pi/2} f(\sin^2 \vartheta) \sin \vartheta \, d\vartheta = \int_{-\pi}^{\pi} f(\sin^2 \vartheta) \sin \vartheta \, d\vartheta = 0, \quad (5.107)$$

where in the second equality we swapped the limits of the integral and did a shift of $\pi/2$, which is allowed since the integrand is periodic. The last equality holds because we are integrating an odd function over an interval centered at the origin. The term that comes from the sine of the optical matrix element is more delicate. It reads

$$\begin{aligned} & \int_0^{2\pi} f(\cos^2 \theta) \cos \theta \, e^{-2i \arctan(J_0(\beta) \arctan \theta)} \, d\theta = \\ &= \int_0^{2\pi} f(\cos^2 \theta) \cos \theta \cos(2 \arctan(J_0(\beta) \arctan \theta)) \, d\theta - \\ & \quad i \int_0^{2\pi} f(\cos^2 \theta) \cos \theta \sin(\arctan(J_0(\beta) \arctan \theta)) \, d\theta. \end{aligned} \quad (5.108)$$

If in the term on the second line of this equation we perform the same change of variables, we end up with an integral of an odd function again, so it does not contribute. For the second term, we simply make a shift of $-\pi$ in the integration limits, since the integrand function is odd from the start, so that term vanishes as well. For the excited state $2y$, all the previous arguments apply as well, so it does not contribute to the conductivity just like the state $2x$. We are left only with the calculation of the optical conductivity deriving from the state $2s$. As in the case of the $1s$ state, the radial integral is analytical, and the angular integral has to be performed numerically. Even though the expression of the radial integral is analytical, it is very complicated, hence we refrain from writing it here.

At the end, we have only the contributions from the $1s$ and $2s$ state to the optical conductivity, that for x polarization reads

$$\sigma_{xx}(\omega) = \frac{e^2}{8\pi^3 i \hbar} \sum_{\nu=1s,2s} E_\nu \frac{|\Omega_\nu^x|^2}{E_\nu - (\hbar\omega + i\Gamma)} + (\omega \rightarrow -\omega)^*, \quad (5.109)$$

where $(\omega \rightarrow -\omega)^*$ is used to refer to the nonresonant term in the sum, which we discard since its contribution is negligible. For y polarization the expression is analogous. We have seen now how to compute the optical conductivity for the system under study. In the next section we discuss the results.

5.5 Results

We organize this section in three parts. In the first one, we discuss the renormalized band structure and the consequence for the excitons, namely the binding energies. With the information from the excitonic energies and states, we can compute the excitonic optical conductivity as we have just discussed in the previous section, which is the subject of the second part of this section. In the third part, we finish with the results for the dispersion

of the exciton–polaritons and the material absorption based on the theory provided in Section 2.3.

5.5.1 Bandgap and excitonic binding energies

As we have seen in Section 5.2, the effective masses renormalize as $m_x = m^*|J_0(\beta)|$ and $m_y = m^*/|J_0(\beta)|$, where $m^* = E_g/2v_F^2$ is the isotropic effective mass of electrons in pristine hBN. This means that, as $J_0(\beta) \rightarrow 0$, the dispersion becomes more and more anisotropic. To study the effects of this anisotropy in the electronic and excitonic properties of the system, we compute the bandgap and the binding energies of the excitonic states $1s$, $2x$, $2y$ and $2s$ for varying β , from the case $\beta = 0$ without potential to $\beta = 2.4$ sufficiently close to the first zero of the Bessel function $J_0(\beta)$. We stress that the bandgap includes the electron–electron exchange correction $\Delta_{\text{ex}}(\beta)$, which also renormalizes with β , and whose calculation is detailed in Section B.2 of the Appendix B. The results are displayed in Fig. 5.2. The value of the total bandgap, also called the fundamental gap, for $\beta = 0$ is the one of pristine monolayer hBN that, with our estimation, is of 6.2 eV, which is not very distant from the values reported in the literature from DFT calculations using hybrid functionals [203], or taking into account many-body effects using the GW +BSE approach. The work by F. Ferreira et al. in Ref. [30] has, in Table 1 therein, a comprehensive list of values for the bandgap of hBN obtained from different works that use the GW and GW +BSE methods. Another report listing different values for the bandgap of hBN in the literature is Ref. [203], where Table I in the main text displays results analogous to those of Ref. [30], also from *ab-initio* calculations at both DFT and GW levels.

The total bandgap $\Delta = \Delta_g(\beta) + \Delta_{\text{ex}}(\beta)$ as a function of β computed with our approach is in Figure 5.2a. As one increases the potential strength V_0 (or the period L), the bandgap of hBN decreases from its initial value to zero, with the bandgap closing at the first zero of $J_0(\beta)$. This information is relevant to calculate the sub-gap optical conductivity, because the excitonic levels E_ν determine the resonant frequencies. Knowing the bandgap Δ and the binding energy $E_{b,\nu}$, we know the exciton energies $E_\nu = \Delta - |E_{b,\nu}|$ ¹. The exciton binding energies obtained in the light of the variational principle are represented in Figure 5.2b. The s states are in red, and the p states $2x$ and $2y$ are in black. While the p states are degenerate in the isotropic case $\beta = 0$, as expected from the Wannier–Mott description, they are not degenerate in the anisotropic case $\beta \neq 0$. Also, even though the binding energies generically decrease with increasing β , the $2y$ state presents a non-monotonic behavior, but it is always the first excited state. On the contrary, the binding energy of the $2x$ state always decreases, and at some point around $\beta \approx 1.6$, the binding energies of the states $2x$ and $2s$ match. After that point, the $2s$ state becomes the second excited state, whereas the $2x$ is the highest one (among these four states). Not surprisingly, the $1s$ state always remains as the ground state. We could use variational wavefunctions for even higher energy states;

¹Most commonly in the literature, the binding energy is presented as a negative value, to stress that it is a bound state. We work with the absolute value of the binding energy, making the analysis indifferent to the convention.

however, the optical response is typically dominated by the lowest energy states. Moreover, as they go higher in energy, successive states become closer and harder to distinguish. In addition to that, as we get closer to the end of the bandgap, the optical response is dominated by the independent-particle approximation (IPA) conductivity, which we do not analyze in this chapter.

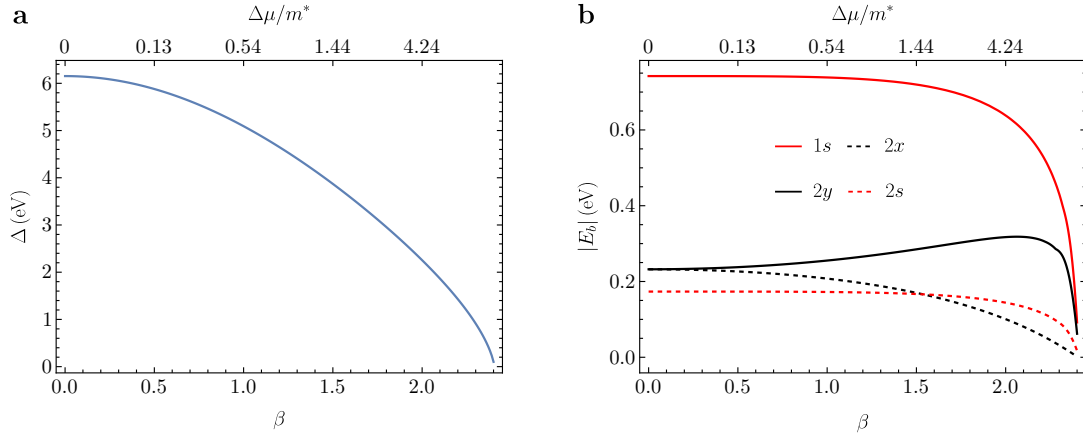


Figure 5.2: Electronic bandgap and excitonic binding energies of a periodically modulated hBN sheet. **a** Bandgap and **b** binding energies of the excitonic states as a function of $\beta = V_0 L / (\hbar v_F \pi)$. In **b** the binding energies of s peaks are represented in red, with a continuous line for the $1s$ state and with a dashed line for the $2s$ state, while the p states $2x$ and $2y$ are represented with dashed black line and with a continuous black line, respectively. The scale in the top horizontal axis in both plots represent the degree of anisotropy in the dispersion, that we define as $\Delta\mu/m^* = (m_y - m_x)/m^*$. For hBN, the parameters are [30]: $E_g = 3.92$ eV, $\hbar v_F = 5.06$ eV $\cdot\text{\AA}$, and $r_0 = 10$ $\text{\AA}$.

To obtain the binding energies, we applied the variational method as explained in Section 5.3, in which we have to find the variational parameters that minimize the binding energy. So, for each value of β , we have to find the roots of equations of the form $\partial E_\nu / \partial a_\nu = \partial E_\nu / \partial b_\nu = 0$, where the forms of the derivatives for each one of the states $1s$, $2x$ and $2y$, and $2s$ are given at the end of the subsections 5.3.1, 5.3.2 and 5.3.3, respectively. This was done using the method **FindRoot** of the software Mathematica [204].

5.5.2 Sub-gap optical conductivity and absorption

In this section, we show the results obtained for the optical conductivity computed through Eq. (5.87). Figure 5.3 displays the results for the excitonic optical conductivity for different realizations of β , and for light polarized along the x and y axes separately. Since the conductivity is complex, we plot separately the real and imaginary parts in the top and bottom panels, respectively. The panels on the left are for $\hat{\mathbf{e}} = \hat{\mathbf{x}}$, while the panels on the right are for $\hat{\mathbf{e}} = \hat{\mathbf{y}}$. We have justified before why the $2x$ and $2y$ states do not contribute to the optical conductivity in Section 5.4.2.2; this is why in panels 5.3a and 5.3b there are only two peaks for each realization of β ; the higher peak corresponds to the $1s$ state, while the smaller peak corresponds to the $2s$ state.

The results for the isotropic case without potential, $\beta = 0$, represented by the black curves in the plots of Fig. 5.3, are in agreement with the literature where the same methods for the excitonic states, energies and conductivity were used [28, 30]. In turn, they are in good qualitative agreement with the results from the $GW+BSE$ approach as discussed in Ref. [30]. Note that the results for the case $\beta = 0$ appear on both sides of Fig. 5.3, where the real and imaginary parts of the conductivity are the same regardless of the polarization. They seem to have different heights for $\hat{\mathbf{e}} = \hat{\mathbf{x}}$ and $\hat{\mathbf{e}} = \hat{\mathbf{y}}$, but the scales of the vertical axes are different. The scales of the vertical axes have to be different, otherwise we would not be able to see the conductivity in the case of $\hat{\mathbf{y}}$ polarization, since it gets attenuated when β increases. On the contrary, the opposite happens for the $\hat{\mathbf{x}}$ polarization; as β increases, the oscillator strength increases as well, and the peaks in the conductivity are enhanced. Besides that, the resonant frequencies, defined by the excitonic levels as $\hbar\omega = E_\nu$, tend to redshift with increasing β .

A particularity of the most extreme case of $\beta = 2.3$, is the existence of only one zero in the imaginary part of the conductivity, as one can see in Fig. 5.3c. This will affect the behavior of the polariton dispersion since, as we discussed in Section 2.3.2, whether polaritons are supported depends on the sign of $\text{Im}\{\sigma(\omega)\}$.

Knowing the optical conductivity, besides being able to calculate the dispersion relation for polaritonic modes, we can also compute the optical absorption for normal incidence by using the expression (2.54), and the definitions of reflected and transmitted radiation fractions in Eqs. (2.53a) and (2.53b), respectively. When $\mathcal{R}$ and $\mathcal{T}$ are given in terms of the permittivities of media 1 and 2, Eq. (2.54) reads

$$\mathcal{A}_{\text{opt}} = 1 - \left| \frac{\sqrt{\varepsilon_1} - \sqrt{\varepsilon_2} - \frac{\sigma(\omega)}{c\varepsilon_0}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2} + \frac{\sigma(\omega)}{c\varepsilon_0}} \right|^2 - \frac{\text{Re}\{\sqrt{\varepsilon_2}\}}{\text{Re}\{\sqrt{\varepsilon_1}\}} \left| \frac{2\sqrt{\varepsilon_1}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2} + \frac{\sigma(\omega)}{c\varepsilon_0}} \right|^2. \quad (5.110)$$

The optical absorption as a function of the frequency is shown in Figures 5.4a and 5.4b, for light polarized along the x - and y -directions, respectively. Each panel has the absorption spectrum for the same values of β as in Fig. 5.3. For $\hat{\mathbf{y}}$ polarization, in the panel 5.4b we can see that initially the 1s peak slightly increases, but after that the material becomes less responsive, with an attenuated absorption as β increases. The behavior of the absorption for $\hat{\mathbf{x}}$ polarization in Fig. 5.4b has more interesting features. For instance, from pristine hBN, where the 1s peak is naturally more intense than the 2s peak, by modulating externally and fine tuning the potential, it is possible to have the 2s higher than the 1s peak in the same material. This can be seen in panel a, in the blue and gray curves more specifically. Even though this behavior is unusual, it has already been observed experimentally [205] and predicted theoretically [202]. However, in the same plot, we can see that the intensity of the 2s peak does not increase monotonically with β .

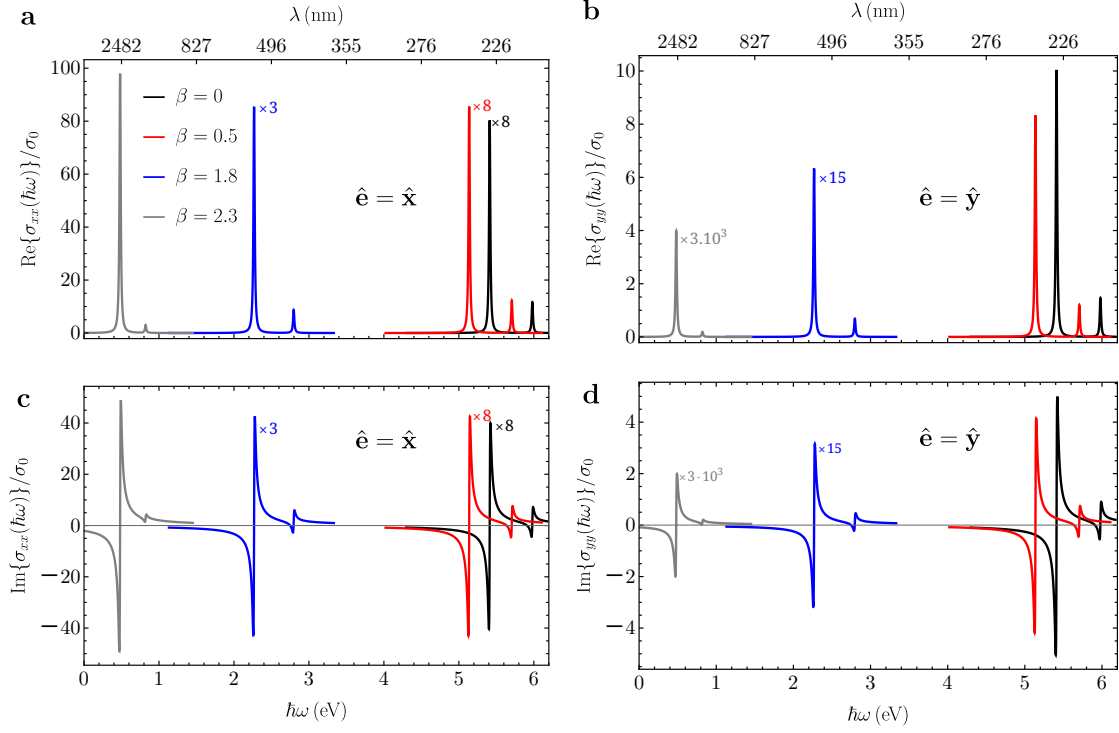


Figure 5.3: Optical conductivity of a periodically modulated hBN sheet. Panels **a** and **b** show the real and imaginary parts of the conductivity, respectively, for light polarized along the x -axis. Panels **c** and **d** show the real and imaginary parts of the conductivity, respectively, for light polarized along the y -axis. In each panel, we plot the conductivity in different colors for different realizations of the parameter β ; black for $\beta = 0$, red for $\beta = 0.5$, blue for $\beta = 1.8$, and gray for $\beta = 2.3$. In all the panels, the conductivities corresponding to each one of these values for β appear in the same order, from right to left. To display all the conductivities for different values of β in the same plots, we applied a scaling factor to the curves when needed. The scaling factor is indicated adjacently to the respective curve. For the broadening parameter in Eq. (5.109), we used $\Gamma = 10$ meV, and $\sigma_0 = 4e^2/\hbar$.

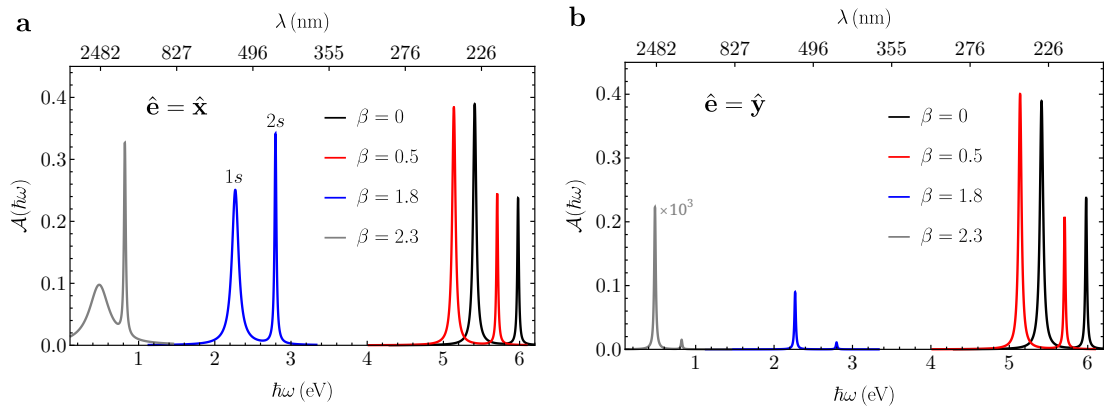


Figure 5.4: Optical absorption of a periodically modulated hBN sheet. Panels **a** and **b** show the absorption for light polarized along the x - and y -directions, respectively. Each panel shows the absorption as a function of frequency for the same realizations of β as in Fig. 5.3, using the same color set. For the material on top we used $\varepsilon_1 = 1$, and for the quartz substrate $\varepsilon_2 = 2.16$, which is an average of the permittivity over a wide frequency window. For simplicity, we do not take into account the full ω dependence of ε_2 .

To better understand the results in Figure 5.4, we use again the expression (5.110), but now computed at the resonant frequencies. We stress that we have the optical conductivity for β varying continuously from zero up to the first zero of $J_0(\beta)$, we just selected a couple of values for β to plot $\sigma(\omega)$. Then, it is possible to compute the maximum of the absorption at the 1s and 2s peaks separately as a function of β , and the result can be seen in Figure 5.5 below. Therein, we can observe on panel **a** how each peak behaves with varying β for both polarizations. For $\hat{\mathbf{x}}$ polarization, the 1s absorption peak (continuous black line) decreases monotonically, while the 2s peak (continuous red line) increases until it surpasses the 1s peak, but it decreases at the end. For $\hat{\mathbf{y}}$ polarization, it is the 1s peak (dashed black line) that has a nonmonotonic behavior, while the 2s peak (dashed red line) always decreases, but unlike the case $\hat{\mathbf{e}} = \hat{\mathbf{x}}$, the 1s peak stays always higher than the 2s peak.

The fact that we can have a 1s peak higher than a 2s peak can be explained by close inspection of equations (2.53a), (2.53b) and (5.110). Writing the conductivity $\sigma = \sigma' + i\sigma''$ in terms of its real and imaginary parts, assuming that the permittivities are real, and working out $\mathcal{R} = |r|^2$ alongside $\mathcal{T} = \sqrt{\varepsilon_2/\varepsilon_1}|t|^2$ (casting the factor $\varepsilon_0 c$ into σ for simplicity) we get

$$\mathcal{R} = \frac{(\sqrt{\varepsilon_2} - \sqrt{\varepsilon_1} + \sigma')^2 + (\sigma'')^2}{(\sqrt{\varepsilon_2} + \sqrt{\varepsilon_1} + \sigma')^2 + (\sigma'')^2}, \quad \text{and} \quad \mathcal{T} = \frac{4\sqrt{\varepsilon_1\varepsilon_2}}{(\sqrt{\varepsilon_2} + \sqrt{\varepsilon_1} + \sigma')^2 + (\sigma'')^2}. \quad (5.111)$$

If we analyze the previous expressions at the frequency of the 1s excitonic resonance, where $\sigma'' = 0$ holds and let $\sigma' \rightarrow \infty$, corresponding to the most extreme case of $\beta = 2.3$, we can easily see that $\mathcal{R} \rightarrow 1$ and $\mathcal{T} \rightarrow 0$. Hence, the system reflects all the incident light, and $\mathcal{A}_{\text{opt}} \rightarrow 0$. When the oscillator strength grows immensely, making σ' grow to infinity, the 2D material behaves like a perfect mirror, similarly to the surface of a very good conducting metal. But if we now consider the resonance at the 2s excitonic state energy, we see that both σ' and σ'' are finite (the imaginary part has only one zero in the most extreme case $\beta = 2.3$, see Fig. 5.3c), which means that $\mathcal{R}$ and $\mathcal{T}$ are finite, and consequently, so is the absorption at the 2s resonance. Initially, without potential, the 1s resonance is still higher than the 2s resonance, both in the conductivity and the absorption. Nonetheless, this does not contradict our reasoning of the perfect mirror when $\beta = 2.3$. The reason is that the absorption peak is not monotonic with σ' . To see this, we work out explicitly the expression for $\mathcal{A}_{\text{opt}}$ in Eq. (5.110) and simplify to yield

$$\mathcal{A}_{\text{opt}} = \frac{4\sqrt{\varepsilon_1}\sigma'}{(\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2} + \sigma')^2 + (\sigma'')^2}. \quad (5.112)$$

Looking always at the same resonant frequency, we have always $\sigma'' = 0$ (unless the 2s resonance is close enough to the 1s resonance so that $\sigma'' \neq 0$), and then the absorption peak is given by

$$\max_{\omega} \mathcal{A}_{\text{opt}} = \frac{4\sqrt{\varepsilon_1}\sigma'}{(\sqrt{\varepsilon_2} + \sqrt{\varepsilon_1} + \sigma')^2}, \quad (5.113)$$

where the real part of the conductivity is computed at a resonant frequency, corresponding either to the excitonic energy E_{1s} or E_{2s} . Plotting the previous expression as a function of σ' in Fig. 5.5b, we observe that indeed it is not monotonous, which explains why we can have in general a $2s$ peak higher than a $1s$ peak, even if the oscillator strength of the $1s$ resonance is higher than that of the $2s$ resonance.

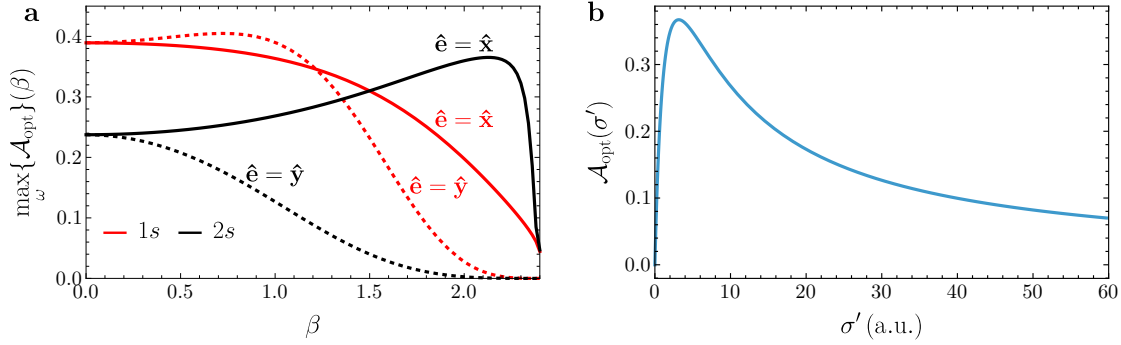


Figure 5.5: Maximum optical absorption for different potential modulations. Panel **a** shows the values of the absorption peaks due to the excitonic states $1s$ (red lines) and $2s$ (black lines) as a function of β for light polarized along the x (continuous lines) and y (dashed lines) directions, respectively. Note that for each value of β the peaks occur at different energies, as one can see in Figs 5.4a and 5.4b. Panel **b** displays the dependence of the absorption as a function of the real part of the conductivity σ' at resonant frequencies, that is, when σ' peaks and the imaginary part $\sigma'' = 0$ is null.

To finish the analysis of the interaction of light with our system, and within the theme of strong-coupling, in the next section we will look at the dispersion relation of the emergent exciton–polaritons our system can support.

5.5.3 Exciton–polaritons

In this section, we analyze the dispersive behavior of the polaritonic modes our system can support. Polaritons have been introduced before in Chapter 2, within a classical description of the theory of light–matter interaction. In that chapter, we derived the equations that provide the dispersion relation of polariton for each type of EM field polarization. The dispersion relation of the polariton differs between the two possible polarizations, s or transverse electric (TE), and p or transverse magnetic (TM) (see Fig. 2.3). In our calculations for the system at hand, we observed that the TE modes, if existent, present either free-light-like dispersion or weak field confinement, something we already expected from the discussion in subsection 2.3.2, and also from the literature [41, 206]. The TM modes are decidedly more interesting, as we will see from our results. Just for the purpose of refreshing the reader’s memory, we present once more their dispersion relation

$$\boxed{\frac{\varepsilon_1}{\kappa_1} + \frac{\varepsilon_2}{\kappa_2} + i \frac{\sigma_{xx}(\omega)}{\varepsilon_0 \omega} = 0}, \quad (5.114)$$

where $\kappa_j = \sqrt{q^2 - \varepsilon_j \omega^2 / c^2}$, $j = 1, 2$, q is the wavenumber of the polariton, and σ_{xx} is the conductivity we obtained in the previous section (see Fig. 5.3). The dispersion relation $\omega(q)$ was obtained as the numerical solutions of the equation just above, which exist whenever $\text{Im}\{\sigma_{xx}(\omega)\} > 0$. We obtained different dispersions for different realizations of β , which are supported in different frequency ranges determined by the excitonic levels E_{1s} and E_{2s} . We selected $\beta = 0$, the case without potential, and two other values of $\beta = 0.5$ and $\beta = 2.3$, this last corresponding to the case where the oscillator strength is the highest in Figs. 5.3 and 5.4. The results are displayed in the three panels of Fig. 5.6, presenting the frequency as a function of the real and imaginary parts of q . In panel **a**, we can see how different the dispersion of the polariton is compared to that of free light in vacuum and in quartz, in continuous and dashed black lines, respectively. In the cases $\beta = 0$ of panel **a** and $\beta = 0.5$ of panel **b**, these black lines cannot be distinguished because the wavenumber of the polariton is much larger than that of free light. The increase in $\text{Re}\{q\}$ is more gradual for lower frequencies, but presents a pronounced kink at higher frequencies alongside $\text{Im}\{q\}$, suggesting these modes can achieve high EM field confinements.

The dispersion relation of the polariton in the $\beta = 2.3$ case of panel **c** is quantitatively distinct from the previous two, as one can see when comparing the scale of the q -axis with the scales of panels **a** and **b**. In panel **c**, the polariton presents wavenumbers of order of the μm , whereas in the first two cases they are of order of the nm. Nonetheless, both real and imaginary parts of q still assume higher values than that of free light, which are represented by the now visible straight black lines. The case of panel **c** is also qualitatively different from the cases of panels **a** and **b**, because in the region of frequencies where the mode is supported, there is a frequency corresponding to E_{2s} where the imaginary part of the conductivity is not zero due to its proximity to E_{1s} (see gray curve in Fig. 5.3c). As a consequence, the dispersion curve back-bends at the frequency where $\text{Re}\{\sigma_{xx}\}$ has a bump resulting from the nonzero oscillator strength of the $2s$ excitation. This phenomenon has been predicted before by C. Wolff and co-authors in Ref. [207], where they showed that this can be seen in any periodic optical system with loss and dispersion, and in any kind of materials strongly coupled to resonances. What we observe here is the particular case of strong exciton–photon coupling in a 2D material. Even though we considered only the real ω and complex q case, the reverse situation has also been considered in Ref. [207] in quite general terms.

To get a feeling for the spatial distribution of the EM field of a surface polariton, we display in Fig. 5.7 a map of the electric and magnetic fields of a TM-polarized mode, represented by the black arrows and the colorful background, respectively. We can see how the EM field decays away from the hBN monolayer along the z -direction, and how these evanescent modes oscillate along the direction of propagation, the x -direction. The region $z > 0$ corresponds to vacuum, while the region $z < 0$ corresponds to the substrate, where the field confinement is higher than in air (stronger colors and bigger arrows). This illustrates that it is possible to engineer the field confinement using different encapsulating materials.

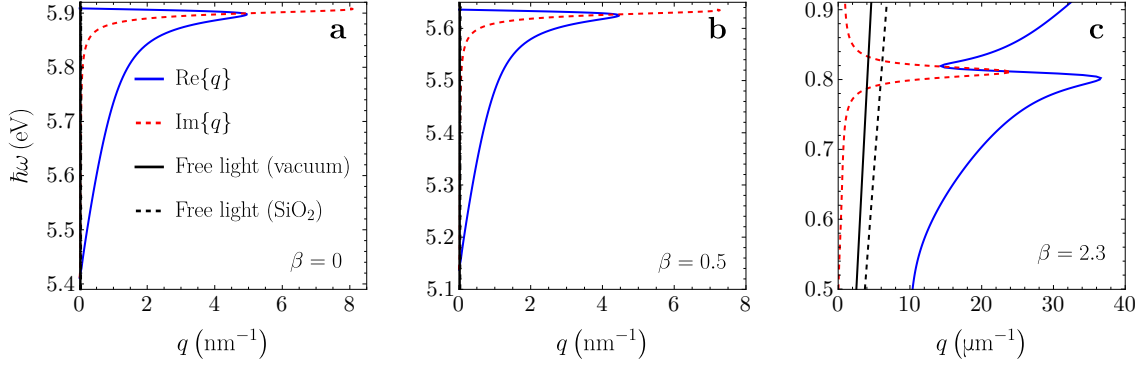


Figure 5.6: Photonic dispersion of exciton-polaritons in modulated hBN. Dispersion relation $\omega(q)$ of exciton-polaritons in the system under study. We solve numerically Eq. (5.114) for different values of β . The solution is the frequency as a function of the wavenumber, that has a real part (solid blue line) and an imaginary part (dashed red line). For comparison, we show the dispersion relation for free propagating light in vacuum (solid black line) and in SiO_2 (dashed black line). The case $\beta = 0$ represents monolayer hBN without potential on top.

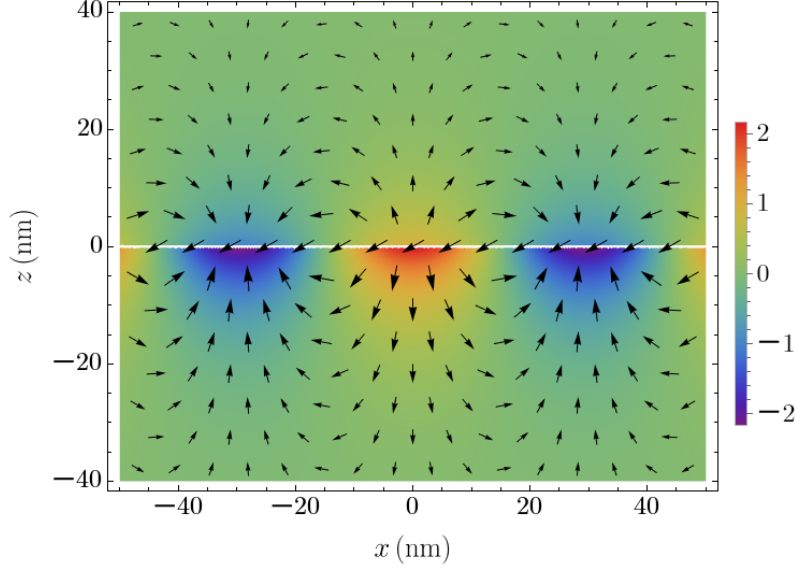


Figure 5.7: Spatial map of the EM field of a TM-polarized polariton. Spatial configuration of the EM field of a TM-polarized polaritonic mode. The color intensity plot in the background represents the intensity of the magnetic field, while the vector plot represents the off-plane components of the electric field, with their respective (relative) intensities represented by the size of the vectors. The top half is the field in air, while the bottom half is in quartz, SiO_2 . We picked the (ω, q) point in the dispersion of Fig. 5.6b of highest (real) wavenumber. This plot does not take into account the imaginary part of the wavenumber. We used the following parameters: $q = 4.45 \text{ nm}^{-1}$, $\hbar\omega = 5.68 \text{ eV}$, $\varepsilon_1 = 1$ and $\varepsilon_2 = 2.16$.

At the time of the publication of the work that this chapter is based on, Ref. [171], the setup proposed initially in Figure 5.1 was a mere suggestion for how one could realise a periodic electrostatic modulation, albeit with a step potential, even though in the derivation of the low-energy Hamiltonian in Section 5.2 we considered a cosine-like potential to arrive at an analytical expression for the Hamiltonian. In the advent of current technologies and developments on the isolation of 2D monolayers, there is actually a recent experiment [194] where the authors realised a similar experiment to the one proposed here also with a hBN layer, and with a periodic potential consisting of a finite number of Fourier components. In this way, with the likelihood of the evolution of experimental techniques, we presume that in a not so distant future one could realise a single-harmonic periodic modulation on a hBN flake to put our theory up to test. This could deem our study relevant for possible applications in nanophotonics or optoelectronics in different ranges of the EM spectrum, since the parameter β is quite tunable, due to its dependence on the amplitude V_0 and the period L of the potential.

Even though our theory based on quantum principles allowed quantifying the excitonic optical response of hBN, it still used a classical model to account for the screening. In the next chapter, we will implement an efficient microscopic approach to screening, which allows to determine in practice the effective Rytova–Keldysh dielectric function we have been using so far, along with the parameter r_0 . Not only that, we can also go beyond the Rytova–Keldysh model, which we will see that is the long-wavelength limit of the macroscopic dielectric function, obtained from a microscopic permittivity function analogous to the one we saw in Section 3.4; the theory of that section applied to 2D materials will be put in practice in the final part of this dissertation.

ELECTROSTATIC SCREENING IN TWO-DIMENSIONAL MATERIALS

Electrostatic screening constitutes a key aspect in this work, as we have seen that it can affect the electronic [208, 209] and optical properties of crystalline solids [210, 211], with a strong emphasis on those of two-dimensional (2D) insulators and semiconductors [26, 212–214]. In the case of gapped materials, the dielectric and screening properties of the crystal have an important role when it comes to electron–electron interactions within the material. The free or bound charges contained in it, either point-like or dipolar, rearrange themselves in response to an external electric potential, which in turn affects how strongly they interact. Within the field of condensed matter physics, one often deals with the many-body interacting problem, where electronic charges communicate with each other via pair-wise interactions mediated by the Coulomb potential. By interacting with each other, their Coulomb repulsion (or attraction) is not the original $v_c(\mathbf{r}) = e/(4\pi\epsilon_0 r)$. It is a weaker Coulomb potential, that we usually call the screened Coulomb potential. However, each charge should be treated equally. This means that the problem of determining the dielectric properties of a crystalline solid is a self-consistent problem. The electronic properties affect how the charges interact, and their interaction in turn affects its electronic properties. The theory of solid state physics which treats the many-body interacting problem is called many-body perturbation theory, which we discussed briefly in Section 3.3.1. In that section, we introduced the dielectric function as a central object that allows to quantify the screened Coulomb potential. However, the dielectric function encompassing all transitions between all the possible states, single-particle or many-particle, cannot be computed in practice. Therefore, we introduced the random-phase approximation (RPA) for the dielectric function in Section 3.4. Within this approximation, one can write a closed expression for the dielectric function considering only transitions between single-particle states. This constitutes a first step in obtaining an estimate for the weaker screened potential, which affects how the electronic properties renormalize (i.e., it affects the emergent quasiparticle band structure). It constitutes as well a first step in obtaining the electronic properties of a material, which in turn can affect its optical properties.

In previous chapters, namely Chapters 4 and 5, we saw how important pair-wise interactions are to determine the sub-gap optical properties of insulators and semiconductors. By the Coulomb interaction between an electron excited to a conduction band and the positively charged hole it left behind in the valence band, the formation of a new composite particle, the exciton, is allowed. And to accurately determine the optical properties induced by excitonic effects, we need to model the interaction between the electron and the hole properly [27, 215]. This is where the RPA dielectric function enters. In exciton calculations for 2D materials, one typically employs the Rytova–Keldysh approximation for the dielectric function [15, 28, 29, 51, 150, 216], including in numerical packages like WANTIBEXOS [217] and *Xatu* [37]. The Rytova–Keldysh dielectric function is a linear model for the permittivity when defined in momentum space. It coincides with the true macroscopic dielectric function in the long-wavelength limit [27, 215], and it has the advantage that the potential has an analytical expression in real space [15, 28, 29, 51], which we have encountered before in this dissertation. It presents some limitations, though. Even if it works for a wide class of 2D semiconductors and insulators, such as hexagonal boron nitride (hBN), a further look into the linear model reveals that taking the $q \rightarrow 0$ limit in the static dielectric function may lead to unphysical behavior. For instance, when taking into account the layer thickness of the material and the dielectric environment in the Rytova–Keldysh model, it may happen that the linearized model has a negative slope [27]. Furthermore, if we take the limit $q \rightarrow \infty$, then the screened potential decays as $1/q^2$, overestimating the screening in the big q or small r limits of the interaction. In this chapter, we make an extension of the *Xatu* code to go beyond the Rytova–Keldysh approximation. A direct application of the implemented model for the screening is the calculation of excitons with the code, which we do for hBN and MoS₂.

This chapter is based on the work developed, in part, during the research stay abroad as part of the PhD program. The development of the code resumed at SDU, always in close collaboration with the hosting one. A manuscript pertaining to this work is contained in the list of publications, but it has not been published yet. The paper is still in preparation, but we expect to timely submit it to a peer-reviewed journal. However, the contents of this chapter and of the final manuscript are not going to be exactly the same. This chapter serves as a complement to the manuscript, and vice-versa. Momentarily, a technical manuscript focusing mainly in the implementation of the 2D dielectric function in the *Xatu* code is not in preparation. Even though we intend to publish the code in a peer-reviewed journal, there are some quantitative analyses and changes in the code to be done. An indepth study of the code’s capabilities and a more logical structure for it are necessary to make it up to the standards of the literature. Nonetheless, we dedicate a section of this chapter to the details about the implementation.

This work serves to close a gap in the literature, in which a strict 2D approach to the numerical implementation of the screening properties of 2D materials has never been done before. However, something similar has been done in analytical calculations for ungapped materials, e.g. graphene [218, 219], and with good results. In the context of excitons in

2D materials, strict 2D approaches for the screening combining analytical with numerical methods have been extensively applied in the literature [24, 26, 37, 143, 171, 220, 221]. However, at the moment of the writing of this dissertation, we are not aware of any numerical package capable of determining the 2D static dielectric function for 2D materials within a 2D picture. So far, only *ab-initio* packages are capable of determining the dielectric function of 2D materials, albeit using a three-dimensional (3D) formulation of the problem [31–36].

This chapter is divided in four main sections. In the first one, we reintroduce the theory for the microscopic dielectric function within RPA, which serves as the basis for the numerical methods herein implemented. The theory is strongly focused on 2D materials. In the second section, we present the numerical outcomes from applying the theory developed in the first section. The case material of choice is once more monolayer hBN. In the third section, we are able to go beyond the strict 2D description for the dielectric function by adopting a quasi-2D picture. We compare the results obtained from both 2D and quasi-2D approaches. In the last section, we provide more details on the implementation of the code, stressing the novelty of our approach and the gain in efficiency. For a change, we test the code for another case material, MoS₂. We finalise the chapter with closing remarks and final conclusions.

6.1 Dielectric Function in Two Dimensions: Theory

In this section we look again into the RPA dielectric function that we have previously introduced in Section 3.4.2. The dielectric function in real-space and time domain, within RPA, which we discussed in Section 3.4, reads

$$\varepsilon(\mathbf{r}, \mathbf{r}'; t, t') = \delta(\mathbf{r} - \mathbf{r}')\delta(t - t') - \int v_c(\mathbf{r} - \mathbf{r}'')\chi^0(\mathbf{r}'', \mathbf{r}'; t - t') d\mathbf{r}'', \quad (6.1)$$

where v_c denotes the bare (without screening) Coulomb potential, and χ^0 is the RPA irreducible polarizability. When Fourier transformed from the $(\mathbf{r}, t)$ domain to the $(\omega, \mathbf{q})$ domain, it has the form

$$\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, \omega) = \delta_{\mathbf{G}\mathbf{G}'} - v_c(\mathbf{q} + \mathbf{G})\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega), \quad (6.2)$$

where $\mathbf{G}$ and $\mathbf{G}'$ are reciprocal lattice vectors, $\mathbf{q}$ is a vector in the first Brillouin zone (BZ), and ω is the frequency. This is nothing more than a recapitulation of what we have discussed in Section 3.4. The novelty in this chapter is that the Fourier transform is not in three dimensions, but in two dimensions. We overlook the z dimension, as if $\mathbf{r} = (x, y)$. Since in this work we deal with 2D materials, we argue that performing a 2D Fourier transform on the dielectric function is the right thing to do, because the material is infinite in the xy plane, but finite along the z -direction. Technically speaking, we are saying that the response to an off-plane varying external potential is local, so that $\varepsilon(\mathbf{r}, \mathbf{r}') = \varepsilon^{2D}(\mathbf{r}_{||}, \mathbf{r}'_{||})\delta(z - z')$, where ε^{2D} is a strictly 2D dielectric function, $\mathbf{r}_{||}$ and $\mathbf{r}'_{||}$ are in-plane vectors, and its Fourier transform is (6.2).

Within the strict 2D framework for the dielectric response we develop here, we can ignore the z coordinate altogether in the dielectric function. This is evidently an approximation that needs to be justified. We postpone doing so to Section 6.3, and see how far we can go with a pure 2D picture. In what follows, we drop the 2D superscript in ε for ease of notation. But still, the reader should keep in mind that it is a 2D quantity.

In what follows, all the geometric quantities like position, reciprocal lattice vectors, and momentum are in-plane. Therefore, the momentum-resolved Coulomb potential in Eq. (6.2) reads

$$v_c(\mathbf{q}) = \frac{e}{2\varepsilon_0 q}, \quad (6.3)$$

which depends on momentum as $\sim 1/q$, unlike its 3D counterpart that goes with $\sim 1/q^2$. As we will explain in a moment, this small detail puts us in advantage over typical first-principle approaches. Naturally, the polarizability $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega)$ is also the 2D counterpart of Eq. (3.75), and reads

$$\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega) = \frac{1}{\mathcal{A}} \sum_{n,n'} \sum_{\mathbf{k}\sigma} (f_{n,\mathbf{k}} - f_{n',\mathbf{k}+\mathbf{q}}) \frac{\langle n, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | n', \mathbf{k} + \mathbf{q} \rangle \langle n', \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | n, \mathbf{k} \rangle}{\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}+\mathbf{q}} + \hbar\omega + i\hbar\alpha}. \quad (6.4)$$

At the end, the expressions (3.75) and (6.4) do not differ by much. The resemblance is clear from their written forms, but the main contrast is in the implementation. First, the sum over momentum $\mathbf{k}$ is a sum over the 2D BZ of the material. Furthermore, even though it is not explicitly evidentiated in the bracket notation, the plane-wave matrix elements in the numerator amount to real-space integrals extending over the plane of the material, and are therefore integrals in 2D space. Such matrix elements have appeared before in Section 4.3.2, when discussing the interaction matrix elements for the Bethe–Salpeter equation (BSE) within an exact diagonalization formalism. To make this chapter as self-contained as possible, we repeat the definition in Eq. (4.60) from that section, but now making it explicit that the integral is in two dimensions. Henceforth, we can write

$$I_{n\mathbf{k},n'\mathbf{k}'}^{\mathbf{G}} \equiv \langle n, \mathbf{k} | e^{-i(\mathbf{k}-\mathbf{k}'+\mathbf{G})\cdot\mathbf{r}} | n', \mathbf{k}' \rangle = \int_{\mathcal{A}} d\mathbf{r} \, \psi_{n,\mathbf{k}}^*(\mathbf{r}) e^{-i(\mathbf{k}-\mathbf{k}'+\mathbf{G})\cdot\mathbf{r}} \psi_{n',\mathbf{k}'}(\mathbf{r}). \quad (6.5)$$

Note that even though the notation is similar to that of Ref. [37], the definition for $I_{n\mathbf{k},n'\mathbf{k}'}^{\mathbf{G}}$ is not exactly the same. We will explain how to compute the matrix elements and the polarizability in practice in the next subsection.

6.1.1 Noninteracting polarizability

Since we are interested in electrostatic screening for 2D insulators and semiconductors (from now on, let us just say insulators to account for all 2D gapped materials), and at very low temperatures, we can simplify the polarizability in Eq. (6.4) to

$$\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, \omega \rightarrow 0) = \frac{1}{\mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \left[\frac{\langle c, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | v, \mathbf{k} + \mathbf{q} \rangle \langle v, \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | c, \mathbf{k} \rangle}{\epsilon_{v\mathbf{k}+\mathbf{q}} - \epsilon_{c\mathbf{k}}} \right] + \frac{1}{\mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \left[\frac{\langle v, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | c, \mathbf{k} + \mathbf{q} \rangle \langle c, \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | v, \mathbf{k} \rangle}{\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}}} \right], \quad (6.6)$$

where the sum over bands n and n' is now over valence bands v and conduction bands c . In the previous expression, if the system is time-reversal symmetric, then the two terms on the right-hand side add up so that we can write the polarizability in a more compact form as

$$\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}) = \frac{2}{\mathcal{A}} \sum_{vc} \sum_{\mathbf{k}\sigma} \frac{I_{c\mathbf{k}, v\mathbf{k}+\mathbf{q}}^{\mathbf{G}} \left(I_{c\mathbf{k}, v\mathbf{k}+\mathbf{q}}^{\mathbf{G}'} \right)^*}{\epsilon_{v\mathbf{k}+\mathbf{q}} - \epsilon_{c\mathbf{k}}}. \quad (6.7)$$

We would like to take this opportunity to make a brief remark. We have confirmed analytically that the expression (6.6) derives from its more general version in (6.4), which is an expression commonly found in textbooks [76]. However, the expression in (6.7) is not commonly found in textbooks, but it has made its appearance in the literature [31, 33]. We stress that the prefactor of 2 and the sum over spins σ are key to get correct screening properties. We prove in Appendix C.1 that expression (6.6) reduces to (6.7) when time-reversal symmetry (TRS) is present. All systems considered in this thesis have TRS, which can be broken, for instance, when a static magnetic field is applied. As we have stated before in Section 2.1.1, we do not study magnetized media.

We have gathered all the necessary knowledge to compute these matrix elements, and thus the matrix elements of the noninteracting polarizability $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$. As we said before, the material is in practice 2D without any repeated off-plane images, unlike in first-principle approaches. Therefore, our implementation is strictly 2D, and we ignore, for the moment, any information pertaining to the out-of-plane degrees of freedom. This means that:

- The sum over momentum in (6.6) is a sum over momentum $\mathbf{k} = (k_x, k_y)$ in the BZ, which is 2D.
- The $\mathbf{G}$ and $\mathbf{G}'$ vectors are 2D vectors of the reciprocal lattice, which is 2D.
- The matrix element $I_{n\mathbf{k}, n'\mathbf{k}'}^{\mathbf{G}}$, which is an integral in real space when written in terms of Bloch state wavefunctions, is an integral in the xy plane. The symbol $d\mathbf{r}$ inside the integrals therefore means $dx dy$, if we do not say otherwise.

With this information in mind, we can work out the plane-wave matrix elements by replacing the Bloch wavefunctions by their expression in terms of localized orbitals, that in the lattice gauge read

$$\psi_{n, \mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}} \sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \phi_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i). \quad (6.8)$$

The plane-wave matrix elements between Bloch states have actually been calculated before in Section 4.2, at the time when we derived the BSE applied to effective models. At this stage, we compute them approximately using once again the point-like orbital approximation, but beyond the envelope function approximation as we did in Section 4.2. Let us then carry out the integral one more time. We have

$$\begin{aligned}
 I_{n\mathbf{k},n'\mathbf{k}'}^{\mathbf{G}} &= \int d\mathbf{r} \psi_{n,\mathbf{k}}^*(\mathbf{r}) e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} \psi_{n',\mathbf{k}'}(\mathbf{r}) = \\
 &= \frac{1}{N} \int d\mathbf{r} \sum_{\mathbf{R}\mathbf{R}'} e^{-i\mathbf{k}\cdot\mathbf{R}} e^{i\mathbf{k}'\cdot\mathbf{R}'} \sum_{i\alpha j\beta} (C_{i\alpha}^{n\mathbf{k}})^* C_{j\beta}^{n'\mathbf{k}'} \underbrace{\phi_{\alpha}^*(\mathbf{r}-\mathbf{R}-\mathbf{t}_i) \phi_{\beta}(\mathbf{r}-\mathbf{R}'-\mathbf{t}_j)}_{\approx \delta_{ij} \delta_{\alpha\beta} \delta_{\mathbf{R}\mathbf{R}'} \delta(\mathbf{r}-\mathbf{R}-\mathbf{t}_i)} e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} = \\
 &= \frac{1}{N} \sum_{\mathbf{R}} \cancel{e^{-i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{R}}} \sum_{i\alpha} (C_{i\alpha}^{n\mathbf{k}})^* C_{i\alpha}^{n'\mathbf{k}'} \cancel{e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}}} e^{-i\mathbf{G}\cdot\mathbf{R}} e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{t}_i} e^{-i\mathbf{G}\cdot\mathbf{t}_i} = \\
 &= \frac{1}{N} \left(\sum_{\mathbf{R}} e^{-i\mathbf{G}\cdot\mathbf{R}} \right) \sum_{i\alpha} (C_{i\alpha}^{n\mathbf{k}})^* C_{i\alpha}^{n'\mathbf{k}'} e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{t}_i} = \\
 &= \sum_{i\alpha} (C_{i\alpha}^{n\mathbf{k}})^* C_{i\alpha}^{n'\mathbf{k}'} e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{t}_i}, \quad \text{since} \quad \sum_{\mathbf{R}} e^{-i\mathbf{G}\cdot\mathbf{R}} = N, \tag{6.9}
 \end{aligned}$$

where from the 2nd line to the 3rd line we performed the point-like orbital approximation, and from the 4th line to the 5th line we used the fact that by definition $e^{-i\mathbf{G}\cdot\mathbf{R}} = 1$. There is a caveat when it comes to using eigenvectors determined numerically, like the ones storing the coefficients $C_{i\alpha}^{n\mathbf{k}}$. The eigenvectors coming from diagonalizing numerically any matrix carry a random phase. This is accounted for in the code. As explained in the paper of the first version of the code in [37], the phase of each eigenvector $\mathbf{C}^{n\mathbf{k}}$ is fixed such that $\sum_{i\alpha} C_{i\alpha}^{n\mathbf{k}} \in \mathbb{R}$ [127]. Furthermore, for density functional theory (DFT) band structures the code performs a Löwdin orthogonalization of the basis after obtaining the eigenvectors $\mathbf{C}^{n\mathbf{k}}$.

To summarize, to compute a single polarizability matrix element $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$ we need:

1. The Bloch Hamiltonian $H(\mathbf{k})$ (regardless of the gauge) and to diagonalize it to obtain the band structure $\{\epsilon_{n\mathbf{k}}\}$ in the set of points $\mathbf{k}$ included in the sum and the eigenvectors $\{\mathbf{C}^{n\mathbf{k}}\}$ storing the tight-binding coefficients $C_{i\alpha}^{n\mathbf{k}}$;
2. The set of energies $\{\epsilon_{n\mathbf{k}+\mathbf{q}}\}$ and eigenvectors $\{\mathbf{C}^{n\mathbf{k}+\mathbf{q}}\}$, where $\mathbf{q}$ is the point in the BZ we are computing the polarizability at;
3. The plane-wave matrix elements of the form $I_{\mathbf{c}\mathbf{k},v\mathbf{k}+\mathbf{q}}^{\mathbf{G}} = \sum_{i\alpha} (C_{i\alpha}^{c\mathbf{k}})^* C_{i\alpha}^{v\mathbf{k}+\mathbf{q}} e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i}$, as we know the position of the atoms in the motif $\mathbf{t}_i$ from the beginning.

In our numerical framework, the Bloch Hamiltonian $H(\mathbf{k})$ is determined by the input model for the code. The code accepts two types of input. One way is by providing an output file from CRYSTAL [222], which contains the Fock matrices $H(\mathbf{R})$ determining the Bloch Hamiltonian through Eq. (3.19), as we saw in Chapter 3. Alternatively, the

code accepts also `.model` files containing a tight-binding model for the material at hand. Quite recently, the code was updated by G. Janone with a utility that transforms an output file from Wannier90 [223] into a `.model` file. The corresponding version of the code can be found in <https://github.com/GuiJanone/xatu-w90>. This utility serves as an intermediary between Wannier90 and the Xatu code.

The Wannier model for the Hamiltonian has to be obtained preferably through the method of maximally localized Wannier functions [81], in order for the point-like orbital approximation to be valid. Then, the orbitals ϕ_α can be perceived as Wannier functions (see footnote 11 in Chapter 3) as well, besides their interpretation as atomic orbitals.¹ Upon diagonalization of the Bloch Hamiltonian $H(\mathbf{k})$ at momenta $\mathbf{k}$ and $\mathbf{k} + \mathbf{q}$, we obtain the energy dispersions $\{\epsilon_{n\mathbf{k}}\}$ and $\{\epsilon_{n\mathbf{k}+\mathbf{q}}\}$ from the eigenvalues, and the tight-binding coefficients $\{\mathbf{C}^{n\mathbf{k}}\}$ and $\{\mathbf{C}^{n\mathbf{k}+\mathbf{q}}\}$ from the eigenvectors. In math notation, and for an orthogonal basis, $H(\mathbf{k})\mathbf{C}^{n\mathbf{k}} = \epsilon_{n\mathbf{k}}\mathbf{C}^{n\mathbf{k}}$. Using this information to compute the polarizability matrix elements, we can compute the elements of the dielectric function, which is the subject of the next subsection.

6.1.2 Dielectric matrix and its inversion

We are interested in the static limit of the dielectric function as per Eq. (6.2), for which we need the static polarizability worked out in the previous section. The static dielectric function, within RPA, reads

$$\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) = \delta_{\mathbf{G}\mathbf{G}'} - \sqrt{v_c(\mathbf{q} + \mathbf{G})}\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})\sqrt{v_c(\mathbf{q} + \mathbf{G}')}, \quad (6.10)$$

if we adopt the symmetrized version of (6.2). The reason why we implement the symmetrized version is that it eases the treatment of numerical divergences in the dielectric function and, subsequently, in the screened potential in the long-wavelength limit [224–226]. Actually, for the head element $\varepsilon_{\mathbf{0}\mathbf{0}}(\mathbf{q} \rightarrow \mathbf{0})$, it is indifferent if we use the symmetrized version as per Eq. (6.10), or the initial one in Eq. (6.2). For this element, we can simply set $\varepsilon_{\mathbf{0}\mathbf{0}}(\mathbf{q} \rightarrow \mathbf{0}) = 1$, because the bare potential goes with $v_c(\mathbf{q}) \sim 1/q$, while the polarizability goes with $\chi_{\mathbf{0}\mathbf{0}}^0(\mathbf{q} \rightarrow \mathbf{0}) \sim q^2$, as we shall confirm in subsection 6.2.2 (see Fig. 6.4). This feature of the head element of the polarizability is universal among insulators, either in two or three dimensions, as long as the material presents a bandgap [24]. This means that, when computing the matrix element $\varepsilon_{\mathbf{0}\mathbf{0}}(\mathbf{q})$ in the limit $\mathbf{q} \rightarrow \mathbf{0}$, we face no numerical instabilities unlike in the 3D case, where $v_c(\mathbf{q}) \sim 1/q^2$.

In *ab-initio* calculations for 2D materials, it is typical to use a truncated Coulomb potential instead of the full one, in order to mitigate the interaction between repeated

¹In general, in a Wannier representation of the Hamiltonian, we have a number of “atomic positions” $\mathbf{t}_i$ higher than the number of atoms in a unit cell. Therefore, the positions $\mathbf{t}_i$, also called Wannier centers, may not coincide with the real positions of the atoms in the unit cell. On the contrary, in a tight-binding model, we have as many atomic positions $\mathbf{t}_i$ as atoms in the motif, and generically more than one orbital per atom. Either way, we do not consider the spatial structure of the orbitals. Hence, we treat both approaches in the same footing.

images, with increasing off-plane separation [225, 227, 228]. Only after performing a 2D Fourier transform on the truncated Coulomb potential and taking the limit $q \rightarrow 0$, can one recover the $1/q$ behavior in the potential, removing any numerical instabilities in that limit. However, it still depends on the off-plane separation between layers $L_\perp$. But even in that case, the value of the static macroscopic dielectric function ε_M at zero momentum is not exactly one [228]. The results have to be converged by increasing $L_\perp$, and only in the limit of infinite separation $\varepsilon_M(0) = 1$ for a 2D material. Besides that, the 2D macroscopic dielectric function is not given by the definition in Eq. (3.80). It can only be obtained after performing an off-plane average of the inverse 3D microscopic dielectric function [144, 145, 228], because the definition in Eq. (3.80) is not directly applicable.

With our framework, the macroscopic dielectric function we get from inverting the microscopic one as per Eq. (6.10) is automatically an effective 2D macroscopic dielectric function. Also, we have $\varepsilon_{\mathbf{0}\mathbf{0}}(\mathbf{0}) = \varepsilon_{\mathbf{0}\mathbf{0}}^{-1}(\mathbf{0}) = 1$ by construction. Regarding the numerical treatment of the wing terms $\varepsilon_{\mathbf{0}\mathbf{G}'}(\mathbf{q})$ and $\varepsilon_{\mathbf{G}\mathbf{0}}(\mathbf{q})$, it is required to look at the asymptotic behavior of the matrix elements of the dielectric matrix and its inverse when $\mathbf{q} \rightarrow \mathbf{0}$. Only in that way can we know how to avoid divergences or undefined values for the screened potential. This is critical for the computation of the excitons, due to how the screened Coulomb potential appears in the calculation of the interaction matrix elements (see Eqs. (4.59) and (4.61)). Issues like this are routine in the literature [31, 127, 225, 229]. To not break the flow of the text, we relegate the details to Appendix C.2.

The calculations are also much more efficient due to the reduced dimensionality of our approach. To compute the dielectric matrix and invert it, we have no off-plane contribution to the reciprocal lattice vectors, so the number of matrix elements to compute, at each point $\mathbf{q} \in \text{BZ}$, scales with N^2 instead of N^3 , if N is the number of cells of the reciprocal lattice along the directions of the two primitive vectors.

After having computed the dielectric matrix, we need to invert it. This poses a difficulty in the problem: there are infinite reciprocal lattice vectors; therefore, we need to apply a cutoff G_c for the vectors $\mathbf{G}$ and truncate the dielectric matrix to invert it in practice. Doing so is routine in the literature [31–33]. Even if we have decided to follow the alternative approach for the RPA inverse dielectric function given by [176]

$$\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}) = \delta_{\mathbf{G}\mathbf{G}'} + v_c(\mathbf{q} + \mathbf{G})\chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}), \quad (6.11)$$

where $\chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q})$ is the reducible polarizability within RPA defined by the Dyson equation (3.69), we would have a similar problem. In that Dyson equation, all quantities when written in momentum space are infinite matrices as well, so we would need to apply a cutoff for the vectors $\mathbf{G}$ either way. This is in great contrast with the jellium model considered by H. Bruus and K. Flensberg in Ref [74], where there is no underlying crystalline structure, and as a consequence, all quantities like the Coulomb potential, the polarizability, and the dielectric function are simply scalars. For now, let us assume that we can in principle compute the dielectric matrix and invert it without any problems. In that way, the screened

Coulomb potential in the static limit reads

$$W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) = \sqrt{v_c(\mathbf{q} + \mathbf{G})} \varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}) \sqrt{v_c(\mathbf{q} + \mathbf{G}')}, \quad (6.12)$$

which is symmetrized as well by virtue of using the symmetrized dielectric function (6.10). Looking at the dynamical expression of the polarizability, we can see that the frequency dependence comes in the denominator as $\hbar\omega - (\epsilon_{n'\mathbf{k}+\mathbf{q}} - \epsilon_{n\mathbf{k}})$. If the bandgap is large enough when compared to the frequency, then we can approximate the screened Coulomb potential by its static limit and avoid an integration in frequency in the interaction matrix elements.

To finish the discussion, we revisit the interaction matrix elements in the BSE for the excitons with the numerical approach. Their expressions derived in Section 4.3.2 remain the same, apart from the prefactor that contains the area $\mathcal{A}$ instead of the volume $\mathcal{V}$, since we follow a strict 2D approach. For the direct term we have

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} (I_{c\mathbf{k}+\mathbf{Q}, c'\mathbf{k}'+\mathbf{Q}}^{\mathbf{G}})^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{k} - \mathbf{k}') I_{v\mathbf{k}, v'\mathbf{k}'}^{\mathbf{G}'}, \quad (6.13)$$

while for the exchange term we have

$$X_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} (I_{c\mathbf{k}+\mathbf{Q}, v\mathbf{k}}^{\mathbf{G}})^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{Q}) I_{c'\mathbf{k}'+\mathbf{Q}, v'\mathbf{k}'}^{\mathbf{G}'}. \quad (6.14)$$

The reader shall notice that in both expressions there is a sum over reciprocal lattice vectors. In fact, it is a double sum over $\mathbf{G}$ and $\mathbf{G}'$. This double sum accounts for local-field effects in the calculation of the excitons, which derives from the translational invariance being discrete and not continuous. If we have enough $(\mathbf{G}, \mathbf{G}')$ matrix elements in the dielectric function, in principle we will not need as many terms in the sums above. In practice, we have two cutoffs, one for the dielectric function G_c , and another one for the excitons G_c^X . To preserve basic symmetries and degeneracies, we hope that G_c^X can be made small enough².

6.2 Dielectric Function in Two Dimensions: Numerical Results

We analyze in this section the numerical results obtained from the 2D implementation of the dielectric function. We will follow the same order as in the previous section; we focus first on the polarizability, then on the dielectric matrix and its inverse, and finally on the excitons. When discussing the dielectric function, we shall make the connection with the macroscopic dielectric function and take the small $\mathbf{q}$ limit when relevant. In principle, we should recover the Rytova–Keldysh model and get an estimate for the screening parameter

²In practice, we always have $G_c^X < G_c$, since we have to compute the dielectric matrix before computing the exciton.

r_0 . The material of choice is hBN once more; but now, we work with a more realistic model as we shall describe, and look at the material from a different perspective.

6.2.1 hBN revisited: DFT band structure and Hamiltonian model

In this section, we describe our model Hamiltonian for hBN, a 2D material quite important in this thesis. We introduced for the first time a minimal tight-binding model for hBN in Section 3.2.2.1, with only two parameters; one parameter for the difference in onsite energies between boron and nitrogen atoms, and the hopping parameter between first neighbors. Then, in Chapter 5, we used a model based on the initial one considered in chapter 3 to study excitons and the excitonic optical response in a hBN nanostructure. We also used the Rytova–Keldysh model to account for the screening. In this chapter, we work out a numerical model for hBN. For that, we need a more realistic model based on *ab-initio* methods. After all, it is with *ab-initio* methods that we can fit the tight-binding band structure to the *ab-initio* one to get the tight-binding parameters.

Herein, the model Hamiltonian for hBN was obtained through DFT calculations (see subsection 3.2.2.2) with the commercial package CRYSTAL [222], which uses a framework of Gaussian basis functions. In our calculations, we used the hybrid functional HSE06 [230], which mixes the exact Hartree–Fock (HF) exchange functional and the Perdew–Burke–Ernzerhof functional [93] within the generalized gradient approximation (GGA)³. For the sampling of $\mathbf{k}$ points, we use a uniform $48 \times 48 \times 1$ Monkhorst–Pack $\mathbf{k}$ -grid [232], with an interlayer distance of 500 Å and $a = 2.51$ Å for the lattice constant. The output file `hBN_base_HSE06.outp` can be found in the official repository for the `Xatu` code: https://github.com/xatu-code/xatu/tree/master/examples/material_models/DFT. This file contains the Fock matrices of the form $H(\mathbf{R})$ in a basis of localized Gaussian functions, sufficient in number to build the Bloch Hamiltonian $H(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} H(\mathbf{R})$ while preserving basic symmetries. In the DFT calculations, 69 atomic orbitals were used. Diagonalizing the Bloch Hamiltonian with the `Xatu` code along the high-symmetry line $\Gamma - K - M - \Gamma$, we obtain the band structure displayed in Fig. 6.1 below. For this model, the bandgap is $\Delta = 6.08$ eV, not very distant from the one reported in Section 5.5.1. With this model Hamiltonian, namely the band structure and the eigenvectors of the Bloch Hamiltonian, we construct a numerical model for the screening properties of hBN, needed later on to compute the excitons through the exact diagonalization method for the BSE, as discussed in Section 4.3.2. That is the subject of the sections to come.

³In typical *ab-initio* calculations within DFT, there is always an ambiguity in the functional one can choose to capture exchange and correlation effects. This is because no one knows the analytical expression for the exact exchange–correlation functional, if it even exists. However, we know that it is a functional of the electron density $n(\mathbf{r})$ by the Hohenberg–Kohn theorems, we just do not have an analytical expression for it. Nonetheless, exchange–correlation contributions to the ground-state energy can be computed numerically. The caveat is that it is very computationally demanding. What one typically does, is mixing the exact numerical functional with an approximated one, say at the local-density approximation (LDA) or GGA level. There is consensus in the community that using hybrid functionals allows better estimating the electronic bandgap than non-hybrid ones when compared with the experimental values [231].

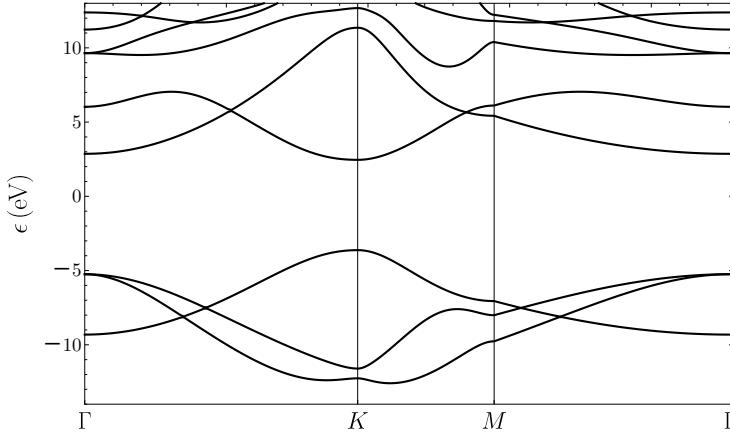


Figure 6.1: Band structure of hBN through first-principle methods. Quasiparticle band structure of monolayer hBN computed within DFT using the CRYSTAL package. The details of the DFT calculations are provided in the paragraph above this figure. The bands were computed using the Xatu code [37]. 169 $H(\mathbf{R})$ matrices were included.

6.2.2 Noninteracting polarizability

Before we proceed with the calculation of the dielectric matrix, we need to study the convergence of the polarizability matrix elements $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$ given by Eq. (6.7) with the number of included conduction bands, for different $(\mathbf{G}, \mathbf{G}')$ pairs, and for different values of momentum $\mathbf{q}$.

We have verified, for this material and others, that the polarizability converges relatively fast with the number of $\mathbf{k}$ points in the BZ mesh that enter in the sum of Eq. (6.7). Typically, between a 15×15 grid up to a 20×20 grid we can converge the polarizability within the first four to five significant digits. We shall look closely into the convergence with the number of conduction bands included in the sum.

In Fig. 6.2 below, we study the convergence of the polarizability with the number of empty bands included in the calculation for hBN, in particular the head element $\chi_{00}^0(\mathbf{q})$. We just advert the reader for the fact that the polarizability presented in the figures is the one in Eq. (6.7) multiplied by the unit cell area $\mathcal{A}_{\text{UC}}$, as in our implementations we cast the area of the unit cell to the Coulomb potential, and leave the factor of $1/N$ in the polarizability for it to converge with the size of the BZ mesh. Therefore, they appear in units of eV^{-1} and scaled by a factor of 10^2 . This choice does not have any physical reasons, it is just how we decided to implement it. At the end, the dielectric function will come out dimensionless as it is supposed to be. We see again that the polarizability converges with the number of empty bands included. Including more than 60 bands makes a difference in the seventh significant decimal place.

Fig. 6.3 does a similar analysis to proof check that the polarizability converges for other elements than the head. We picked a wing element to check the convergence of both real and imaginary parts. Wing elements are those of the form $\chi_{\mathbf{G}0}^0$ or $\chi_{0\mathbf{G}}^0$, with $\mathbf{G} \neq 0$. This nomenclature applies to the dielectric matrix as well. We can see that both real and imaginary parts of the chosen matrix element converge with the number of conduction bands. Something similar has been seen before in the literature, for example in Fig. 4 of Ref. [31].

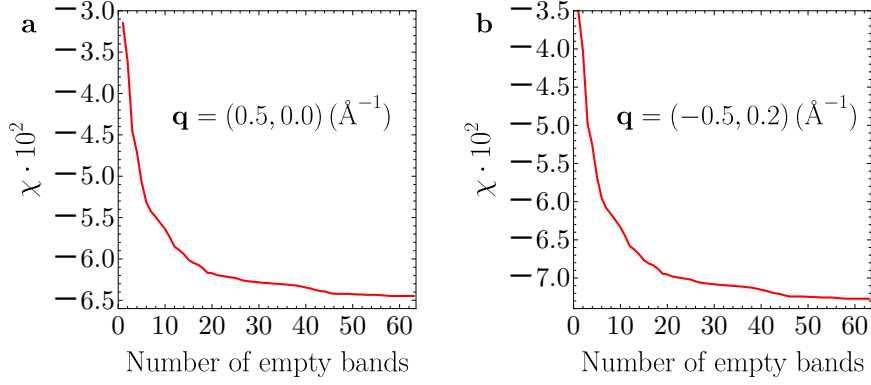


Figure 6.2: Convergence of the polarizability with the number of conduction bands: head element. Matrix element $\chi_{00}^0(\mathbf{q})$ of the polarizability for two different values of momentum $\mathbf{q}$ as a function of included conduction bands for hBN. There are in total 6 full bands and 63 empty bands for this model. 169 Fock matrices were included.

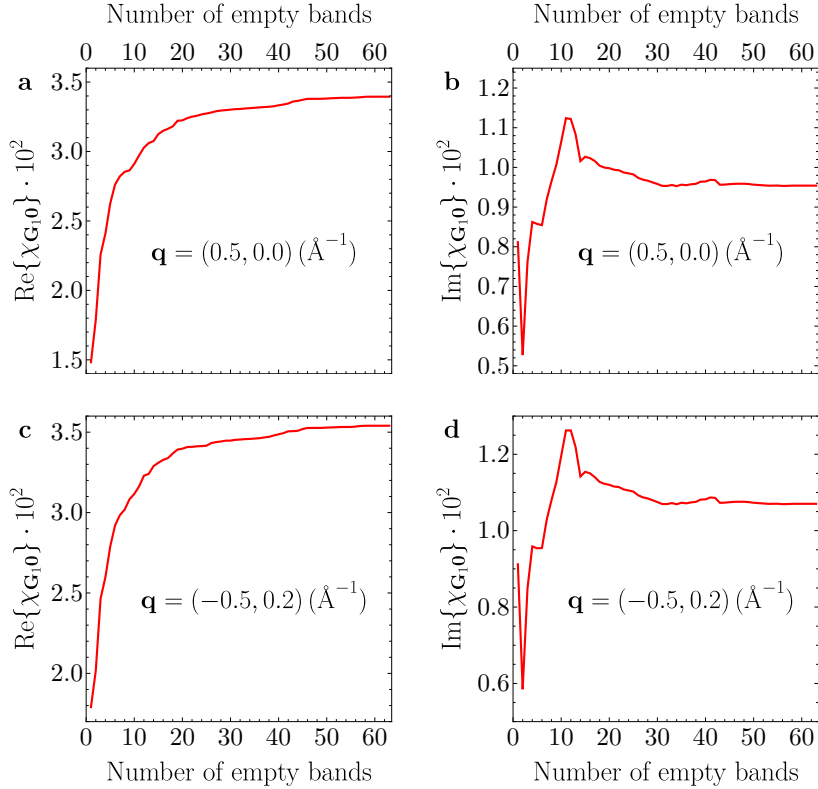


Figure 6.3: Convergence of the polarizability with the number of conduction bands: a wing element. Matrix element $\chi_{\mathbf{G}_1 0}^0(\mathbf{q})$ of the polarizability, where $\mathbf{G}_1$ is a primitive vector of the reciprocal lattice, for two different values of momentum $\mathbf{q}$ as a function of included conduction bands for hBN. The left panels display the real part, while the right panels display the imaginary part. The model for hBN is the same as in Fig. 6.2. Herein, $\mathbf{G}_1 = (-1.45, 2.50) \text{ \AA}^{-1}$. 169 Fock matrices were included.

Next, we study how the polarizability behaves for any momentum $\mathbf{q} \in \text{BZ}$. We start with the head matrix element. In Fig. 6.4 below, we plot $\chi_{00}^0(\mathbf{q})$ in the BZ mesh, more precisely in panel **b**. Panel **a** displays the quantity $\chi_{00}^0(\mathbf{q})$ as a function of $q = |\mathbf{q}|$, with

each dot representing a point in the BZ mesh. What we see from that plot, is that the head element of the polarizability:

- does not depend on the direction of $\mathbf{q}$, but only on q (apart from small numerical fluctuations). This is expected for hBN, an isotropic material.
- approaches zero with a quadratic q -dependence when $q \rightarrow 0$, as expected for a 2D insulator [24].

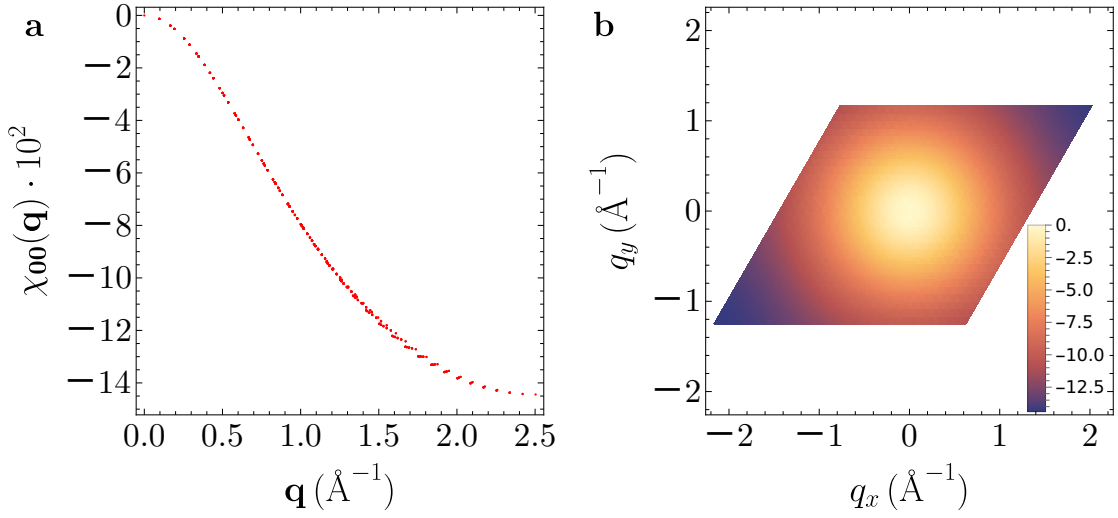


Figure 6.4: Polarizability in the BZ: the head element. Matrix element $\chi_{00}^0(\mathbf{q})$ for hBN in the BZ. Panel **a** displays the polarizability as a function of the magnitude of the momentum. Each red dot is a $\mathbf{q}$ point in the mesh, and they are sorted by increasing order of $|\mathbf{q}|$. Panel **b** displays a color map of the (absolute) value of the polarizability in the BZ mesh. The bar legend in panel **b** is inside the plot frame. In both plots, the polarizability is multiplied by 10^2 .

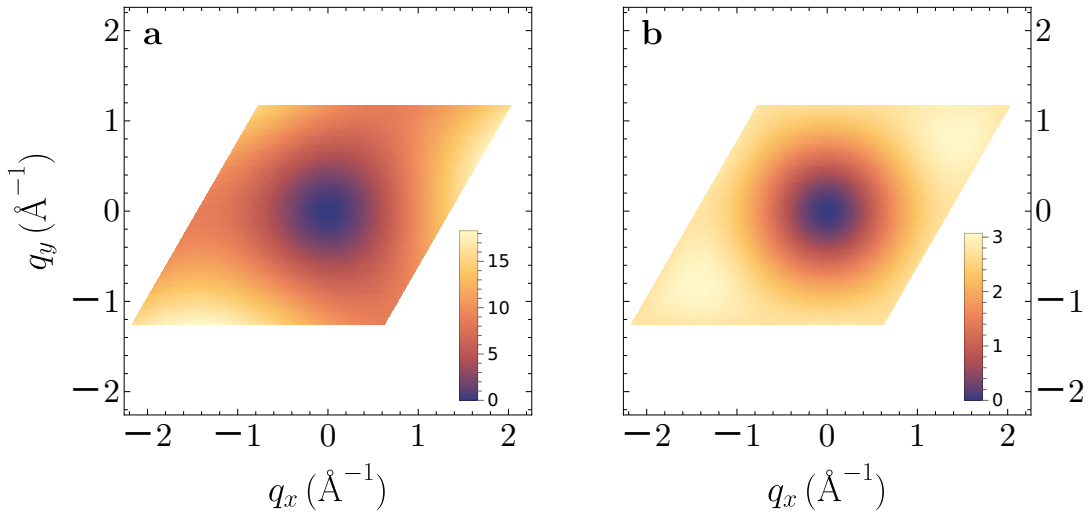


Figure 6.5: Polarizability in the BZ: a wing element. Matrix element $\chi_{G_{10}}^0(\mathbf{q})$ of the polarizability in the BZ for hBN. Panel **a** displays the real part of the polarizability, while panel **b** displays the imaginary part. The bar legends are inside the frames. The polarizability is multiplied by 10^2 in both plots.

As we already worked out the polarizability, we can compute the matrix elements of the static dielectric function following Eq. (6.10). The analysis of the dielectric matrix and its inverse is reserved for the next subsection.

6.2.3 Inverse dielectric matrix

6.2.3.1 Local-field effects and convergence

We study the convergence of the inverse dielectric matrix with the number of included reciprocal lattice vectors. It is well known that including reciprocal lattice vectors in screening calculations is necessary due to local-field effects [233, 234]. In the box below, we explain more precisely what we mean by local-field effects.

Box 6.2.1: Local-field effects

As the name indicates, local-field effects pertain to the local details of the electric polarization, or equivalently of the electric field, that are often ignored when adopting phenomenological descriptions of dielectric properties of materials, often designated as macroscopic. This is usually the case in Classical Electrodynamics, where the macroscopic Maxwell equations (2.1) reign. Those equations are obtained from their microscopic counterparts, involving the microscopic electric and magnetic fields, after performing a spatial average of the rapidly varying local fields over distances big compared to the inter-atomic lattice spacing, but small compared to the overall size of the crystal ^a. More technically, applying a slowly varying external potential perturbation (weak enough so that we stay within the linear regime) of the form

$$\phi^{\text{ext}}(\mathbf{q}, \omega) = \phi_0(\mathbf{q}, \omega) e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \quad (6.15)$$

to the crystal, where $2\pi/|\mathbf{q}|$ is much smaller than its size. The crystal responds and the total potential is the external one plus the induced one:

$$\phi^{\text{tot}} = \phi^{\text{ext}} + \phi^{\text{ind}} = \sum_{\mathbf{G}} \phi(\mathbf{q}, \mathbf{G}, \omega) e^{i[(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r} - \omega t]}. \quad (6.16)$$

It has all the $\mathbf{G}$ Fourier components with wavevector $\mathbf{q} + \mathbf{G}$, where $\mathbf{G}$ is a vector of the reciprocal lattice. The bigger $|\mathbf{G}|$ is, the faster that component oscillates. This potential is the microscopic one. To extract the macroscopic or averaged one, we simply set $\mathbf{G} = \mathbf{0}$, as $\exp\{i\mathbf{q} \cdot \mathbf{r}\}$ varies slowly over the averaging region, while on the contrary the components $\exp\{i(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r}\}$ vary rapidly over regions of the size of the unit cell, hence local-field effects are discarded on average (but does not mean that they do not affect the results, as we see in the results we obtain for the dielectric function after this box). This discussion is adapted from [233].

^aConsidering the full equations including dynamics and time evolution, an average over long time scales is performed as well besides the spatial one, which is pointed out by J. D. Jackson in his textbook in Ref. [38], but that is not relevant.

This is a sensitive point—choosing an appropriate cutoff G_c to compute the inverse dielectric matrix $\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q})$. For the matrix elements to converge, we expect a high enough number of reciprocal lattice vectors is needed. We will see how high. We compute the matrix $\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q})$ for $\mathbf{q} \neq \mathbf{0}$ with an increasing number of reciprocal lattice vectors included, and inspect four matrix elements: the head element, two elements in the body, a diagonal one and an off-diagonal one, and a wing element. We call the submatrix $\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$ with $\mathbf{G} \neq \mathbf{0}$ and $\mathbf{G}' \neq \mathbf{0}$ the body of the dielectric matrix. In Fig. 6.6 below, we showcase four matrix elements of the inverse dielectric matrix, one in each panel. For each point in each curve, we applied a cutoff G_c in integer multiples of $3 \text{ \AA}^{-1}$, although the horizontal axis shows the number of vectors included, and not the cutoff. We can notice a convergence tendency with the number of reciprocal lattice vectors included, albeit quite slow. Nonetheless, in the next subsection we study the dispersive behavior of the inverse dielectric matrix elements for a given cutoff G_c assumed to be high enough. The discussion will include an analysis on the macroscopic 2D dielectric function, alongside its long-wavelength limit. As promised at the end of the previous chapter, we will estimate r_0 .

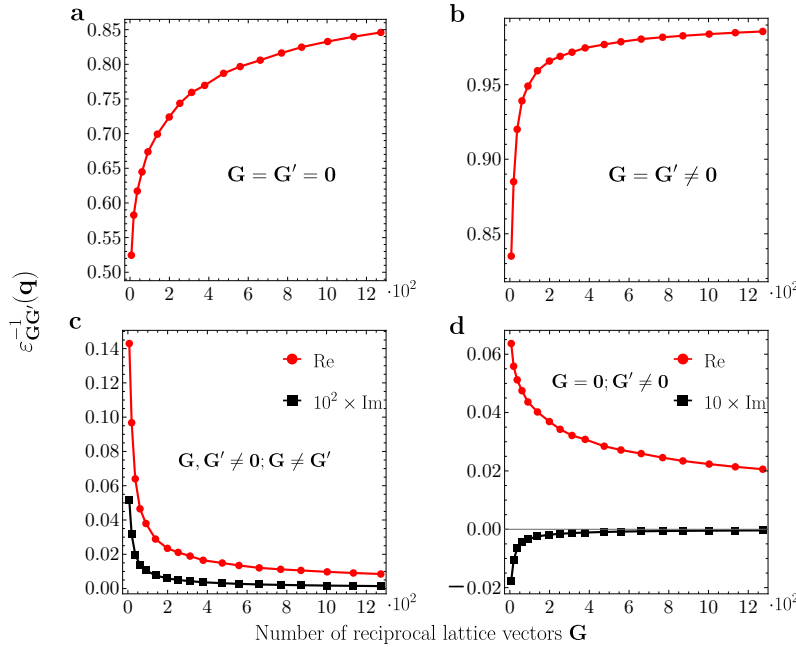


Figure 6.6: Convergence of the inverse dielectric matrix. Selection of inverse dielectric function matrix elements $\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q})$ computed at a point $\mathbf{q} \neq \mathbf{0}$ in the BZ for hBN. We display the head element in panel a, a diagonal body element in panel b, an off-diagonal body element in panel c, and a wing element in panel d. The cutoff G_c increases by $3 \text{ \AA}^{-1}$ for each point, from $G_c = 3 \text{ \AA}^{-1}$ up to $G_c = 54 \text{ \AA}^{-1}$.

6.2.3.2 Dispersion of the inverse dielectric function

In this section, we start with looking at few matrix elements of the inverse dielectric matrix as a function of momentum, similarly to what we did for the polarizability in Fig. 6.4a. Fig. 6.7 below shows how some matrix elements of the inverse dielectric matrix depend on the momentum $q = |\mathbf{q}|$, with $\mathbf{q}$ taken along the symmetry line $\Gamma - K$ in the BZ, and for a cutoff of $G_c = 54 \text{ \AA}^{-1}$. We did not go higher due to constraints in the computational power. In panel a, we plot the head element $\varepsilon_{\mathbf{0}\mathbf{0}}^{-1}(\mathbf{q})$ in red, and the first diagonal body element $\varepsilon_{\mathbf{G}_1\mathbf{G}_1}^{-1}(\mathbf{q})$ in blue, using the same vector $\mathbf{G}_1$ from Fig. 6.5. In panel b, we plot the

wing element $\varepsilon_{\mathbf{G}\mathbf{10}}^{-1}(\mathbf{q})$, with the real part and imaginary part plotted separately in red and black, respectively. The head element presents a sharp linear q -dependence near the origin, similarly to results previously reported in the literature from *ab-initio* calculations [31, 227–229]. Furthermore, as we stated in the previous subsection, we have $\varepsilon_{\mathbf{00}}^{-1}(\mathbf{0}) = 1$. We verified numerically that this identity holds for any cutoff G_c . Even though we have done a convergence study similar to the one in Fig. 6.6 for $\mathbf{q} = \mathbf{0}$, we do not include it in the text for brevity. We can say though that the convergence curves with the number of reciprocal lattice vectors included are similar to the ones we report here (apart from the head element, which is always one).

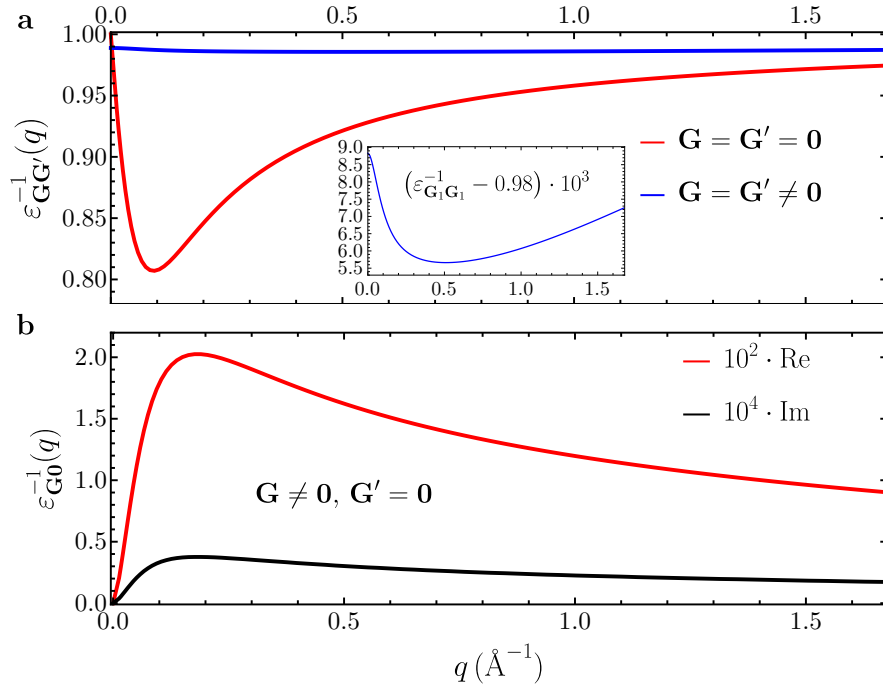


Figure 6.7: Elements of the inverse dielectric matrix as a function of momentum. Selected elements of the inverse dielectric matrix $\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q})$ as a function of momentum q . Panel **a** displays the head element in red and the first diagonal body element $\varepsilon_{\mathbf{G}_1\mathbf{G}_1}^{-1}$, where the vector $\mathbf{G}_1$ is the same as in Fig. 6.5. The inset in panel **a** shows a zoomed in view of the blue curve (subtracted by 0.98 and scaled by a factor of 10^3). This body element is dispersive, but within a much smaller scale than the head element. Panel **b** displays the real and imaginary parts of the wing element $\varepsilon_{\mathbf{G}\mathbf{0}}^{-1}$ in red and black, respectively. This element presents also a q -dependence at a smaller scale than the head element, since both real and imaginary parts are scaled by a power of ten. The momentum $\mathbf{q}$ followed the symmetry line $\Gamma - K$ in the BZ, and the applied cutoff is $G_c = 54 \text{ \AA}^{-1}$.

As we will see in a moment, the sharp behavior of the head element of the inverse dielectric matrix in the long-wavelength limit leads to a sharp linear dependence of the macroscopic dielectric function for small q . This behavior is universal for 2D insulators, and is problematic for converging exciton calculations with the number of $\mathbf{k}$ points in the BZ mesh, but we leave that concern for later. On the contrary, the body element in panel **a** does not present a strong q -dependence along the whole $\Gamma - K$ line when compared to the head element. This has also been seen in the literature, specifically in Fig. 1 of the

work by A. Guandalini and co-workers in Ref. [229]. Nonetheless, when we zoom in to take a closer look at the blue curve as inset in Fig. 6.7a, it is in truth dispersive, but within a scale three orders of magnitude smaller than that of the head element. If we also want to see a detailed dispersion in the wing element of panel **b**, scaling the real and imaginary parts is also necessary.

6.2.4 Macroscopic dielectric function and the screening length r_0

In this section, and with the results from the previous subsection, we compute the macroscopic dielectric function for hBN, through the usual definition $\varepsilon_M(\mathbf{q}) = (\varepsilon_{00}^{-1}(\mathbf{q}))^{-1}$. Our framework allows for directly obtaining the static 2D effective dielectric function from this definition, which comes as a striking advantage when compared to typical *ab-initio* methods in three dimensions. We display in Fig. 6.8 the effective 2D dielectric function for hBN as a function of momentum $q = |\mathbf{q}|$, where $\mathbf{q}$ follows the same line as in Fig. 6.7. All the parameters from that figure carry over to the one below. The numerically obtained dielectric function corresponds to the red curve, while the one in black is the dielectric function according to the Rytova–Keldysh model $\varepsilon_{\text{RK}}(q) = 1 + r_0 q$, where the parameter r_0 was estimated from the low- q behavior of the red curve. With this model for hBN, and for the parameters we used to make Fig. 6.8, we obtained the screening parameter $r_0 \approx 5.084 \text{ \AA}$ by computing the incremental difference between the first and second data points. This value is actually not close to the one we used in the previous chapter. However, in the previous chapter the hBN flake was supported by a quartz substrate, which affects the screening properties of the material [27]. The value $r_0 = 10 \text{ \AA}$ that we used has been used in previous works [28, 30], showing good agreement with experiments with monolayer hBN deposited on quartz [107]. In Ref. [28], from the arguments of P. Cudazzo, I. V. Tokatly, and A. Rubio in Ref. [26], the authors estimated $r_0 = 6.5 \text{ \AA}$ for hBN, knowing the material off-plane and in-plane bulk dielectric constants [235]. Still in the same work, with the help of Refs. [27, 215], the authors in Ref. [28] estimated $r_0 = 6.8 \text{ \AA}$. Both values estimated in Ref. [28] are not that far from the one we get.

In Fig. 6.8 one can notice as well a feature of the macroscopic dielectric function in the opposite limit of $q \rightarrow 0$. When $q \rightarrow \infty$, the curve tends asymptotically to one, which is a universal characteristic shared among all 2D insulators. That is indeed what one sees in all the dielectric functions reported in the literature [24, 215, 225, 228], regardless of the material. Physically, having $\varepsilon_M(q) \rightarrow 1$ when $q \rightarrow \infty$ means that, in the short-wavelength limit, the spatial variations of the external potential vary so fast (in space), that the system does not respond to the perturbation. Therefore, test charges feel a total potential that coincides with the external potential—the internal charges do not realign in response to the external potential, which is completely unscreened. With symbols, $\phi^{\text{tot}} = \phi^{\text{ind}} + \phi^{\text{ext}} \approx \phi^{\text{ext}}$ for $q \rightarrow \infty$. This is in sharp contrast with the prediction from the Rytova–Keldysh model, for which the external potential is largely overscreened, decaying with $\sim 1/q^2$ and not $\sim 1/q$.

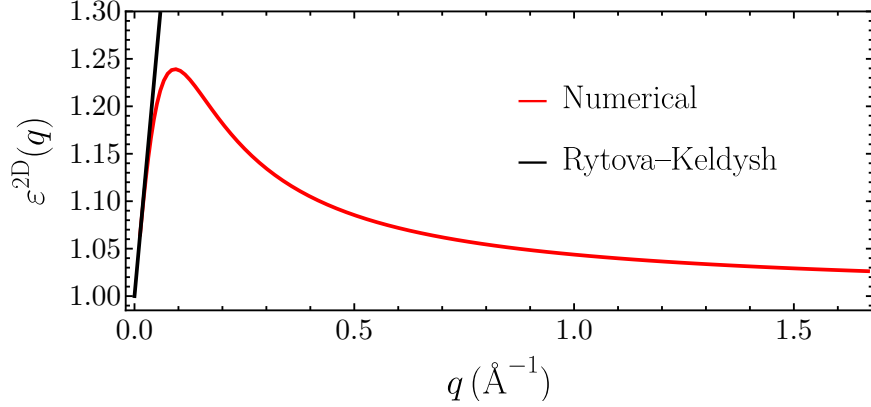


Figure 6.8: Effective 2D dielectric function for hBN. Macroscopic effective 2D dielectric function of monolayer hBN. We compare the full numerical q -dependence with the Rytova–Keldysh model, using the value for the screening parameter $r_0 = 5.084 \text{ \AA}$ estimated by us.

During the course of this work, we also found out that the parameter r_0 can actually be estimated directly from the head element of the dielectric function $\epsilon_{\mathbf{00}}(\mathbf{q})$, by taking the $\mathbf{q} \rightarrow \mathbf{0}$ limit. This is not very surprising if we take into account that P. Li and I. Appelbaum, in Ref. [236], showed that the screening length can be calculated by

$$r_0 = \frac{e^2 \hbar^2}{4\pi^2 \epsilon_0 m_0^2} \sum_{vc} \int \frac{|\mathbf{P}_{cv}(\mathbf{k})|^2}{(\epsilon_{c\mathbf{k}} - \epsilon_{v\mathbf{k}})^3} d\mathbf{k}, \quad (6.17)$$

where the integral in momentum is in-plane, m_0 is the free electron mass, and $\mathbf{P}_{cv}(\mathbf{k}) = \langle c, \mathbf{k} | \mathbf{p} | v, \mathbf{k} \rangle$ is the momentum matrix element. This was proved, in the supplemental material of Ref. [236], by taking the static and $L_\perp \rightarrow \infty$ limits in the dynamical dielectric function of the respective bulk material (after taking the $q \rightarrow 0$ limit). The expression in Eq. (6.17) has been used before in the literature for exciton calculations in 2D systems [237].

To settle the debate and get a better feeling for the validity of our calculations, let us compare our dielectric function directly with the one obtained from first-principles methods, instead of looking only at the small- q limit behavior of the dielectric function. We compare our results with those from S. Latini, T. Olsen, and K. S. Thygesen in Ref. [215], where the authors plot the effective 2D dielectric function for hBN and MoS₂, obtained from *ab-initio* methods. Therein, the cutoff applied for the reciprocal lattice vectors is given in terms of energy, and it is 150 eV. The cutoff for including local-field effects is usually given in terms of energy in the literature [31, 215, 229], and the conversion from energy to wavevector is done through $E_c = \hbar^2 G_c^2 / 2m_0$, where m_0 is the free electron mass. When $E_c = 150 \text{ eV}$, we get $G_c \approx 6.27 \text{ \AA}^{-1}$. In principle, this means that to compare results, we have to apply the same cutoff. However, we will see that truncating the dielectric matrix at a smaller cutoff is sufficient to make comparisons. For our model for hBN, the smallest cutoff possible for including reciprocal lattice vectors other than the null one, is $G_c \approx 3.0 \text{ \AA}^{-1}$. Only above $G_c = 5.0 \text{ \AA}^{-1}$ are more reciprocal lattice vectors included. We stress that we only have in-plane $\mathbf{G}$ vectors.

We computed the dielectric function first for $G_c \approx 3.0 \text{ \AA}^{-1}$, and then for $G_c \approx 5.01 \text{ \AA}^{-1}$, choosing again the symmetry line $\Gamma - K$ for the momentum points $\mathbf{q}$. The results are displayed in Fig. 6.9 below. It comes as a surprise for us, that the agreement between our results, in the colored dashed lines, and theirs, in the continuous black line, is so good. Not only the curves overlap in the small- q limit, but they overlap as well for higher values of momentum q inside the BZ and further apart from the origin. Naturally, the agreement between the curves ceases for a given value of momentum q , which is related with the finite thickness of the material $d_\perp$. When $qd_\perp \lesssim 1$, we expect for our strict 2D approach to agree with the *ab-initio* one, while for $qd_\perp \gtrsim 1$ we expect for the two approaches to disagree.

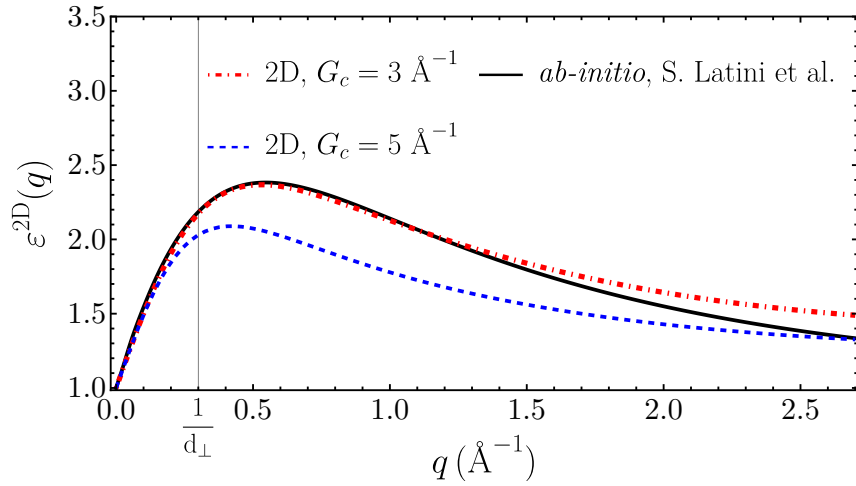


Figure 6.9: *Ab-initio* and strict 2D effective dielectric function for hBN. Macroscopic effective 2D dielectric function of monolayer hBN, computed by our strict 2D approach in comparison to the *ab-initio* approach. To get the red and blue curves, we applied the cutoffs $G_c = 3.0 \text{ \AA}^{-1}$ and $G_c = 5.01 \text{ \AA}^{-1}$, respectively. The black curve was obtained by S. Latini et al. in Ref. [215] through *ab-initio* methods, applying a cutoff of 150 eV for the reciprocal lattice vectors. In Ref. [215], the band structure of hBN was obtained through DFT calculations within LDA. Afterward, the dielectric function was calculated as a post-processing step. Just like in previous cases, we computed the dielectric function along the symmetry line $\Gamma - K$. A vertical line indicating $qd_\perp = 1$ is included, with $d_\perp = 3.33 \text{ \AA}$.

Even though the effective 2D dielectric function through *ab-initio* methods can only be obtained after performing an off-plane average on the inverse dielectric matrix, which can be a bit cumbersome, first-principles approaches have the advantage of naturally incorporating the finite thickness of the material. Apparently, taking the finite thickness into account is not required to have a good estimate of the low- q behavior of the dielectric function, and therefore, to have an estimate of r_0 . Not only that, but Fig. 6.9 highlights an interesting aspect of the low- q dielectric properties of a 2D material; the very low- q dielectric response seems to not depend on the local-field effects. We see this by comparing the red dot-dashed curve and the black curve with the blue one, which was obtained by applying a slightly higher cutoff of G_c . On the other hand, one can conclude that local-field effects are crucial for higher q . We have numerically verified, but do not present in this dissertation, that totally ignoring local-field effects overestimates the screening. In math

notation, we mean that $\varepsilon_{\mathbf{00}}(q) \gg (\varepsilon_{\mathbf{00}}^{-1}(q))^{-1}$ for intermediate q . If the cutoff is not high enough, then the dielectric response comes out overscreened. For us, applying the lowest possible cutoff (above zero) is sufficient to replicate the *ab-initio* results, at least for $qd_{\perp} \lesssim 1$. By increasing more the cutoff, our dielectric function gets underestimated when compared to the *ab-initio* one, as it can be seen in Fig. 6.9, when comparing our blue dashed curve and the *ab-initio* one in black. At this point, we cannot say if the disagreement for higher q is also in part due to the approximation of point-like orbital we used to compute the plane-wave matrix elements between Bloch states, which are typically computed exactly in *ab-initio* codes via fast Fourier transform algorithms [31]. At least for low momenta, this approximation does not set us back.

Now that we have successfully computed the dielectric function, it is now time to compute the excitons using the numerical model for screening. Knowing the excitonic states and energies, just like in the previous chapter in Section 5.4, we can compute the excitonic optical conductivity. The excitons and the excitonic optical response are the subject of the next subsection.

6.2.5 Excitonic states and conductivity

To see the effect the numerical screening has on the excitons, we computed the excitons using the **Xatu** code with the two different models for screening possible. In both calculations, we used the same set of parameters, apart from the way the interaction matrix elements are computed. We used a 60×60 $\mathbf{k}$ -grid to sample the BZ, included one valence band, $N_v = 1$, and one conduction band, $N_c = 1$, applied a cutoff for the reciprocal lattice vectors of $G_c = 3.0 \text{ \AA}^{-1}$, and for the regularization radius we used a percentage of 1.3%. This means for the regularization radius $q_0 = 1.3k_0$, where k_0 is the norm of the BZ momentum closest to the origin. The same regularization scheme as explained in Appendix C.2 is applied to the calculations using the Rytova–Keldysh potential. For the sake of completeness, we repeat here the expression used to regularize the term $W_{\mathbf{00}}(\mathbf{0})$.

The regularization scheme we use is inspired by a previous work by F. A. Rasmussen et al. in Ref. [225], where the authors suggest (but do not implement it themselves) to regularize this term with an average of the screened Coulomb potential over a small region around the origin (the authors implement instead an average of the bare potential). In the later work in Ref. [225], also by F. A. Rasmussen in collaboration with others, the authors concluded that including the regularized singular term speeds up the convergence of many-body calculations against the number of $\mathbf{k}$ points in the BZ mesh. In this work, we adopt the scheme of the averaged screened potential, taking advantage of its behavior for low q , within which the macroscopic dielectric function behaves like the Rytova–Keldysh potential. This deems the expression for the regularization analytical (see Eq. (C.21) in Appendix C.2). Therefore, we implement the singular term $W_{\mathbf{00}}(\mathbf{0})$ as

$$W_{\mathbf{00}}(\mathbf{0}) = \frac{1}{\mathcal{A}_{\Gamma}} \int_{\mathcal{A}_{\Gamma}} d\mathbf{q} v_c(\mathbf{q}) [1 + \mathbf{q} \cdot \nabla_{\mathbf{q}} \varepsilon_{\mathbf{00}}^{-1}(\mathbf{q})|_{\mathbf{q}=\mathbf{0}}], \quad (6.18)$$

where $\mathcal{A}_\Gamma$ is a small, circular region around the origin Γ . The gradient of the head element of the inverse dielectric matrix is computed with the code before the initialization of the Bethe–Salpeter Hamiltonian. Although we can compute it just as described, we use the fact that, for low $q = |\mathbf{q}|$, $(\varepsilon_{\mathbf{00}}^{-1}(\mathbf{q}))^{-1} \approx \varepsilon_{\mathbf{00}}(\mathbf{q})$, as verified in Fig. 6.9. If the system is isotropic, $\mathbf{q} \cdot \nabla_{\mathbf{q}} \varepsilon_{\mathbf{00}}^{-1}(\mathbf{q})|_{\mathbf{q}=\mathbf{0}} \approx r_0 q$, where r_0 is estimated as $r_0 \approx (\varepsilon_{\mathbf{00}}(\mathbf{q}) - 1)/|\mathbf{q}|$, for a very small $\mathbf{q}$. On the other hand, if the system is anisotropic, then the derivative is computed along two perpendicular directions for $\mathbf{q}$, but conserving the norm $|\mathbf{q}|$.

In both calculations, and just like we did for hBN, we set $\mathbf{Q} = \mathbf{0}$ and exclude the exchange term in the interaction matrix elements. When using the Rytova–Keldysh model for screening, the direct interaction matrix element is computed through Eq. (4.62a), where for the screening length we used the value obtained by us of $r_0 = 5.084 \text{ \AA}$. Using the numerical (inverse) dielectric function, the same interaction matrix element is computed through Eq. (4.59), but with the volume $\mathcal{V}$ replaced by the area $\mathcal{A}$.

The numerical results for the excitons are displayed in Table 6.1 below, showing the excitonic levels of the first eight excitons. The values obtained with different potentials are displayed side by side. The values on the left were obtained using the Rytova–Keldysh dielectric function, while the ones on the right were obtained using the numerical dielectric matrix. First, one can notice that for the Rytova–Keldysh calculation, the states are not as close to the expected degeneracies as they are for the numerical screening calculation. For the Rytova–Keldysh calculation, this does not come as a surprise, since it is known that the binding energies for hBN excitons are the highest among all 2D insulators. This means that they are localized in real space, therefore they are delocalized in momentum space. Henceforth, one needs a high enough cutoff G_c^X for the interaction matrix elements to recover the expected degeneracies and symmetries. This fact, that we touched on at the end of subsection 4.3.2, has been discussed before in the literature by E. Ridolfi et al. in Ref. [150], and verified again more recently in Ref. [37].

In Table 6.1, we can also verify that using the Rytova–Keldysh model the excitonic states are higher in energy when compared to the ones obtained using $\varepsilon_{\mathbf{GG}'}$, by looking at the second and third columns. If we consider only the head term $W_{\mathbf{00}}(\mathbf{q})$ of the numerical screened potential and compare it with the Rytova–Keldysh potential $V_{\text{RK}}(\mathbf{q})$, knowing that the Rytova–Keldysh model overestimates the screening, then the attraction between the electron and the hole is weaker than the one coming from $W_{\mathbf{00}}(\mathbf{q}) = \varepsilon_{\mathbf{00}}^{-1}(\mathbf{q})v_c(\mathbf{q}) = v_c(\mathbf{q})/\varepsilon_M(\mathbf{q})$, since $\varepsilon_M(\mathbf{q}) < \varepsilon_{\text{RK}}(\mathbf{q})$. This means that the excitons obtained with $V_{\text{RK}}(q)$ should be higher in energy than the ones from using $W_{\mathbf{GG}'}(\mathbf{q})$. In Fig. 6.10, we illustrate the q -space Coulomb attraction between the electron and the hole by plotting the bare Coulomb potential $v_c(q)$ in black, the Rytova–Keldysh potential $V_{\text{RK}}(q)$ in blue, and the head element of the screened potential $W_{\mathbf{00}}(q)$ in red. Since the Rytova–Keldysh dielectric function is obtained from $1/\varepsilon_{\mathbf{00}}^{-1}(\mathbf{q})$, in the long-wavelength limit the two screened potentials coincide. However, for intermediate q , the Rytova–Keldysh potential underestimates the electrostatic Coulomb interaction, decaying to zero faster than $W_{\mathbf{00}}(q)$ and than the Coulomb potential, as we have already pointed out in subsection 6.2.4. For high q , the head element $W_{\mathbf{00}}(q)$

Table 6.1: Exciton spectrum of hBN. Exciton spectrum of hBN using the DFT model from CRYSTAL, using the Rytova–Keldysh potential and the numerical inverse dielectric matrix to compute the interaction matrix elements. The second and fourth columns pertain to the Rytova–Keldysh model, while the third and fifth columns pertain to the numerical screening. The second and third columns show the exciton energies E_X , while the fourth and fifth columns show the corresponding binding energies E_b (in absolute value). States labeled with the same number but different letters are states for which degeneracy is expected. We observe that the states that are supposed to be degenerate, are more proximate in energy for the numerical screening calculation, and are in general lower in energy compared to those obtained from the Rytova–Keldysh model. We used $N = 60^2$, $N_c = N_v = 1$, $G_c = 3.0 \text{ \AA}^{-1}$, $q_0 = 1.3k_0$, and $r_0 = 5.084 \text{ \AA}$ for the Rytova–Keldysh potential. In the numerical model for the screening, we included all the $\mathbf{G}$ vectors. We excluded the exchange interaction term in both calculations like in Ref. [215], and set $\mathbf{Q} = \mathbf{0}$. Increasing the number of bands included to $N_c = N_v = 2$ resulted in a change in binding energies less than 0.02 eV for all states, except for the state 2, for which the binding energy was of 2.1 eV. This was tested only for the numerical screening calculation.

State	E_X (RK) (eV)	E_X ($\epsilon_{\mathbf{G}\mathbf{G}'}$) (eV)	E_b (RK) (eV)	E_b ($\epsilon_{\mathbf{G}\mathbf{G}'}$) (eV)
1a	3.68	3.61	2.40	2.47
1b	3.84	3.62	2.24	2.46
2	4.37	4.11	1.71	1.97
3a	4.86	4.83	1.21	1.25
3b	4.90	4.84	1.18	1.24
4	4.93	4.92	1.15	1.16
5a	5.19	5.181	0.89	0.899
5b	5.21	5.182	0.87	0.897

of the screened potential approaches the unscreened potential $v_c(q)$.

For this model for hBN, with the electronic bandgap of $\Delta = 6.08$ eV, we obtain for the binding energy of the excitonic ground state $E_b = 2.47$ eV. Honestly, it is not very close to other values previously obtained in the literature using full BSE *ab-initio* methods. For instance, in Ref. [215] the authors obtained a binding energy of $E_b = 2.05$ eV, while in Ref. [238] a value of $E_b = 2.08$ eV was reported. However, in none of those works do the authors use a hybrid functional for the DFT calculations as we do. Even though with different models one can get similar dielectric functions, the excitonic properties may differ more from model to model. Even if it is the same material, having a different band structure, specially a different bandgap, may naturally lead to different exciton binding energies. Our bandgap of 6.08 eV is underestimated with respect to the one reported in Ref. [238], where the authors obtained a bandgap of 8.03 eV at the *GW* level. In Ref. [215] the bandgap is not reported, but we can infer an estimated value of approximately 5.0 eV from the reported band structure. The difference between their bandgap and ours is due to the different approximations used for the exchange–correlation functional. We used a hybrid functional, while they computed the band structure within LDA.

Before we analyse the optical conductivity, we plot in Fig. 6.11 below the first two excitonic states, 1a and 1b, whose energies can be consulted in Table 6.1. The colors represent the values assumed by the wavefunction in momentum space, after taking its absolute value squared. The top panels reflect the 1a states, and the bottom panels reflect

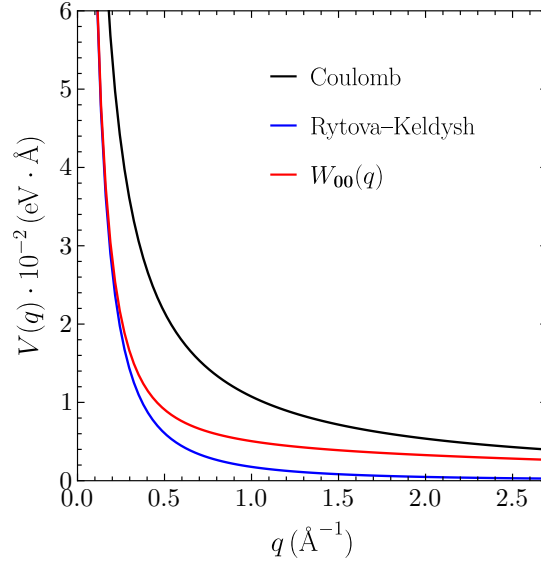


Figure 6.10: Coulomb, Rytova–Keldysh and numerical macroscopic potentials in momentum space. Comparison between the bare Coulomb potential (red continuous line) and the Rytova–Keldysh potential (dashed blue line) as a function of the in-plane distance r . The dotted black curve represents the small r limit of the Rytova–Keldysh potential in Eq. (2.21). We used $r_0 = 10 \text{ Å}$.

the 1b states. The left side panels correspond to the Rytova–Keldysh model for the dielectric function, while the right side panels correspond to the numerical approach for the screening. As expected from what we discussed about the values in Table 6.1, the states on the right panels are slightly more delocalized in momentum space than the ones represented on the left panels (note the different amplitude scales).

At last, we compare the optical conductivity of hBN computed with both methods. Unlike the BSE calculations, the optical conductivity is obtained always in the same way. We use it just as it was implemented in the first version of the **Xatu** code, in Ref. [37]. Therein, the conductivity is computed in terms of velocity matrix elements, instead of using the dipole operator approach from the previous chapter. The expressions (4.71) and (4.74) are the ones we use now to compute the conductivity at the independent-particle approximation (IPA) and excitonic levels. In the last chapter, we focused solely on the excitonic conductivity. On the contrary, as both noninteracting and interacting conductivities are implemented in the code, we plot both for the sake of completion. The noninteracting one does not depend on how one computes the excitons, logically.

Before we comment on the conductivity obtained with the **Xatu** code, for the sake of having a discussion as self-contained as possible, we briefly explain how the conductivity is computed, in particular the velocity matrix elements. We start by recalling the expression for the real part of the elements of the excitonic conductivity tensor from Chapter 4, which reads

$$\text{Re}\{\sigma_{\alpha\beta}(\omega)\} = \frac{\pi e^2 \hbar}{\mathcal{A}} \sum_{\nu} \frac{1}{E_{\nu}} V_{\nu}^{\alpha} (V_{\nu}^{\beta})^* \delta(\hbar\omega - E_{\nu}), \quad (6.19)$$

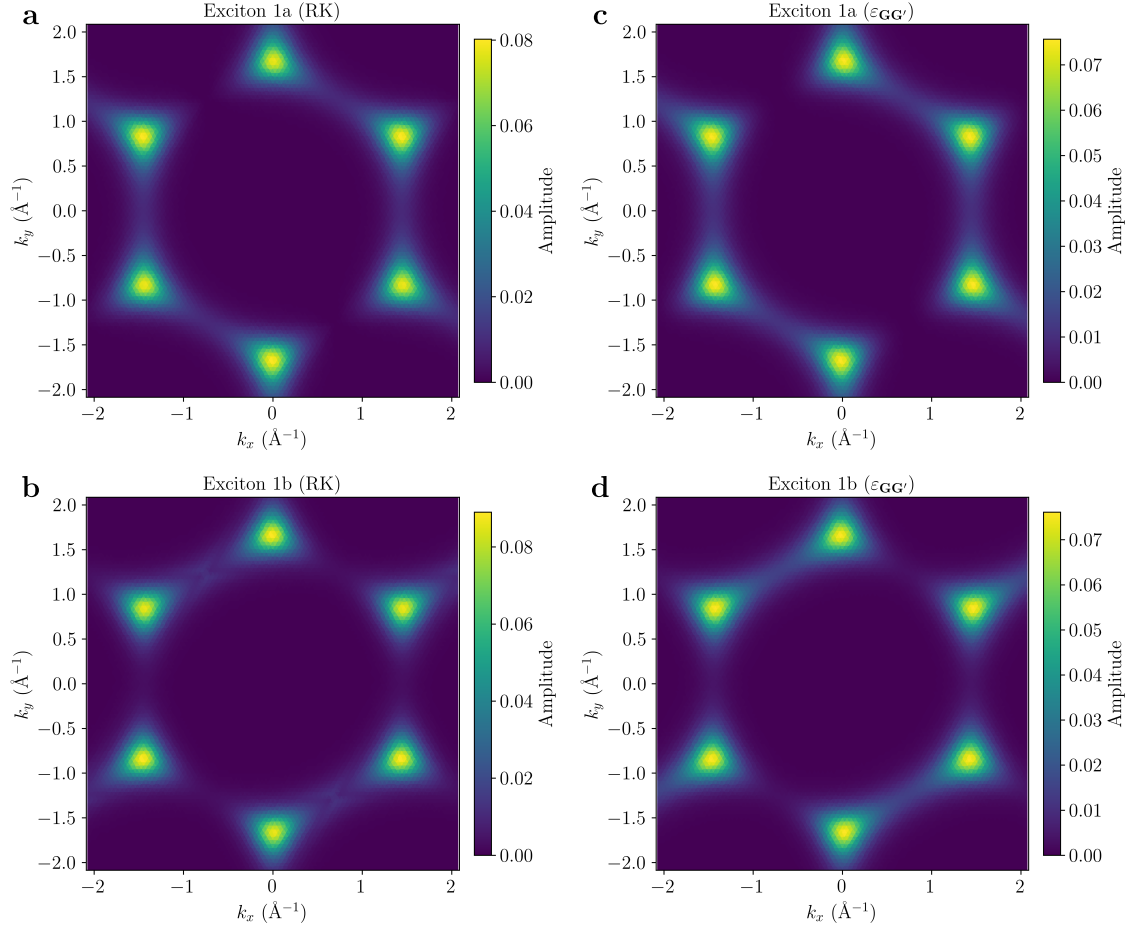


Figure 6.11: Excitonic ground state wavefunction for hBN: Rytova–Keldysh vs. numerical screening. Momentum-resolved wavefunctions (in absolute value squared) of the first two excitonic states of hBN, using the Rytova–Keldysh potential on the left panels, and using the numerical dielectric function on the right panels. The top panels correspond to the 1a state from Table 6.1, while the bottom panels correspond to the 1b state in the same table. The states obtained using the numerical dielectric function present a lower maximum for the probability density, therefore those states are more delocalized in reciprocal space than the ones on the left. This is consistent with higher binding energies for the states obtained using $\varepsilon_{\mathbf{G}\mathbf{G}'}$, as we have seen in Table 6.1.

with the volume $\mathcal{V}$ replaced by the area $\mathcal{A}$, as the material is 2D, with the interacting velocity matrix element for the excitonic state ν given in terms of the velocity matrix elements $v_{vc}^\alpha(\mathbf{k}) = \langle v, \mathbf{k} | v^\alpha | c, \mathbf{k} \rangle^4$ by

$$V_\nu^\alpha = \langle \text{GS} | v^\alpha | X_\nu \rangle = \sum_{vc, \mathbf{k}} A_{vc}^\mathbf{Q}(\mathbf{k}) v_{vc}^\alpha(\mathbf{k}). \quad (6.20)$$

By virtue of the identity $\mathbf{v} = i/\hbar[H_0, \mathbf{r}]$ (see Box 4.4.1), it is possible to evaluate the velocity matrix elements in a Bloch basis. They are given by [37, 159, 163]

⁴In the light of what was discussed in Box 4.4.1 and to be more rigorous, we can say that this matrix element is an integral in real space extending over the whole material, considered finite and obeying periodic boundary conditions.

$$\mathbf{v}_{vc}(\mathbf{k}) = \sum_{i\alpha i'\alpha'} \left(C_{i\alpha}^{m\mathbf{k}} \right)^* C_{i'\alpha'}^{m'\mathbf{k}} \left[\nabla_{\mathbf{k}} H_{i\alpha i'\alpha'}(\mathbf{k}) + i\epsilon_{n\mathbf{k}} \xi_{i\alpha i'\alpha'}(\mathbf{k}) - \epsilon_{n'\mathbf{k}} \xi_{i\alpha i'\alpha'}^*(\mathbf{k}) \right], \quad (6.21)$$

where the quantity $\xi_{i\alpha i'\alpha'}(\mathbf{k}) = i \langle u_{i\alpha\mathbf{k}} | \nabla_{\mathbf{k}} | u_{i'\alpha'\mathbf{k}} \rangle$ is the Berry connection (see footnote 12), and evaluates to [37, 159, 163]

$$\xi_{i\alpha i'\alpha'}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \langle i\alpha \mathbf{0} | \mathbf{r} | i'\alpha' \mathbf{R} \rangle + i \nabla_{\mathbf{k}} S_{i\alpha i'\alpha'}(\mathbf{k}), \quad (6.22)$$

where we recall that $S_{i\alpha i'\alpha'}(\mathbf{k}) = \langle i\alpha \mathbf{k} | i'\alpha' \mathbf{k} \rangle$ is a matrix element of the momentum-defined overlap matrix (see Eq. 3.17). In the tight-binding approximation, or in the limit of maximally localized Wannier functions, only the intra-atomic matrix elements of the position operator are nonzero, and it is not hard to show that the matrix elements in Eq. 6.21 evaluate to

$$\langle n\mathbf{k} | v^a | n'\mathbf{k} \rangle = \sum_{i\alpha i'\alpha'} \left(C_{i\alpha}^{m\mathbf{k}} \right)^* C_{i'\alpha'}^{m'\mathbf{k}} \left[\frac{\partial H_{i\alpha i'\alpha'}(\mathbf{k})}{\partial k_a} + i H_{i\alpha i'\alpha'}(\mathbf{k}) (t_{i'}^a - t_i^a) \right], \quad (6.23)$$

which involves k -derivatives of the Bloch Hamiltonian matrix elements; typically, one needs a more refined, or dense, BZ mesh to make the conductivity converge. For the Dirac delta part, we use typically use a Lorentzian broadening of the form $\delta(x) \rightarrow \frac{1}{\pi} \frac{\Gamma}{x^2 + \Gamma^2}$, where Γ is the broadening parameter.

For the sake of brevity, we did not rederive the expression for the noninteracting optical conductivity in Eq. (6.19) from the dielectric function, which is now 2D, as we did in subsection (4.4.1). It is an instructive exercise to do so, but, we need to have a caveat in mind. The equivalence between the dielectric function and the optical conductivity from Chapter 2 does not apply, because we implicitly assumed three dimensions for deriving it. One can see that the same expression cannot be used, because the potential now goes with $\sim 1/(2\epsilon_0 q)$ instead of $\sim 1/(\epsilon_0 q^2)$. Then, we need the 2D counterpart of the same relation, which is derived in Appendix D. As one can see in that appendix, the new permittivity–conductivity equivalence relation reads $\epsilon = 1 + iq\sigma/(2\epsilon_0\omega)$, instead of the old one $\epsilon = 1 + i\sigma/(\epsilon_0\omega)$. The extra $q/2$ factor going with the conductivity compensates for the conversion of the Coulomb potential from $v_c^{3D}(q) = e/(\epsilon_0 q^2)$ to $v_c^{2D}(q) = e/(2\epsilon_0 q)$.

We can observe in Fig. 6.12 below the real part of the optical conductivity of hBN computed with the **Xatu** code. In comparison with each other, we have in blue the excitonic conductivity obtained using the Rytova–Keldysh potential, and in red the conductivity obtained using the numerical dielectric function. Besides the expected red-shift of the resonant peaks, we can see that the two peaks around the energies 3.8 eV and 4.2 eV are more pronounced for the case of the numerical screened potential. However, the difference in the peak around 3.8 eV may be due to the energy splitting of the first two states in the case of the Rytova–Keldysh potential, which results in a shoulder together with that

peak. If the two energies of the states 1a and 1b were closer, then the peaks around 3.8 eV would probably be similar for both models. However, the peak around 4.2 eV is clearly more intense in the numerical screening case. The peak around 4.8 eV seems to be lower, though, but after that the two optical responses, in red and in blue, approach each other as they get closer to the end of the bandgap. Above the bandgap, both excitonic responses coincide.

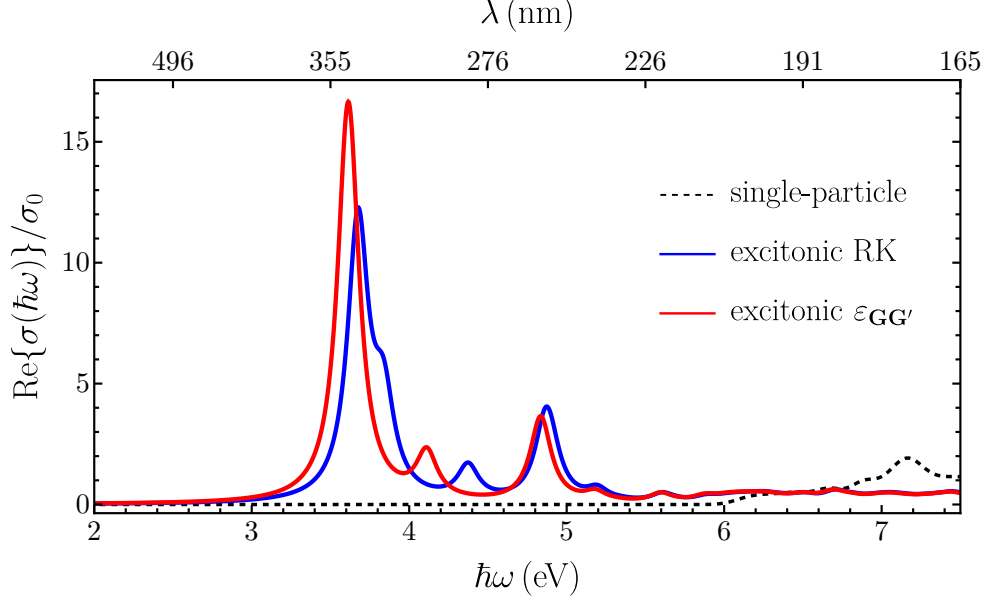


Figure 6.12: Optical conductivity of hBN: analytical and numerical screening. Optical conductivity of freestanding hBN computed with the `Xatu` code. The IPA conductivity is represented by the dashed black curve. The excitonic conductivities corresponding to the Rytova–Keldysh model and the numerical model for screening are represented in blue and in red curves, respectively. We used for both excitonic conductivities a Lorentzian broadening with $\Gamma = 0.08$ eV.

From all the numerical results obtained so far in this chapter, we would say that the main one is the agreement between our effective 2D dielectric function and the *ab-initio* one, as we saw in Fig. 6.9. However, the agreement is far from perfect for intermediate and higher values of the momentum. This is for sure a consequence of the zero thickness approximation we introduced in the beginning of Section 6.1. To be more precise, we assumed that the response of the monolayer was local in the off-plane direction, so that the dielectric function depending on all the Cartesian coordinates is approximately given by $\varepsilon(\mathbf{r}, \mathbf{r}') \approx \delta(z - z')\varepsilon_{2D}(\mathbf{r}_{\parallel}, \mathbf{r}'_{\parallel})$, and the dielectric function ε_{2D} is the one we have been working with thus far. In other words, we ignored altogether any physics coming from the z -dependence of the permittivity. So we have to pose two questions: (i) How do we show that our strict 2D implementation is valid in the zero thickness limit, and that the permittivities with zero and finite thickness can only agree when $qd_{\perp} < 1$?; (ii) Can we not do better, and somehow take the z -dependence into account? We answer both questions in the next section.

6.3 Dielectric Function in the Quasi-2D approach

In the numerical approach to the dielectric function within a strict 2D framework, we have been taking advantage of the reduced dimensionality of the implementation to produce results efficiently. Our results are comparable with the *ab-initio* ones, at least in the regime of low momentum $q < 1/d_\perp$, where $d_\perp$ is the thickness of the monolayer. However, we would like to see where does this limit come from with a more technical point of view. For that, the z -dependence of the dielectric function must be accounted for. We do so in the quasi-2D approach, or the mixed representation approach, for the dielectric function we will introduce now. Once again, the point-like orbital approximation will be key.

We start from the general expression for the static dielectric function in real space in Eq. (6.1), but without the time dependence. The position vectors are now 3D. On the one hand, the systems we deal with are infinite in the x and y directions. Therefore, we would like to work in momentum space. On the other hand, the systems are finite in the z direction. Unlike in typical first-principle approaches, where the systems are infinite in all directions, we cannot perform a Fourier transform including the z coordinate. Nonetheless, nothing stops us from performing an in-plane 2D Fourier transform, and keeping the z -dependence at the same time. For that, we need the 2D Fourier transform of the bare Coulomb potential, but having in mind that the position vector is 3D. The bare Coulomb potential in the mixed $(\mathbf{q}, z)$ -representation reads

$$v_c(\mathbf{q}, z - z') = \frac{e}{2\varepsilon_0 q} e^{-q|z-z'|}, \quad (6.24)$$

where $\mathbf{q}$ is in-plane. The previous expression is a key ingredient in our quasi-2D approach. The $1/q$ -dependence of the potential is kept intact, and, at the same time, the z -dependence is not ignored. For the static dielectric function in the mixed representation, we have

$$\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z') = \delta_{\mathbf{G}\mathbf{G}'}\delta(z - z') - \frac{e}{2\varepsilon_0|\mathbf{q} + \mathbf{G}|} \int_{-\infty}^{\infty} dz'' e^{-|\mathbf{q} + \mathbf{G}||z-z''|} \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z'', z'), \quad (6.25)$$

where we introduced the noninteracting polarizability in the mixed representation defined by

$$\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') = \int_{\mathcal{A}} d\mathbf{r}_\parallel \int_{\mathcal{A}} d\mathbf{r}'_\parallel e^{-i(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r}_\parallel} e^{i(\mathbf{q}' + \mathbf{G}') \cdot \mathbf{r}'_\parallel} \chi^0(\mathbf{r}, \mathbf{r}'). \quad (6.26)$$

Working with a polarizability in the mixed $(\mathbf{q}, z)$ -representation is not a complete novelty. As far as we know, this idea dates back to 1995, in a work by X. Blase et al. [239]. However, their formulation is not exactly the same as ours, as their mixed-space formulation does not mix in-plane momentum coordinates with off-plane spatial positions. They use

simultaneously 3D momentum and 3D spatial coordinates, the latter restricted to a single unit cell. More recently, the object $\chi(\mathbf{q}, z, z')$, mixing the in-plane $\mathbf{q}$ -dependence with off-plane z -dependence, is treated by K. Andersen et al. in Ref. [176], where the authors develop a model, called the quantum electrostatic heterostructure (QEH) model, to compute the dielectric properties of van der Waals (vdW) structures in an efficient way, without taking into account the detailed quantum influence of the layers on each other, and using a classical electrostatic model for the interlayer coupling. The name of the approach gives the name of the QEH package for Python developed in Ref. [176]. The isolated monolayer system is treated within a full quantum picture, like we do. However, we keep track of the local-field effects introduced by the (in-plane) inhomogeneity of the crystal. Furthermore, we consider only monolayer materials, which keeps the computational effort to the minimum. A dielectric function in the mixed $(\mathbf{q}, z)$ -representation, more precisely its inverse, has made an appearance in the work by S. Latini et al. in Ref. [215] (the same reference for the data we display in Fig. 6.9), but within the context of *ab-initio* calculations. In the *ab-initio* case, the z -dependence in the head element can be recovered by performing an inverse Fourier transform along the z -direction on the microscopic inverse dielectric function as

$$\varepsilon_{\mathbf{0}\mathbf{0}}^{-1}(\mathbf{q}, z, z') = \frac{1}{L_{\perp}} \sum_{G_z, G'_z} e^{iG_z z} e^{-iG'_z z'} \varepsilon_{G_z \hat{\mathbf{z}} G'_z \hat{\mathbf{z}}}^{-1}(\mathbf{q}), \quad (6.27)$$

where $L_{\perp}$ is the distance between consecutive images in the off-plane direction, the sum is over reciprocal lattice vectors with no in-plane component, and the momentum vector $\mathbf{q}$ is in-plane. Then, an effective 2D macroscopic dielectric function can be extracted from the 3D microscopic one. Other similar approaches for defining an effective 2D potential exist in the literature [144, 145]. Our approach is different. We start from the dielectric function in the mixed representation, and then invert it.

To obtain the final expression for the dielectric function in the mixed representation, we need first to work out the quantity $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z')$. Doing so is not very hard, but it is somewhat tedious. We briefly explain key steps of the derivation, as the fine details are not relevant. In the expression for the RPA irreducible polarizability in Eq. (3.72), in a basis of Bloch states, at zero temperature, and in the static limit, we replace the state functions by the Bloch wavefunctions in the linear combination of atomic orbitals as per Eq. (3.11). Then, when evaluating the real space 2D integrals, we introduce the point-like orbital approximation. The difference now, is that we write $\phi_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) \phi_{\beta}(\mathbf{r}' - \mathbf{R}' - \mathbf{t}_j) \approx \delta_{\alpha\beta} \delta_{ij} \delta_{\mathbf{R}\mathbf{R}'} \delta(\mathbf{r} - \mathbf{R} - \mathbf{t}_i) \delta(\mathbf{r}_{\parallel} - \mathbf{R} - \mathbf{t}_{i,\parallel}) \delta(z - t_{i,z})$, with $\mathbf{t}_i = \mathbf{t}_{i,\parallel} + t_{i,z} \hat{\mathbf{z}}$. The last equality is valid, since we can split a 3D delta function as $\delta(\mathbf{r} - \mathbf{r}') = \delta(\mathbf{r}_{\parallel} - \mathbf{r}'_{\parallel}) \delta(z - z')$, and the vector $\mathbf{R}$ has no off-plane component, but the atomic positions $\mathbf{t}_i$ in general do. The integrals in real space are therefore trivial, and the expression for the polarizability in the mixed representation reads

$$\begin{aligned} \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') = \frac{1}{\mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \frac{1}{(\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}})} \Bigg\{ & (6.28) \\ \sum_{i\alpha} \left[C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \left(C_{i\alpha}^{v\mathbf{k}} \right)^* e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i} \delta(z - t_{i,z}) \right] \sum_{j\beta} \left[C_{j\beta}^{v\mathbf{k}} \left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_j} \delta(z' - t_{j,z}) \right] + \\ \sum_{i\alpha} \left[\left(C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \right)^* C_{i\alpha}^{v\mathbf{k}} e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i} \delta(z - t_{i,z}) \right] \sum_{j\beta} \left[\left(C_{j\beta}^{v\mathbf{k}} \right)^* C_{j\beta}^{c\mathbf{k}+\mathbf{q}} e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_j} \delta(z' - t_{j,z}) \right] \Bigg\}. \end{aligned}$$

We can finally present the expression for the dielectric function in the $(\mathbf{q}, z)$ -representation. For that, the integral in z'' in Eq. (6.25) is also trivial to work out, as the z -dependence in the polarizability is modeled with Dirac deltas. Working out the integral and simplifying, the mixed dielectric function reads

$$\begin{aligned} \varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z') = \delta_{\mathbf{G}\mathbf{G}'} \delta(z - z') - \frac{e}{2\varepsilon_0 |\mathbf{q} + \mathbf{G}| \mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \frac{1}{(\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}})} \Bigg\{ & (6.29) \\ \sum_{i\alpha} \left[C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \left(C_{i\alpha}^{v\mathbf{k}} \right)^* e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i} e^{-|\mathbf{q}+\mathbf{G}||z-t_{i,z}|} \right] \sum_{j\beta} \left[C_{j\beta}^{v\mathbf{k}} \left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_j} \delta(z' - t_{j,z}) \right] + \\ \sum_{i\alpha} \left[\left(C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \right)^* C_{i\alpha}^{v\mathbf{k}} e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i} e^{-|\mathbf{q}+\mathbf{G}||z-t_{i,z}|} \right] \sum_{j\beta} \left[\left(C_{j\beta}^{v\mathbf{k}} \right)^* C_{j\beta}^{c\mathbf{k}+\mathbf{q}} e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_j} \delta(z' - t_{j,z}) \right] \Bigg\}. \end{aligned}$$

An effective 2D dielectric function $\bar{\varepsilon}_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$ can be defined as the quasi-2D dielectric function $\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z')$ averaged along the thickness of the material $d_\perp$. Performing the average explicitly yields

$$\begin{aligned} \bar{\varepsilon}_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) &\equiv \frac{1}{d_\perp} \int_{-d_\perp/2}^{d_\perp/2} \int_{-d_\perp/2}^{d_\perp/2} \varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z') dz dz' = & (6.30) \\ &= \delta_{\mathbf{G}\mathbf{G}'} - \frac{e}{2\varepsilon_0 |\mathbf{q} + \mathbf{G}| \mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \frac{1}{(\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}})} \Bigg\{ \\ &\sum_{i\alpha} \left[C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \left(C_{i\alpha}^{v\mathbf{k}} \right)^* \frac{2e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i} (1 - e^{-|\mathbf{q}+\mathbf{G}|d_\perp/2} \cosh[|\mathbf{q} + \mathbf{G}|t_{i,z}])}{|\mathbf{q} + \mathbf{G}|d_\perp} \right] \sum_{j\beta} \left[C_{j\beta}^{v\mathbf{k}} \left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_j} \right] - \\ &\sum_{j\beta} \left[\left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* C_{j\beta}^{v\mathbf{k}} \frac{2e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_j} (1 - e^{-|\mathbf{q}+\mathbf{G}|d_\perp/2} \cosh[|\mathbf{q} + \mathbf{G}|t_{j,z}])}{|\mathbf{q} + \mathbf{G}|d_\perp} \right] \sum_{i\alpha} \left[\left(C_{i\alpha}^{v\mathbf{k}} \right)^* C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_i} \right] \Bigg\}, \end{aligned}$$

where we have used the identities

$$\begin{aligned} \int_{-d_\perp/2}^{d_\perp/2} e^{-|\mathbf{q}+\mathbf{G}||z-t_{i,z}|} dz &= \frac{2(1 - e^{-|\mathbf{q}+\mathbf{G}|d_\perp/2} \cosh[|\mathbf{q} + \mathbf{G}|t_{i,z}])}{|\mathbf{q} + \mathbf{G}|} \quad \text{and} \\ \int_{-d_\perp/2}^{d_\perp/2} \delta(z' - t_{j,z}) dz' &= 1. \end{aligned} \quad (6.31)$$

Taking in the final expression of Eq. (6.30) the limit $d_\perp, t_{i,z} \rightarrow 0$, expanding the real exponential to leading order as $e^{-|\mathbf{q}+\mathbf{G}|d_\perp/2} \approx 1 - |\mathbf{q} + \mathbf{G}|d_\perp/2$, and considering that $\cosh(x) \approx 1$ for low x , we find

$$\begin{aligned} \bar{\varepsilon}_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) \approx & \delta_{\mathbf{G}\mathbf{G}'} - \frac{e}{2\varepsilon_0|\mathbf{q} + \mathbf{G}|\mathcal{A}} \sum_{vc,\mathbf{k}\sigma} \frac{1}{(\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}})} \left\{ \right. \\ & \sum_{i\alpha} \left[C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \left(C_{i\alpha}^{v\mathbf{k}} \right)^* e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_i} \right] \sum_{j\beta} \left[C_{j\beta}^{v\mathbf{k}} \left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_j} \right] - \\ & \left. \sum_{j\beta} \left[\left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* C_{j\beta}^{v\mathbf{k}} e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{t}_j} \right] \sum_{i\alpha} \left[\left(C_{i\alpha}^{v\mathbf{k}} \right)^* C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{t}_i} \right] \right\}. \end{aligned} \quad (6.32)$$

In this way, we can identify in the previous expression the strict 2D dielectric function we implemented in the code, $\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$, as the zero thickness limit of the averaged quasi-2D dielectric function $\bar{\varepsilon}_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$. In math notation, we can write

$$\boxed{\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) = \lim_{d_\perp \rightarrow 0} \frac{1}{d_\perp} \int_{-d_\perp/2}^{d_\perp/2} \int_{-d_\perp/2}^{d_\perp/2} \varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z') dz dz'}, \quad (6.33)$$

just as we expected. Question (i) posed at the end of the previous section is answered. Now we understand how, in the long-wavelength limit, the results coming from our strict 2D implementation agree so well with the *ab-initio* ones. The only thing left to do is to put up to test our theory by computing the effective 2D dielectric function from the quasi-2D approach, $\bar{\varepsilon}_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$, and compare once more with the *ab-initio* results. Fortunately, the answer to question (ii) posed at the end of the previous section is positive.

We have found that our quasi-2D dielectric function using the cutoff $G_c = 3 \text{ \AA}^{-1}$ is underestimated with respect to the *ab-initio* one from Ref. [215]. Luckily, as one can see in Fig. 6.13 below, we get a very good agreement by using the QEH package [176]. For completeness, we plot in the same figure the dielectric functions from Fig. 6.9. The package QEH works for any vdW structure, but since we are only interested in the monolayer case, we wrote a short Python script to compute the dielectric function of monolayer hBN. For each monolayer material, the package uses data from *ab-initio* calculations previously done with the GPAW code [32]. Therefore, the computational effort to get the dielectric function using the QEH package was minimal. The *ab-initio* curve from the QEH package is represented by the dashed black line, while our quasi-2D curve is represented with a continuous blue line. Although the two curves do not exactly overlap, the overall agreement is excellent along the whole $\Gamma - K$ line. For completion, we included in the plot below the curves previously displayed in Fig. 6.9.

In the following section, we discuss details about the implementation in the **Xatu** code of the theory presented in sections 6.1 and 6.3. We shall focus on the implementation of the dielectric function, since that is the main contribution from this work. Details about the exciton calculation are also in order, the ones relevant for our implementation of the screening.

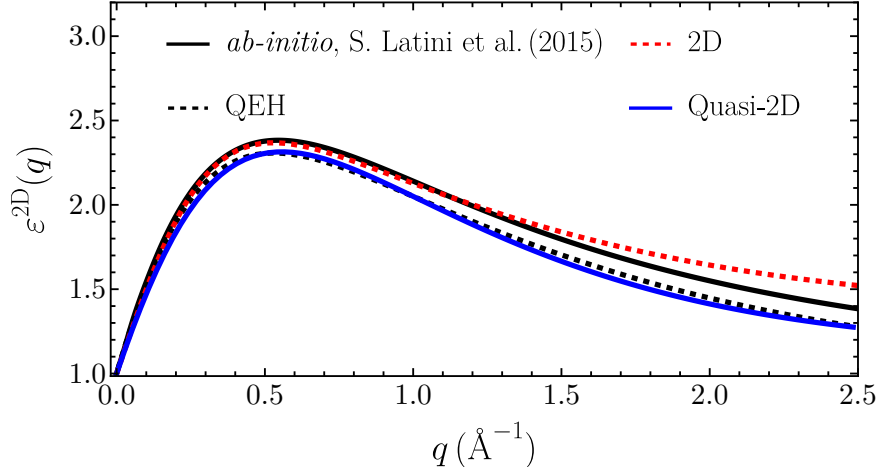


Figure 6.13: *Ab-initio* and quasi-2D effective dielectric function for hBN. 2D effective dielectric function of monolayer hBN, computed by our strict 2D and quasi-2D approaches, in dashed red and continuous blue lines, respectively. For comparison, the *ab-initio* ones from Ref. [215] and Ref. [176] are represented by the continuous black line and by the dashed black line, respectively. We applied the cutoff $G_c = 3.0 \text{ \AA}^{-1}$ to both 2D and quasi-2D dielectric matrices. The dashed black curve was obtained through the QEH package for Python [176]. For the monolayer thickness in the quasi-2D approach, we used the value provided by the QEH package, $d_{\perp} = 3.33 \text{ \AA}$.

6.4 Efficient Computation of the Static Dielectric Function with the Xatu Code

In this section, we explore the novelties in the code `Xatu` [37] that allowed the implementation of the 2D dielectric function we have been investigating. We keep the discussion as brief as possible, and will strongly focus on the numerical implementation of the screening. Apart from a couple of novelties in the functionality of the code, the class structure and previous algorithms stays the same, whose details can be consulted in the PhD thesis by A. J. Uría-Álvarez in Ref. [132].

6.4.1 The new version of the Xatu code: general remarks

Before we start with the more technical discussion, some general remarks are in order. The `Xatu` code [37] is a library, and at the same time a program, written mostly in C++ and with the objective of determining the excitonic and optical properties of insulators, with focus on one dimensional (1D), 3D, molecular, and 2D materials, with a strong emphasis on the latter. The code works essentially as a solver for the BSE with an exact diagonalization approach, and adapted for the tight-binding approximation. The efficiency of the code relies on the point-like orbital approximation, which we have discussed extensively by now, for the calculation of the interaction matrix elements. In concomitance with this code, another one was being developed in parallel, and later published in Ref. [217]. Both codes implement identical theories. With the Rytova–Keldysh approximation for the potential initially implemented in the code `Xatu`, one can get a good estimate for the exciton energies

and states, alongside the excitonic optical conductivity, with a fraction of the computational power demanded by *ab-initio* calculations [31, 33, 144, 145, 229]. With this work, we enable efficient calculations for the microscopic dielectric function in the static limit, closing a gap in the literature of computational condensed matter physics. As far as we know, our numerical implementation of the screening properties of 2D insulators is the first one working with a 2D formulation of the dielectric response problem. To implement the dielectric function numerically, we edited the code considerably. Nonetheless, some general aspects of the code were left intact. One in particular is how the code handles linear algebra calculations. All matrices and related properties are manipulated through a C++ library called ARMADILLO [240, 241], alongside the typical ones for linear algebra, BLAS and LAPACK. Another one, is how the optical conductivity calculated after the excitons is implemented. It is not written in C++, but in Fortran.

The **Xatu** code is versatile when it comes to the Hamiltonian models. It supports tight-binding models, and Hamiltonian models resulting from DFT calculations with the CRYSTAL package, which works with Gaussian basis functions. Therefore, there are two types of input model files: files with a `.model` extension, which contain the properties of the tight-binding Hamiltonian and basic geometrical aspects of the material at hand; alternatively, output files from CRYSTAL, which have the extension `.outp`. Both types of models have been tested; at a first instance to calculate excitons with the first version of the code [37], and now for the numerical implementation of the screening. Another interesting aspect of **Xatu**, is that it can work as a program through the script called `xatu`, or as a library if the end user feels comfortable enough with writing scripts in C++. In the latter case, the library can be used as an application programming interface (API), without the need for any kind of input files. Otherwise, the user can have access to the main functionalities with the `xatu` script. This script is previously written in the `xatu.cpp` file from the repository (inside the folder called “main”). Another aspect we did not change is the class hierarchy of the code. All the properties related with the dielectric function consist of a set of objects that are members of the class `ExcitonTB`. The code follows the object-oriented programming (OOP) paradigm, where the developer defines their own types of objects, called classes, and operates with them. The class `ExcitonTB` is just one example among all the classes we can find in the code. One can inspect the structure of the code in the official repository, and consult the documentation in <https://xatu-documentation.readthedocs.io/en/latest/> for a tutorial-like guide. The code developed in this work can be found in the `screening_reciprocal` branch of the official repository in <https://github.com/Peteracn96/xatu/>. At the moment of the writing of this dissertation, the dielectric function is not yet a class, even though we envision to create one for it in the future. However, a new class was created, called `ScreeningConfiguration`, responsible from parsing the input parameters pertaining to the dielectric function, which are contained in an extra input file with the `.txt` extension. The extra input file, called a screening file, was not comprised in the first version of the code. In the next section, we describe the structure of a general screening input file in

more detail (see Box 6.4.1).

Each parameter in an exciton or screening file starts with the cardinal symbol `#`. Whenever we use it in this text, we are referring to an input parameter. The new version of the `# reciprocal` parameter, called now `# gcutoff`, is a number defined by the user to set the reciprocal vectors cutoff for the exciton, G_c^X , determining which reciprocal lattice vectors $\mathbf{G}$ are included in the interaction matrix elements (see Eqs. (4.62), (6.13) and (6.14)). When the `# gcutoff` is parsed, it sets the program to the reciprocal space mode, just as in the first version of the code with the parameter `# reciprocal`. Doing so enables the calculation of the interaction matrix elements using the Coulomb potential resolved in momentum space. Additionally, the user can choose which potential to use (bare Coulomb, Rytova–Keldysh, or numerical). If the reciprocal space mode is set, then the user can additionally provide another parameter, ς , to determine the size of the regularization radius (see Appendix C.2). The parameter is called `# percentage`, and it should be a number of the order of the unit. The radius q_0 of the regularization region, which is a circle, is then $q_0 = \varsigma k_0$, where k_0 is the length of the vector in the BZ mesh closest to the origin. Since we use the Rytova–Keldysh analytical scheme for the regularization [225], the parameter ς has to be of the order of the unity, so that the low- q limit of the dielectric function remains valid.

Before we dive into the technical aspects of the implementation, we would like to dwell on the new functionalities, in particular the new algorithm of the `xatu` program, and useful new methods for the calculation of the dielectric function. For the latter, we implemented the method `ExcitonTB::compute_2D_DielectricMatrix`⁵, which calculates the dielectric function in the entire BZ mesh. It is called in the `xatu` program if the screening calculations are enabled. Doing so is possible by passing the `-z` flag to the program. After concluding the calculation of the dielectric matrix, its inversion follows. Finally, the inverse dielectric matrix is written into a `.dat` file. These last two tasks are performed in separate methods of their own, and in turn separate from `ExcitonTB::compute_2D_DielectricMatrix`. This last method method computes only the strict 2D dielectric function in the BZ. At this point, we have not implemented a counterpart for the quasi-2D dielectric function.

The user can call the method `ExcitonTB::readInverseDielectricMatrix` to read a previously generated `.dat`, giving its name as an argument. In this way, it is possible to write separate scripts using the `Xatu` library to compute the excitons without the need to repeat the screening calculations. However, if the reader is only interested in obtaining the macroscopic dielectric function in a specific set of $\mathbf{q}$ points, we implemented two identical functions, `compute_2D_DielectricMatrix(std::string filename)` and `compute_quasi2D_DielectricMatrix(std::string filename)`, computing the strict 2D and quasi-2D dielectric functions, respectively. Both functions receive as an argument the name of file containing all the $\mathbf{q}$ points predefined by the user. The file should contain only one momentum point per line, and each line should have the three momentum components in the following way: `qx qy qz` (naturally, the user has to set `qz` to zero). If the

⁵Having `ExcitonTB::` before the name of the method helps the reader remembering that the method belongs to the `ExcitonTB` class. It is also correct C++ syntax, but that is not of relevance for the reader.

user is interested in the dielectric function over the entire BZ mesh, we recommend using the `xatu.cpp` script instead, as it is optimized for calculating $\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$ in the BZ. With the `xatu.cpp` script, the inverse dielectric matrix is calculated and stored to be used later when initializing Bethe–Salpeter Hamiltonian. While doing so, the interaction matrix elements are computed on the fly, regardless of the chosen potential. To better visualize what we have just discussed, a flowchart for the `xatu` program is provided in Fig. 6.14 in the next page. The flowchart is adapted from its first version elaborated by A. J. Uría-Álvarez, which can be consulted in Figure 3.2 of his PhD thesis in Ref. [132].

If the reader is interested in using the code for themselves, we provide in Box 6.4.1, after the flowchart, an explanation on how to use the new screening functionalities. If the reader is more interested in the technical discussion of the implementation, the flowchart and the box can be read at a later instance.

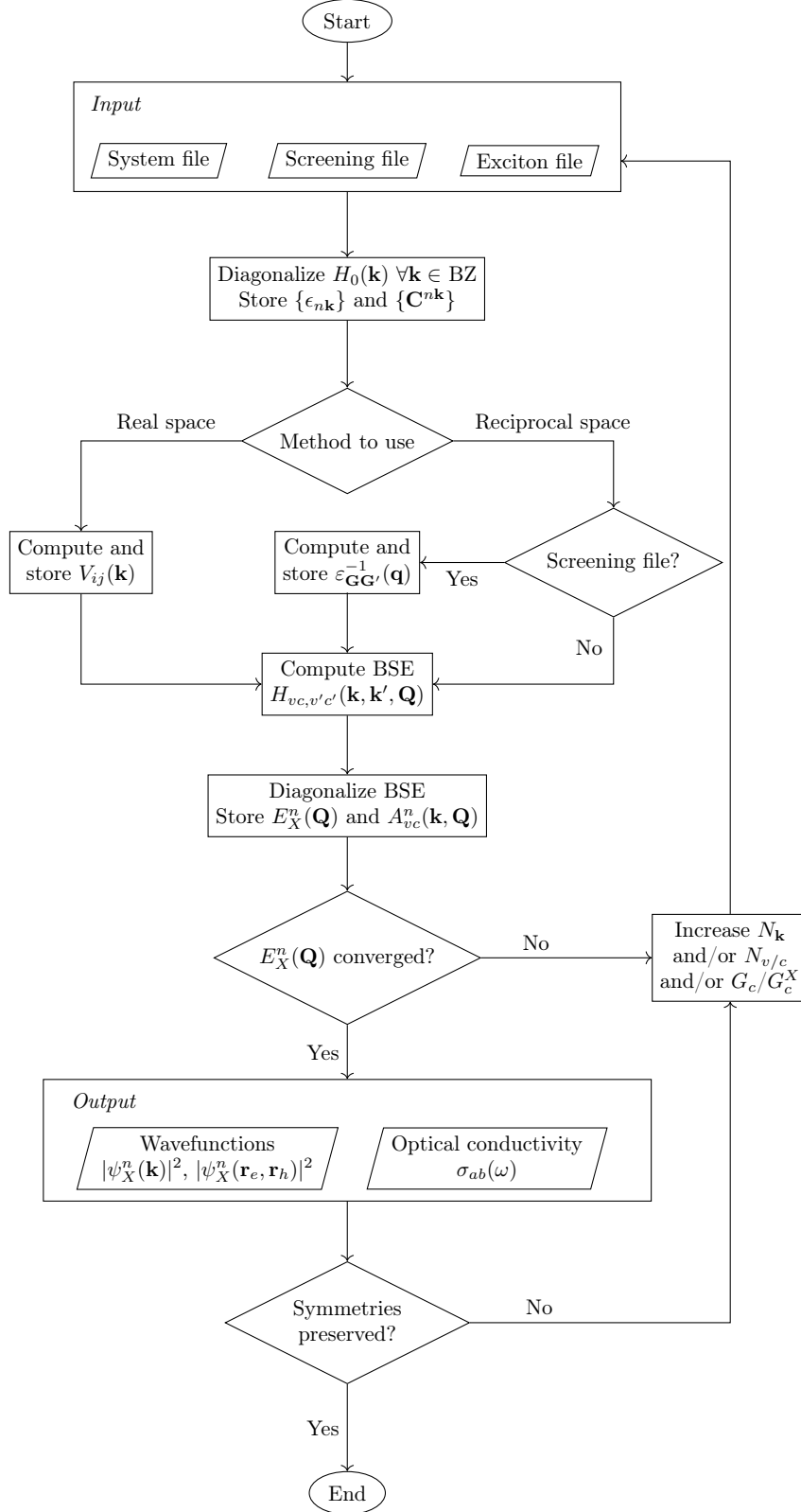


Figure 6.14: Flow diagram of an exciton calculation with xatu, including the numerical model for the screening. Flow diagram showing the typical steps in an exciton calculation with Xatu. The calculation of the dielectric matrix is included in the algorithm. The screening file is optional, and has to be provided only if the screening calculations are enabled. If they are not, and the user chose the reciprocal space approach then the Rytova–Keldysh potential is used.

Box 6.4.1: Using the screening functionalities of Xatu

In this box we give instructions on how to use the new functionalities in **Xatu** pertaining to the screening when one uses **Xatu** as a script. The **Xatu** script is a binary resulting from compiling the file **xatu.cpp**. To use it, when running the binary on the command line, one must provide the flag **-z** together with the name of the screening file, whose structure we discuss in the next. In general, we must do

```
$ ./xatu [OPTIONS] systemfile excitonfile -z screeningfile
```

At this point, if a screening file is provided, then an exciton file must be provided as well. The cutoff G_c^X for the reciprocal lattice vectors in the interaction matrix elements has to be provided in the exciton file in the parameter **s** called **# gcutoff** (previously called reciprocal). The cutoff G_c for the reciprocal lattice vectors in the *dielectric matrix* has to be provided in the screening file (See Eq. (6.10)). If not provided, they are set to a default value.

The cutoffs G_c and G_c^X can be different. In principle, a smaller G_c^X is needed to make the interaction matrix elements converge than G_c , which justifies our choice. We always have $G_c^X \leq G_c$ in practice.

Still pertaining to the exciton file, the functionality of choosing the desired potential to compute the interaction matrix elements has been extended to the calculations using the reciprocal space approach. In the first version of the code [37], this option was only implemented for the real space approach. Now the user can also choose for the interaction the bare Coulomb potential, in addition to the new option of using the numerical model for the screening. To do so, the option **rpa** has to be provided to the parameter **# potential**, that enables the computation of the interaction matrix elements using the numerical dielectric function. The potentials for the direct and exchange terms can be chosen separately, just as before.

Furthermore, we have added another parameter to the exciton configuration file, called **# percentage**, which is a percentage defining the size of the regularization region (see Appendix C.2). For instance, if the value ς is provided, then the radius of the regularization region is $q_0 = \varsigma k_0$, where k_0 is the norm of the wavevector(s) closest to the origin. Typically, $\varsigma \sim 0.5 - 1.5$. This parameter is optional, and if not provided, the default value $\varsigma = 0.5$ is used. If the user does not want a regularization and simply set it to zero, then the value 0 must be provided.

The structure of the screening file is identical to that of an exciton file. As possible input parameters we have:

- **# ncell_aux**: Number of unit cells, in each direction, of the auxiliary BZ mesh to compute the polarizability matrix element(s)
- **# valence.bands**: Number of valence bands included in the calculations

- **# conduction.bands**: Number of conduction bands included in the calculations
- **# spin**: It assumes the values **true** or **false**, indicating if the spin degree of freedom has been considered in the system model or not. If not, usually the case for systems with no spin–orbit coupling (SOC), then all bands have a spin degeneracy $g_s = 2$, so the sum over spins in the polarizability translates into a mere prefactor of two, $g_s = 2$. If the spin degree of freedom has been accounted for, then $g_s = 1$. For example, if the system is MoS_2 , and the SOC has not been included in the model even though this material has SOC in reality, then we have to parse the string **false**, so that $g_s = 2$. If a similar model for the same material is used, but now including SOC, then the number of bands doubles. Therefore, the sum over spins is accounted for in the sum over the bands. Hence, we have to provide the string **true**, so that $g_s = 1$.
- **# gcutoff**: Defines the cutoff G_c for the reciprocal lattice vectors to be included in the calculation of the dielectric matrix, as explained above.
- **# [isotropic]**: Indicates if the system is isotropic or not. If yes, then the value **true** must be provided, otherwise **false**. This is relevant if the material is not isotropic, as in that case the regularization needed for the exciton calculation has to account for the anisotropy of the (head element of the) dielectric function. The default value is **true**.
- **# function**: Used to specify which functionality of the screening the user intends. It can be **dielectric**, **polarizability**, **inverse.dielectric** or **exciton**. Each of these options does the following, respectively:
 1. **dielectric**. Computes the dielectric function at a specified momentum and for a chosen matrix element $(\mathbf{G}, \mathbf{G}')$. The values of the polarizability as a function of included valence and conduction bands are computed and printed to the **polarizability_convergence.dat**. Each line of the file has the form
`[# valence bands] [# conduction bands] [Re{ $\chi$ }] [Im{ $\chi$ }]`
 2. **polarizability**. Computes the polarizability matrix element $\chi_{\mathbf{G}\mathbf{G}'}$ for the specified pair $(\mathbf{G}, \mathbf{G}')$ in the BZ mesh. The values are printed to the **polarizability_mesh.dat** file, and each line of the file has the structure `[ $k_x$ ] [ $k_y$ ] [ $k_z$ ] [Re{ $\chi$ }] [Im{ $\chi$ }]`. At the moment, the screening works only for 2D materials, whence $k_z = 0$ at all times.
 3. **inverse.dielectric**. Computes the inverse of the dielectric matrix $\varepsilon^{-1}(\mathbf{q})$ at the specified momentum in the BZ mesh. The cutoff for the

reciprocal lattice vectors is specified by the parameter `# gcutoff`. The matrix elements are printed to the file `epsilon.dat`. The order of the $\mathbf{G}$ vectors is the same as the one they appear by when printed to the terminal/output file when this option is used. Each row in the file has the real and imaginary parts of each matrix element printed separately separated by a space (and conserving the order of the $\mathbf{G}$ vectors).

4. **exciton**. Computes the inverse of the dielectric matrix in the BZ mesh, and proceeds with the calculation of the exciton. A file containing all the points in the BZ mesh is generated. The file has a name of the form `kgrid_<ncells>.dat`, where `ncells` is the number of cells in each direction, specified in the exciton file. The object ε^{-1} is printed to the file `<label>_invepsilon.dat`, where `<label>` is the value of the label parameter specified in the exciton file [37]. All the individual matrices $\varepsilon^{-1}(\mathbf{q})$ are printed by the same order of momentum points as that of the file containing the $\mathbf{k}$ points.

- **# vectors**: Indicates which pair of reciprocal lattice vectors $(\mathbf{G}, \mathbf{G}')$ the polarizability is computed for. Two indices have to be provided in the same line like this: `<index1> <index2>`. If this input is omitted, then the calculations are done by default for the head element. This entry is valid for both functions `dielectric` and `polarizability`.
- **# momentum**: Specifies which momentum the polarizability, or inverse of the dielectric matrix, is to be computed at. A vector in the form `qx qy qz` has to be given. By default, the computation is done at $\mathbf{q} = (0.2, 0, 0)$, if this block is not provided.

6.4.2 Complexity analysis

In this part of the chapter, we provide a brief complexity analysis of the calculation of the dielectric function. Our goal is to estimate the memory complexity and the time complexity, that we will denote by $\mathcal{O}_m$ and $\mathcal{O}_t$, respectively, of the full calculation of the dielectric matrix in the entire BZ mesh. This is done before initializing the Bethe–Salpeter Hamiltonian with the `xatu` program, by the `ExcitonTB::compute_2D_DielectricMatrix` method. For brevity, we do not discuss the space complexity, since the time complexity is the one we are more interested in. Since one can improve the processing time by engineering the storage of information, it makes sense to address the memory complexity as well. By memory complexity, we mean the memory requirements in order to *compute* the dielectric matrix, and *not* to *store* it. Regardless of the way the program computes the dielectric matrix, it is stored always in the same way, since the size of the BZ and the number of reciprocal lattice vectors included are pre-determined by the user. Another aspect of the implementation we do not discuss, is the complexity in accessing the elements stored in memory to compute the dielectric matrix and to compute the exciton. We consider that memory accessing does not affect considerably the processing time. We believe that the aspect affecting the processing time the most is the calculation of all matrix elements. Moreover, we suspect that the calculation of the plane-wave matrix elements does not cause a substantial slowdown of the program. At the end, we want to estimate the time complexity of a full calculation of the dielectric matrix in the entire BZ mesh, which comes before the calculation of the excitons using the `xatu` program. For that, we analyse first the complexity of computing a single polarizability matrix element $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$.

The polarizability matrix elements as per Eq. (6.7) are computed through a sum over valence bands, conduction bands, and momentum (apart from an eventual sum over the spin). In the code, they are implemented as three nested `for` loops. If we would like to keep the memory requirements to the minimum, then we could implement the `for` loops such that the one for momentum is the innermost one. Then, for each term in the sum, we would need to diagonalize the Bloch Hamiltonian $H(\mathbf{k})$ twice, at the points $\mathbf{k}$ and $\mathbf{k} + \mathbf{q}$. In this way, we get the sets of energies $\{\epsilon_{n\mathbf{k}}\}$ and $\{\epsilon_{n\mathbf{k}+\mathbf{q}}\}$, alongside their respective eigenvectors $\{\mathbf{C}^{n\mathbf{k}}\}$ and $\{\mathbf{C}^{n\mathbf{k}+\mathbf{q}}\}$. To remind the reader, the eigenvectors are used to compute the plane-wave matrix elements as per Eq. (6.9). However, this process to compute the sum for the polarizability would be extremely slow. There is no point in having to diagonalize the Bloch Hamiltonian for each iteration in the sum. What we do instead, is diagonalize the Bloch Hamiltonian before computing the polarizability matrix element, and store the energies and eigenvectors in the stack. If we denote the number of total bands by N_B , and the number of $\mathbf{k}$ points in the BZ mesh by N_k , then the memory complexity of computing a single polarizability matrix element is $\mathcal{O}_m(2N_B N_k + 2N_B^2 N_k)$, where we get a contribution from the $2N_B N_k$ eigenvalues to store, $\{\epsilon_{n\mathbf{k}}\}$ and $\{\epsilon_{n\mathbf{k}+\mathbf{q}}\}$, and from the $2N_B N_k$ eigenvectors $\{\mathbf{C}^{n\mathbf{k}}\}$ and $\{\mathbf{C}^{n\mathbf{k}+\mathbf{q}}\}$ (with each eigenvector containing N_B components). The code allows for using a number of valence bands N_v and conduction

bands N_c provided by the user, which can be smaller than the total ones to speed up the calculations. Nonetheless, to simplify the discussion, we assume that all the bands are included in the calculation. We have $N_B = N_v + N_c$. Unlike in *ab-initio* approaches [31–33], the calculation of the plane-wave matrix elements does not considerably affect the time complexity, nor the memory complexity. This is because for those matrix elements, we only need the eigenvectors $\mathbf{C}^{n\mathbf{k}}$ of the Bloch Hamiltonian and the atomic positions $\mathbf{t}_i$, which are already stored in memory at the time of the calculation of the dielectric matrix. Those matrix elements are computed on the fly at each iteration of the nested **for** loops, and are not calculated and stored *a priori*, unlike other packages in the literature, such as the BERKELEYGW package [31]. In this way, the time complexity of computing a single polarizability matrix element is $\mathcal{O}_t(N_v N_c N_k)$. Then, the calculation of the whole polarizability matrix at a single point $\mathbf{q}$ in the BZ has time complexity $\mathcal{O}_t(N_G^2 N_v N_c N_k)$, where N_G is the number of reciprocal lattice vectors included.

In the calculation of the dielectric matrix in the entire BZ mesh, we have to compute all the N_G^2 matrix elements $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$, for all the $\mathbf{q}$ points in the BZ mesh. We need to pay attention to the momentum $\mathbf{q} + \mathbf{k}$, and use the fact that the quasiparticle dispersion has the periodicity of the reciprocal lattice. There is a method, which was already implemented in the first version of the code, that can find for any vector $\mathbf{p}$ in the reciprocal lattice the corresponding momentum in the first BZ. In this way, at any time when doing the sum over momentum in the polarizability matrix element, we could let the program find the momentum $\mathbf{p}$ that verifies $\mathbf{q} + \mathbf{k} = \mathbf{p} + \mathbf{g}$, where $\mathbf{p} \in \text{BZ}$. Then, we would have $\epsilon_{n\mathbf{q}+\mathbf{k}} = \epsilon_{n\mathbf{p}}$, without having the need to diagonalize the Bloch Hamiltonian for each point $\mathbf{q} + \mathbf{k}$. This method of implementing the calculation of the polarizability matrix elements would keep the memory complexity at $\mathcal{O}_m(N_B N_k + N_B^2 N_k)$, without the factors of two, since with this approach having all energies $\{\epsilon_{n\mathbf{k}}\}$ and eigenvectors $\{\mathbf{C}^{n\mathbf{k}}\}$ is enough. It would be the same as in the computation of a single matrix element, but without the factors of two. For the time complexity, we find $\mathcal{O}_t(N_G^2 N_v N_c N_k^2)$. However, this method has a drawback in the time complexity. As we stated before, in the beginning of subsection 6.2.2, the polarizability converges quite fast with the size of the BZ mesh. This is in sharp contrast with the exciton. Henceforth, we made a different choice while developing the code, because we noticed that the calculation of a single element slows down drastically with the number of $\mathbf{k}$ points in the BZ. We have implemented an auxiliary BZ, that we can denote by BZ' , which can be coarser than the original one. The number of points $\mathbf{q}$ where we have to compute the dielectric matrix is the same, though. Let us denote the size of the initial BZ by N_q . We need to obtain the band structure $\{\epsilon_{n\mathbf{k}+\mathbf{q}}\}$ for each point $\mathbf{q}$, using a coarser mesh for the momentum $\mathbf{k}$. Let us denote the size of BZ' by N_k . This allows reducing the time complexity from $\mathcal{O}_t(N_G^2 N_v N_c N_k^2)$ to $\mathcal{O}_t(N_G^2 N_v N_c N_k N_q)$. The price to pay is an increase in the memory complexity from $\mathcal{O}_m(N_B N_k + N_B^2 N_k)$ to $\mathcal{O}_m(N_B N_q N_k + N_B^2 N_q N_k)$. This increase in memory requirement is not critical, because the end user can (and should) choose a value for N_k quite smaller than N_q . We make sure that the diagonalization of $H(\mathbf{k}+\mathbf{q})$ for all $\mathbf{q} \in \text{BZ}$ and all $\mathbf{k} \in \text{BZ}'$ is performed before the calculation of the dielectric

matrix elements begins.

We have analysed the complexity of a full calculation for the dielectric matrix in the entire BZ, which is implemented in the method `ExcitonTB::compute_2D_DielectricMatrix`. Fortunately, it is possible to reduce the time complexity considerably by exploiting the symmetry properties of the polarizability. Taking them into account is essential to optimize the code, and is also routine in the literature [31, 229], since the time to calculate all the polarizability matrix elements is what affects the processing time the most. It does not mean, however, that we actually need to call the method that computes a single polarizability matrix element that many times. First of all, it is not hard to show that $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}) = (\chi_{\mathbf{G}'\mathbf{G}}^0(\mathbf{q}))^*$. This means that we only need to calculate the upper triangular part of $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q})$. Instead of computing, for each momentum $\mathbf{q}$, N_G^2 matrix elements, we only need to compute $(N_G^2 - N_G)/2 + N_G = N_G(N_G + 1)/2$ elements. Furthermore, it is also not hard to show (but not as straightforward as the previous property) that $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}) = \chi_{-\mathbf{G}'-\mathbf{G}}^0(-\mathbf{q})^*$. With this information, as long as for every vector $\mathbf{G}$ we also include the vector $-\mathbf{G}$, which we do, then we only have to compute the polarizability matrix in half of the BZ. These symmetry properties of the polarizability are general, and are not related to the way it is herein implemented. There is a small caveat, though. If the number of cells on each direction, $\sqrt{N_q}$, is even, then the BZ mesh is not centrosymmetric. In that case, it includes part of the boundary of the BZ. This is due to using the Monkhorst-Pack algorithm [232] to generate a uniform $\mathbf{q}$ -grid, while imposing that the origin must always be included in the BZ mesh. For simplicity, let us say that we can reduce the number of $\mathbf{q}$ -points where we need to compute the polarizability matrix from N_q to $N_q/2$, approximately. Therefore, we can reduce the time complexity from $\mathcal{O}_t(N_G^2 N_v N_c N_k N_q)$ to $\mathcal{O}_t(\frac{N_G(N_G+1)}{2} N_v N_c N_k \frac{N_q}{2})$. To stress the efficiency gained in using a 2D (or quasi-2D) approach for the dielectric function of 2D materials, we remind the reader that, by virtue of the reduced dimensionality of the problem, the number of $\mathbf{G}$ vectors necessary to be included in the calculations is much lower when compared to 3D approaches; our $\mathbf{G}$ vectors are always in-plane. As a consequence, we can also claim that the problem of inverting the dielectric matrix $\epsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$ at each point $\mathbf{q}$ in the BZ mesh has a vanishing contribution to the time complexity. For example, we tested matrices as big as 1200×1200 in Fig. 6.7, where we applied the maximum cutoff $G_c = 54 \text{ \AA}^{-1}$. Looking at Fig. 6.6, it corresponds to including a bit more than 1200 vectors in the calculations. We verified that calculating the entire dielectric matrix for each $\mathbf{q}$ was, by far, a much bigger problem than inverting it.

There are a couple more remarks we need to make regarding the implementation. As the calculation of the dielectric matrix constitutes the bottleneck of the code, it is optimized by parallelizing the calculation of each matrix element. The code generates *a priori* all the $\frac{N_G(N_G+1)}{2} \frac{N_q}{2}$ possible combinations of the $(\mathbf{q}, \mathbf{G}, \mathbf{G}')$ variables that are strictly necessary to compute the dielectric matrix in the entire BZ mesh. Then, all those elements are computed in parallel. For that, we used the OPENMP package. At the moment, all parallelism is based on shared-memory, while distributed parallelism is to be implemented in the future.

A thorough study on the processing time would be necessary in order to better quantify

the time complexity of the new methods implemented in the **Xatu** library. However, we can say that we obtained the dielectric function for hBN used to calculate the excitons in a relatively short amount of time. The calculation of the dielectric matrix itself lasted for approximately 47 minutes (a fraction of an hour) on a single node, which has in total 48 CPU cores, and sensibly 250 GB of RAM memory. An exciton calculation using the same dielectric function and parameters as in Table 6.1, on a similar node with the same amount of CPU cores lasts for a bit less than three minutes. Our discussion on the complexity using the $\mathcal{O}$ notation gives a qualitative idea of what one would expect in a deeper quantitative study, which cannot be completely replaced by ours. Finally, now with respect to the calculation of the excitons, there is a detail on the calculation of the interaction matrix elements that should not be overlooked. To compute the interaction matrix element D , we need to sum over reciprocal lattice vectors as

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} (I_{c'\mathbf{k}'+\mathbf{Q}, c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}})^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{k} - \mathbf{k}') I_{v'\mathbf{k}', v\mathbf{k}}^{\mathbf{G}'}, \quad (6.34)$$

where the screened potential matrix element $W_{\mathbf{G}\mathbf{G}'}(\mathbf{k} - \mathbf{k}') = \varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{k} - \mathbf{k}') v_c(\mathbf{k} - \mathbf{k}')$ is needed (we use the nonsymmetrized version of the dielectric matrix just to make the point, as it is not implemented like this). We do not compute the dielectric function for each combination of $\mathbf{k} - \mathbf{k}'$. That would make the time complexity scale faster with N_q . Instead, we find the vector $\mathbf{q}$ such that $\mathbf{k} - \mathbf{k}' = \mathbf{q} + \mathbf{g}$, where $\mathbf{q} \in \text{BZ}$ and $\mathbf{g}$ is a vector of the reciprocal lattice. If $\mathbf{g}$ is not null, then

$$\begin{aligned} D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) &= \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} (I_{c'\mathbf{k}'+\mathbf{Q}, c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}})^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{q} + \mathbf{g}) I_{v'\mathbf{k}', v\mathbf{k}}^{\mathbf{G}'} = \\ &= \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} (I_{c'\mathbf{k}'+\mathbf{Q}, c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}})^* W_{\mathbf{G}+\mathbf{g}\mathbf{G}'+\mathbf{g}}(\mathbf{q}) I_{v'\mathbf{k}', v\mathbf{k}}^{\mathbf{G}'}. \end{aligned} \quad (6.35)$$

In this way, we only need to have the inverse dielectric matrix inside the first BZ⁶. However, a naive implementation of the previous expression as is would incur the program in a memory error at runtime. This is so, because we apply a cutoff for the dielectric function, which is defined prior to runtime. In the sum over reciprocal lattice vectors, we would eventually try to access elements of the dielectric matrix that do not exist in memory (but that exist physically). Therefore, by shifting the vectors $\mathbf{G}$ and $\mathbf{G}'$, we implement instead

$$D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} \left(I_{c'\mathbf{k}'+\mathbf{Q}, c\mathbf{k}+\mathbf{Q}}^{\mathbf{G}-\mathbf{g}} \right)^* W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}) I_{v'\mathbf{k}', v\mathbf{k}}^{\mathbf{G}'-\mathbf{g}}. \quad (6.36)$$

The 2D nature of our implementation proved its worth, not only by its efficiency, but also by its effectiveness; with a fraction of the computational effort, we reproduced results in very good agreement with those obtained via *ab-initio* calculations (see Figs. 6.9 and 6.13). To further test and validate our results, we reserve the next section for testing the code, this time on a different material, MoS₂.

⁶Note that $\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q} + \mathbf{g}) = \varepsilon^{-1}(\mathbf{q} + \mathbf{G} + \mathbf{g}, \mathbf{q} + \mathbf{G}' + \mathbf{g}) = \varepsilon_{\mathbf{G}+\mathbf{g}\mathbf{G}'+\mathbf{g}}^{-1}(\mathbf{q})$.

6.4.3 Validation and benchmarks

At this stage, we have extensively discussed the theory behind our implementation for the screening properties of 2D materials, and we have tested our theory against *ab-initio* calculations using monolayer hBN. We provided also insights on the implementation itself. To validate our theory and implementation even more, it would make sense to try it on a different material. We do so in this subsection as the closing of this chapter. The material of choice is monolayer molybdenum disulfide, MoS_2 , which is another prototypical 2D material to study excitons. We mentioned it for the first time in this dissertation in subsection 3.2.2.1 as a material where SOC has a nonvanishing contribution on its material properties (unlike in hBN). It was also introduced in Section 4.1 as an excitonic material presenting a direct electronic bandgap, belonging to big family of 2D materials called transition metal dichalcogenides (TMDs).

Materials like MoS_2 have given traction to a recent field of research within condensed matter physics and materials science, called valleytronics [242, 243]. It consists in the study of the optoelectronic properties of 2D materials with honeycomb structure, with the bandgap located at the K and K' points of the BZ (see Fig. 3.2). Their interesting optical and electronic properties come from the coupling between the spin and valley degrees of freedom allowed by the SOC; the coupling between the spin and the orbital electronic motion, together with inversion symmetry breaking, lifts the degeneracy of the bands at the K and K' points, which permits tuning the material optical properties by spin and valley control [242]. This deems these materials promising for applications in electronics, spintronics, and new ways of storing and processing information [243]. An interesting optical effect they display is circular dichroism—distinct absorption spectra for circularly polarized light with right- and left-handedness [120, 244, 245]. As a gapped material, we have an interest in its optical properties coming from excitonic effects. Even though it has a bandgap much smaller than that of hBN, making it a semiconductor instead of an insulator, it still exhibits strong absorption resonances for sub-gap frequencies. Before we study excitons in MoS_2 , we would like to look at its crystallographic structure and electronic single-particle properties, specifically the band structure. After that, we analyse its dielectric properties. Only then, will we look at excitonic and optical effects in this material.

Just like other TMDs, hBN and graphene, monolayer MoS_2 presents a honeycomb lattice with Mo atoms alternating with S atoms. The main difference from hBN and graphene is that MoS_2 is not atomically thin, presenting three atomic planes. The sublattice of the Mo atoms lives in the middle plane, while the bi-atomic S sublattice has two planes, one above and one below the Mo plane. The atomic arrangement we have just described can be seen in Fig. 6.15. In the same figure, we have the monolayer seen from three different views: a top view in panel **a**, a side view in panel **b**, and a perspective view in panel **c**.

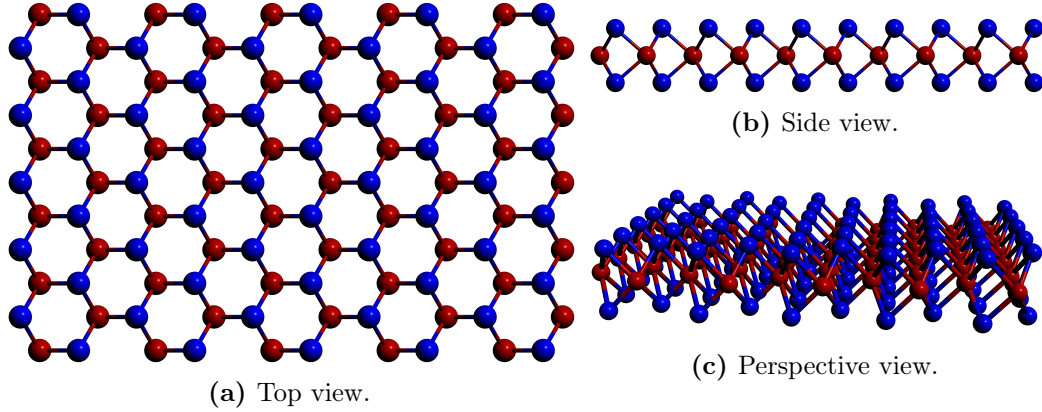


Figure 6.15: Crystalline structure of monolayer MoS_2 . Atomic structure of a MoS_2 monolayer. Three different points of view are displayed, a top view, a side view, and a perspective view, in panels **a**, **b**, and **c**, respectively. The atomic radii of the Mo and S atoms are in real proportion, but the length of the bonds are not in the correct proportion with respect to the atomic radii to facilitate the visualization. This figure was made with the coding language *Asymptote*, with the help from the *asy-atoms* package [78].

To model the single-particle Hamiltonian, we use a Wannierized Hamiltonian after performing the DFT calculations to obtain the band structure. The calculations were done by G. Janone, and their details follow. The DFT calculations were performed using the QUANTUM ESPRESSO package [246], with exchange and correlation effects modeled by the screened hybrid functional HSE06 [230] (see discussion before Fig. 6.1, and the footnote in the same page). The fraction of exact Hartree–Fock exchange was 0.30, and the BZ was sampled in a $2 \times 2 \times 1$ q -grid, for each point in the $21 \times 21 \times 1$ Monkhorst–Pack k -grid [232], the sufficient to converge the total energy and electronic properties. For the physics of the atomic cores, the KJPAW pseudopotentials were used. The plane-wave cutoff applied was of 50 Ry for the wavefunctions, and 310 Ry for the electronic density. An atomic relaxation was performed to determine the lattice constant, with a threshold of 10^{-4} for the residual forces. The interlayer separation used was $L_{\perp} = 20 \text{ \AA}$. Afterward, the WANNIER90 package [223] was used to obtain the tight-binding Hamiltonian. Usually, one applies the method of maximally localized Wannier functions [81]. Instead, the method of closest Wannier functions [247, 248] was employed. Then, a projection onto s , p , and d orbitals of Mo, and onto s and p orbitals of S was performed. Defining the temperatures $T_1 = 1 \cdot 10^{-6} \text{ K}$ and $T_2 = 6.0 \text{ K}$, the window of energies for the Fermi–Dirac smearing was set to $T_1 - T_2$ to disentangle the bands. No maximal localization step was done. Moreover, SOC effects were not accounted for. We verified numerically, by using similar models for MoS_2 with and without SOC, that it does not affect the polarizability matrix elements considerably.

The band structure of the Wannier model for MoS_2 is shown in Fig. 6.16. It was obtained using a `.model` file as input to *Xatu*. The `.model` file was generated by a utility developed by G. Janone and N. N. Batista, which converts an output file from WANNIER90 with the format `_tb.dat` into an input file for *Xatu* with the format `.model`. This is the web

link for the official repository of the `Xatu` code including the utility: <https://github.com/GuiJanone/xatu-w90>. The structure of a `.model` file is explained in Ref. [37]. It contains the tight-binding (or Wannier) model defining the single-particle Hamiltonian. For this model, we get a bandgap of $\Delta = 2.57$ eV, which is the value we use to obtain the binding energies. In band structures for the same material including SOC, one typically sees a splitting in the bands due to spin flipping induced by the SOC. Including those effects in the exciton calculations is required if one wants to compare the theoretical results with the experimental observations. In particular, including the SOC term in the Hamiltonian leads to a lift in the degeneracy of the excitonic states, whose imprinting in the absorption spectrum is the splitting of the absorption peaks [245].

We proceed with the study of the dielectric properties of MoS_2 , namely the polarizability and the microscopic dielectric function. To keep the discussion short, we will actually skip the details of the polarizability. The features in the polarizability we saw for this model for MoS_2 are similar to those seen for hBN in subsection 6.2.2. The sanity check we always have to do before continuing to the dielectric function, is to evaluate the isotropy of the head element. If a system is isotropic, which is the case of both hBN and MoS_2 , then we should expect to obtain an isotropic $\mathbf{q}$ -dependence of the element $\chi_{00}^0(\mathbf{q})$. We tested with other Wannier models for MoS_2 , and observed that the head element of the polarizability when plotted as a function of $q = |\mathbf{q}|$ with $\mathbf{q}$ covering the whole BZ, like in Fig. 6.4a for hBN, may present considerable numerical fluctuations. Those models were obtained without forcing the symmetry preservations when running WANNIER90. Therefore, we suspect that the symmetries of the model Hamiltonian play a role in the dielectric properties of the material, most likely symmetries related to the space group of the crystalline material. However, it is not clear for us how do these symmetries exactly influence the polarizability and its symmetries. There is just one small caveat in this model, or Wannier models in general. Typically, Wannier Hamiltonians describe correctly the system in a range of energies close

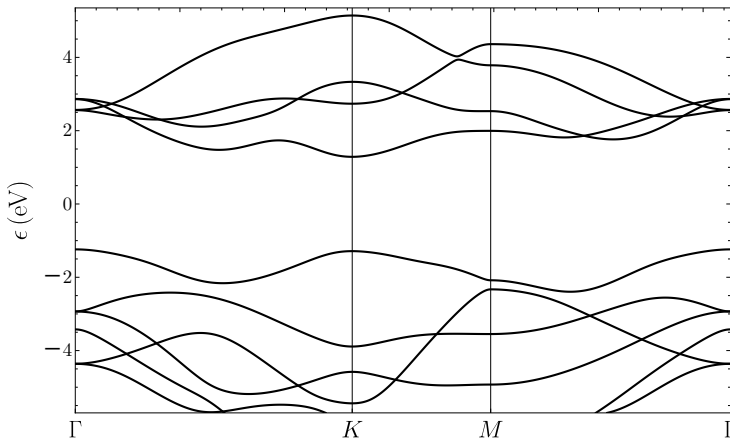


Figure 6.16: Band structure of MoS_2 with a Wannier model. Quasiparticle band structure of monolayer MoS_2 computed through DFT calculations with the QUANTUM ESPRESSO, followed by a Wannierization to change the Hamiltonian to a basis of localized Wannier functions using the WANNIER90 code. The bands were generated with `Xatu`, in similarity to what we did for hBN in Fig. 6.1. Details of the calculation are in the second paragraph of the previous page.

to the Fermi level, which means that it is hard to obtain a Wannier model containing a substantial number of bands. This model has thirteen valence bands, and four conduction bands. These are not sufficient to see a convergence of the polarizability with the number of empty bands as we saw in Figs. 6.2 and 6.3. Nonetheless, we will see that this does not come as a setback in the results we will show in a moment. After all, using a model with so few bands comes as a numerical advantage, so we will keep it.

Before we discuss the results obtained for the dielectric function, we shall say that we did a study on the convergence of the inverse dielectric function with the cutoff G_c , just like for hBN. We identified similar futures to what we saw in Fig. 6.6. Specifically, we observed that the convergence with the number of reciprocal lattice vectors included was just as slow. But once more, keeping a low G_c cutoff was enough to reproduce the *ab-initio* results. The comparison of our effective 2D dielectric function with the *ab-initio* one is shown in Fig. 6.17. The quasi-2D dielectric function is also represented in the same plot. In Fig. 6.17, we can see that the strict 2D static dielectric function, in the dashed red line, underestimates the screening for low q in comparison with the *ab-initio* curve, in the continuous black line, but overestimates the screening for intermediate and higher q . Nonetheless, the strict 2D curve, the *ab-initio* curve, and the quasi-2D one, in the dashed blue line, agree for very low q ; this confirms our hypothesis from Section 6.3, that the layer thickness does not influence the dielectric properties in the very small q limit. On the contrary, the agreement between the quasi-2D permittivity and the *ab-initio* one is very good. Even though we still underestimate the screening for small q , for intermediate and higher momenta the agreement between the dashed blue and continuous black lines are excellent. The agreement is not as good as in the case of hBN (see Fig. 6.13) when looking at the whole q -range, perhaps because monolayer hBN is atomically thin, unlike MoS₂, meaning that the 2D and quasi-2D pictures work better for hBN than for MoS₂.

Before testing the excitonic optical response of MoS₂, we need to compute the excitonic energies and eigenstates using the **Xatu** code. Just like we did for hBN in subsection 6.2.5, we do the calculation using the Rytova–Keldysh potential and the numerical inverse dielectric matrix, and then compare. The only difference from one method to the other, was the type of potential used to compute the interaction matrix elements. We used a 60×60 $\mathbf{k}$ -grid to sample the BZ, included one valence band, $N_v = 1$, and one conduction band, $N_c = 1$, applied a cutoff for the reciprocal lattice vectors of $G_c = 3.0 \text{ \AA}^{-1}$ (in which we included all the seven vectors), and for the regularization radius we used a percentage of 0.6%. The same regularization scheme as explained in Appendix C.2 is applied to the calculations using the Rytova–Keldysh potential. In both calculations, we set $\mathbf{Q} = \mathbf{0}$. For the screening length in the Rytova–Keldysh potential, we used the value $r_0 = 31.74 \text{ \AA}$, determined from the 2D dielectric function shown in Fig. 6.17. It does not exactly match the value determined from the 2D dielectric function we used for the calculations with the numerical screening, which is approximately $r_0 = 30.87 \text{ \AA}$. We do not expect a big difference between using one value or the other though, since they are relatively close. Moreover, both values are within the *ab-initio* interval for r_0 in MoS₂ of $30 - 40 \text{ \AA}$ reported in the literature [249].

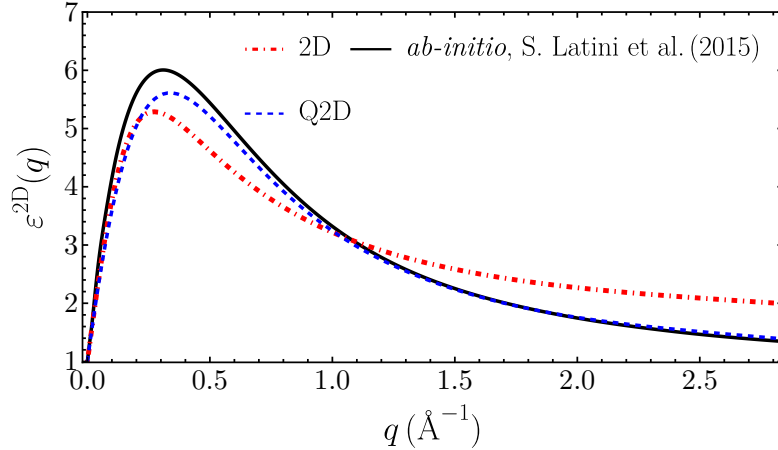


Figure 6.17: *Ab-initio* and quasi-2D effective dielectric function for MoS₂. Effective dielectric function of monolayer MoS₂ calculated with the strict 2D and quasi-2D approaches, in dashed red and dashed blue lines, respectively, in comparison with the *ab-initio* one, in the continuous black line. Our results agree better with the *ab-initio* one if we include only five reciprocal lattice vectors out of the seven verifying $|\mathbf{G}| < G_c = 3 \text{ \AA}^{-1}$ (including the origin). Two of them were discarded. The black curve was obtained by S. Latini et al. in Ref. [215] through *ab-initio* methods, applying a cutoff of 150 eV for the size of the dielectric matrix. In Ref. [215], the band structure of MoS₂ was obtained through DFT calculations within LDA. The dielectric function was calculated as a post-processing step. For the monolayer thickness, we used the value $d_{\perp} = 6.15 \text{ \AA}$ provided by the QEH package [176]. We compared the *ab-initio* dielectric function in black with the corresponding dielectric function computed through the QEH package, in similarity with what was done in Fig. 6.13, to observe that the two curves overlap almost exactly. Therefore, we do not present the curve from the QEH package for MoS₂, as it provides redundant information. For this model and set of parameters, we found $r_0 = 31.57 \text{ \AA}$.

The results for the excitons are displayed in Table 6.2, where one can see the excitonic levels and binding energies obtained with the two methods (in the caption, N is the size of the BZ mesh used to compute the excitons, and not the size of the auxiliary one BZ' to compute the polarizability matrix elements). We can see that using one potential or the other does not affect much the degeneracy of the states. Unlike in hBN (see Table 6.1), all the states have the expected degeneracy when using the Rytova–Keldysh potential. It is so because, as one can see in Table 6.2, the binding energies of MoS₂ are smaller than those of hBN in Table 6.1. This means that the excitons in MoS₂ are more extended in real space, and as a consequence, they are more localized in reciprocal space. Therefore, one needs fewer reciprocal lattice vectors to capture the expected symmetries and degeneracies for the excitonic states in MoS₂, at least in comparison with hBN. This is in agreement with the discussion and results by E. Ridolfi et al. in Ref. [150], and also with the results in Ref. [37].

From the information provided in the table below, we can compare our results with the *ab-initio* ones. Once again, the binding energy we get for the ground state of around 0.80 eV differs from the full *ab-initio* BSE result of 0.43 eV from Ref. [215]. They have used a different model for MoS₂ with a different band structure, though, so comparing the binding energies might not be fair. Nonetheless, the values we obtain for the ground state

energy of approximately 1.78 eV and 1.75 eV, for the Rytova–Keldysh potential and the numerical screened potential, respectively, are very close to the experimentally measured optical gap of 1.90 eV [20, 250]. Therefore, we believe that our results are quite reasonable.

Table 6.2: Exciton spectrum of MoS₂. Exciton spectrum of MoS₂ using the same Wannier model from 6.16, using the Rytova–Keldysh potential and the numerical inverse dielectric matrix to compute the interaction matrix elements. The second and fourth columns pertain to the Rytova–Keldysh model, while the third and fifth columns pertain to the numerical screening. The second and third columns show the exciton energies E_X , while the fourth and fifth columns show the corresponding binding energies E_b (in absolute value). States labeled with the same number but different letters are states for which degeneracy is expected. We used $N = 60^2$, $N_c = N_v = 1$, $G_c = 3.0 \text{ \AA}^{-1}$, $q_0 = 0.6k_0$, and $r_0 = 31.74 \text{ \AA}$ for the Rytova–Keldysh potential. In the numerical model for the screening, we included all the $\mathbf{G}$ vectors with $|\mathbf{G}| < G_c$. We excluded the exchange interaction term in both calculations, just like in Ref. [215], and set $\mathbf{Q} = \mathbf{0}$. When increasing the number of bands included to $N_c = N_v = 2$, the highest difference in binding energies is around 0.011 eV, and occurs for the state 2. All the other energies change by less than 0.01 eV. This was tested only for the numerical screening calculation.

State	E_X (RK) (eV)	E_X ($\varepsilon_{\mathbf{G}\mathbf{G}'}$) (eV)	E_b (RK) (eV)	E_b ($\varepsilon_{\mathbf{G}\mathbf{G}'}$) (eV)
1a	1.78	1.75	0.796	0.82
1b	1.78	1.76	0.793	0.82
2a	2.03	2.01	0.546	0.56
2b	2.03	2.03	0.542	0.55
3a	2.07	2.07	0.502	0.508
3b	2.07	2.07	0.502	0.508
4a	2.13	2.12	0.446	0.455
4b	2.13	2.12	0.445	0.455

To show the distribution of the excitonic states in momentum space, we plot in Fig. 6.18 below the squared absolute value of the $\mathbf{k}$ -resolved excitonic wavefunction for MoS₂. We plot the ground and first excited states in the top and bottom panels, respectively. We observed that the wavefunctions do not present any noticeable differences in their spreading when comparing degenerate eigenstates; 1a with 1b, and 2a with 2b. That is why we decided to display an excited state instead. This was the case for both calculations using the Rytova–Keldysh potential and the numerical one. The former are represented on the left side of the figure, while the latter are represented on the right side. The ones on the right side are more delocalized than the ones on the left side, which one can notice by looking at the amplitude scales (the plots themselves do not show any noticeable differences to the naked eye).

Finally, we shall examine the optical response of monolayer MoS₂ at the IPA and excitonic levels. In Fig. 6.19 below, we display the real part of the conductivity of MoS₂ when using the Wannier model Hamiltonian we have been using so far. For the excitonic conductivity, we display both conductivities obtained by using the Rytova–Keldysh and numerical models for screening, in the continuous blue and red lines, respectively. The noninteracting conductivity is shown in the dashed black line. Identical features to the ones already seen for hBN in Fig. 6.12 can be identified. For instance, the excitonic and independent-particle responses differ drastically. In the excitonic conductivity, we identify

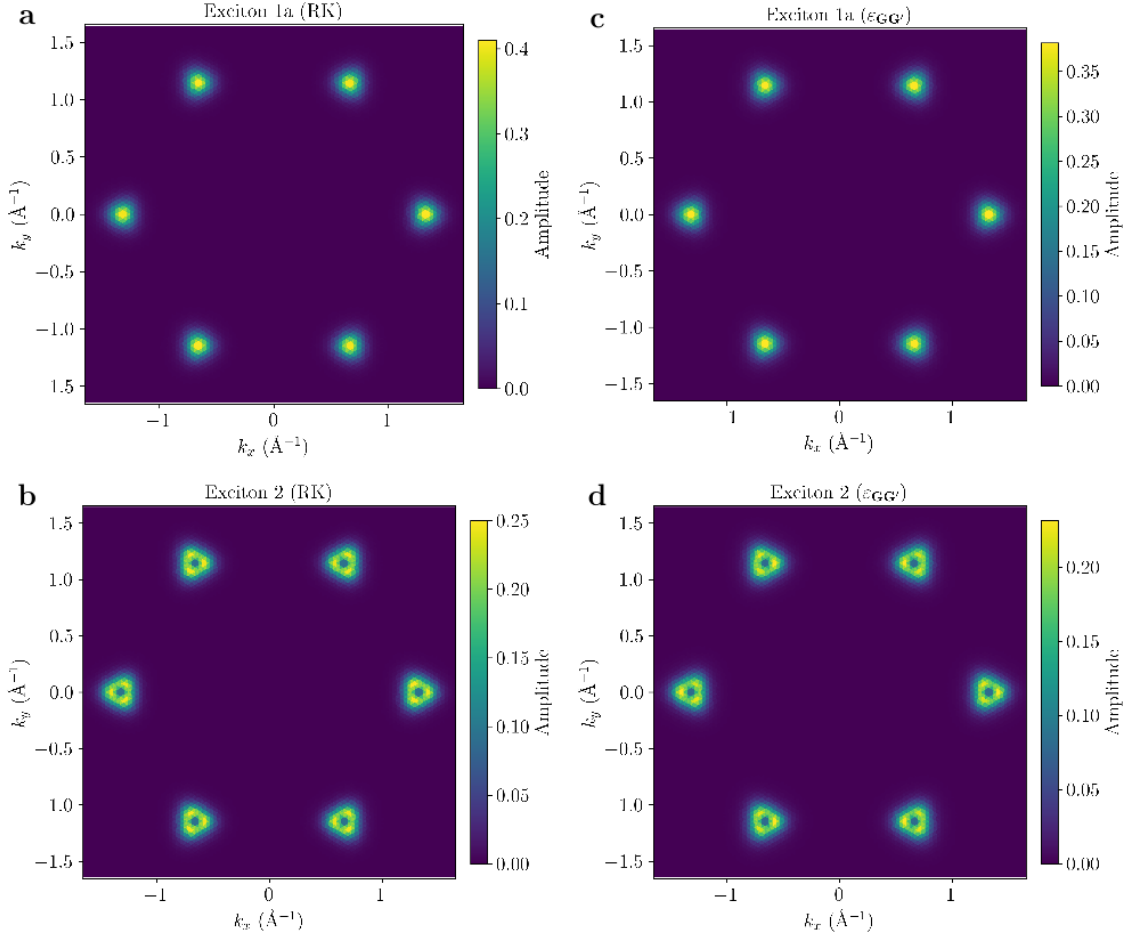


Figure 6.18: Wavefunctions of the ground and first excited excitonic states for MoS₂: Rytova–Keldysh vs. numerical screening. Momentum-resolved wavefunctions (in absolute value squared) of the ground and first excited excitonic states of MoS₂, using the Rytova–Keldysh potential on the left panels, and using the numerical dielectric function on the right panels. The top panels correspond to the 1a state from Table 6.2, while the bottom panels correspond to the 2a state in the same table. The states obtained using the numerical dielectric function present a lower maximum for the probability density, therefore those states are more delocalized in reciprocal space than the ones on the left. This is consistent with higher binding energies for the states obtained using $\epsilon_{\mathbf{GG}'}$, as we have seen in Table 6.2. Something identical has been seen for hBN in Fig. 6.11.

absorption peaks at resonant frequencies that correspond to the excitonic energy levels from Table 6.2. Just like in the case of hBN, the lowest peaks in energy display an enhancement from the Rytova–Keldysh model to the numerical one, while the third peak diminishes. At higher energies closer to the bandgap, both excitonic responses approach each other. We give just a final word for the noninteracting conductivity. One can clearly see that there is no response for energies higher than ~ 4.4 eV. This can be explained by looking at the band structure for MoS₂ in Fig. 6.16. Above an energy of ~ 4 eV, there are no allowed energies for single-particle (Bloch) states. We confirmed that this energy gap continues above the highest energy shown in the plot of the bands, even though we do not display higher energies. The electrons cannot be excited from the valence bands to Bloch states with those energies, because those states do not exist. Therefore, there is no optical

response at those frequencies, at least the response the IPA picture could capture.

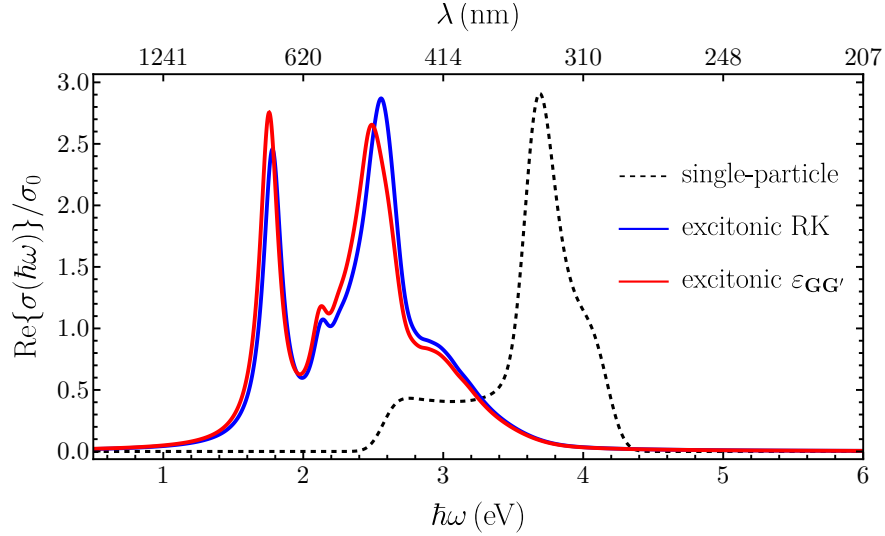


Figure 6.19: Optical conductivity of MoS₂: analytical and numerical screening. Optical conductivity of freestanding MoS₂ computed with the `Xatu` code. The IPA conductivity is represented by the dashed black curve. The excitonic conductivities corresponding to the Rytova–Keldysh model and the numerical model for screening are represented in the blue and red curves, respectively. We used for both excitonic conductivities a Lorentzian broadening with $\Gamma = 0.08$ eV.

All in all, we have verified that the numerical results obtained with the `Xatu` code agree quite well with the literature. The agreement between the 2D, quasi-2D, and *ab-initio* effective dielectric functions for MoS₂ has proved that our theory and implementation are in equal footing with first-principle approaches. The dielectric functions do not present an exact agreement, but that is expected because, even though the dielectric properties should be approximately the same for the same material, using different models will naturally lead to different numerical results. There must be an overall qualitative agreement, however.

Besides the validation of our theory against the *ab-initio* results, we would like to emphasize that the efficiency of the implementation constitutes another figure of merit of this work. In summary, we have seen with our numerical approach to the dielectric properties of gapped 2D materials that:

- the point-like (and orthogonal) orbital approximation allowed for computing the plane-wave matrix elements between Bloch states with negligible computational effort;
- an implementation that is 2D (or quasi-2D) in nature permits reducing the dimensionality and computational complexity of the problem;
- using a 2D (or quasi-2D) form for the Coulomb potential brings striking numerical benefits; the product $v_c(q)\chi_{00}^0(q)$ in the long-wavelength limit goes with $\sim \mathcal{O}(q)$, presenting thus no numerical instabilities; Furthermore, there is no need to use any tricks to curb spurious interactions between images repeated in the off-plane directions, such as truncating schemes for the bare Coulomb potential $v_c(\mathbf{r})$, and

increasing the amount of vacuum between layers; we do not have repeated images in our approach;

- the strict 2D approach agrees very well with the *ab-initio* (and quasi-2D) approaches in the very low q limit, or when $qd_{\perp} \lesssim 1$; this allows for a very fast and direct estimation of the screening length r_0 that agrees with the literature;
- in a 2D approach like ours, the size of the dielectric matrix $\varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q})$ scales much slower than in typical 3D approaches; all our $\mathbf{G}$ vectors are in-plane, since the system is not periodic along the z -direction;
- the previous point has the consequence that the problem of inverting the dielectric matrix poses no hindrance in the processing time;
- quite remarkably, by simply inverting the microscopic dielectric function, we can obtain directly the effective 2D dielectric function of the material; off-plane averaging schemes to convert the 3D inverse dielectric matrix into an effective 2D one are not needed;
- finally, the $1/q$ behavior of the unscreened Coulomb potential facilitates the regularization of the divergent elements of the dielectric function and the screened potentials (see Appendix C.2); the implementation of the interaction matrix elements in the BSE for obtaining the excitons is quite direct, since we can avoid numerical averages to treat the singular terms, contrary to Ref. [225].

It is surprising how notable our results are when compared to those from *ab-initio* approaches for extended systems, which are pretty much established in the literature [31–36], and typically very heavy computationally. Our implementation requires much less computational power. As seen before with the first version of the code in Ref. [37], the point-like orbital approximation did not bring any setback in the calculation of the excitons, at the time using only the Rytova–Keldysh potential. Herein, we realized that the same can be said about the dielectric function, as our results differ from the *ab-initio* ones by a small margin. However, for a better agreement for the excitonic states between our approach and *ab-initio* ones, a more in-depth study should be in order. In particular, the convergence of the energies with both cutoffs G_c and G_c^X , and also with the size of BZ, N . The latter is crucial, due to the sharp dependence of the dielectric function for low q , which hinders the convergence against N_k . The bigger r_0 is, the bigger is the problem.

We hope that this work will respark the interest of the computational condensed matter physics community in 2D materials. It also helped us to identify gaps in the literature when it comes to first-principle approaches to 2D materials. In principle, an extension of the code to perform efficient *GW* calculations could be done. Such efficient implementations could cause a speed up on the output from *ab-initio* calculations, and allow the treatment of excitons in bigger systems, like moiré materials [251]. Eventually, it could also cause the birth of a new research topic: 2D implementations of *ab-initio* methods. As far as we know,

and at the moment of the writing of this thesis, only one example for exciton calculations can be found in the literature [252].

Regarding research output and contributions to the literature, we have reached the end. It is time now for wrapping up in the next and final chapter of this dissertation, where we will take a look back, discuss novel ideas for future work, and draw main conclusions.

OUTLOOK AND CONCLUSIONS

As a final say in this dissertation, after all the concepts from the literature reviewed throughout the thesis, and applying them for studying light–matter interactions in semiconducting and insulating two-dimensional (2D) materials, there is pertinence in providing an overview of everything that was discussed. Throughout the course of the PhD studies, and as a byproduct of the research undertaken, many novel ideas naturally arose. However, there was not enough time to pursue them, but we provide a brief discussion.

Chapter 2 | Classical Electrodynamics of Continuous Media

This chapter had the purpose of providing a simple, but complete, technical introduction for the reader, consisting of the classical framework for electrodynamics in solids and continuous media. Several key concepts in this work were introduced, specifically those that can quantify light–matter interactions through material response functions. The physics related to electromagnetic phenomena at planar interfaces between different media was also discussed, with the intent of contextualizing the topic of 2D materials, the focus of this thesis. The dielectric function, optical conductivity, and their interrelation in optically responsive systems were specially emphasized and particularized for 2D materials. As these materials present many phenomena of interest at nanometric scales coming from quantum effects, the tools herein introduced were deemed insufficient. The necessary tools were provided in the two chapters that followed.

Chapter 3 | Quantum Mechanics of Electrons in Crystalline Solids

Chapter 3 dwelled on the basic principles of the quantum mechanics machinery, and applied to solids, allowing for describing physical properties at length scales of the order of the nanometer, or even smaller. Central concepts in the quantum theory of solids were mentioned, such as the Bloch theorem and Bloch states, first-principles and second-principles approaches for band structure calculations, and many-body perturbation theory. This last one offered a different point of view on the dielectric function. Within the formalism of quantum many-particle physics, one could describe response functions at the microscopic level, shedding light on the concepts introduced in Chapter 2. Nonetheless, an approximation had to be introduced for the dielectric function, the random-phase approximation (RPA), which permits computing the dielectric function in practice, albeit

at the noninteracting level. Doing so was postponed for the end of the thesis, only after introducing and studying excitons in the subsequent chapters.

Chapter 4 | Excitonic effects in Two-dimensional Semiconductors

The purpose of the previous chapter was actually twofold; provide the required concepts of quantum mechanics for a later stage in the thesis, and to set a background to prepare the reader for this chapter. It dealt with the exciton—a composite particle formed by a tightly bound electron–hole pair. With the tools from the previous chapter, we could describe excitons at different degrees of complexity. The simplest description was that of the Wannier equation, which is nothing more than the Schrödinger equation for the electron–hole pair. Although used later in this thesis, we already had in mind that this approach is not accurate enough for more complicated systems. Therefore, we introduced the Bethe–Salpeter equation (BSE) formalism. Two different degrees of approximation for it were given, both informed by the Hamiltonian of the material: first, using a low-energy description of the Hamiltonian, that allowed for recovering the Wannier equation; second, we introduced the exact diagonalization method, which is informed by the precise details of the Hamiltonian. In this way, the reader was well-informed to understand the research included in the next two chapters.

Chapter 5 | Optical Properties of Nanostructured hBN

In this chapter, we investigated the optical and excitonic properties of a nanostructured hexagonal boron nitride (hBN) monolayer, a 2D material simple enough so that we can easily describe excitons in it, but also interesting in its properties in bulk or single-layer form as the active element in experiments. This material has already been widely studied in the literature, both in theory and in experiments. It is so simple that we can infer its excitonic properties by using semi-analytical calculations. We proposed a way of taking advantage of its strong optical resonances by engineering the electronic dispersion, so that hBN can be made useful in a wide range of frequencies of interest. By applying an electrostatic periodic potential on hBN, we could induce simultaneously an increasing anisotropy in the low-energy dispersion and a diminished bandgap when one increases the strength of the potential. This could be predicted with an easy analytical expression for the renormalized low-energy Hamiltonian. Using the variational principle to get the excitonic energies and states with the Wannier equation, we saw that the excitons become less and less tightly bound as the strength of the potential grows. As a consequence for the optical conductivity, the oscillator strength for the excitonic resonances is enhanced with the increasing anisotropy. At the same time, the energies of the excitonic resonances in the optical absorption spectrum of hBN red-shift. Lastly, we could make predictions on the sub-gap photonic dispersion for the modified hBN monolayer, resultant from the strong coupling of the exciton with the electromagnetic (EM) field—the exciton–polariton. Interestingly, we saw that this light–matter hybrid presents a dispersion quite different from that of free light. The field confinement is much higher, of the order of the nanometer, and in the case of extreme anisotropy, the dispersion changes qualitatively, backbending at the resonance corresponding to the first excited state. This is in agreement with theoretical

predictions in the existing literature. We concluded that by nanostructuring hBN, we can tailor the nanophotonic properties of this material, making it more versatile for practical applications.

Chapter 6 | Electrostatic Screening in Two-dimensional materials

In this chapter, we started by revisiting the dielectric function from Chapter 3, but in two dimensions. We explained in detail how to compute it in practice. This had to be done numerically, since the problem of calculating the dielectric function cannot be done analytically, apart from some simple cases. In this thesis, we work with semiconducting and insulating materials, where local-field effects have to be taken into account. That makes this problem challenging. The dielectric function is now an infinite matrix, which has to be truncated and inverted to compute the screened Coulomb potential. Although these issues are not trivial, we found that the reduced dimensionality of the implementation allowed for obtaining results comparable with those from the literature in *ab-initio* methods, much more established in the literature, but for a fraction of the computational effort. Noticeably, the agreement between the static macroscopic dielectric function from our 2D approach agreed very well with the *ab-initio* one in the long-wavelength limit. By extending the 2D model to a quasi-2D one, where the thickness of the layer is accounted for in the dielectric function by adopting a mixed momentum-space representation, the results agree even better with *ab-initio* ones. The nonlocal behavior of the 2D effective dielectric function agrees with the Rytova–Keldysh one in the low momentum limit, and is accurately captured beyond that limit. These results were applied to the calculation of the excitons with the code, which was extended to incorporate the numerical screening in the Bethe–Salpeter Hamiltonian. The excitons did not completely agree with the literature, which reports ambiguous results coming from models and/or methods. Nonetheless, the results were still reasonable, and not very distant from any available experimental ones.



We investigated interesting problems that have natural extensions that sparked our curiosity. As always, we think more could have been done if we had more time. One question we wonder ourselves with is related to the interaction of the exciton with light in the quantum picture. The electron–hole complex eventually decouples after a certain time, and the exciton decays while emitting a photon. This could have been studied in the context of Chapter 5 [171], similarly to how it was done in Refs. [28, 253]. The radiative decay time can be calculated using Fermi’s golden rule [40, 69, 153], which involves a matrix element of the dipole operator as seen in Chapter 5, where the dipole approximation was assumed. However, in the literature there is an interest in the strong-coupling regime of light–matter interactions in cavities [254, 255]. In a cavity, the exciton radiative decay time depends on the exciton center-of-mass momentum [253], and we need to go beyond the dipole approximation. This has been done previously in Ref. [256], albeit for finite-size

systems. As far as we know, this has never been done for extended systems. In that case, one has to use the velocity gauge, and the matrix element reads $\langle X | \exp\{i\mathbf{Q} \cdot \mathbf{r}\} \mathbf{v} | 1_{\text{photon}} \rangle$. With our code we could compute it, by inserting a resolution of the identity in a Bloch basis between $\exp\{i\mathbf{Q} \cdot \mathbf{r}\}$ and $\mathbf{v}$, because we have the matrix elements of each factor. In turn, this idea also allows for computing the noninteracting and excitonic transverse optical conductivities for finite momentum, something we have never seen as well.

Naturally, with a quasi-2D approach for the dielectric function, we could extend the calculation of the screening and the excitons to the multilayer case. How to do that is explained in Appendix E. One can also implement the screening in real space with the noninteracting polarizability in real space as per Eq. (3.72), and compute the screened Coulomb potential directly in real space. This idea is discussed in detail in the PhD thesis of A. J. Uría-Álvarez in Ref. [132], Section 2.2.3 therein. At last but not the least, it is also possible to extend the code to compute the dielectric function in the dynamic limit and at finite temperatures, for both metallic and insulating 2D materials. A striking application could be graphene, studied extensively in the literature [41, 59, 61], a good material to compare with the results already known within plasmonics. We hope our research has sparked the curiosity of the reader on these topics. Moreover, we hope that it can serve as an inspiration for the community in general and reignite the research on 2D materials to push further the boundaries of knowledge.

This thesis was written using an adapted version of the original novathesis template, which can be found in Ref. [257].

THE FOURIER TRANSFORM: DEFINITIONS AND IDENTITIES

In this appendix, we summarize the definitions used for the Fourier transform and its inverse used throughout this thesis. The discussion presented here is adapted from Appendix A of Ref [74]. In the sense that at the end we consider as well the case of discrete translation symmetry, this discussion is also a generalization of the content presented by H. Bruus and K. Flensberg in [74].

A.1 Fourier transforms in time and frequency domains

We define the direct and inverse Fourier transforms between time and frequency domains as

$$f(t) = \mathcal{F}^{-1}\{f(\omega)\}(t) = \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} f(\omega) e^{-i\omega t}, \quad \text{and} \quad f(\omega) = \mathcal{F}\{f(t)\}(\omega) = \int_{-\infty}^{+\infty} dt f(t) e^{i\omega t}, \quad (\text{A.1})$$

respectively, where we used the symbols $\mathcal{F}$ and $\mathcal{F}^{-1}$ to denote Fourier transform and its inverse, respectively. The following relations involving the Dirac delta function are also valid:

$$\int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} f(\omega) e^{-i\omega t} = \delta(t), \quad \text{and} \quad \int_{-\infty}^{+\infty} dt f(t) e^{i\omega t} = \delta(\omega). \quad (\text{A.2})$$

A two-point function in time $f(t, t')$ can be written in terms of its Fourier components as

$$f(t, t') = \int_{-\infty}^{+\infty} d\omega \int_{-\infty}^{+\infty} d\omega' f(\omega, \omega') e^{-i\omega t} e^{i\omega' t'}. \quad (\text{A.3})$$

If this function enjoys continuous time translation symmetry as $f(t, t') = f(t - t')$, which is always the case in this work, then one can show that the Fourier transformed function can only be nonzero if $\omega = \omega'$. More than that, it is of the form $f(\omega, \omega') = \tilde{f}(\omega) \delta(\omega - \omega')$, and $\tilde{f}(\omega)$ is the Fourier transform of $f(t - t')$ when we consider the time difference $t - t'$ as a single variable. Therefore, we are allowed to write

$$f(t - t') = \int_{-\infty}^{+\infty} d\omega \tilde{f}(\omega) e^{-i\omega(t-t')}. \quad (\text{A.4})$$

A.2 Fourier transforms in real and reciprocal spaces

A.2.1 One-point functions

Let us consider first a system of finite volume $\mathcal{V}$ in real space with periodic, or Born-von Karman [56], boundary conditions. Then, any function $f(\mathbf{r})$ can be written in a Fourier series as

$$f(\mathbf{r}) = \frac{1}{\mathcal{V}} \sum_{\mathbf{k}} f_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}, \quad (\text{A.5})$$

with the momentum $\mathbf{k}$ assuming discrete values, and

$$f_{\mathbf{k}} = \int_{\mathcal{V}} d\mathbf{r} f(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}. \quad (\text{A.6})$$

The following relations hold

$$\int_{\mathcal{V}} d\mathbf{r} f(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}} = \mathcal{V} \delta_{\mathbf{k},0} \quad \text{and} \quad \frac{1}{\mathcal{V}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta(\mathbf{r}), \quad (\text{A.7})$$

where $\delta_{\mathbf{k},\mathbf{q}}$ is the Kronecker delta.

If we let the volume of the region $\mathcal{V} \rightarrow \infty$ tend to infinity, then $\mathbf{k}$ assumes continuous values, and the sums over momentum transform to integrals by the rule $\sum_{\mathbf{k}} \rightarrow \frac{\mathcal{V}}{(2\pi)^3} \int d\mathbf{k}$. In this way, relations (A.5) and (A.6) can be written as

$$f(\mathbf{r}) = \int \frac{d\mathbf{k}}{(2\pi)^3} f(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} \quad \text{and} \quad f(\mathbf{k}) = \int d\mathbf{r} f(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}, \quad (\text{A.8})$$

respectively, and both relations below

$$\int \frac{d\mathbf{k}}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta(\mathbf{r}) \quad \text{and} \quad \int d\mathbf{r} e^{-i\mathbf{k}\cdot\mathbf{r}} = (2\pi)^3 \delta(\mathbf{k}), \quad (\text{A.9})$$

hold as well. If the system under consideration is two-dimensional (2D) and not three-dimensional (3D), then instead of the volume $\mathcal{V}$ we have the area $\mathcal{A}$, the momentum $\mathbf{k}$ is a 2D vector, the integrals over volume become integrals over area, and whenever we have $(2\pi)^3$, we would have $(2\pi)^2$ instead.

A.2.2 Two-point functions

Just like in the time domain example, there can also be real space resolved two-point functions. For a general function $f(\mathbf{r}, \mathbf{r}')$ where the position is 3D and defined in a region of infinite volume, its inverse Fourier transform writes

$$f(\mathbf{r}, \mathbf{r}') = \int \frac{d\mathbf{k}}{(2\pi)^3} \int \frac{d\mathbf{k}'}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} f(\mathbf{k}, \mathbf{k}') = \int \frac{d\mathbf{k}}{(2\pi)^3} \tilde{f}(\mathbf{k}) e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}, \quad (\text{A.10})$$

where the second equality holds for translation invariant systems, and imposing $f(\mathbf{k}, \mathbf{k}') = \delta(\mathbf{k} - \mathbf{k}') \tilde{f}(\mathbf{k})$. The direct Fourier transform then writes

$$\tilde{f}(\mathbf{k}) = \int d\mathbf{r} e^{-i\mathbf{k}\cdot\mathbf{r}} f(\mathbf{r}), \quad (\text{A.11})$$

by choosing $\mathbf{r}' = \mathbf{0}$ without loss of generality. The finite volume case writes

$$f(\mathbf{r}, \mathbf{r}') = \frac{1}{\mathcal{V}^2} \sum_{\mathbf{k}, \mathbf{k}'} e^{i\mathbf{k}\cdot\mathbf{r}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} f(\mathbf{k}, \mathbf{k}') = \frac{1}{\mathcal{V}} \sum_{\mathbf{k}} \tilde{f}(\mathbf{k}) e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}, \quad (\text{A.12})$$

where the second equality is for systems with translation invariance, and identifying $f(\mathbf{k}, \mathbf{k}') = \mathcal{V} \delta_{\mathbf{k}, \mathbf{k}'} \tilde{f}(\mathbf{k})$.

We give a typical, and very important example; the Fourier transform of the Coulomb potential. The derivations are skipped, and only the final results will be stated. The Fourier transform of the Coulomb potential v_c , that in electron-Volt and in real space reads

$$v_c(\mathbf{r} - \mathbf{r}') = \frac{1}{4\pi\epsilon_0} \frac{e}{|\mathbf{r} - \mathbf{r}'|}, \quad (\text{A.13})$$

has different expressions in reciprocal space, depending on the dimensionality of the system. Designating by $k = |\mathbf{k}|$ the length of the vector $\mathbf{k}$, in 3D, the momentum-resolved Coulomb potential is given by

$$v_c(\mathbf{k}) = \int d^3\mathbf{r} \frac{e}{4\pi\epsilon_0|\mathbf{r}|} e^{-i\mathbf{k}\cdot\mathbf{r}} = \frac{e}{\epsilon_0 k^2}, \quad (\text{A.14})$$

while in 2D it assumes the form

$$v_c(\mathbf{k}) = \int d^2\mathbf{r} \frac{e}{4\pi\epsilon_0|\mathbf{r}|} e^{-i\mathbf{k}\cdot\mathbf{r}} = \frac{e}{2\epsilon_0 k}. \quad (\text{A.15})$$

We remark that due to the definitions used in this work, the Coulomb potential, unlike in real space, does not have units of energy in reciprocal space. Nonetheless, the final quantities to compute will always have the right units.

If the system enjoys discrete translation symmetry instead of continuous translation symmetry, when we want to describe solids microscopically, an expression similar to (A.12) can still be worked out. But for that, we need to use the concepts from Chapter 3, namely that of Brillouin zone (BZ) and reciprocal lattice vectors. If the system enjoys discrete translational symmetry, one can prove that $f(\mathbf{k}, \mathbf{k}') \neq 0$ only if $\mathbf{k}' = \mathbf{k} + \mathbf{G}$, where $\mathbf{G}$ is a vector of the reciprocal lattice. Besides that, using the fact that any momentum $\mathbf{q}$ in the reciprocal lattice can be written as $\mathbf{q} = \mathbf{k} + \mathbf{G}$, where $\mathbf{k}$ is a vector in the BZ, we can rewrite the sums over momentum in (A.12) as

$$\begin{aligned}
 f(\mathbf{r}, \mathbf{r}') &= \frac{1}{\mathcal{V}^2} \sum_{\mathbf{k}, \mathbf{k}' \in \text{BZ}} \sum_{\mathbf{G}, \mathbf{G}'} f(\mathbf{k} + \mathbf{G}, \mathbf{k}' + \mathbf{G}') e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} e^{i\mathbf{G} \cdot \mathbf{r}} e^{-i\mathbf{G}' \cdot \mathbf{r}'} = \\
 &= \frac{1}{\mathcal{V}} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{G}, \mathbf{G}'} f(\mathbf{k} + \mathbf{G}, \mathbf{k} + \mathbf{G}') e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} e^{i\mathbf{G} \cdot \mathbf{r}} e^{-i\mathbf{G}' \cdot \mathbf{r}'},
 \end{aligned} \tag{A.16}$$

if we identify $f(\mathbf{k} + \mathbf{G}, \mathbf{k}' + \mathbf{G}') = \mathcal{V} \delta_{\mathbf{k}, \mathbf{k}'} f(\mathbf{k} + \mathbf{G}, \mathbf{k}' + \mathbf{G}')$.

SUPPORT INFORMATION FOR CHAPTER 5

This appendix serves as supplemental material for the work in Chapter 5 of the main text. Just like in that chapter, the contents of this appendix are adapted from the publication in Ref. [171].

B.1 Recovering the original Hamiltonian

To conciliate the renormalized low-energy Hamiltonian (5.29) with the original one (3.32), we take in the former the two limits $V_0 \rightarrow 0$ and $L \rightarrow \infty$ separately. In both cases, we should recover the latter. In this appendix, we choose for the momentum q as in the Chapter 3, instead of k as in the Chapter (5). We take first the limit $V_0 \rightarrow 0$ to obtain

$$\bar{H}_m = \frac{m}{2}G_0\hbar v_F + \hbar v_F\sigma_x q_x \quad (\text{B.1})$$

for $m \neq 0$, and

$$\bar{H} = \hbar v_F[\sigma_x q_x + \sigma_y q_y] - \sigma_z M \quad (\text{B.2})$$

for $m = 0$. The latter case is simply the original low-energy Hamiltonian of hBN in the absence of potential as expected. There is only a difference in the sign of the mass term, but that is not critical. The result for $m \neq 0$ is not null, but we can interpret the two terms on the right-hand side as follows. The first term is purely a global shift in energies resultant from shifting the momentum along the x direction by $mG_0/2$, a consequence of using basis states with the spatial dependent part of the form $\exp\{i(\mathbf{q} \pm mG_0\hat{\mathbf{x}}/2)\}$ used when deriving the renormalized low-energy Hamiltonian. The second term comes from the measurement of momentum from the edges of the mini Brillouin zone along the x direction, which no longer exists since $V_0 = 0$. Therefore, it is not relevant.

We take now the limit $L \rightarrow \infty$, that is, the limit of a constant potential. For each value of m we have a Hamiltonian, but if we gather them all by summing over m , we get

$$\begin{aligned}
 H_{\text{total}} &= \sum_m \overline{H}_m = \\
 &= \sum_m \frac{m}{2} G_0 \hbar v_F + \hbar v_F [\sigma_x q_x + \sigma_y q_y \sum_m J_m(\beta)] - \sigma_z M \sum_m J_m(\beta) = \\
 &\stackrel{L \rightarrow \infty}{=} \hbar v_F [\sigma_x q_x + \sigma_y q_y] - \sigma_z M,
 \end{aligned} \tag{B.3}$$

where, to arrive at the final result, we used the fact that $G_0 \rightarrow 0$ when $L \rightarrow \infty$ and the rule $\sum_m J_m(\beta) = 1$, valid for the Bessel functions of the first kind. In this way, we are able to recover the original low-energy Hamiltonian of hBN (with the small caveat of the sign in the mass term once more).

B.2 Exchange energy correction to the bandgap

We recall the Bethe–Salpeter equation:

$$\begin{aligned}
 E_\nu \frac{1}{\sqrt{A}} \phi_\nu(\mathbf{k}) &= \phi_\nu(\mathbf{k}) (E_{\mathbf{k},c} - E_{\mathbf{k},v}) - \frac{1}{A} \sum_{\mathbf{p}} \phi_\nu(\mathbf{p}) V_{\mathbf{k}-\mathbf{p}} F_{cvvc}(\mathbf{k}, \mathbf{p}, \mathbf{k}-\mathbf{p}) + \\
 &+ \frac{1}{A} \phi_\nu(\mathbf{k}) \sum_{\mathbf{p}} V_{\mathbf{p}} [F_{vvvv}(\mathbf{p} + \mathbf{k}, \mathbf{k}, \mathbf{p}) - F_{vcvc}(\mathbf{p} + \mathbf{k}, \mathbf{k}, \mathbf{p})].
 \end{aligned} \tag{B.4}$$

We wish to compute the contribution to the bandgap due to the exchange energy coming from the third term in the right-hand side of the previous equation. We assume that this term renormalizes strongly the electronic bandgap but has no effect on the curvature of the bands. So we start from that term with $\mathbf{k} = 0$. We denote the exchange energy by Δ_{ex} and it writes

$$\Delta_{\text{ex}} = \frac{1}{A} \sum_{\mathbf{p}} V_{\mathbf{p}} [F_{vvvv}(\mathbf{p}, 0, \mathbf{p}) - F_{vcvc}(\mathbf{p}, 0, \mathbf{p})]. \tag{B.5}$$

The first form factor is

$$F_{vvvv}(\mathbf{p}, 0, \mathbf{p}) = \langle v, 0 | v, \mathbf{p} \rangle \langle v, \mathbf{p} | v, 0 \rangle = |\langle v, 0 | v, \mathbf{p} \rangle|^2. \tag{B.6}$$

The overlap of the spinors (see Eqs. (5.37) and (5.36)) reads

$$\langle v, 0 | v, \mathbf{p} \rangle = \cos \frac{\xi_p}{2}, \quad \text{with} \quad \xi_p = \arctan \frac{2\hbar v_F p}{E_g}. \tag{B.7}$$

This gives for the form factor

$$F_{vvvv}(\mathbf{p}, 0, \mathbf{p}) = \cos^2 \frac{\xi_p}{2}. \tag{B.8}$$

The second form factor is

$$F_{vcvc}(\mathbf{p}, 0, \mathbf{p}) = \langle c, 0 | v, \mathbf{p} \rangle \langle v, \mathbf{p} | c, 0 \rangle = |\langle c, 0 | v, \mathbf{p} \rangle|^2. \quad (\text{B.9})$$

The spinors overlap (up to a phase) is now

$$\langle c, 0 | v, \mathbf{p} \rangle = \sin \frac{\xi_p}{2}, \quad (\text{B.10})$$

hence

$$F_{vcvc}(\mathbf{p}, 0, \mathbf{p}) = \sin^2 \frac{\xi_p}{2}. \quad (\text{B.11})$$

Now we take the difference between the form factors

$$\begin{aligned} F_{vvvv}(\mathbf{p}, 0, \mathbf{p}) - F_{vcvc}(\mathbf{p}, 0, \mathbf{p}) &= \cos^2 \frac{\xi_p}{2} - \sin^2 \frac{\xi_p}{2} = \cos \xi_p = \cos \left(\arctan \frac{2\hbar v_F p(\beta)}{J_0(\beta) E_g} \right) = \\ &= \frac{1}{\sqrt{1 + \frac{p^2 \gamma^2(\theta)}{k_0^2}}}, \end{aligned} \quad (\text{B.12})$$

where we defined the characteristic wavenumber

$$k_0 \equiv |J_0(\beta)| \frac{E_g}{2\hbar v_F}. \quad (\text{B.13})$$

Using the auxiliary function $\gamma(\theta) = \sqrt{\cos^2 \theta + J_0^2(\beta) \sin^2 \theta}$, the exchange energy now reads

$$\Delta_{\text{ex}} = \frac{1}{\mathcal{A}} \sum_{\mathbf{p}} V_{\mathbf{p}} \frac{1}{\sqrt{1 + \left(\frac{p\gamma(\theta)}{k_0} \right)^2}} = \frac{e}{8\pi^2 \varepsilon_0 \kappa} \int_0^{2\pi} d\theta \int_0^\infty dp \frac{1}{(1 + r_0 p / \kappa) \sqrt{1 + \left(\frac{p\gamma(\theta)}{k_0} \right)^2}}. \quad (\text{B.14})$$

where we converted the sum into an integral as $\sum_{\mathbf{p}} \rightarrow \mathcal{A}/(2\pi)^2 \int d\mathbf{p}$, and used for the Rytova–Keldysh potential

$$V_{\mathbf{p}} = \frac{e}{2\varepsilon_0 p(\kappa + r_0 p)}. \quad (\text{B.15})$$

with κ being the arithmetic average between the dielectric constants of the substrate and the cladding material.

To perform the final integral, we make the change of variable $p\gamma(\theta)/k_0 = \tan u$. The differential becomes $dp = k_0 d(\tan u) / \gamma(\theta) = k_0 \frac{1}{\cos^2 u} / \gamma(\theta) du$, and the square root in the denominator transforms into $\sqrt{1 + p^2 \gamma^2(\theta) / k_0^2} = \sqrt{1 + \tan^2 u} = 1 / \cos u$. When $p = 0$ we have $u = 0$, and when $p = \infty$ we have $u = \pi/2$. Furthermore, we introduce a new dimensionless parameter $\gamma \equiv r_0 k_0$, so that the integral becomes now

$$k_0 \int_0^{2\pi} \frac{1}{\gamma(\theta) \sqrt{1 + \alpha^2(\theta)}} \ln \left(\frac{1 + \alpha(\theta) + \sqrt{1 + \alpha^2(\theta)}}{1 + \alpha(\theta) - \sqrt{1 + \alpha^2(\theta)}} \right) d\theta. \quad (\text{B.16})$$

Finally, we arrive at the expression for the exchange energy

$$\Delta_{\text{ex}} = \frac{ek_0}{8\pi^2\varepsilon_0\kappa} \int_0^{2\pi} \frac{1}{\gamma(\theta)\sqrt{1+\alpha^2(\theta)}} \ln\left(\frac{1+\alpha(\theta)+\sqrt{1+\alpha^2(\theta)}}{1+\alpha(\theta)-\sqrt{1+\alpha^2(\theta)}}\right) d\theta, \quad (\text{B.17})$$

which can be evaluated numerically without any computational effort.

SUPPORT INFORMATION TO CHAPTER 6

C.1 Static Noninteracting Polarizability and Time-Reversal Symmetry

In this section we derive expression (6.7) in Section 6.1.1 from its more general form (6.6), assuming the system is time-reversal symmetric. In a periodic system with no contribution to the Hamiltonian that violates time-reversal symmetry (TRS), there exists a unitary transformation T such that

$$TH^*(-\mathbf{k})T^\dagger = H(\mathbf{k}) \quad (\text{C.1})$$

is verified for the Bloch Hamiltonian $H(\mathbf{k})$. Apart from systems with spin-orbit coupling (SOC), our Hamiltonian is diagonal in the spin. If there is spin, then the transformation of TRS is slightly more complicated. However, we refrain from including the spin and will simply ignore it. Either way, we have verified numerically that for models where SOC is present, considering or ignoring the spin makes no relevant difference in the calculation of the polarizability. Also, for time-reversal invariant systems, the quasiparticle energies naturally obey $\epsilon_{n\mathbf{k}} = \epsilon_{n-\mathbf{k}}$.

The expression (6.6) in the main text for the polarizability reads

$$\begin{aligned} \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}) = & \frac{1}{\mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \left[\frac{\langle c, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | v, \mathbf{k} + \mathbf{q} \rangle \langle v, \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | c, \mathbf{k} \rangle}{\epsilon_{v\mathbf{k}+\mathbf{q}} - \epsilon_{c\mathbf{k}}} \right] + \\ & \frac{1}{\mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \left[\frac{\langle v, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | c, \mathbf{k} + \mathbf{q} \rangle \langle c, \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | v, \mathbf{k} \rangle}{\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}}} \right]. \end{aligned} \quad (\text{C.2})$$

Now, we show that the second term on the right-hand side of the previous expression equals the first one. We first recall that the sum over the momentum $\mathbf{k}$ is over momenta in the BZ, which is centrosymmetric. Therefore, we can replace $\mathbf{k} \rightarrow -\mathbf{k}$ and then shift $\mathbf{k} \rightarrow \mathbf{k} + \mathbf{q}$ to obtain

$$\begin{aligned}
 \sum_{vc, \mathbf{k}\sigma} \frac{\langle v, -\mathbf{k} - \mathbf{q} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | c, -\mathbf{k} \rangle \langle c, -\mathbf{k} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | v, -\mathbf{k} - \mathbf{q} \rangle}{\epsilon_{v-\mathbf{k}-\mathbf{q}} - \epsilon_{c-\mathbf{k}}} = \\
 \sum_{vc, \mathbf{k}\sigma} \frac{\langle v, -\mathbf{k} - \mathbf{q} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | c, -\mathbf{k} \rangle \langle c, -\mathbf{k} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | v, -\mathbf{k} - \mathbf{q} \rangle}{\epsilon_{v\mathbf{k}+\mathbf{q}} - \epsilon_{c\mathbf{k}}}, \quad (C.3)
 \end{aligned}$$

where we used TRS to write the equality, namely $\epsilon_{n\mathbf{k}} = \epsilon_{n-\mathbf{k}}$. Now, we only need some sort of symmetry for the plane-wave matrix elements of the form $\langle n, \mathbf{k} | e^{-i\mathbf{q}\cdot\mathbf{r}} | n', \mathbf{k}' \rangle$. For that, we recall here their expression in terms of the eigenvectors of the Bloch Hamiltonian that we denote by $\mathbf{C}^{n\mathbf{k}}$. Being eigenvectors of the Bloch Hamiltonian, the tight-binding coefficients obey

$$\begin{aligned}
 H(\mathbf{k})\mathbf{C}^{n\mathbf{k}} = \epsilon_{n\mathbf{k}}\mathbf{C}^{n\mathbf{k}} &\Leftrightarrow H(-\mathbf{k})\mathbf{C}^{n-\mathbf{k}} = \epsilon_{n-\mathbf{k}}\mathbf{C}^{n-\mathbf{k}} \Leftrightarrow \\
 &\Leftrightarrow H^*(-\mathbf{k})(\mathbf{C}^{n-\mathbf{k}})^* = \epsilon_{n\mathbf{k}}(\mathbf{C}^{n-\mathbf{k}})^* \Leftrightarrow TH^*(-\mathbf{k})T^\dagger T(\mathbf{C}^{n-\mathbf{k}})^* = \epsilon_{n-\mathbf{k}}T(\mathbf{C}^{n-\mathbf{k}})^* \Leftrightarrow \\
 &\Leftrightarrow H(\mathbf{k})T(\mathbf{C}^{n-\mathbf{k}})^* = \epsilon_{n\mathbf{k}}T(\mathbf{C}^{n-\mathbf{k}})^*, \quad (C.4)
 \end{aligned}$$

where, to get the last equality, we used the identities verified by $H(\mathbf{k})$ and $\epsilon_{n\mathbf{k}}$ introduced above. So, in principle, we have $T(\mathbf{C}^{n-\mathbf{k}})^* = \mathbf{C}^{n\mathbf{k}}$. For the matrix elements of interest, we introduce the following notation:

$$\langle n, \mathbf{k} | e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} | n', \mathbf{k}' \rangle = \sum_{i\alpha} (\mathbf{C}_{i\alpha}^{n\mathbf{k}})^* C_{i\alpha}^{n'\mathbf{k}'} e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{t}_i} \equiv (\mathbf{C}^{n\mathbf{k}})^\dagger \cdot (\mathbf{C}^{n'\mathbf{k}'} \odot \underline{\varphi}), \quad (C.5)$$

where the vector $\underline{\varphi}$ has by entries the phases $e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{t}_i}$, that is, $(\underline{\varphi})_{i\alpha} = e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{t}_i}$, and $\odot$ is used to denote element-wise product, usually called Hadamard product in the literature. Then, we only need to employ the identity $T(\mathbf{C}^{n-\mathbf{k}})^* = \mathbf{C}^{n\mathbf{k}}$ to obtain

$$\begin{aligned}
 \langle n, \mathbf{k} | e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} | n', \mathbf{k}' \rangle &= (\mathbf{C}^{n\mathbf{k}})^\dagger \cdot (\mathbf{C}^{n'\mathbf{k}'} \odot \underline{\varphi}) = \left((\mathbf{C}^{n-\mathbf{k}})^T T^\dagger \right) \cdot \left((T(\mathbf{C}^{n'-\mathbf{k}'})^*) \odot \underline{\varphi} \right) = \\
 &= (\mathbf{C}^{n-\mathbf{k}})^T (T^\dagger T) \left((\mathbf{C}^{n'-\mathbf{k}'})^* \odot \underline{\varphi} \right) = (\mathbf{C}^{n'-\mathbf{k}'})^\dagger \cdot (\mathbf{C}^{n-\mathbf{k}} \odot \underline{\varphi}) = \\
 &= \langle n', -\mathbf{k}' | e^{-i(\mathbf{k}'-\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} | n, -\mathbf{k} \rangle, \quad (C.6)
 \end{aligned}$$

with T denoting transpose. In this way, the second term in the polarizability can be rewritten as

$$\begin{aligned}
 \sum_{vc, \mathbf{k}\sigma} \frac{\langle v, -\mathbf{k} - \mathbf{q} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | c, -\mathbf{k} \rangle \langle c, -\mathbf{k} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | v, -\mathbf{k} - \mathbf{q} \rangle}{\epsilon_{v\mathbf{k}+\mathbf{q}} - \epsilon_{c\mathbf{k}}} = \\
 \sum_{vc, \mathbf{k}\sigma} \frac{\langle c, \mathbf{k} | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | v, \mathbf{k} + \mathbf{q} \rangle \langle v, \mathbf{k} + \mathbf{q} | e^{i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}} | c, \mathbf{k} \rangle}{\epsilon_{v\mathbf{k}+\mathbf{q}} - \epsilon_{c\mathbf{k}}}, \quad (C.7)
 \end{aligned}$$

which concludes our proof. This identity has also been verified numerically.

C.2 Regularization of the dielectric function

In this appendix, we explain in detail how the divergences of the dielectric matrix in the long-wavelength limit $\mathbf{q} \rightarrow \mathbf{0}$ are treated in this work. As commonly done in the literature, different matrix elements of the dielectric matrix have different treatments, namely the head $\varepsilon_{\mathbf{00}}$, the wing $\varepsilon_{\mathbf{G0}}$ and $\varepsilon_{\mathbf{0G}}$, and body $\varepsilon_{\mathbf{GG}'}$ matrix elements, with $\mathbf{G}, \mathbf{G}' \neq \mathbf{0}$. As discussed in the main text, we adopt the symmetrized dielectric matrix defined by

$$\varepsilon_{\mathbf{GG}'}(\mathbf{q}) = \delta_{\mathbf{GG}'} - \sqrt{v_c(\mathbf{q} + \mathbf{G})} \chi_{\mathbf{GG}'}^0(\mathbf{q}) \sqrt{v_c(\mathbf{q} + \mathbf{G}')}, \quad (\text{C.8})$$

which eases the technical issues related to the long-wavelength limit, as we explain next. We have already seen, in subsection 6.1.2, that for the head element it does not matter whether we symmetrize the dielectric function or not. This is because

$$\begin{aligned} \varepsilon_{\mathbf{00}}(\mathbf{q} \rightarrow \mathbf{0}) &= 1 - \sqrt{v_c(\mathbf{q} \rightarrow \mathbf{0})} \chi_{\mathbf{00}}^0(\mathbf{q} \rightarrow \mathbf{0}) \sqrt{v_c(\mathbf{q} \rightarrow \mathbf{0})} = 1 - v_c(\mathbf{q} \rightarrow \mathbf{0}) \chi_{\mathbf{00}}^0(\mathbf{q} \rightarrow \mathbf{0}) = \\ &= 1 - \frac{e}{2\varepsilon_0 q} \mathbf{q} \cdot \mathbf{B} \mathbf{q} \rightarrow 1, \end{aligned} \quad (\text{C.9})$$

where the last equality holds for very small $\mathbf{q}$, and $\mathbf{B}$ is the Hessian matrix of the function $\chi_{\mathbf{00}}^0(q_x, q_y)$ at the point $(q_x, q_y) = (0, 0)$. The Hessian, here a two by two matrix, does not depend on the small vector $\mathbf{q}$, because the function $\chi_{\mathbf{00}}^0(\mathbf{q})$ is analytic in $\mathbf{q}$ around the origin according to Pick, Cohen, and Martin in [258]. In the same paper, they explain that all the functions $\chi_{\mathbf{GG}'}^0(\mathbf{q})$, for any $\mathbf{G}$ and $\mathbf{G}'$ vectors, are analytic at $\mathbf{q} = \mathbf{0}$. In other words, the functions $\chi_{\mathbf{GG}'}^0(\mathbf{q})$ can be expanded around the origin in powers of $\mathbf{q}$. For our analysis, we only need to expand it up to second order in Taylor series. Therefore, we can write for the wing terms

$$\begin{aligned} \varepsilon_{\mathbf{G0}}(\mathbf{q} \rightarrow \mathbf{0}) &= -\sqrt{v_c(\mathbf{G})} \chi_{\mathbf{G0}}^0(\mathbf{q} \rightarrow \mathbf{0}) \sqrt{v_c(\mathbf{q} \rightarrow \mathbf{0})} = \\ &= -\frac{e}{2\varepsilon_0 \sqrt{|\mathbf{G}|} \sqrt{q}} \mathbf{q} \cdot \mathbf{A}(\mathbf{G}) \propto \sqrt{q} \rightarrow 0, \end{aligned} \quad (\text{C.10})$$

$$\begin{aligned} \varepsilon_{\mathbf{0G}}(\mathbf{q} \rightarrow \mathbf{0}) &= -\sqrt{v_c(\mathbf{q} \rightarrow \mathbf{0})} \chi_{\mathbf{0G}}^0(\mathbf{q} \rightarrow \mathbf{0}) \sqrt{v_c(\mathbf{G})} = \\ &= -\frac{e}{2\varepsilon_0 \sqrt{|\mathbf{G}|} \sqrt{q}} \mathbf{q} \cdot \mathbf{A}(\mathbf{G})^* \propto \sqrt{q} \rightarrow 0, \end{aligned} \quad (\text{C.11})$$

where $\mathbf{A}(\mathbf{G})$ is the Jacobian matrix (a vector, actually) of the function $\chi_{\mathbf{G0}}^0(\mathbf{q})$ at $\mathbf{q} = \mathbf{0}$, which does not depend on $\mathbf{q}$ due to $\chi_{\mathbf{G0}}^0(\mathbf{q})$ being analytic, as mentioned above.

If we were to use the nonsymmetric dielectric matrix, then the wing terms $\varepsilon_{\mathbf{G0}}$ would also tend to zero as $\mathbf{q} \rightarrow \mathbf{0}$, but with q instead. On the contrary, the terms $\varepsilon_{\mathbf{0G}}$ would tend to $\varepsilon_{\mathbf{0G}}(\mathbf{q} \rightarrow \mathbf{0}) \rightarrow -e \hat{\mathbf{q}} \cdot \mathbf{A}(\mathbf{G})^* / 2\varepsilon_0$, which is finite and depends on the direction along which one takes the limit $\mathbf{q} \rightarrow \mathbf{0}$. One can see the usefulness of using the symmetrized dielectric matrix, making it easier to implement numerically. We just set $\varepsilon_{\mathbf{0G}}(\mathbf{0}) = \varepsilon_{\mathbf{G0}}(\mathbf{0}) = 0$, leaving eventual long-wavelength regularizations for later when dealing with the screened potential.

Now, we investigate the behavior of the inverse dielectric matrix for near-zero $\mathbf{q}$, in the light of Refs. [225] and [258], by writing the dielectric matrix as a matrix by blocks. We separate the head, the wing, and the body terms, which we represent by $\varepsilon_{\mathbf{00}}$, $\varepsilon_{\mathbf{G0}}$ and $\varepsilon_{\mathbf{0G}}$, and $B_{\mathbf{GG}'}$, respectively. Skipping the details of the inversion, we write directly

$$\varepsilon_{\mathbf{00}}^{-1} = \left[\varepsilon_{\mathbf{00}} - \sum_{\mathbf{G}, \mathbf{G}' \neq \mathbf{0}} \varepsilon_{\mathbf{0G}} B_{\mathbf{GG}'}^{-1} \varepsilon_{\mathbf{G'0}} \right]^{-1}, \quad (\text{C.12})$$

$$\varepsilon_{\mathbf{G0}}^{-1} = -\varepsilon_{\mathbf{00}}^{-1} \sum_{\mathbf{G}' \neq \mathbf{0}} B_{\mathbf{GG}'}^{-1} \varepsilon_{\mathbf{G'0}}, \quad (\text{C.13})$$

$$\varepsilon_{\mathbf{0G}}^{-1} = -\varepsilon_{\mathbf{00}}^{-1} \sum_{\mathbf{G}' \neq \mathbf{0}} \varepsilon_{\mathbf{0G}'} B_{\mathbf{G'G}}^{-1}, \quad (\text{C.14})$$

$$\varepsilon_{\mathbf{GG}'}^{-1} = B_{\mathbf{GG}'}^{-1} + \varepsilon_{\mathbf{00}}^{-1} \left(\sum_{\mathbf{G}'' \neq \mathbf{0}} B_{\mathbf{GG}''}^{-1} \varepsilon_{\mathbf{G''0}} \right) \left(\sum_{\mathbf{G}'' \neq \mathbf{0}} \varepsilon_{\mathbf{0G}''} B_{\mathbf{G''G}'}^{-1} \right). \quad (\text{C.15})$$

Note that in the previous relations, $B_{\mathbf{GG}'}^{-1}$ designates the $(\mathbf{G}, \mathbf{G}')$ matrix element of the inverse of the body, and not the inverse of the body element $B_{\mathbf{GG}'}$. Given the properties of the wing matrix elements derived above when taking the long-wavelength limit, we can infer that

$$\varepsilon_{\mathbf{00}}^{-1}(\mathbf{q} \rightarrow \mathbf{0}) \rightarrow 1, \quad (\text{C.16})$$

$$\varepsilon_{\mathbf{GG}'}^{-1}(\mathbf{q} \rightarrow \mathbf{0}) \rightarrow B_{\mathbf{GG}'}^{-1}(\mathbf{q} \rightarrow \mathbf{0}), \quad (\text{C.17})$$

as one could expect for the head element of a strictly 2D dielectric matrix, while the body of the inverse dielectric matrix is simply the inverse of the body of the dielectric matrix. However, the analysis of the wing elements needs more care, since we need to know how they behave for small $\mathbf{q}$. Using the very same properties of the wing elements $\varepsilon_{\mathbf{G0}}$ and $\varepsilon_{\mathbf{0G}}$, we can write

$$\varepsilon_{\mathbf{G0}}^{-1}(\mathbf{q} \rightarrow \mathbf{0}) \rightarrow \sum_{\mathbf{G}' \neq \mathbf{0}} B_{\mathbf{GG}'}^{-1} \frac{e}{2\varepsilon_0 \sqrt{\mathbf{G}'}} \sqrt{q} \hat{\mathbf{q}} \cdot \mathbf{A}(\mathbf{G}') \rightarrow 0, \quad (\text{C.18})$$

$$\varepsilon_{\mathbf{0G}}^{-1}(\mathbf{q} \rightarrow \mathbf{0}) \rightarrow \sum_{\mathbf{G}' \neq \mathbf{0}} B_{\mathbf{G'G}}^{-1} \frac{e}{2\varepsilon_0 \sqrt{\mathbf{G}'}} \sqrt{q} \hat{\mathbf{q}} \cdot \mathbf{A}(\mathbf{G}')^* \rightarrow 0, \quad (\text{C.19})$$

so the wing terms of the inverse dielectric matrix tend to zero when $\mathbf{q}$ approaches zero, just as expected for a 2D system [229]. We are in position to address the troublesome matrix elements of the screened potential at zero momentum.

We know now that $\varepsilon_{\mathbf{00}}^{-1}(\mathbf{q} \rightarrow \mathbf{0}) \rightarrow 1$, but the bare Coulomb potential diverges with $1/q$, hence, the head term of the screened potential $W_{\mathbf{00}}(\mathbf{q})$ diverges with $1/q$ in the long-wavelength limit. Nonetheless, the $\mathbf{q} = \mathbf{0}$ term in reciprocal space calculations is known to be of upmost relevance to make calculations converge faster [225]. Fortunately,

we know that $\varepsilon_{\mathbf{00}}^{-1}(\mathbf{q}) \rightarrow 1$ has a linear behavior for small $\mathbf{q}$. Therefore, we can use an analytical expression to regularize the term $W_{\mathbf{00}}(\mathbf{0})$, as suggested by F. Hüser, T. Olsen and K.S. Thygesen in Ref. [228]. Their suggestion is approximating the divergent term by

$$W_{\mathbf{00}}(\mathbf{0}) = \frac{1}{\mathcal{A}_\Gamma} \int_{\mathcal{A}_\Gamma} d\mathbf{q} v_c(\mathbf{q}) [1 + \mathbf{q} \cdot \nabla_{\mathbf{q}} \varepsilon_{\mathbf{00}}^{-1}(\mathbf{q})|_{\mathbf{q}=\mathbf{0}}], \quad (\text{C.20})$$

where $\mathcal{A}_\Gamma$ is a small circular region around the origin of the BZ, the point Γ . This integral has an analytical expression. For an anisotropic material, we can simply take $\varepsilon_{\mathbf{00}}^{-1}(\mathbf{q})|_{\mathbf{q}=\mathbf{0}} = (-r_{0,x}, -r_{0,y})$, where the parameters $r_{0,x}$ and $r_{0,y}$ are the screening parameters computed numerically along the x and y directions, respectively. The expression evaluates to

$$W_{\mathbf{00}}(\mathbf{0}) = \frac{e}{\varepsilon_0 q_0} \left[1 - \frac{1}{4} q_0 (r_{0,x} + r_{0,y}) \right] = \frac{e}{\varepsilon_0 q_0} \left(1 - \frac{q_0 r_0}{2} \right), \quad (\text{C.21})$$

where the second equality holds for isotropic materials, and q_0 is the radius of $\mathcal{A}_\Gamma$. Typically, for q_0 we choose a percentage of the magnitude of the $\mathbf{k}$ point in the BZ mesh closest to the origin, which depends on the number of $\mathbf{k}$ points used, and of course the geometry of the BZ. The exact value for the percentage may also differ from material to material, but we have it as an input parameter for Xatu.

Regarding the wing terms of the screened potential, using the symmetrized dielectric function we get

$$\begin{aligned} W_{\mathbf{G0}}(\mathbf{q} \rightarrow 0) &= \sqrt{v_c(\mathbf{G})} \varepsilon_{\mathbf{G0}}^{-1}(\mathbf{q} \rightarrow 0) \sqrt{v_c(\mathbf{q} \rightarrow 0)} = \\ &= -\frac{e}{2\varepsilon_0 \sqrt{|\mathbf{G}|}} \hat{\mathbf{q}} \cdot \sum_{\mathbf{G}' \neq 0} \frac{1}{\sqrt{\mathbf{G}'}} \mathbf{A}(\mathbf{G}') B_{\mathbf{G}'\mathbf{G}}^{-1}, \quad \text{and} \end{aligned} \quad (\text{C.22})$$

$$\begin{aligned} W_{\mathbf{0G}}(\mathbf{q} \rightarrow 0) &= \sqrt{v_c(\mathbf{q} \rightarrow 0)} \varepsilon_{\mathbf{0G}}^{-1}(\mathbf{q} \rightarrow 0) \sqrt{v_c(\mathbf{G})} = \\ &= -\frac{e}{2\varepsilon_0 \sqrt{|\mathbf{G}|}} \hat{\mathbf{q}} \cdot \sum_{\mathbf{G}' \neq 0} \frac{1}{\sqrt{\mathbf{G}'}} \mathbf{A}(\mathbf{G}')^* B_{\mathbf{G}\mathbf{G}'}^{-1}, \end{aligned} \quad (\text{C.23})$$

with both results depending on the direction of the vector $\mathbf{q}$ as it approaches zero. Even though they do not tend to zero, they can be averaged around the origin over a center-symmetric region, such that they can be deemed null. This is due to the dependence on the direction of the vector $\mathbf{q}$ is linear in its versor $\hat{\mathbf{q}} = \mathbf{q}/|\mathbf{q}|$. This contrasts with the case in Rasmussen et al.'s work [225], where the angular dependence of the wing terms in the screened potential is not trivial, in such a way that the average has to be evaluated numerically. Therein, for the head term the dependence on $\hat{\mathbf{q}}$ is also not trivial, but the simplicity of the expression allows for an approximate analytical evaluation. Our expression for the regularization of $W_{\mathbf{00}}(\mathbf{0})$ is analytical, as shown in Eq. (C.21).

To summarize the results of this appendix, we provide Table C.1. To finish the appendix, we emphasize that the simplicity of the numerical treatment of the divergences stems essentially from the slower divergence of the Coulomb potential in two dimensions, which goes with $1/q$, unlike the 3D case commonly found in the literature where $v_c(q \rightarrow 0) \sim 1/q^2$, making our numerical methods easier to implement and interpret.

Table C.1: Summary of how the sensitive elements of the dielectric matrix, inverse dielectric matrix, and screened potential behave in the long-wavelength limit $q \rightarrow 0$. The head term of the screened potential craves a regularization scheme, while its wing terms will not contribute when averaged over a center-symmetric region.

	Head $(\mathbf{0}, \mathbf{0})$	Wing $(\mathbf{G}, \mathbf{0})$ or $(\mathbf{0}, \mathbf{G})$
ε	1	$\sqrt{q}$
ε^{-1}	1	$\sqrt{q}$
W	reg.	0

2D DIELECTRIC FUNCTION AND OPTICAL CONDUCTIVITY EQUIVALENCE

In this appendix, we relate the dielectric function with the optical conductivity in 2D systems. The derivation presented here is adapted from unpublished notes by N. M. R. Peres.

Let us consider the Poisson equation

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = \frac{e}{\varepsilon_0} \delta(\mathbf{r}) + \frac{e}{\varepsilon_0} n_{2D}(\mathbf{r}, t) \delta(z), \quad (\text{D.1})$$

and the continuity equation

$$-e \frac{\partial n_{2D}}{\partial t} + \nabla \cdot \mathbf{J}_{2D} = 0, \quad (\text{D.2})$$

that in the frequency–momentum domain is equivalent to

$$\omega e n_{2D}(\mathbf{k}, \omega) = -\mathbf{k} \cdot \mathbf{J}_{2D}(\mathbf{k}, \omega). \quad (\text{D.3})$$

If we invoke Ohm's law $\mathbf{J}_{2D}(\mathbf{k}, \omega) = \overset{\leftrightarrow}{\sigma}(\mathbf{k}, \omega) \mathbf{E}(\mathbf{k}, \omega)$ taking into account the tensor nature of the conductivity, and write the electric field in terms of a potential ϕ as $\mathbf{E}(\mathbf{k}, \omega) = -i\mathbf{k} \phi(\mathbf{k}, z=0, \omega)$, such that we obtain

$$e n_{2D}(\mathbf{k}, \omega) = \frac{i}{\omega} \mathbf{k} \cdot \overset{\leftrightarrow}{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k} \phi(\mathbf{k}, z=0, \omega). \quad (\text{D.4})$$

Then, we perform a Fourier transform on Eq. (D.1), while separating the in-plane wavevector from the off-plane one, that we shall denote by $\mathbf{k}$ and κ_z , respectively. Doing so, we arrive at

$$\begin{aligned} -k^2 \phi(\mathbf{k}, \kappa_z, \omega) - \kappa_z^2 \phi(\mathbf{k}, \kappa_z, \omega) &= \frac{e}{\varepsilon_0} + \frac{i}{\varepsilon_0 \omega} \mathbf{k} \cdot \overset{\leftrightarrow}{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k} \phi(\mathbf{k}, z=0, \omega) \Leftrightarrow \\ \Leftrightarrow \phi(\mathbf{k}, \kappa_z, \omega) &= -\frac{1}{k^2 + \kappa_z^2} \left[\frac{e}{\varepsilon_0} + \frac{i}{\varepsilon_0 \omega} \mathbf{k} \cdot \overset{\leftrightarrow}{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k} \phi(\mathbf{k}, z=0, \omega) \right]. \end{aligned} \quad (\text{D.5})$$

In the previous expression, if we do now the inverse Fourier transform only in the off-plane coordinate, we get

APPENDIX D. 2D DIELECTRIC FUNCTION AND OPTICAL CONDUCTIVITY EQUIVALENCE

$$\phi(\mathbf{k}, z, \omega) = \frac{e}{2\varepsilon_0 k} e^{-kz} - i \frac{\mathbf{k} \cdot \overleftrightarrow{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k}}{\varepsilon_0 \omega} \phi(\mathbf{k}, z=0, \omega) \frac{e^{-kz}}{2k}, \quad (\text{D.6})$$

which at $z=0$ implies

$$\begin{aligned} \phi(\mathbf{k}, z=0, \omega) \left(1 + i \frac{\mathbf{k} \cdot \overleftrightarrow{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k}}{2k\varepsilon_0 \omega} \right) &= \frac{e}{2\varepsilon_0 k} \Leftrightarrow \\ \Leftrightarrow \phi(\mathbf{k}, z=0, \omega) &= \frac{v_c(k)}{\varepsilon(\mathbf{k}, \omega)}, \end{aligned} \quad (\text{D.7})$$

so that we can identify the dielectric function as

$$\varepsilon(\mathbf{k}, \omega) = 1 + i \frac{\mathbf{k} \cdot \overleftrightarrow{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k}}{2k\varepsilon_0 \omega} = 1 + i \frac{k\sigma(\mathbf{k}, \omega)}{2\varepsilon_0 \omega}, \quad (\text{D.8})$$

where we considered $\mathbf{k} \cdot \overleftrightarrow{\sigma}(\mathbf{k}, \omega) \cdot \mathbf{k} = k^2 \sigma(\mathbf{k}, \omega)$ to write the second equality. This expression contrasts with the expression $\varepsilon(\mathbf{k}, \omega) = 1 + i \frac{\sigma(\mathbf{k}, \omega)}{\varepsilon_0 \omega}$ that we encountered in the first chapter. Therein, we have implicitly assumed that the material was 3D, explaining the discrepancy between the two expressions.

QUASI-2D APPROACH FOR THE SCREENING AND EXCITONS: THE MULTILAYER CASE

In this appendix, we extend the formalism of the dielectric function in the mixed $(\mathbf{q}, z)$ -representation from Chapter 6 in the main text to the multilayer case. As we need the expressions for the quasi-2D polarizability and the quasi-2D dielectric function, we recall their expressions that read

$$\begin{aligned} \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') = \frac{1}{\mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \frac{1}{(\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}})} \Bigg\{ & \quad (E.1) \\ \sum_{i\alpha} \left[C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \left(C_{i\alpha}^{v\mathbf{k}} \right)^* e^{-i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{t}_i} \delta(z - t_{i,z}) \right] \sum_{j\beta} \left[C_{j\beta}^{v\mathbf{k}} \left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* e^{i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{t}_j} \delta(z' - t_{j,z}) \right] + \\ \sum_{i\alpha} \left[\left(C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \right)^* C_{i\alpha}^{v\mathbf{k}} e^{-i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{t}_i} \delta(z - t_{i,z}) \right] \sum_{j\beta} \left[\left(C_{j\beta}^{v\mathbf{k}} \right)^* C_{j\beta}^{c\mathbf{k}+\mathbf{q}} e^{i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{t}_j} \delta(z' - t_{j,z}) \right] \Bigg\}, \end{aligned}$$

and

$$\begin{aligned} \varepsilon_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z') = \delta_{\mathbf{G}\mathbf{G}'} \delta(z - z') - \frac{e}{2\varepsilon_0 |\mathbf{q} + \mathbf{G}| \mathcal{A}} \sum_{vc, \mathbf{k}\sigma} \frac{1}{(\epsilon_{v\mathbf{k}} - \epsilon_{c\mathbf{k}+\mathbf{q}})} \Bigg\{ & \quad (E.2) \\ \sum_{i\alpha} \left[C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \left(C_{i\alpha}^{v\mathbf{k}} \right)^* e^{-i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{t}_i} e^{-|\mathbf{q}+\mathbf{G}| |z-t_{i,z}|} \right] \sum_{j\beta} \left[C_{j\beta}^{v\mathbf{k}} \left(C_{j\beta}^{c\mathbf{k}+\mathbf{q}} \right)^* e^{i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{t}_j} \delta(z' - t_{j,z}) \right] + \\ \sum_{i\alpha} \left[\left(C_{i\alpha}^{c\mathbf{k}+\mathbf{q}} \right)^* C_{i\alpha}^{v\mathbf{k}} e^{-i(\mathbf{q}+\mathbf{G}) \cdot \mathbf{t}_i} e^{-|\mathbf{q}+\mathbf{G}| |z-t_{i,z}|} \right] \sum_{j\beta} \left[\left(C_{j\beta}^{v\mathbf{k}} \right)^* C_{j\beta}^{c\mathbf{k}+\mathbf{q}} e^{i(\mathbf{q}+\mathbf{G}') \cdot \mathbf{t}_j} \delta(z' - t_{j,z}) \right] \Bigg\}, \end{aligned}$$

respectively. Now, we work out an expression for the inverse of the dielectric function, necessary for the screened Coulomb potential to compute the interaction matrix elements in the Bethe–Salpeter Hamiltonian. According to Eq. (3.62) in the main text,

$$\varepsilon^{-1}(\mathbf{r}, \mathbf{r}'; \omega) = \delta(\mathbf{r} - \mathbf{r}') + \int v_c(\mathbf{r} - \mathbf{r}'') \chi(\mathbf{r}'', \mathbf{r}; \omega) d\mathbf{r}'', \quad (E.3)$$

APPENDIX E. QUASI-2D APPROACH FOR THE SCREENING AND EXCITONS: THE MULTILAYER CASE

where χ is the interacting reducible polarizability, that relates to the irreducible one through the Dyson equation in Eq. (3.64) from the main text, which reads

$$\chi = \chi^{\text{irr}} + \chi^{\text{irr}} \star v_c \star \chi, \quad (\text{E.4})$$

if we use math symbols instead of diagrams, and where the symbol $\star$ is an abbreviation for the convolution operation. Computing the full polarizability at all interacting levels is an impossible endeavor; approximations are necessary. First, for the irreducible polarizability, we can work in the random-phase approximation (RPA) (see Sections 3.3.1 and 3.4.1). In this way, we have

$$\chi^{\text{RPA}} = \chi^0 + \chi^0 \star v_c \star \chi^{\text{RPA}}, \quad (\text{E.5})$$

where the interacting reducible polarizability is approximated to a noninteracting one at the single-particle level (see Eq. (3.69)). Making now explicit the $(\mathbf{q}, z)$ -dependence in the previous expression, we write

$$\begin{aligned} \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z, z') = & \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') + \\ & \sum_{\mathbf{G}_1} \int \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \chi_{\mathbf{G}_1\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z_2, z') dz_1 dz_2. \end{aligned} \quad (\text{E.6})$$

Let us assume that we can determine $\chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z, z')$. Then, for the inverse dielectric function within RPA, we would have

$$\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}, z, z') = \delta_{\mathbf{G}\mathbf{G}'} \delta(z - z') + \int v_c(\mathbf{q} + \mathbf{G}, z - z'') \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z'', z') dz''. \quad (\text{E.7})$$

In principle, assuming we have the inverse of the dielectric function, we can compute the interaction matrix elements. We skip the derivation, as the fine details of the calculation are not relevant. In the definition for the direct interaction term in Eq. (4.54), we proceed as usual, and replace the Bloch state wavefunctions $\psi_{n,\mathbf{k}}(\mathbf{r})$ by their expression in Eq. (3.11), apply the point-like orbital approximation while keeping track of the z -dependence (see Eq. (E.1)), and simplify the integrals and the sums to obtain

$$\begin{aligned} D_{vc,v'c'}(\mathbf{k}, \mathbf{k}', \mathbf{Q}) = & \int d\mathbf{r} \int d\mathbf{r}' \psi_{c,\mathbf{k}+\mathbf{Q}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') W(\mathbf{r}, \mathbf{r}') \psi_{c',\mathbf{k}'+\mathbf{Q}}(\mathbf{r}) \psi_{v,\mathbf{k}}(\mathbf{r}') = \\ = & \frac{1}{\mathcal{A}} \sum_{\mathbf{G}, \mathbf{G}'} \sum_{i\alpha} \left(C_{i\alpha}^{c\mathbf{k}+\mathbf{Q}} \right)^* C_{i\alpha}^{c'\mathbf{k}'+\mathbf{Q}} e^{i(\mathbf{k}-\mathbf{k}'+\mathbf{G}) \cdot \mathbf{t}_i} \sum_{j\beta} \left(C_{j\beta}^{v'\mathbf{k}'} \right)^* C_{j\beta}^{v\mathbf{k}} e^{-i(\mathbf{k}-\mathbf{k}'+\mathbf{G}') \cdot \mathbf{t}_j} W_{\mathbf{G}\mathbf{G}'}(\mathbf{k} - \mathbf{k}', t_i^z, t_j^z). \end{aligned} \quad (\text{E.8})$$

To evaluate this expression numerically, we need to compute $W_{\mathbf{G}\mathbf{G}'}(\mathbf{k}, t_i^z, t_j^z)$ for every $\mathbf{k} \in \text{BZ}$. For that, we already have the inverse of the dielectric function from above. For any pair of points (z, z') , we can use the expression

$$W_{\mathbf{G}\mathbf{G}'}(\mathbf{k}, z, z') = \int \varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}, z, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - z') dz'', \quad (\text{E.9})$$

where $\varepsilon_{\mathbf{G}\mathbf{G}'}^{-1}(\mathbf{q}, z, z')$ is the inverse dielectric matrix in the mixed $(\mathbf{q}, z)$ -representation within RPA that we previously determined. Replacing it by the expression (E.7) above, one arrives at

$$W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, z, z') = \delta_{\mathbf{G}\mathbf{G}'} v_c(\mathbf{q} + \mathbf{G}', z - z') + \int v_c(\mathbf{q} + \mathbf{G}, z - z_1) \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z_1, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - z') dz_1 dz''. \quad (\text{E.10})$$

Since in the point-like orbital approximation we only need to compute the screened potential between z points located at the heights $t_{i,z}$, we can adapt the previous expression to use it in the interaction matrix elements as

$$W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, t_i^z, t_j^z) = \delta_{\mathbf{G}\mathbf{G}'} v_c(\mathbf{q} + \mathbf{G}, t_i^z - t_j^z) + \int v_c(\mathbf{q} + \mathbf{G}, t_i^z - z_1) \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z_1, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z) dz_1 dz''. \quad (\text{E.11})$$

However, in practice, one cannot solve the Dyson equation for $\chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z_1, z'')$ directly, as we model the z -dependence in the polarizability $\chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z')$ with Dirac delta functions. One possible solution is working with a basis of potential functions $\{\phi_\alpha(z)\}$, which assume the form $\phi_\alpha(z) \sim (z - z_0)^\alpha$, and compute the respective matrix elements of the polarizability in that basis (which was done in Ref. [176]). As a workaround, one can develop an iterative process, in which the interaction matrix elements are computed with a self-consistently determined RPA polarizability. To do this, let us first recall the Dyson equation for the polarizability within RPA

$$\chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z, z') = \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') + \sum_{\mathbf{G}_1} \int dz_1 dz_2 \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \chi_{\mathbf{G}_1\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z_2, z'), \quad (\text{E.12})$$

for which an iterative process looks like

$$\text{Step 0) } \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z, z') = \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z');$$

$$\text{Step 1) } \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z, z') = \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') +$$

$$\sum_{\mathbf{G}_1} \int dz_1 dz_2 \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \chi_{\mathbf{G}_1\mathbf{G}'}^0(\mathbf{q}, z_2, z');$$

$$\text{Step 2) } \chi_{\mathbf{G}\mathbf{G}'}^{\text{RPA}}(\mathbf{q}, z, z') = \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z') +$$

$$\sum_{\mathbf{G}_1} \int dz_1 dz_2 \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \chi_{\mathbf{G}_1\mathbf{G}'}^0(\mathbf{q}, z_2, z') +$$

$$\sum_{\mathbf{G}_1, \mathbf{G}_2} \int dz_1 dz_2 dz_3 dz_4 \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \chi_{\mathbf{G}_1\mathbf{G}_2}^0(\mathbf{q}, z_2, z_3) \times$$

$$v_c(\mathbf{q} + \mathbf{G}_2, z_3 - z_4) \chi_{\mathbf{G}_2\mathbf{G}'}^0(\mathbf{q}, z_4, z'); \quad (\text{E.13})$$

$\vdots$

Along this line of thought, we can evaluate numerically the screened potential in the mixed representation within a self-consistent picture as

$$\text{Step 0) } W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, t_i^z, t_j^z) = \delta_{\mathbf{G}\mathbf{G}'} v_c(\mathbf{q} + \mathbf{G}, t_i^z - t_j^z) +$$

$$\int dz dz'' v_c(\mathbf{q} + \mathbf{G}, t_i^z - z) \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z);$$

$$\text{Step 1) } W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, t_i^z, t_j^z) = \delta_{\mathbf{G}\mathbf{G}'} v_c(\mathbf{q} + \mathbf{G}, t_i^z - t_j^z) +$$

$$\int dz dz'' v_c(\mathbf{q} + \mathbf{G}, t_i^z - z) \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z) +$$

$$\sum_{\mathbf{G}_1} \int dz dz'' dz_1 dz_2 v_c(\mathbf{q} + \mathbf{G}, t_i^z - z) \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \times$$

$$\chi_{\mathbf{G}_1\mathbf{G}'}^0(\mathbf{q}, z_2, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z);$$

$$\text{Step 2) } W_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, t_i^z, t_j^z) = \delta_{\mathbf{G}\mathbf{G}'} v_c(\mathbf{q} + \mathbf{G}, t_i^z - t_j^z) +$$

$$\int dz dz'' v_c(\mathbf{q} + \mathbf{G}, t_i^z - z) \chi_{\mathbf{G}\mathbf{G}'}^0(\mathbf{q}, z, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z) +$$

$$\sum_{\mathbf{G}_1} \int dz dz'' dz_1 dz_2 v_c(\mathbf{q} + \mathbf{G}, t_i^z - z) \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \times$$

$$\chi_{\mathbf{G}_1\mathbf{G}'}^0(\mathbf{q}, z_2, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z) +$$

$$\sum_{\mathbf{G}_1, \mathbf{G}_2} \int dz dz'' dz_1 dz_2 dz_3 dz_4 v_c(\mathbf{q} + \mathbf{G}, t_i^z - z) \chi_{\mathbf{G}\mathbf{G}_1}^0(\mathbf{q}, z, z_1) v_c(\mathbf{q} + \mathbf{G}_1, z_1 - z_2) \times$$

$$\chi_{\mathbf{G}_1\mathbf{G}_2}^0(\mathbf{q}, z_2, z_3) v_c(\mathbf{q} + \mathbf{G}_2, z_3 - z_4) \chi_{\mathbf{G}_2\mathbf{G}'}^0(\mathbf{q}, z_4, z'') v_c(\mathbf{q} + \mathbf{G}', z'' - t_j^z);$$

$$(\text{E.14})$$

$\vdots$

Hopefully, the elements of the screened potential converge. This method consists of a

quasi-2D treatment that allows for computing with rigor excitons in multilayered systems. An interesting and direct application is the calculation of excitons in moiré materials.

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